cSlope

Verification Manual

Limit equilibrium slope stability analysis — comparison with published results

This document presents factors of safety computed by cSlope for 171 verification problems from the geotechnical literature and from the verification manuals of other slope stability programs, next to the values the sources publish and the range of other programs' results.

Program version	cSlope 0.1.0-beta.1 (commit 4164cdc)
Report generated	2026-10-28
Problems	171
Values published by the sources	199
cSlope analyses	334

Generated automatically by packages/desktop/scripts/verification-report.ts from the engine's verification test cases.

1 Introduction

Each problem below is defined once in the engine's verification data (packages/engine/src/verification/problems.ts). The same definitions are run by the automated test suite on every change, so the values in this report are the values the program computes.

Analysis settings

Setting	Value
Slip surfaces	Circular, or the non-circular surface the source gives. Fixed surfaces where the source gives one, otherwise a critical-circle search (with non-circular optimisation where stated).
Number of slices	50 unless the problem states otherwise
Search	4000 entry/exit combinations plus 2000 refined around the best surfaces
Convergence tolerance	0.001 on the factor of safety; 0.001 with non-circular optimisation
Methods	Bishop simplified, Janbu simplified (uncorrected unless stated), Morgenstern-Price with half-sine interslice function; Spencer = Morgenstern-Price with a constant interslice function
Pore pressure	Piezometric line; head = vertical distance below the line (default, the usual convention). Checks marked "phreatic correction" use vertical distance × $\cos^2\theta$.
Unit weight of water	9.81 kN/m³

Units

Problems published in imperial units without water are entered in their own units (ft, psf, pcf); the factor of safety is dimensionless. Problems with water are converted to SI because the program uses γw = 9.81 kN/m³ (62.4 pcf = 9.80 kN/m³, < 0.1 % difference).

Source of the input data

The input data were transcribed from the publications and from the verification manuals of other slope stability programs, most of them commercial, and cross-checked against an independent open-source transcription of the same problems. For each problem the report lists the values the source publishes, with the method where the source names one; some are hand calculations, charts, trial wedges or finite elements rather than the method of slices. For each analysis cSlope runs, it gives the range of results that other programs publish in their verification manuals, without naming the programs, and cSlope's result. The report presents the results; it does not mark them as passed or failed. Problem ids are cSlope's own: a number is one slope, a letter a variant of it.

2 Summary of results

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V001	Arai & Tagyo (1985) example 1	1.451; 1.265; 1.357	Bishop simplified	1.409–1.417	1.404
			Morgenstern-Price (half-sine)	1.414	1.401
			Spencer, non-circular	1.386	1.384
			Janbu simplified, non-circular	–	1.252
			Janbu corrected, non-circular	–	1.345
V002	Arai & Tagyo (1985) example 2, three layers	0.417; 0.405; 0.43; 0.43; 0.44; 0.40–0.45; 0.39	Bishop simplified	0.420	0.419
			Spencer, non-circular	0.414	0.411
			Janbu simplified, non-circular	–	0.393
			Janbu corrected, non-circular	–	0.413
V003	Arai & Tagyo (1985) example 3, water table	1.138; 0.995; 1.071	Bishop simplified	1.112–1.118	1.113
			Spencer	1.118	1.113
			Janbu corrected	1.131	1.124
			Janbu simplified, non-circular	–	0.966
			Janbu corrected, non-circular	–	1.047
V004a	4:1 slope, Mohr-Coulomb fit	2.66	Spencer	2.794	2.802
			Janbu simplified	2.662	2.677
V004b	Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters	2.64; 2.66	Janbu simplified	2.610	2.609
			Spencer	2.696	2.666
V005	Baker (1993) dam, end of construction, infinite-slope face	2.50; 2.41	Bishop simplified	–	2.524
			Spencer	–	2.524
			Spencer, non-circular	–	2.502
V006	Baker & Leshchinsky (2001) planar surface	1.35	Spencer	1.351	1.352
			Janbu simplified	1.351	1.352
V007a	Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle	1.50	Janbu simplified	1.469	1.468
V007b	Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters	0.97; 1.50	Janbu simplified	0.957	0.955
			Spencer	0.981	0.978
V007c	Baker (2003) ex. 3, Janbu simplified	1.35	Janbu simplified	1.291	1.290
V008a	Baker (2003) example 1, Mohr-Coulomb fit	1.50	Spencer	1.536	1.519
V008b	Baker (2003) London clay slope, Mohr-Coulomb fit	1.35	Spencer	1.366	1.365
V009a	Baker & Leshchinsky clay-core dam, reservoir and crest crack	–	Spencer	1.925	1.926
V009b	Baker & Leshchinsky dam, Baker's non-circular surface	1.91	Spencer	–	1.882
V009c	Baker & Leshchinsky clay-core dam, critical surface search	1.91	Spencer, non-circular	1.868	1.867
V010	Borges & Cardoso embankment 2 (geosynthetic)	–	Morgenstern-Price (half-sine)	1.171	1.162
			Bishop simplified	1.170	1.164

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V010b	Borges & Cardoso embankment 2 (geosynthetic), the paper's circle B	1.15; 1.19; 1.22	Bishop simplified	–	1.159
V011a	Borges & Cardoso embankment 3, H = 7 m, circle A	1.25	Bishop simplified	1.230	1.219
V011b	Borges & Cardoso embankment 3, H = 7 m, circle B	1.19	Bishop simplified	1.220	1.216
V012	Borges & Cardoso embankment 3, H = 8.75 m, circle C	0.99	Bishop simplified	0.980	0.982
V013a	Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile rows)	1.13	Bishop simplified	1.140	1.143
V013b	Cai & Ugai slope, pile row D1/D = 2	1.54	Bishop simplified	1.540	1.541
V013c	Cai & Ugai slope, pile row D1/D = 3	1.37	Bishop simplified	1.430	1.437
V013d	Cai & Ugai slope, pile row D1/D = 4	1.31	Bishop simplified	1.330	1.331
V013e	Cai & Ugai slope, pile row D1/D = 6	1.25	Bishop simplified	1.250	1.253
V014	Nailed wall, 14 rows, deep wedge	–	Janbu corrected	1.448–1.460	1.448
			Spencer	1.576	1.575
V015	Soldier-pile tieback wall	–	Janbu corrected	1.479	1.476
			Janbu simplified	1.446	1.442
V016a	Stepped gabion wall, equivalent cohesion	–	Bishop simplified	1.787–1.790	1.791
			Spencer	1.791	1.797
V016b	Stepped gabion wall, mesh as geosynthetic supports	–	Bishop simplified	1.830–1.835	1.830
			Spencer	1.839	1.835
V016c	Gabion wall with weak joint layers	–	Bishop simplified	1.790	1.791
V017a	Ontario gabion wall (Cao et al. 2016), equivalent cohesion	–	Bishop simplified	1.373	1.382
			Spencer	1.386	1.398
V017b	Ontario gabion wall (Cao et al. 2016), mesh as geosynthetic supports	–	Bishop simplified	1.378	1.382
V018	Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism	1.05	Spencer	1.051	1.052
			Morgenstern-Price (specified f(x))	1.009–1.017	1.008
V019	Cheng et al. 11 m homogeneous slope	1.383	Spencer (40 slices, vertical pore-pressure head)	1.380–1.398	1.398
			Spencer (vertical pore-pressure head)	–	1.398
			Bishop simplified	1.401	1.401
V020	Cheng et al. benched slope, global minimum	1.33	Spencer	1.336–1.338	1.333
			Bishop simplified	1.342	1.337
V021a	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 2	1.1178	Morgenstern-Price (half-sine)	1.132	1.128
			Bishop simplified	–	1.128
			Bishop simplified (60 slices)	1.128	1.128
V021b	Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the base of clay 3	1.1088	Morgenstern-Price (half-sine)	1.116	1.110
			Bishop simplified	–	1.110
			Bishop simplified (60 slices)	1.109	1.110

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V021c	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 2	1.1096	Morgenstern-Price (half-sine)	1.110	1.108
			Bishop simplified	1.108	1.108
V021d	Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the base of clay 3	1.0639	Morgenstern-Price (half-sine)	1.063	1.058
			Bishop simplified	1.058	1.058
V021e	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 2	2.2343	Morgenstern-Price (half-sine)	2.248	2.245
			Bishop simplified	2.245	2.245
V021f	Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base of clay 3	2.1396	Morgenstern-Price (half-sine)	2.138	2.128
			Bishop simplified	2.128	2.128
V021g	Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the base of clay 2	1.4239	Morgenstern-Price (half-sine)	1.422	1.428
			Bishop simplified	–	1.428
V022a	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 15	1.1625	Morgenstern-Price (half-sine)	1.158	1.157
			Bishop simplified	–	1.159
			Bishop simplified (60 slices)	1.160	1.159
V022b	Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y = 5	1.1479	Morgenstern-Price (half-sine)	1.178	1.174
			Bishop simplified	1.185	1.177
V023	LASH terminal trench, Duncan's Taylor series example	1.17	Spencer	1.145	1.138
V024	Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)	1.57	Bishop simplified	1.563–1.572	1.572
			Spencer	1.557	1.562
V025	Duncan & Wright fig. 6.38, piezometric line	1.141	Bishop simplified	1.142	1.132
			Spencer	1.142	1.132
V026	Duncan & Wright fig. 7.12, embankment on clay	1.22; 1.19	Bishop simplified	1.228	1.220
			Spencer	–	1.195
			Spencer, non-circular	1.190	1.170
V027	Duncan & Wright fig. 14.3, undrained clay	1.124	Bishop simplified	1.126	1.122
			Spencer	–	1.132
V028	Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation	1.124	Bishop simplified	1.126	1.122
			Spencer	–	1.122
V029	Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation	1.119	Bishop simplified	1.125	1.117
			Spencer	–	1.117
V030	Duncan & Wright fig. 14.4	1.40	Bishop simplified	1.412	1.408
			Spencer	–	1.399
V031a	Duncan & Wright fig. 14.5, R = 87 ft	2.56	Bishop simplified	2.549	2.534
			Spencer	2.545	2.531
V031b	Duncan & Wright fig. 14.5, R = 102 ft	1.35	Bishop simplified	1.398	1.389
			Spencer	1.359	1.352
V032a	Duncan & Wright fig. 14.7, deep circle	1.21	Bishop simplified	1.230	1.224

Problem	Description	Published	cSlope analysis	Other programs	cSlope
			Spencer	1.209	1.205
V032b	Duncan & Wright fig. 14.7, infinite-slope mechanism	1.15	Bishop simplified	1.155	1.156
			Spencer	1.155	1.156
V033	Duncan & Wright fig. 14.20a, water table	1.529	Bishop simplified	1.533	1.521
			Spencer	–	1.533
			Spencer, non-circular	1.534	1.522
V034a	Duncan & Wright fig. 14.20b, Su = 200 + 15·depth	1.276	Bishop simplified	1.313	1.306
			Spencer	–	1.276
			Spencer, non-circular	1.262	1.260
V034b	Duncan & Wright fig. 14.20b, Su = 300 psf	1.328	Bishop simplified	1.335	1.329
			Spencer	–	1.328
			Spencer, non-circular	1.182	1.168
V035a	Duncan & Wright tieback, active	1.51	Morgenstern-Price (half-sine)	1.575	1.568
V035b	Duncan & Wright tieback, passive	1.32	Bishop simplified	1.324	1.319
V036a	Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)	0.75	Bishop simplified	0.761	0.757
			Spencer	0.756	0.752
			Spencer, non-circular	0.740	0.736
V036b	Embankment on an undrained foundation, cz = 5 psf/ft	0.90	Bishop simplified	0.909	0.906
			Spencer	0.898	0.898
V036c	Embankment on an undrained foundation, cz = 10 psf/ft	1.03	Bishop simplified	1.045	1.042
			Spencer	1.032	1.028
			Spencer, non-circular	1.018	1.017
V036d	Embankment on an undrained foundation, cz = 15 psf/ft	1.13	Bishop simplified	1.154	1.152
			Spencer	1.134	1.132
			Spencer, non-circular	1.116	1.119
V037	Geogrid-reinforced fill on rock (D&W fig. 7.28)	1.62	Bishop simplified	1.629	1.613
			Spencer	1.620	1.607
V038	Homogeneous dam with a piezometric line (D&W fig. 7.19)	1.16	Bishop simplified	1.070–1.090	1.069
			Spencer	1.100	1.077
V038b	Homogeneous dam with a piezometric line, Duncan & Wright's input	1.16; 1.19	Spencer	–	1.157
V039	James Bay dyke (Duncan & Wright fig. 7.16)	1.45; 1.16	Spencer	–	1.447
			Spencer, non-circular	1.167	1.162
V040	James Bay dyke, reference model (deterministic)	1.46	Bishop simplified	1.461	1.459
V041a	Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)	1.60	Bishop simplified	1.603	1.597
			Spencer	1.599	1.594
			Spencer, non-circular	1.592	1.591
V041b	Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)	1.60	Bishop simplified	1.603	1.597

Problem	Description	Published	cSlope analysis	Other programs	cSlope
			Spencer	1.599	1.594
			Spencer, non-circular	1.590	1.589
V042	Thick-core dam with a piezometric line (D&W fig. 7.24)	1.67	Bishop simplified	1.584	1.582
			Spencer	1.648	1.649
V043	Syncrude tailings dyke (El-Ramly et al.), composite circle	1.31	Bishop simplified	1.320	1.320
V044	Fredlund & Krahn (1977), dry	2.08; 2.076; 2.073	Bishop simplified	–	2.076
			Morgenstern-Price (half-sine)	–	2.072
			Spencer	–	2.072
V045	Fredlund & Krahn (1977), with water table	1.834; 1.832; 1.83	Bishop simplified	–	1.829
			Morgenstern-Price (half-sine)	–	1.827
			Spencer	–	1.827
V046a	Fredlund & Krahn slope on a weak seam, composite surface	1.37; 1.377; 1.373	Bishop simplified	1.382	1.380
			Spencer	1.382	1.379
			Morgenstern-Price (half-sine)	–	1.370
V046b	Fredlund & Krahn slope on a weak seam, with a piezometric line	1.248; 1.245	Bishop simplified	–	1.241
			Spencer	–	1.242
			Morgenstern-Price (half-sine)	–	1.235
V047a	ACADS 1(a), homogeneous slope, dry	1.00	Bishop simplified	0.987–0.992	0.985
			Spencer	0.986	0.984
			Janbu corrected	0.990	0.986
			Morgenstern-Price (half-sine)	0.991	0.984
V047b	ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry crack)	1.65–1.70	Bishop simplified	1.664	1.659
			Morgenstern-Price (half-sine)	1.660	1.656
V048	ACADS 1(b), water-filled tension crack	1.65	Bishop simplified	1.596	1.587
			Spencer	1.592	1.582
			Janbu corrected	1.489	1.475
V049	ACADS 1(c), three-layer slope	1.39	Bishop simplified	1.403–1.405	1.403
			Spencer	1.375	1.372
			Janbu corrected	1.357	1.355
			Morgenstern-Price (half-sine)	1.382	1.372
V050a	ACADS 3(b), weak layer, specified non-circular surface	1.34	Janbu corrected	1.294	1.294
			Spencer	1.277	1.276
			Morgenstern-Price (half-sine)	1.261	1.260
V050b	ACADS 3(a), weak layer, non-circular search	1.24–1.27	Spencer, non-circular	1.246	1.225
V051	ACADS 4, weak layer, water table and surcharges	0.78	Spencer, non-circular	0.707	0.714
V052a	Talbingo Dam, end of construction	1.95	Bishop simplified	1.948–1.951	1.964
			Spencer	1.948	1.964
			Janbu corrected	1.949	1.974
V052b	Talbingo Dam, specified circle	2.29	Bishop simplified	2.208	2.206

Problem	Description	Published	cSlope analysis	Other programs	cSlope
			Spencer	2.292	2.290
			Janbu corrected	2.073	2.073
			Morgenstern-Price (half-sine)	2.299	2.299
V053	Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry	1.009–1.025; 0.97; 1.08	Spencer, non-circular	1.010	1.006
			Spencer	–	1.091
V054	Greco (1996) example 5 / Chen & Shao (1988)	1.08	Bishop simplified	1.087	1.086
			Spencer	1.093	1.091
V055	Griffiths & Lane example 2, slope with a foundation layer	1.40	Spencer	1.366	1.366
V056	Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2	0.485	Janbu corrected	–	0.499
V057a	Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)	1.46; 1.47; 1.45	Spencer	1.468	1.471
V057b	Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)	2.04; 2.10; 2.03	Spencer	2.022	2.023
V058	Griffiths & Lane example 6, earth dam with full reservoir	1.90	Spencer	1.915	1.915
V059a	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4	–	Spencer, non-circular	1.215	1.214
V059b	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5	–	Spencer, non-circular	1.290	1.289
V059c	Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6	–	Spencer, non-circular	1.315	1.312
V060a	Cannon Dam end of construction (Hassan & Wolff), critical circle	–	Bishop simplified	2.529	2.530
V060b	Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle	–	Bishop simplified	2.973	2.974
V061	Clarence Cannon Dam (Wolff & Harr), specified surface	2.36	Morgenstern-Price (half-sine)	2.343	2.348
			Spencer	2.383	2.388
V062a	Clarence Cannon Dam, surface 7A	2.753	Bishop simplified	2.551–2.560	2.549
V062b	Clarence Cannon Dam, surface 7B	2.352	Bishop simplified	2.806–2.820	2.815
V062c	Clarence Cannon Dam, surface 7C	2.523	Bishop simplified	2.771–2.777	2.779
V062d	Clarence Cannon Dam, surface 7D	2.457	Bishop simplified	2.583–2.589	2.577
V062e	Clarence Cannon Dam, surface 7E	2.602	Bishop simplified	2.692–2.703	2.690
V062f	Clarence Cannon Dam, surface 8B	2.995	Bishop simplified	2.672–2.673	2.673
V062g	Clarence Cannon Dam, surface 8F	3.916	Bishop simplified	3.586–3.598	3.592
V062h	Clarence Cannon Dam, surface 8G	10.576	Bishop simplified	6.069–6.074	6.063
V062i	Clarence Cannon Dam, surface 8H	6.293	Bishop simplified	11.583	11.567
V063	Huang & Jia earth dam, dry	2.43; 2.50	Spencer	2.455	2.451
			Bishop simplified	–	2.453
V064	Leshchinsky & Han MSE wall, five tiers	1.00	Bishop simplified	1.020	1.017
V065	Leshchinsky & Han MSE wall, weak foundation	1.00; 0.86; 0.98; 0.92	Spencer	0.964	0.959
			Bishop simplified	–	0.977
V066a	Leshchinsky & Han three-tier MSE wall, baseline	1.00; 0.99	Bishop simplified	1.040	1.031
V066b	Leshchinsky & Han MSE wall, L = 4.2 m	1.00; 0.98	Spencer	1.011	1.011

Problem	Description	Published	cSlope analysis	Other programs	cSlope
			Bishop simplified	–	1.013
V066c	Leshchinsky & Han MSE wall, two reinforcement types	1.00; 1.01	Bishop simplified	–	1.012
V066d	Leshchinsky & Han MSE wall, water hw = 3 m	1.00; 1.01	Bishop simplified	1.010	1.010
V066e	Leshchinsky & Han MSE wall, surcharge	1.00; 1.02	Bishop simplified	0.958	0.961
V067	Low (1989), layered undrained clay	1.44	Bishop simplified	1.439	1.436
V068	Low (1989), stiff crust over soft clays, published circle	1.14; 1.36; 1.31; 1.38; 1.17; 1.13–1.27	Bishop simplified	–	1.134
			Spencer, non-circular	–	–
V069a	Morgenstern (1963) complete drawdown, 3:1	1.20	Bishop simplified	–	1.200
			Spencer	–	1.205
V069b	Morgenstern complete drawdown, the reference circle	1.20	Bishop simplified	1.212	1.209
V070	Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)	1.41; 1.52; 1.48	Bishop simplified	1.417	1.415
			Spencer	1.422	1.421
V071a	Pockoski & Duncan (2000) slope 1	–	Bishop simplified	1.293	1.290
			Spencer	1.300	1.296
V071b	Pockoski & Duncan slope 1, circular search	–	Bishop simplified	1.293	1.288
			Spencer	1.300	1.294
			Janbu simplified	1.151	1.147
V072a	Pockoski & Duncan test slope 2, dry tension crack	–	Bishop simplified	1.285	1.282
			Spencer	1.290	1.287
V072b	Pockoski & Duncan slope 2, circular search	–	Bishop simplified	1.285	1.282
			Spencer	1.290	1.287
			Janbu simplified	1.141	1.135
V073	Pockoski & Duncan test slope 3, clay seam and dry crack	–	Bishop simplified	1.417	1.410
			Spencer	1.422	1.415
V074	Soil-nailed wall with tension crack and surcharges	–	Spencer	1.009	1.010
			Janbu corrected	1.073	1.075
V075	Tieback wall in sand, drawdown water table	–	Spencer	0.570	0.578
V076	Tied-back wall in layered soil	–	Bishop simplified	1.147	1.142
			Spencer	1.145	1.140
V077	Prandtl bearing capacity on level ground	1.00	Spencer	0.940	0.943
V078	Prandtl bearing capacity, fully specified Prandtl surface (reference values)	1.00	Morgenstern-Price (half-sine)	0.960	0.958
V079	Priest (1993) rigid block on a plane, water-filled crack	1.049	Janbu simplified	1.049	1.048
			Spencer	1.049	1.048
V080	Priest (1993) rigid block, reference geometry (60° face)	1.049; 1.049	Morgenstern-Price (half-sine)	1.049	1.049
			Spencer	–	1.049
V081a	Sharma (1996) reinforcement example, unreinforced	–	Bishop simplified	0.764	0.764

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V081b	Sharma (1996) slope with a reinforced zone (φ = 54.93°)	–	Bishop simplified	1.504	1.501
V082a	Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap	–	Bishop simplified	1.532	1.534
			Spencer	1.532	1.533
			Morgenstern-Price (half-sine)	1.532	1.533
			Janbu corrected	1.544	1.546
V082b	Sharma (1996) slope, 11 ft tension crack with 6 ft of water	–	Bishop simplified	1.511	1.513
			Spencer	1.510	1.512
			Morgenstern-Price (half-sine)	1.510	1.512
			Janbu corrected	1.520	1.522
V082c	Sharma (1996) slope over undulating bedrock, zero-strength cap	–	Bishop simplified	1.396	1.396
			Spencer	1.402	1.402
			Janbu corrected	1.391	1.392
			Morgenstern-Price (half-sine)	1.398	1.398
V082d	Sharma (1996) slope over undulating bedrock, circular search with exit limits	–	Bishop simplified	1.376	1.369
			Spencer	1.382	1.376
			Morgenstern-Price (half-sine)	1.378	1.371
			Janbu corrected	1.345	1.343
V083	Soil-nailed wall in clay (Amherst test wall)	0.887	Janbu corrected	0.890	0.885
V084a	Soil-nailed wall in sand (Clouterre test wall)	0.989	Janbu corrected	0.989	0.988
V084b	Soil-nailed wall in sand (Clouterre test wall), 45° plane	1.176	Janbu corrected	1.123	1.124
V084c	Soil-nailed wall in sand (Clouterre test wall), 50° plane	1.07	Janbu corrected	1.043	1.043
V084d	Soil-nailed wall in sand (Clouterre test wall), 60° plane	0.929	Janbu corrected	0.945	0.944
V084e	Soil-nailed wall in sand (Clouterre test wall), 65° plane	0.893	Janbu corrected	0.922	0.920
V084f	Soil-nailed wall in sand (Clouterre test wall), 70° plane	0.887	Janbu corrected	0.923	0.920
V085	Bradwell excavated slope, stiff-fissured London clay	1.76; 1.63	Bishop simplified	1.762	1.759
			Spencer	1.758	1.759
			Janbu corrected	1.736	1.735
			Janbu simplified	1.628	1.627
V086a	Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m	1.35	Spencer	1.332–1.350	1.333
V086b	Tandjiria embankment, clay fill, water-filled crack	0.981	Spencer (2 m tension crack)	0.975	0.968
			Spencer (tension crack of automatic depth)	0.975	0.966
			Morgenstern-Price (half-sine)	0.975	0.968
V086c	Tandjiria embankment, sand fill, dry crack	1.219	Spencer	1.209	1.206
			Morgenstern-Price (half-sine)	1.218	1.215
V086d	Tandjiria embankment, clay fill, the paper's geometry	0.981	Spencer	–	0.973

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V086e	Tandjiria embankment, sand fill, the paper's geometry	1.219	Spencer	–	1.205
V087	USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)	2.30	Janbu corrected	2.290	2.248
V088	USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)	2.44	Spencer	2.445	2.442
			Bishop simplified	2.447	2.442
			Janbu corrected	2.430	2.444
			Morgenstern-Price (half-sine)	2.447	2.442
V089	USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)	2.71	Bishop simplified	2.716	2.698
			Spencer	2.736	2.721
			Janbu corrected	2.650	2.648
			Morgenstern-Price (half-sine)	2.744	2.727
V090	USACE EM 1110-2-1902, example F-5	1.33	Bishop simplified	1.332	1.320
			Spencer	1.328	1.317
			Janbu corrected	1.345	1.341
			Morgenstern-Price (half-sine)	1.327	1.316
V091	USACE example E-2 (fig. E-10), ponded water	1.24	Bishop simplified	1.241	1.235
			Spencer	1.241	1.235
			Morgenstern-Price (half-sine)	1.244	1.235
			Janbu corrected	1.385	1.380
V092	USACE example F-6, steady-seepage dam	2.01; 2.07	Bishop simplified	2.011	1.990
			Spencer	2.026	2.005
			Janbu corrected	1.830	1.824
			Morgenstern-Price (half-sine)	2.027	2.005
V093	Yamagami & Ueta (1988)	1.348; 1.282; 1.185; 1.339; 1.33	Bishop simplified	1.344	1.343
			Spencer, non-circular	1.325	1.322
			Janbu simplified, non-circular	1.178	1.176
V094	Yamagami & Ueta (1988) four-layer slope, non-circular	1.40–1.42	Spencer, non-circular	1.398	1.400
			Morgenstern-Price (half-sine), non-circular	1.389	1.389
V095a	Yamagami micro-pile slope, no pile	1.10	Bishop simplified	1.102	1.101
V095c	Homogeneous slope with a micro-pile row	1.20	Bishop simplified	1.193	1.196
V096a	Zhu & Lee heterogeneous slope, dry, deep circle	–	Bishop simplified	1.804	1.793
			Spencer	1.804	1.794
			Morgenstern-Price (half-sine)	1.790	1.780
V096b	Zhu & Lee heterogeneous slope with a water table, deep circle	–	Bishop simplified	1.176	1.173
			Spencer	1.189	1.187
			Morgenstern-Price (half-sine)	1.174	1.172
V096c	Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack	–	Spencer	1.804	1.799
			Bishop simplified	1.804	1.799

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V096d	Zhu & Lee slope, the paper's surface 3, dry	1.836; 1.823; 1.744; 1.429; 1.229	Spencer	–	1.824
			Morgenstern-Price (half-sine)	–	1.808
			Bishop simplified	–	1.823
V096e	Zhu & Lee slope, the paper's surface 3, with the water table	1.211; 1.197; 1.166; 1.079; 0.922	Spencer	–	1.212
			Morgenstern-Price (half-sine)	–	1.197
			Bishop simplified	–	1.199
V097a	Eurocode 7 cutting with surcharge	–	Bishop simplified	1.173	1.172
V097b	Eurocode 7 cutting, partial factors applied by cSlope	–	Bishop simplified	1.173	1.172
			Spencer	1.174	1.174
V098	Example slope without seismic load	–	Spencer, non-circular	1.360	1.357
			Spencer	1.372	1.372
V021a	Congress Street cut, probabilistic (tangent to clay 2)	–	Pf, share FOS < 1 (Monte Carlo)	0.246	0.239
V021a	Congress Street cut, probabilistic (tangent to clay 2)	–	β lognormal (Monte Carlo)	0.650	0.671
V021a	Chowdhury & Xu (1995) example 1, probability of failure	–	Pf, share FOS < 1 (Monte Carlo)	0.253	0.245
V021b	Congress Street cut, probabilistic (tangent to clay 3)	–	Pf, share FOS < 1 (Monte Carlo)	0.279	0.282
V021b	Congress Street cut, probabilistic (tangent to clay 3)	–	β lognormal (Monte Carlo)	0.537	0.553
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	–	Pf, share FOS < 1 (Monte Carlo)	0.206–0.212	0.206
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	–	β lognormal (Monte Carlo)	0.799–0.800	0.812
V022a	Embankment on soft clay, probabilistic (tangent to the interface)	–	β lognormal (Taylor series)	0.787	0.788
V022a	Chowdhury & Xu (1995) example 5, probability of failure	–	Pf, share FOS < 1 (Monte Carlo)	0.210	0.208
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	–	Pf, share FOS < 1 (Monte Carlo)	0.199–0.209	0.212
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	–	β lognormal (Monte Carlo)	0.789–0.820	0.789
V022b	Embankment on soft clay, probabilistic (tangent to the clay base)	–	β lognormal (Taylor series)	0.798	0.802
V022b	Chowdhury & Xu (1995) example 5, probability of failure	–	Pf, share FOS < 1 (Monte Carlo)	0.214	0.216
V060b	Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β circle	–	β lognormal (Taylor series)	3.353	3.354
V023	LASH terminal trench, Duncan's Taylor series example	0.900	β lognormal (Taylor series)	–	0.892
V023	LASH terminal trench, Duncan's Taylor series example	0.180	Pf lognormal (Taylor series)	–	0.186
V062a	Clarence Cannon Dam, surface 7A	–	β lognormal (Monte Carlo)	10.953	11.034
V062b	Clarence Cannon Dam, surface 7B	–	β lognormal (Monte Carlo)	4.351	4.402
V062c	Clarence Cannon Dam, surface 7C	–	β lognormal (Monte Carlo)	4.263	4.325
V062d	Clarence Cannon Dam, surface 7D	–	β lognormal (Monte Carlo)	11.092	11.076
V062e	Clarence Cannon Dam, surface 7E	–	β lognormal (Monte Carlo)	10.281	10.329

Problem	Description	Published	cSlope analysis	Other programs	cSlope
V062f	Clarence Cannon Dam, surface 8B	–	β lognormal (Monte Carlo)	4.858	4.933
V062g	Clarence Cannon Dam, surface 8F	–	β lognormal (Monte Carlo)	5.485	5.528
V062h	Clarence Cannon Dam, surface 8G	–	β lognormal (Monte Carlo)	5.563	5.490
V062i	Clarence Cannon Dam, surface 8H	–	β lognormal (Monte Carlo)	6.394	6.431

3 Verification problems

3.1 V001 — Arai & Tagyo (1985) example 1

Arai & Tagyo (1985), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1.5:1 slope, 20 m high; critical circle search.

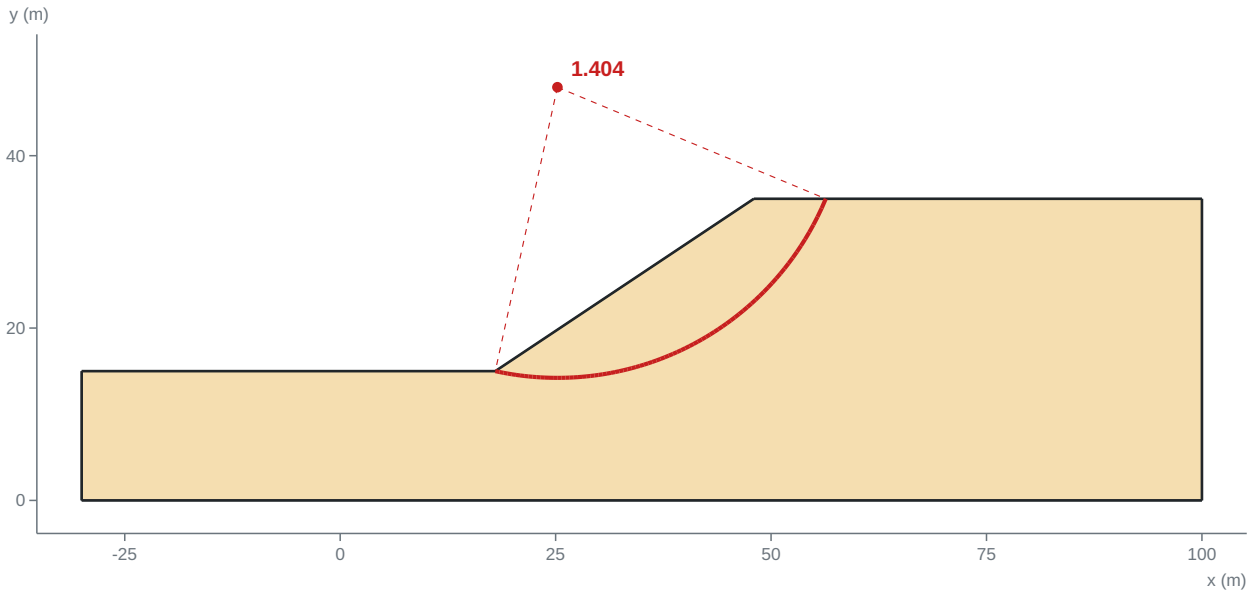


Figure 1. V001: critical circle (Bishop simplified, FOS = 1.404).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.82	c' = 41.65	15.00°

Published value	Method	Source	cSlope
1.451	Bishop simplified	Arai & Tagyo (1985)	1.404
1.265	Janbu simplified, non-circular	Arai & Tagyo (1985)	1.252
1.357	Janbu corrected, non-circular	Arai & Tagyo (1985)	1.345

cSlope analysis	Other programs	cSlope
Bishop simplified	1.409–1.417	1.404
Morgenstern-Price (half-sine)	1.414	1.401
Spencer, non-circular	1.386	1.384
Janbu simplified, non-circular	–	1.252
Janbu corrected, non-circular	–	1.345

Arai & Tagyo's Bishop value 1.451 is 2–3 % above the other programs (1.409–1.417) and cSlope on the same slope; their non-circular Janbu values are within 1 % of cSlope's.

3.2 V002 — Arai & Tagyo (1985) example 2, three layers

Arai & Tagyo (1985), example 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

Slope with a weak middle layer between a stronger upper layer and a very strong base; critical circle search.

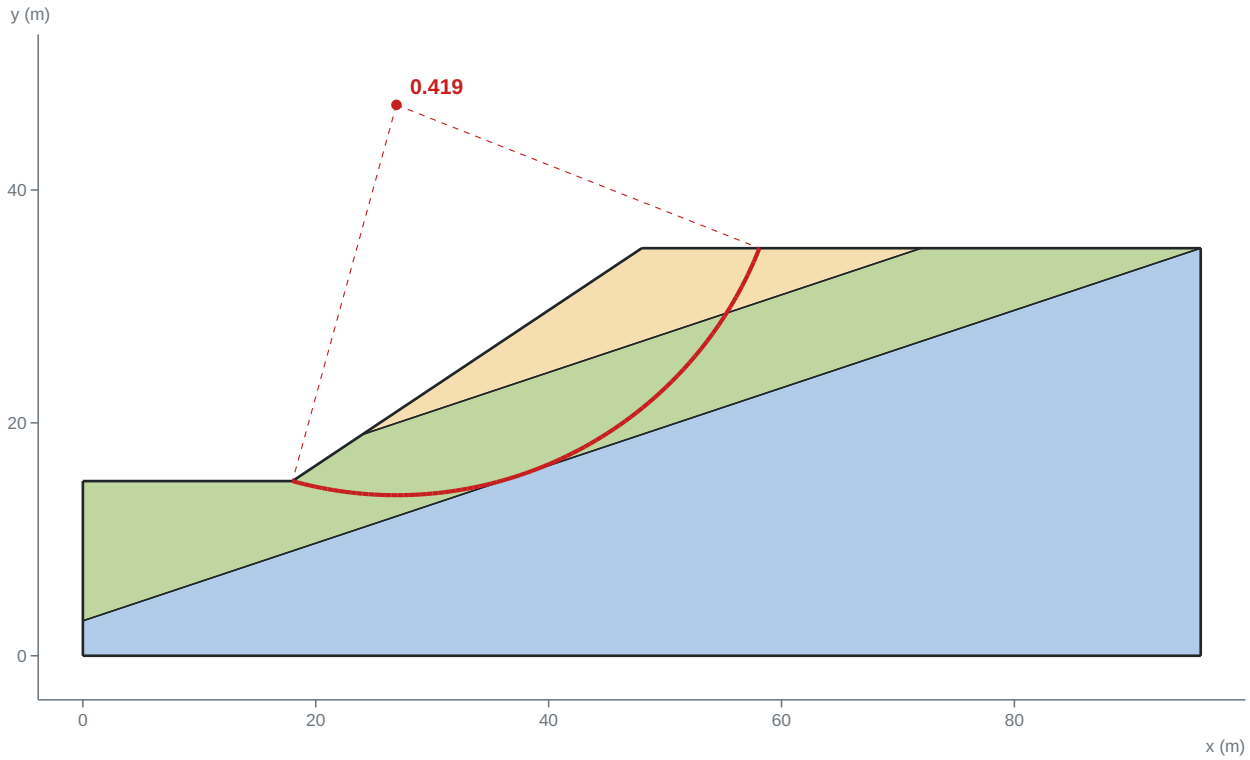


Figure 2. V002: critical circle (Bishop simplified, FOS = 0.419).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	18.82	c' = 29.40	12.00°
Middle	Mohr-Coulomb	18.82	c' = 9.80	5.00°
Lower	Mohr-Coulomb	18.82	c' = 294.0	40.00°

Published value	Method	Source	cSlope
0.417	Bishop simplified	Arai & Tagyo (1985)	0.419
0.405	Janbu simplified, non-circular	Arai & Tagyo (1985)	0.393
0.43	Janbu corrected, non-circular	Arai & Tagyo (1985)	0.413
0.43	Bishop simplified	Kim et al. (2002)	0.419
0.44	Spencer, non-circular	Kim et al. (2002)	0.411
0.40–0.45	limit analysis, lower and upper bound	Kim et al. (2002)	–
0.39	Spencer, non-circular	Greco (1996)	0.411

cSlope analysis	Other programs	cSlope
Bishop simplified	0.420	0.419
Spencer, non-circular	0.414	0.411
Janbu simplified, non-circular	–	0.393
Janbu corrected, non-circular	–	0.413

3.3 V003 — Arai & Tagyo (1985) example 3, water table

Arai & Tagyo (1985), Soils and Foundations 25(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a water table; critical-circle search.

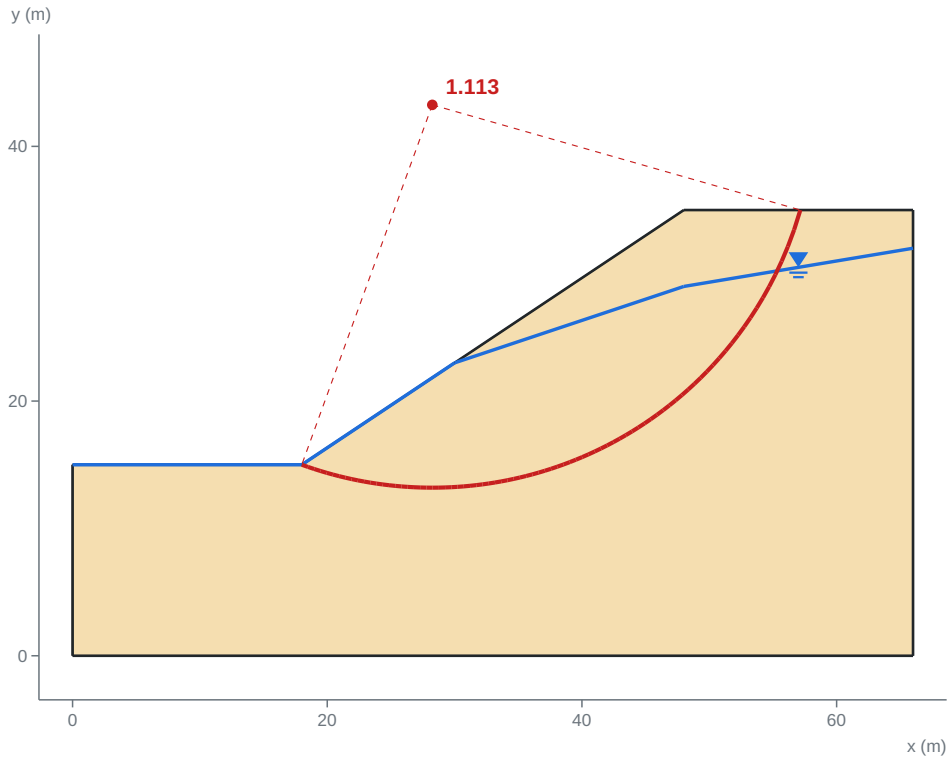


Figure 3. V003: critical circle (Bishop simplified, FOS = 1.113).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A&T ex3 soil	Mohr-Coulomb	18.82	c' = 41.65	15.00°

Published value	Method	Source	cSlope
1.138	Bishop simplified	Arai & Tagyo (1985)	1.113
0.995	Janbu simplified, non-circular	Arai & Tagyo (1985)	0.966
1.071	Janbu corrected, non-circular	Arai & Tagyo (1985)	1.047

cSlope analysis	Other programs	cSlope
Bishop simplified	1.112–1.118	1.113
Spencer	1.118	1.113
Janbu corrected	1.131	1.124
Janbu simplified, non-circular	–	0.966
Janbu corrected, non-circular	–	1.047

Arai & Tagyo's Bishop value 1.138 is about 2 % above the other programs (1.112–1.118 with Bishop).

3.4 V004a — 4:1 slope, Mohr-Coulomb fit

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

12 m high 4:1 slope with the same soil as V008a; critical circle search.

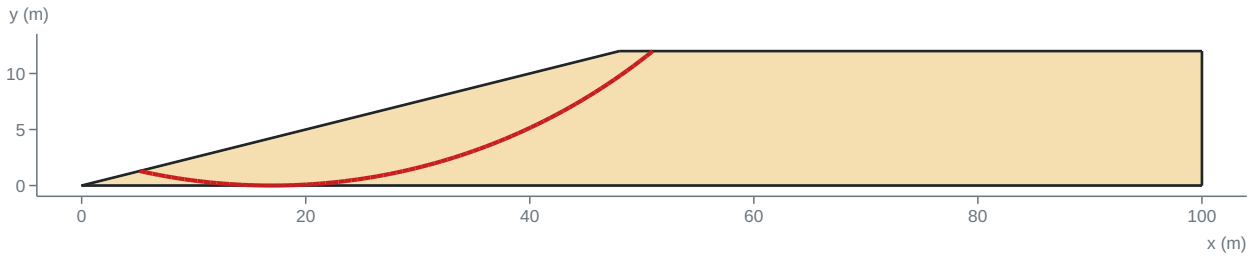


Figure 4. V004a: critical circle (Spencer, FOS = 2.802).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Published value	Method	Source	cSlope
2.66	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Spencer	2.794	2.802
Janbu simplified	2.662	2.677

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.5 V004b — Baker (2003) ex. 2 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V004a 4:1 slope with Baker's iterated Mohr-Coulomb parameters (c′ = 2.439 kPa, φ′ = 30.392°); critical-circle search.

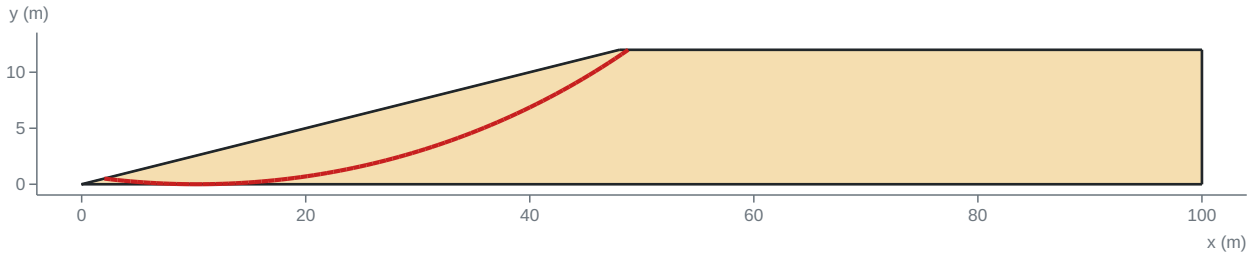


Figure 5. V004b: critical circle (Janbu simplified, FOS = 2.609).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 2.44	30.39°

Published value	Method	Source	cSlope
2.64	non-linear strength envelope	Baker (2003)	–
2.66	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Janbu simplified	2.610	2.609
Spencer	2.696	2.666

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.6 V005 — Baker (1993) dam, end of construction, infinite-slope face

Baker, R., Frydman, S. & Talesnick, M. (1993). Slope stability analysis for undrained loading conditions. Int. J. Numer. Anal. Meth. Geomech. 17, 15–43.

Critical circle search · Units: SI (m, kPa, kN/m³)

Compacted-clay embankment (c′ = 6.5 kPa, φ′ = 40°) on a deep natural-clay foundation (c′ = 0, φ′ = 32°) with a 6 m crest and a straight 4H:1V downstream face, whose natural-clay part governs; dry; circular search. Theoretical (infinite slope) FOS tan 32°/tan 14.04° = 2.50.

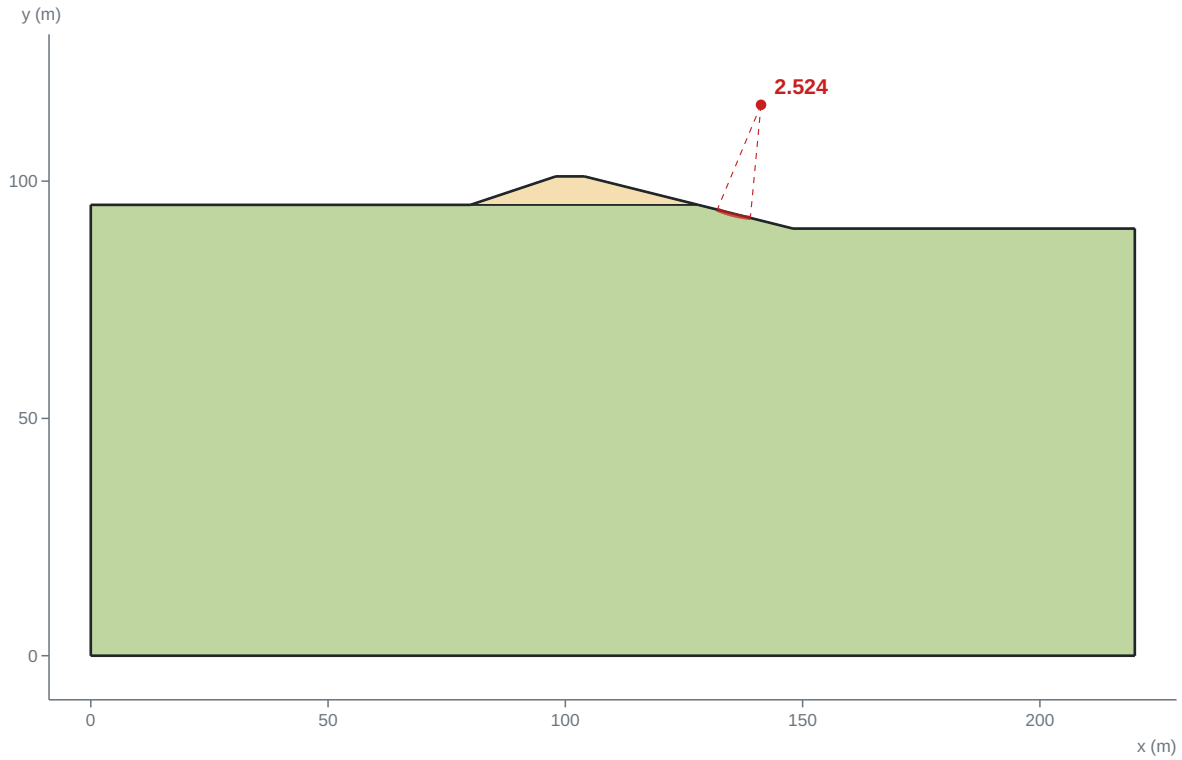


Figure 6. V005: critical circle (Bishop simplified, FOS = 2.524).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Compacted clay	Mohr-Coulomb	18.00	c' = 6.50	40.00°
Natural clay	Mohr-Coulomb	18.00	c' = 0.00	32.00°

Published value	Method	Source	cSlope
2.50	infinite slope	Theory	2.524
2.41	Spencer, non-circular (dynamic programming)	Baker, Frydman & Talesnick (1993)	2.502

cSlope analysis	Other programs	cSlope
Bishop simplified	–	2.524
Spencer	–	2.524
Spencer, non-circular	–	2.502

3.7 V006 — Baker & Leshchinsky (2001) planar surface

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Homogeneous 10 m slope on the critical plane from the toe to the crest at 49.6°.

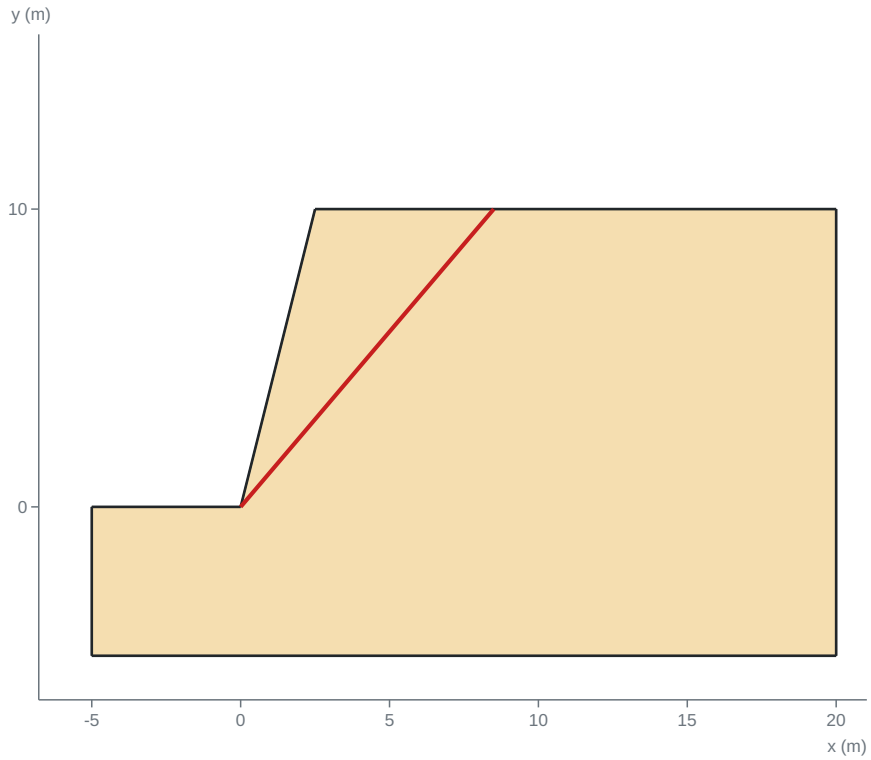


Figure 7. V006: analysed surface (Spencer, FOS = 1.352).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 30.00	30.00°

Published value	Method	Source	cSlope
1.35	Culmann (planar wedge)	Baker & Leshchinsky (2001)	–

cSlope analysis	Other programs	cSlope
Spencer	1.351	1.352
Janbu simplified	1.351	1.352

3.8 V007a — Baker (2003) ex. 1, Mohr-Coulomb, Janbu simplified on the reference circle

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V008a slope (Mohr-Coulomb fit c′ = 11.64 kPa, φ′ = 24.7°) with Janbu simplified (uncorrected) on the reference critical circle (Fig. 44.3).

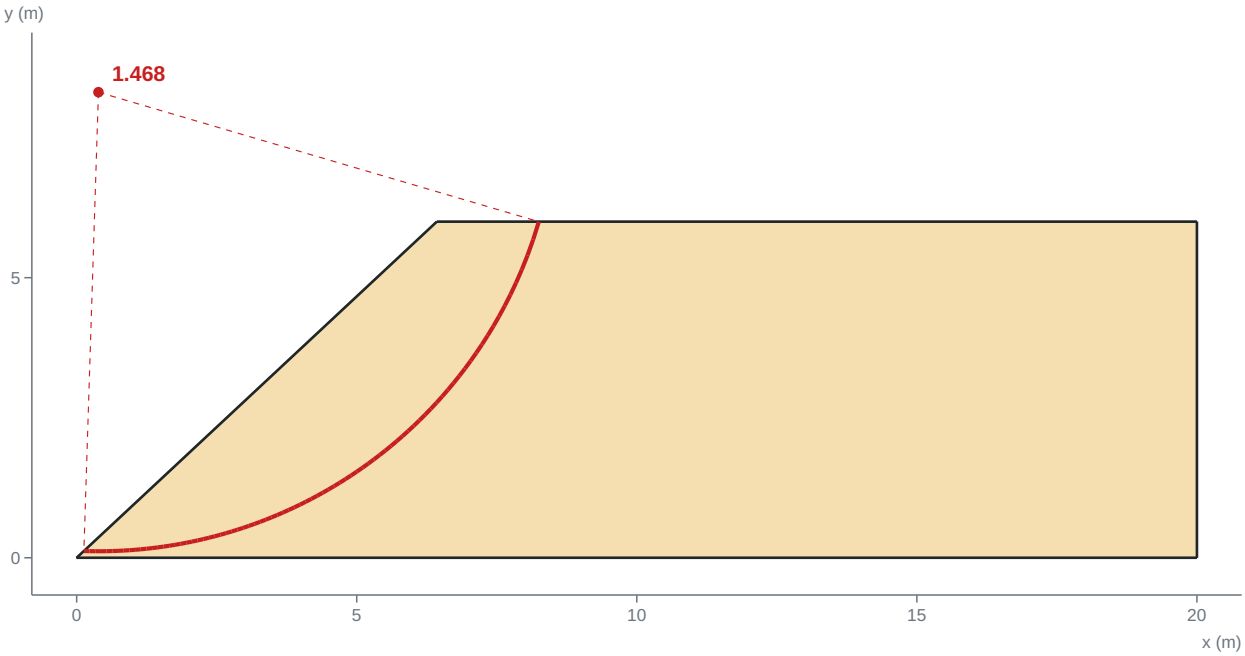


Figure 8. V007a: analysed circle (Janbu simplified, FOS = 1.468).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Published value	Method	Source	cSlope
1.50	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Janbu simplified	1.469	1.468

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.9 V007b — Baker (2003) ex. 1 slope, Mohr-Coulomb iteration parameters

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008a slope with Baker's Mohr-Coulomb parameters from his iterative linearisation (c′ = 0.39 kPa, φ′ = 38.6°); critical-circle search.

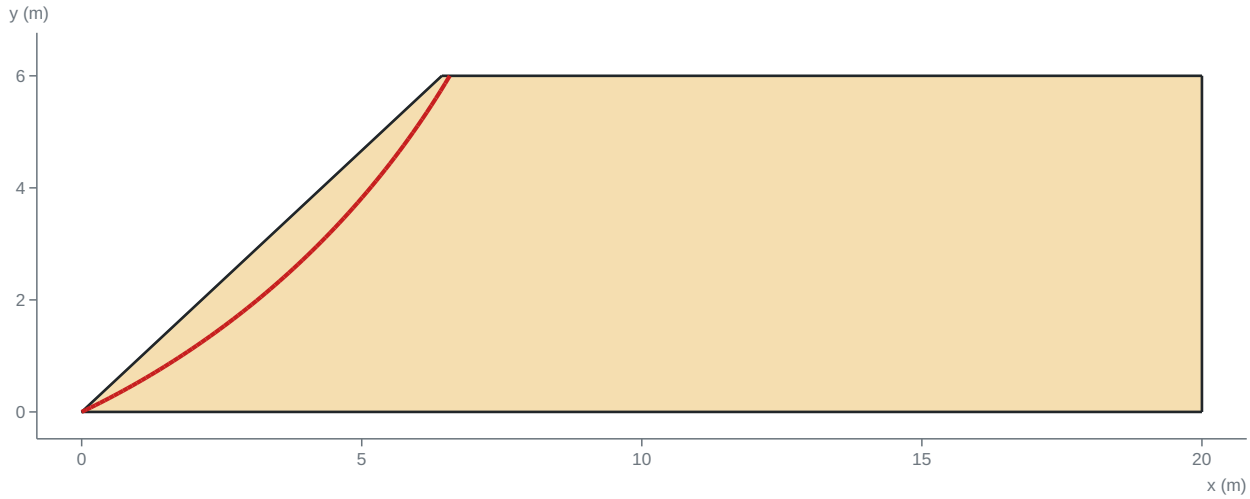


Figure 9. V007b: critical circle (Janbu simplified, FOS = 0.955).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 0.39	38.60°

Published value	Method	Source	cSlope
0.97	non-linear strength envelope	Baker (2003)	–
1.50	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Janbu simplified	0.957	0.955
Spencer	0.981	0.978

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit and for the non-linear envelope with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387); 1.50 is his Mohr-Coulomb value for the same slope.

3.10 V007c — Baker (2003) ex. 3, Janbu simplified

Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V008b London clay slope (c′ = 6 kPa, φ′ = 32°) with Janbu simplified (uncorrected); critical-circle search.

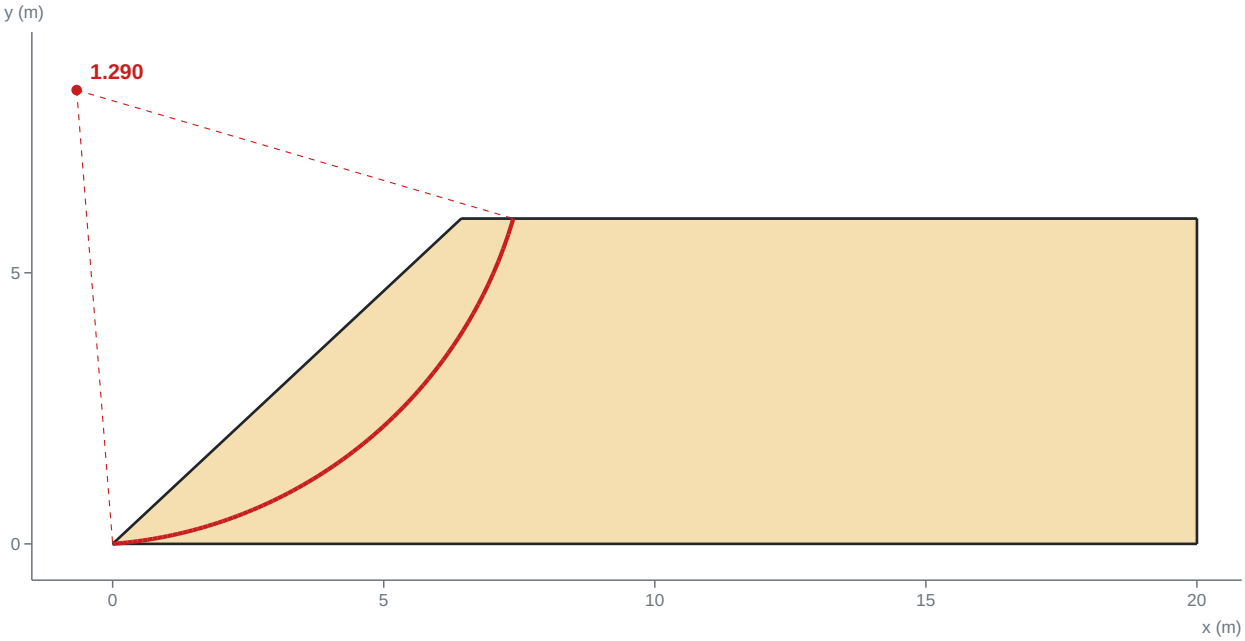


Figure 10. V007c: critical circle (Janbu simplified, FOS = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 6.00	32.00°

Published value	Method	Source	cSlope
1.35	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Janbu simplified	1.291	1.290

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.11 V008a — Baker (2003) example 1, Mohr-Coulomb fit

Baker (2003), example 1.

Critical circle search · Units: SI (m, kPa, kN/m³)

6 m high slope at about 43° in soil with a linear Mohr-Coulomb fit to a curved envelope; critical circle search.

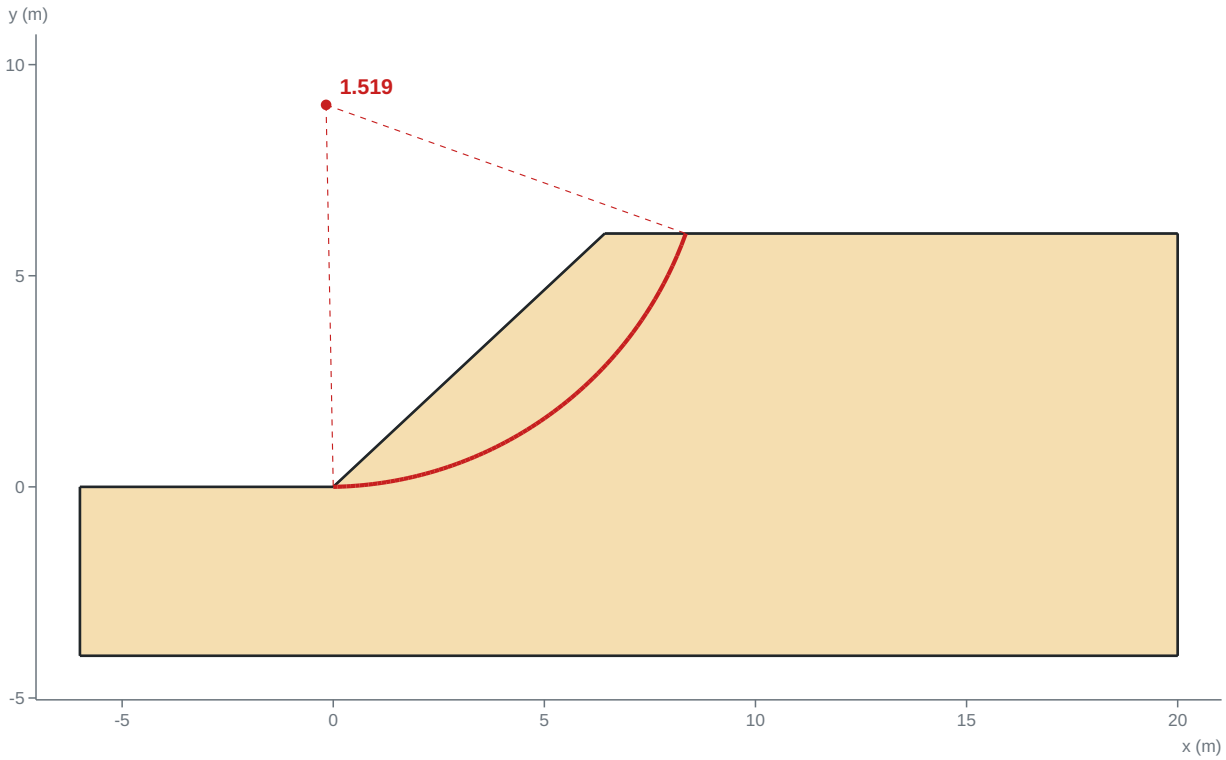


Figure 11. V008a: critical circle (Spencer, FOS = 1.519).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.00	c' = 11.64	24.70°

Published value	Method	Source	cSlope
1.50	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Spencer	1.536	1.519

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.12 V008b — Baker (2003) London clay slope, Mohr-Coulomb fit

Baker (2003) example 3.

Critical circle search · Units: SI (m, kPa, kN/m³)

43° slope, 6 m high, with the Mohr-Coulomb envelope fitted to London clay data (c' = 6 kPa, φ' = 32°); critical circle search.

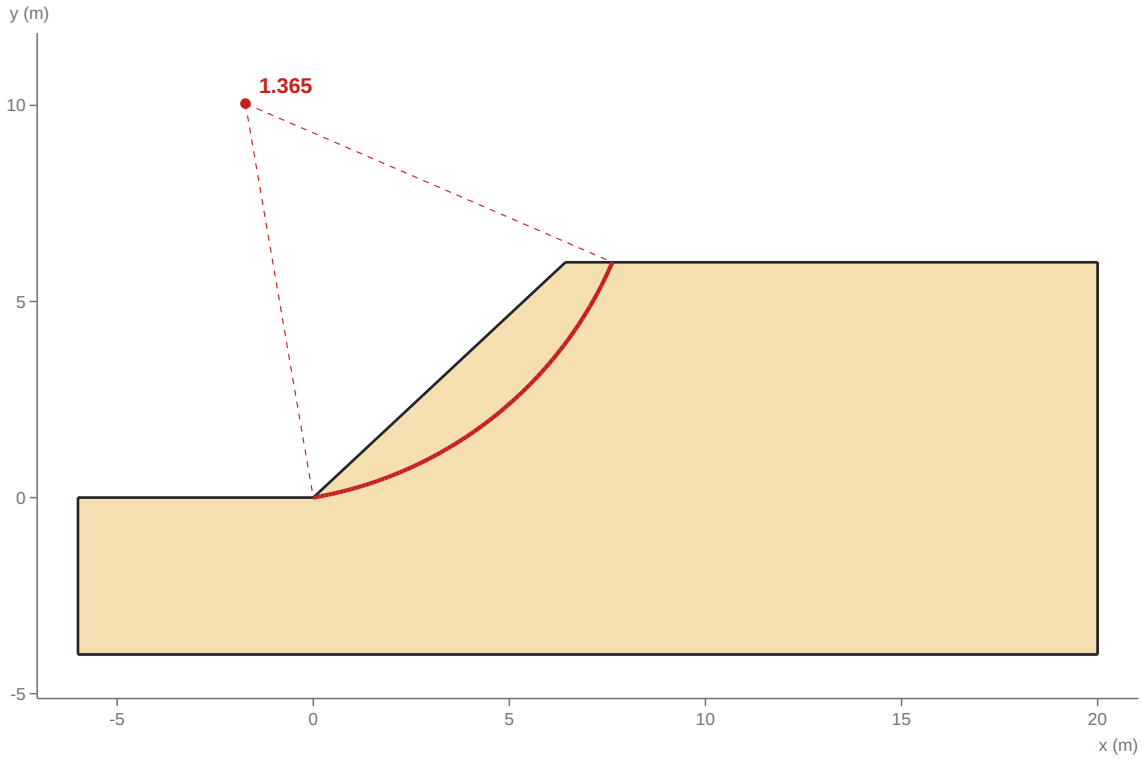


Figure 12. V008b: critical circle (Spencer, FOS = 1.365).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
London clay	Mohr-Coulomb	18.00	c' = 6.00	32.00°

Published value	Method	Source	cSlope
1.35	Mohr-Coulomb strength	Baker (2003)	–

cSlope analysis	Other programs	cSlope
Spencer	1.366	1.365

Baker (2003) gives the factor of safety for the Mohr-Coulomb fit with Taylor's stability chart (friction-circle method, critical circle per Baker 2002, Janbu's normal-stress approximation; Baker 2003 pp. 385, 387).

3.13 V009a — Baker & Leshchinsky clay-core dam, reservoir and crest crack

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular dam with a clay core (a hexagon reaching the base) on a hard base, half-full reservoir standing on the face (free water), phreatic surface through the core and a 5 m dry tension crack at the crest; a commercial program's critical circle.

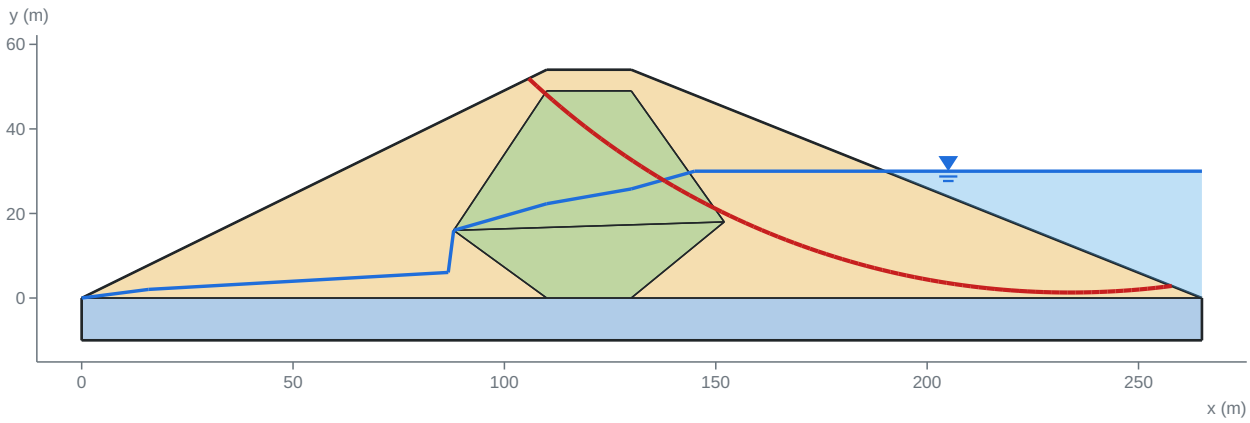


Figure 13. V009a: analysed circle (Spencer, FOS = 1.926).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer	1.925	1.926

The published 1.91 belongs to Baker & Leshchinsky's own non-circular surface (V009b), not to this circle.

3.14 V009b — Baker & Leshchinsky dam, Baker's non-circular surface

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

As V009a, on the non-circular surface published by Baker & Leshchinsky.

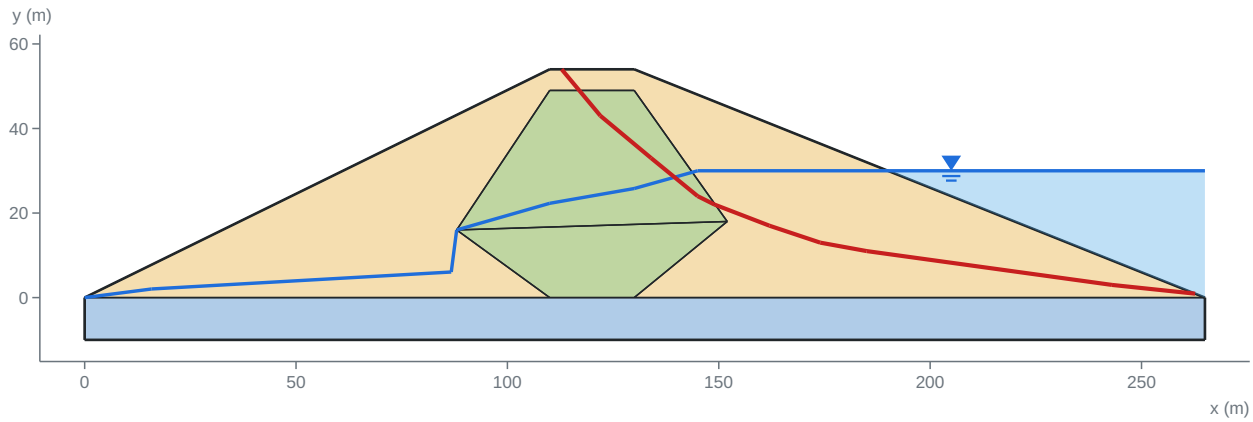


Figure 14. V009b: analysed surface (Spencer, FOS = 1.882).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

Published value	Method	Source	cSlope
1.91	Spencer	Baker & Leshchinsky	1.882

cSlope analysis	Other programs	cSlope
Spencer	–	1.882

3.15 V009c — Baker & Leshchinsky clay-core dam, critical surface search

Baker, R. & Leshchinsky, D. (2001). Spatial distribution of safety factors. J. Geotech. Geoenviron. Eng. 127(2), 135–145.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

The V009a clay-core dam (reservoir on the face, 5 m dry crest crack) with a circular search and non-circular optimisation.

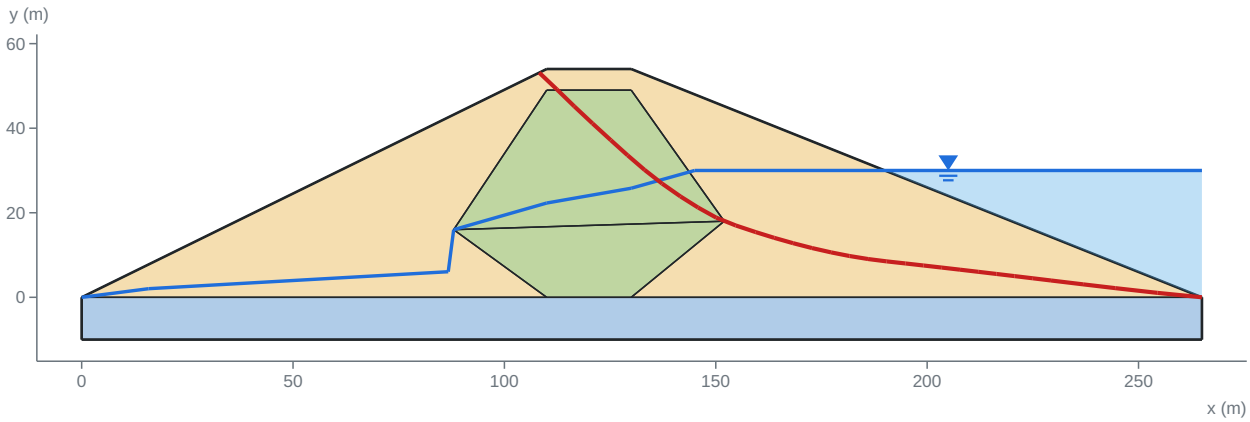


Figure 15. V009c: critical surface (Spencer, non-circular, FOS = 1.867).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular fill	Mohr-Coulomb	21.50	c' = 0.00	40.00°
Clay core	Mohr-Coulomb	20.00	c' = 20.00	20.00°
Hard base	Mohr-Coulomb	24.00	c' = 200.0	45.00°

Published value	Method	Source	cSlope
1.91	Spencer, non-circular	Baker & Leshchinsky (2001)	1.867

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.868	1.867

Baker & Leshchinsky's 1.91 is for their own surface (V009b); searches find lower surfaces.

3.16 V010 — Borges & Cardoso embankment 2 (geosynthetic)

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

5 m embankment (φ′ = 35°) on four soft clays with Su increasing with depth (Borges & Cardoso Table 6, p. 408: clay 3 from 16 to 18.4 kPa, clay 4 from 18.4 to 55.1 kPa), a 200 kN/m geosynthetic at y = 1 m (axial, active, not anchored, interface friction 33.7° on both faces); a commercial program's critical circle.

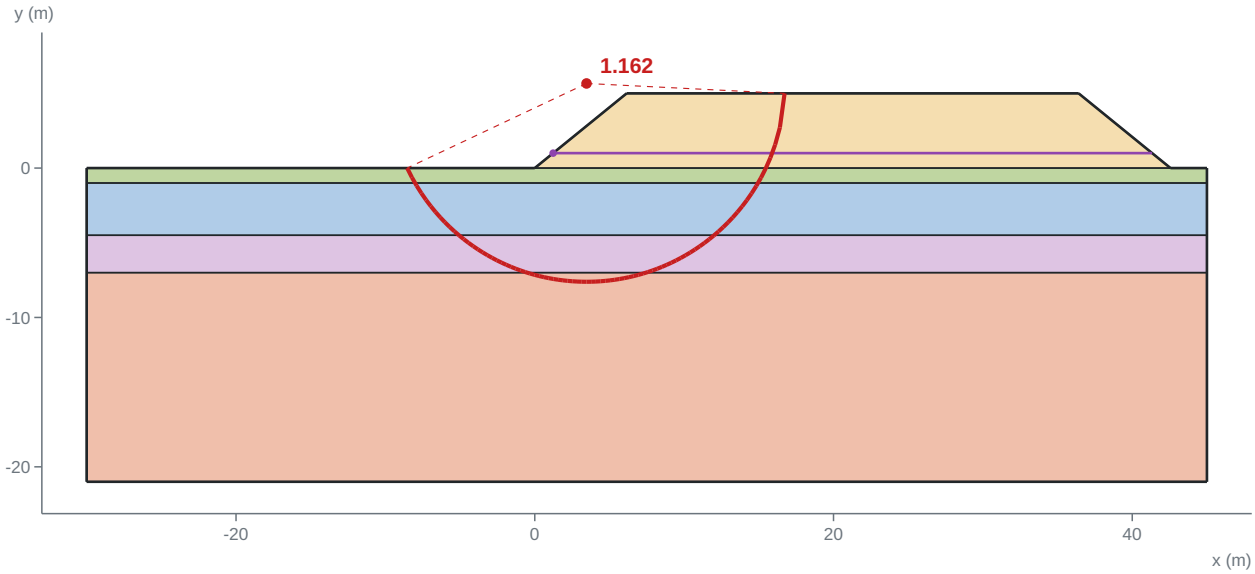


Figure 16. V010: analysed circle (Morgenstern-Price (half-sine), FOS = 1.162).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c' = 0.00	35.00°
Clay 1	Undrained (φ = 0)	17.00	Su = 33.00	0°
Clay 2	Undrained (φ = 0)	17.00	Su = 16.00	0°
Clay 3	Undrained (φ = 0)	17.00	Su = 16.00 at y = -4.50, +0.95 kPa/m below	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 18.38 at y = -7.00, +2.63 kPa/m below	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.171	1.162
Bishop simplified	1.170	1.164

3.17 V010b — Borges & Cardoso embankment 2 (geosynthetic), the paper's circle B

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421; case 2, Table 11, Fig. 14.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V010 embankment as Borges & Cardoso analyse it: the 200 kN/m geosynthetic at 0.9 m height (p. 407) with its full horizontal force in the resisting moment (passive), on their circle B (3.0, 8.0, R = 15.04).

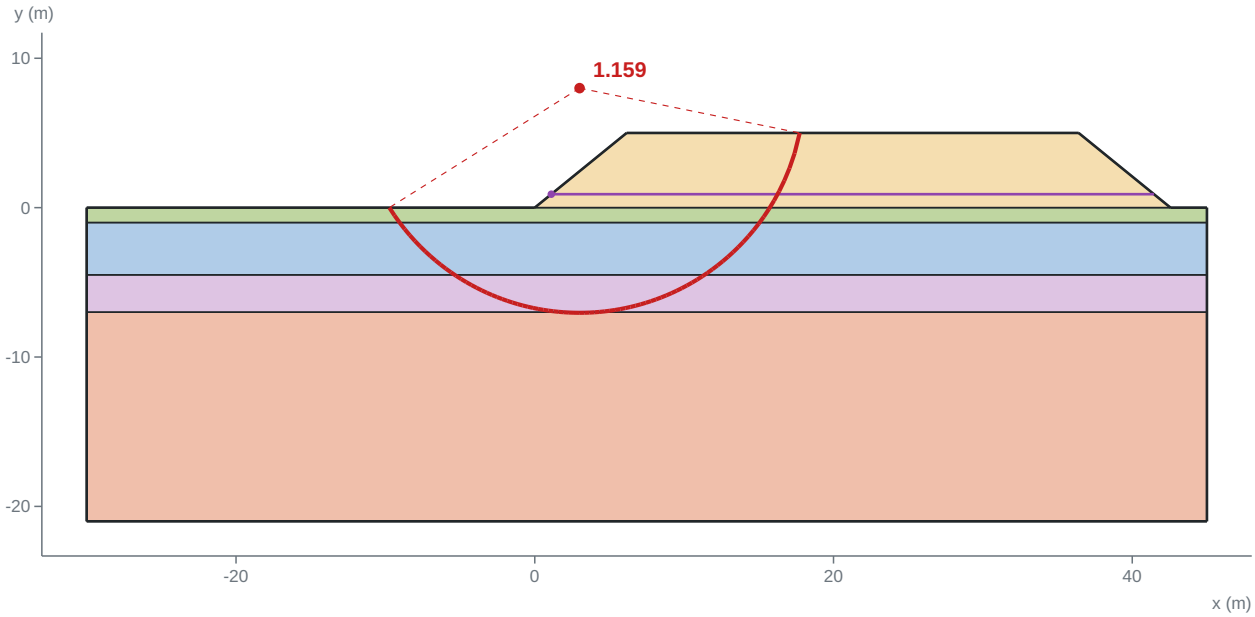


Figure 17. V010b: analysed circle (Bishop simplified, FOS = 1.159).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c' = 0.00	35.00°
Clay 1	Undrained (φ = 0)	17.00	Su = 33.00	0°
Clay 2	Undrained (φ = 0)	17.00	Su = 16.00	0°
Clay 3	Undrained (φ = 0)	17.00	Su = 16.00 at y = -4.50, +0.95 kPa/m below	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 18.38 at y = -7.00, +2.63 kPa/m below	0°

Published value	Method	Source	cSlope
1.15	Bishop simplified	Borges & Cardoso	1.159
1.19	authors' moment method, circle A	Borges & Cardoso (2002)	–
1.22	finite elements, circle A	Borges & Cardoso (2002)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.159

Borges & Cardoso use their own moment-equilibrium method of slices on circles (Kaniraj & Abdullah 1993 with their improvements); simplified Bishop is the nearest of cSlope's methods. Circle B's coordinates and the geosynthetic level are printed in the paper (Table 11, p. 407).

3.18 V011a — Borges & Cardoso embankment 3, H = 7 m, circle A

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

Granular fill (two stages) on five soft clay layers with a T = 200 kN/m geosynthetic at y = 0.9 m (axial, passive as in a commercial program, pullout over 1.3 m from each end); Bishop on circle A of the paper.

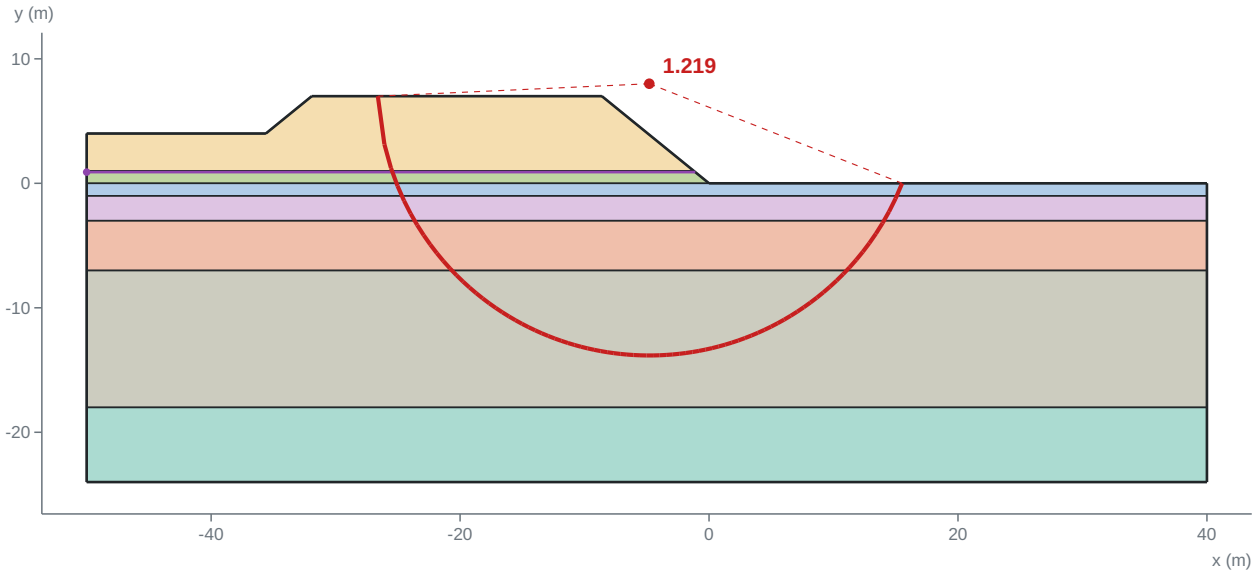


Figure 18. V011a: analysed circle (Bishop simplified, FOS = 1.219).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Published value	Method	Source	cSlope
1.25	not stated	Borges & Cardoso (2002)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.230	1.219

Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.

3.19 V011b — Borges & Cardoso embankment 3, H = 7 m, circle B

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V011a on circle B.

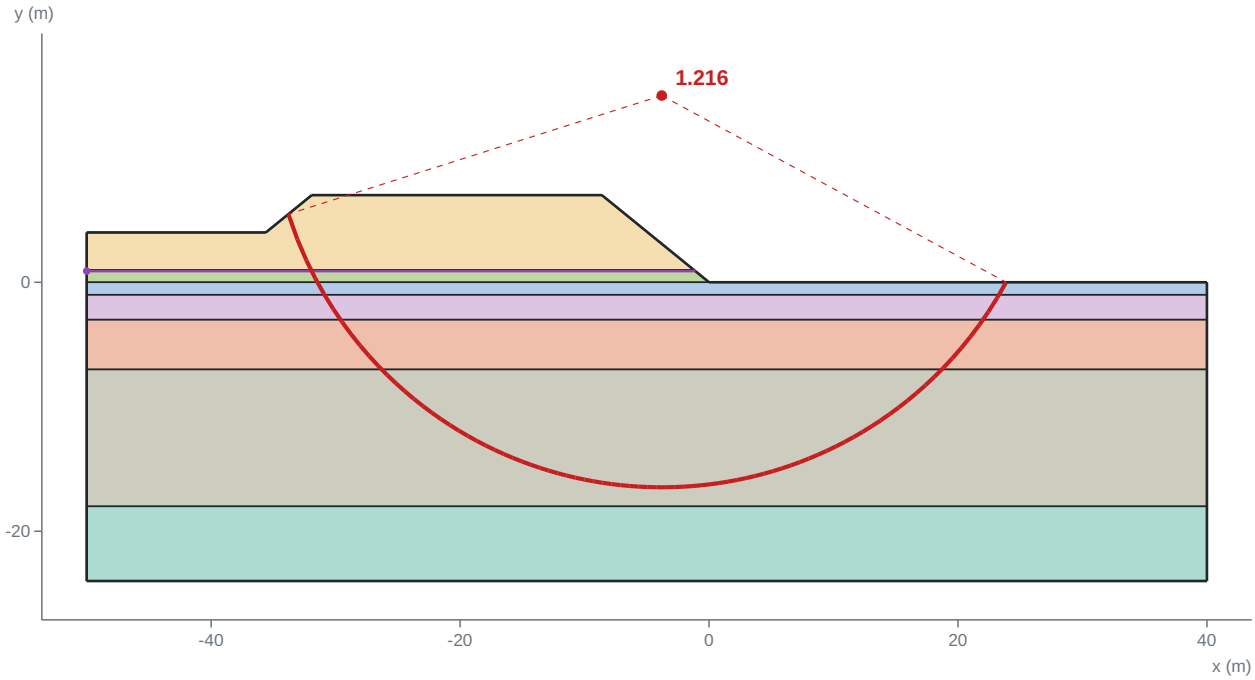


Figure 19. V011b: analysed circle (Bishop simplified, FOS = 1.216).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Published value	Method	Source	cSlope
1.19	not stated	Borges & Cardoso (2002)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.220	1.216

Borges & Cardoso print two decimals and do not name the method; the other programs differ from them by up to 2.5 %.

3.20 V012 — Borges & Cardoso embankment 3, H = 8.75 m, circle C

Borges, J.L. & Cardoso, A.S. (2002). Overall stability of geosynthetic-reinforced embankments on soft soils. Geotextiles and Geomembranes 20, 395–421.

Fixed circle · Units: SI (m, kPa, kN/m³)

The second fill stage (8.75 m crest) on circle C.

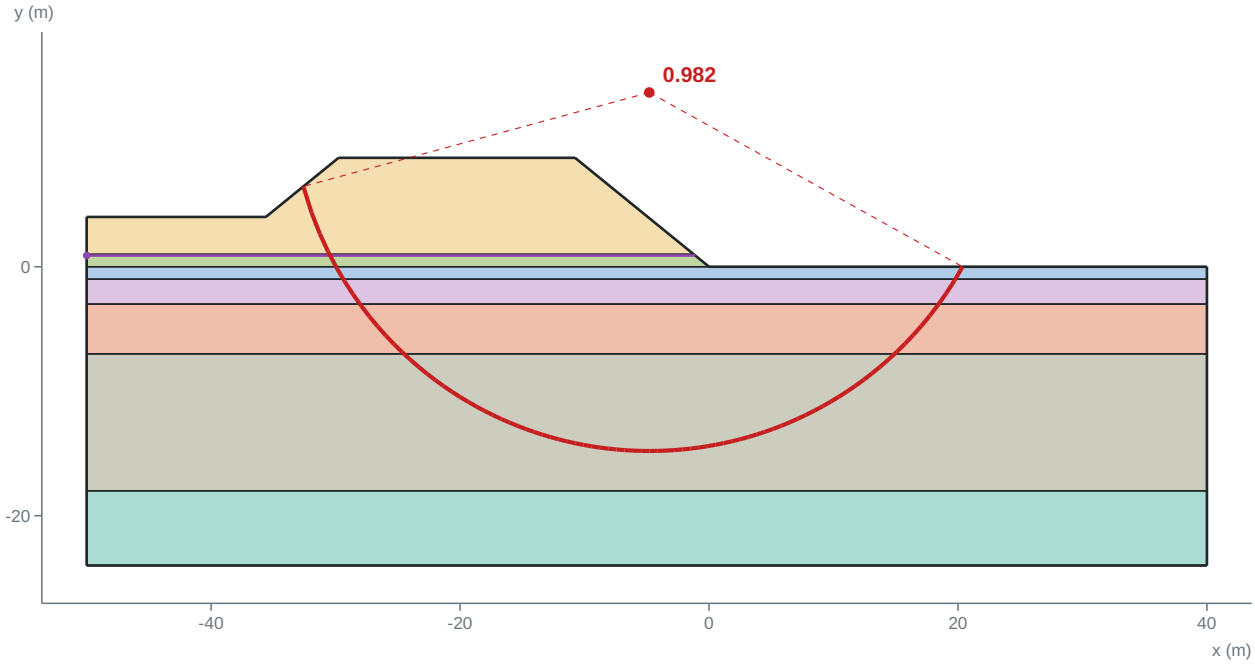


Figure 20. V012: analysed circle (Bishop simplified, FOS = 0.982).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper embankment	Mohr-Coulomb	21.90	c' = 0.00	35.00°
Lower embankment	Mohr-Coulomb	17.20	c' = 0.00	33.00°
Clay 1	Undrained (φ = 0)	18.00	Su = 43.00	0°
Clay 2	Undrained (φ = 0)	16.60	Su = 31.00	0°
Clay 3	Undrained (φ = 0)	13.50	Su = 30.00	0°
Clay 4	Undrained (φ = 0)	17.00	Su = 32.00	0°
Clay 5	Undrained (φ = 0)	17.50	Su = 32.00	0°

Published value	Method	Source	cSlope
0.99	not stated	Borges & Cardoso (2002)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	0.980	0.982

Borges & Cardoso print two decimals and do not name the method.

3.21 V013a — Cai & Ugai (2000) slope, no pile (see V013b, V013c, V013d and V013e for the pile r

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 1:1.5 slope, 10 m high, without piles; critical circle search.

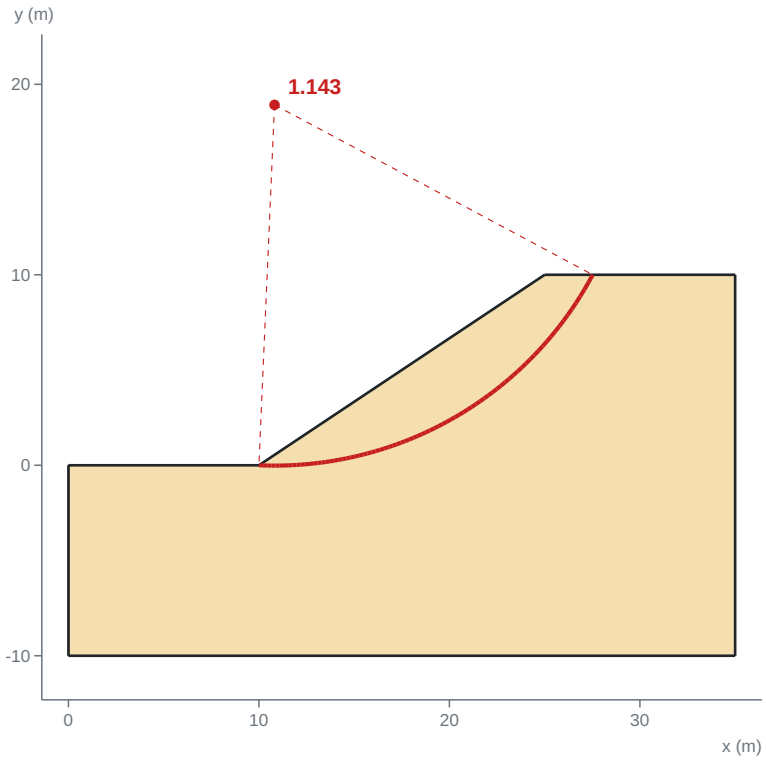


Figure 21. V013a: critical circle (Bishop simplified, FOS = 1.143).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Published value	Method	Source	cSlope
1.13	Bishop simplified	Cai & Ugai	1.143

cSlope analysis	Other programs	cSlope
Bishop simplified	1.140	1.143

3.22 V013b — Cai & Ugai slope, pile row D1/D = 2

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

A row of 0.8 m piles at mid-slope, centre-to-centre 1.6 m, with the Ito & Matsui force (passive); Bishop search.

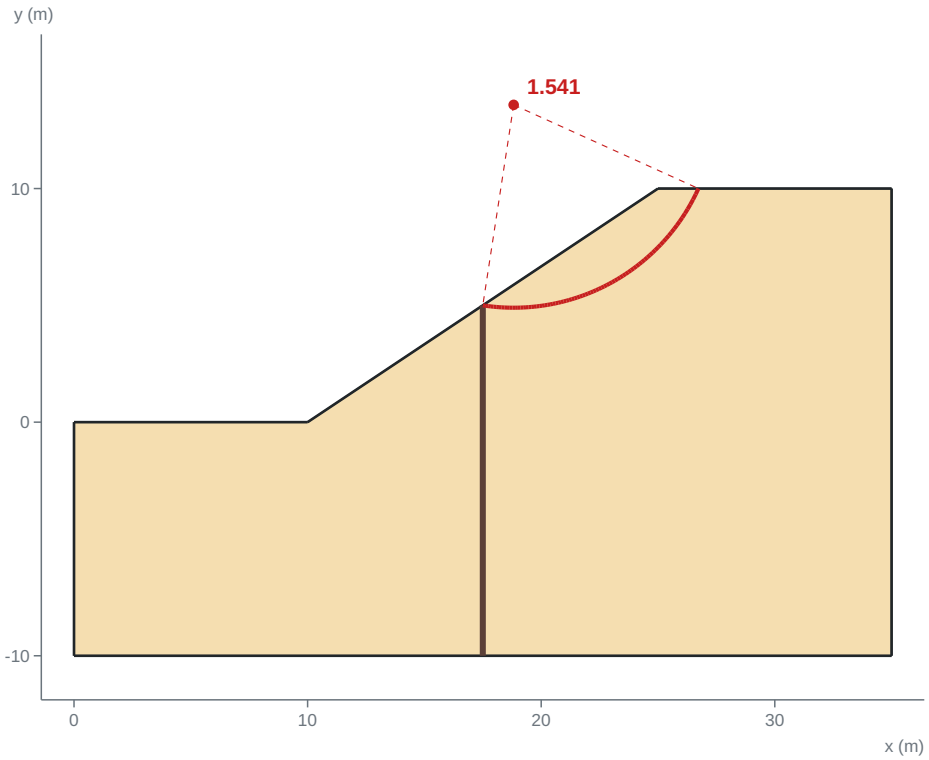


Figure 22. V013b: critical circle (Bishop simplified, FOS = 1.541).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Published value	Method	Source	cSlope
1.54	not stated	Cai & Ugai (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.540	1.541

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.23 V013c — Cai & Ugai slope, pile row D1/D = 3

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 2.4 m spacing.

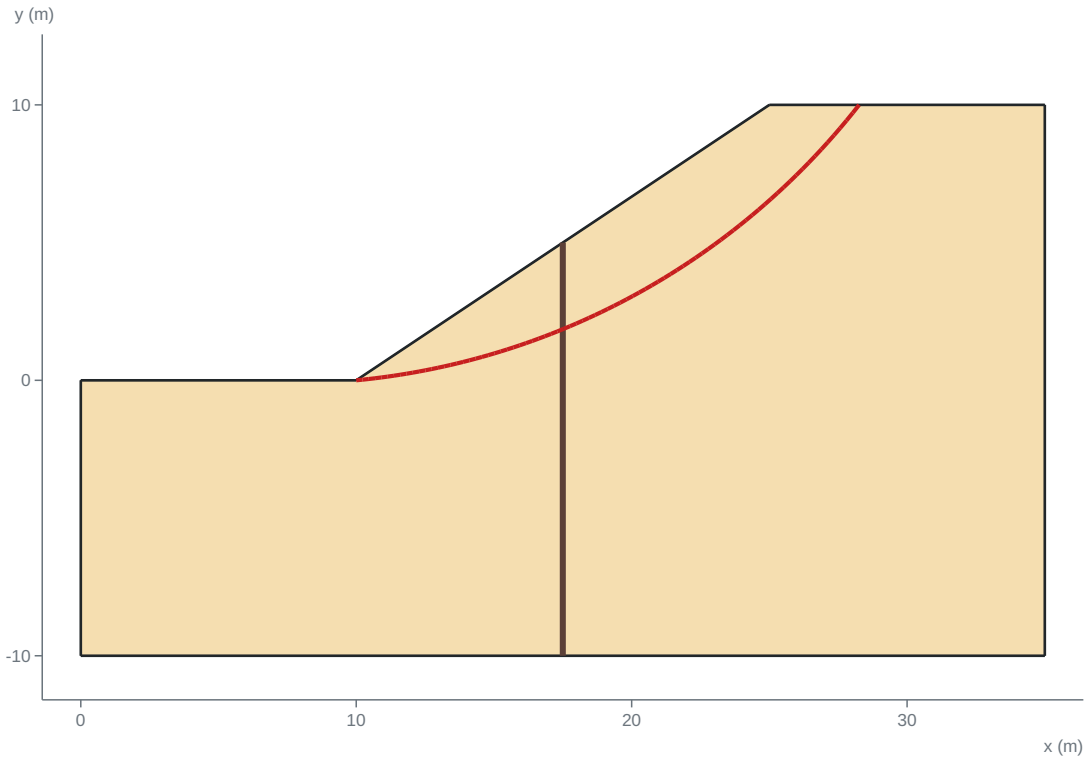


Figure 23. V013c: critical circle (Bishop simplified, FOS = 1.437).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Published value	Method	Source	cSlope
1.37	not stated	Cai & Ugai (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.430	1.437

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.24 V013d — Cai & Ugai slope, pile row D1/D = 4

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 3.2 m spacing.

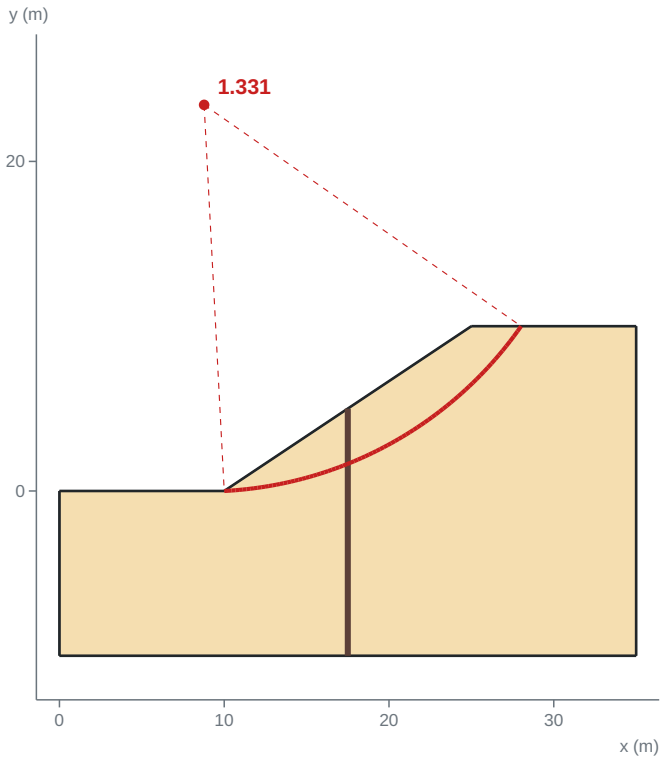


Figure 24. V013d: critical circle (Bishop simplified, FOS = 1.331).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Published value	Method	Source	cSlope
1.31	not stated	Cai & Ugai (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.330	1.331

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.25 V013e — Cai & Ugai slope, pile row D1/D = 6

Cai & Ugai (2000), Soils and Foundations 40(1).

Critical circle search · Units: SI (m, kPa, kN/m³)

As V013b at 4.8 m spacing.

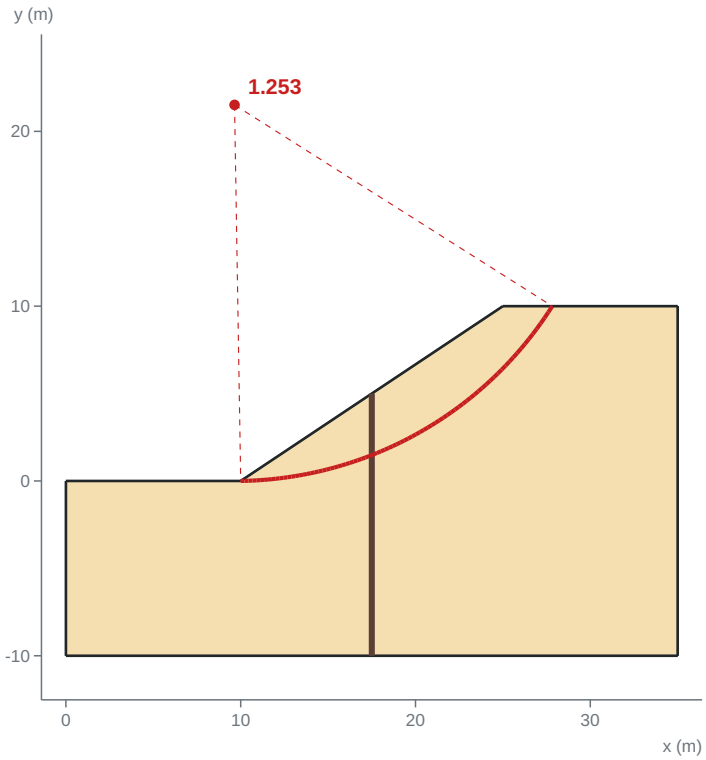


Figure 25. V013e: critical circle (Bishop simplified, FOS = 1.253).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 10.00	20.00°

Published value	Method	Source	cSlope
1.25	not stated	Cai & Ugai (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.250	1.253

Cai & Ugai's values are for surfaces through the pile row; the programs search their own critical surfaces, which can differ.

3.26 V014 — Nailed wall, 14 rows, deep wedge

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Nail-reinforced wall with 14 horizontal rows (per-row length, capacity and bond; fully anchored at the wall, tangent, passive) on the printed bilinear wedge. Converted to SI.

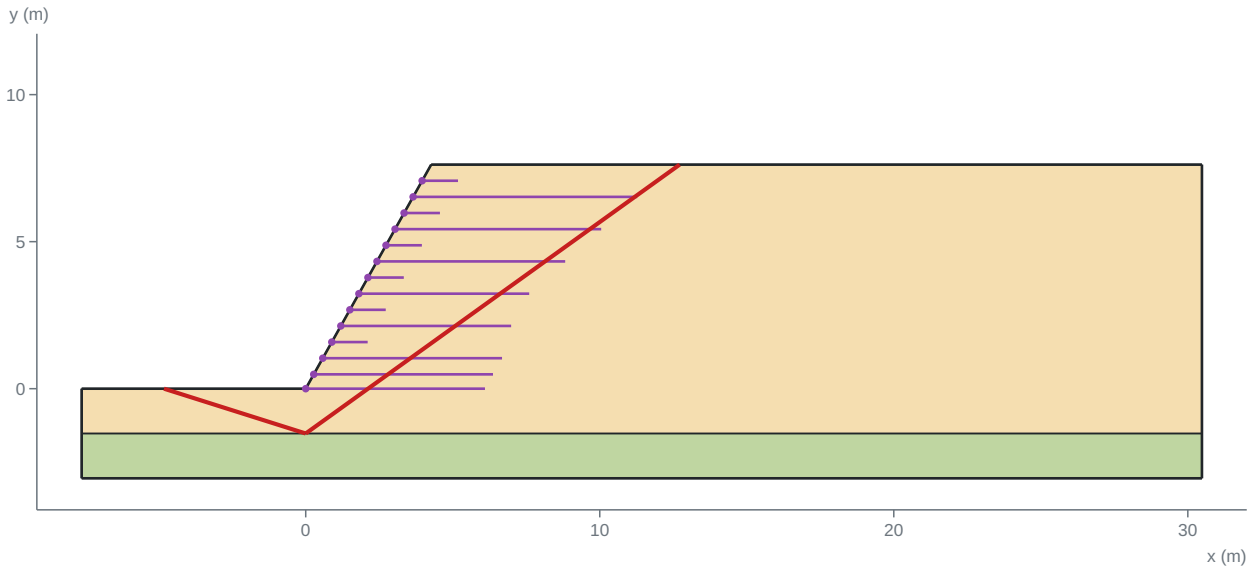


Figure 26. V014: analysed surface (Janbu corrected, FOS = 1.448).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	19.64	c' = 0.00	32.00°
Layer 2	Mohr-Coulomb	20.11	c' = 23.94	35.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Janbu corrected	1.448–1.460	1.448
Spencer	1.576	1.575

The reference results come from design programs quoted in the soil nail manual; no independent value is published.

3.27 V015 — Soldier-pile tieback wall

Caltrans soil nail reference manual.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Two-layer slope cut by a soldier-pile wall with two grouted tieback rows (8 ft spacing); the soldier pile is a shear force of 5900 lb/ft where the surface passes, as a commercial program models it; the manual's bilinear wedge from the wall toe. Converted to SI.

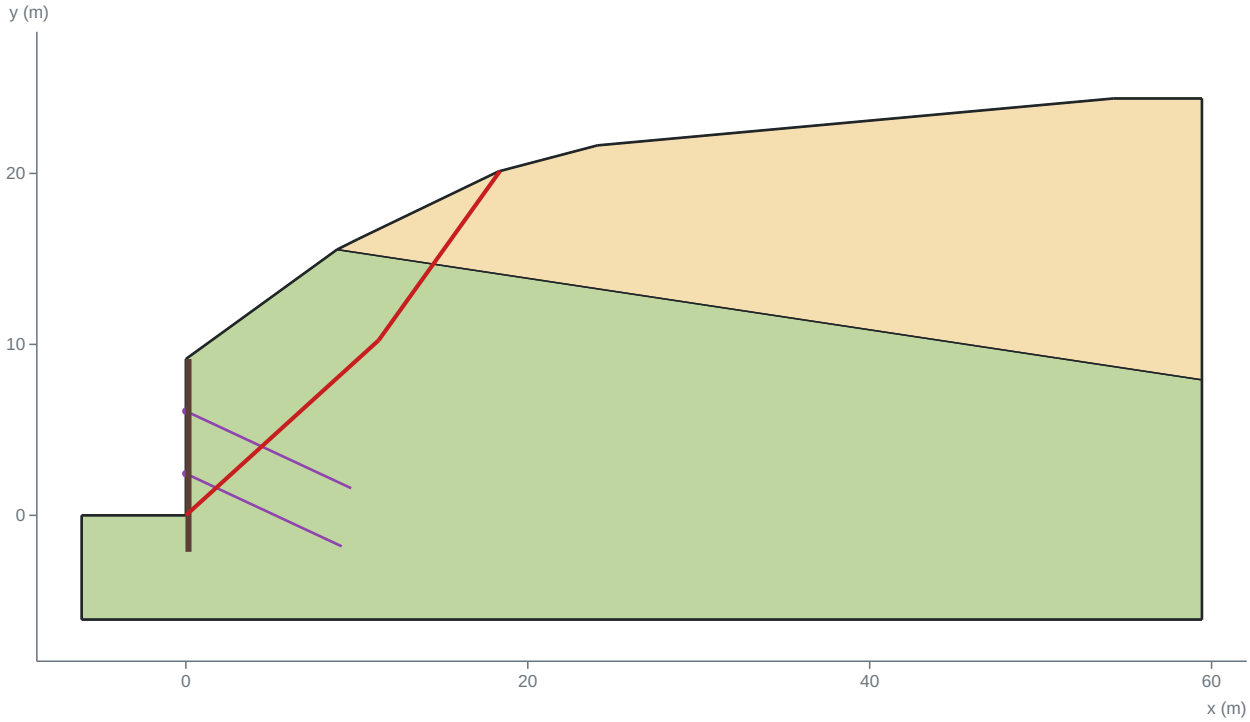


Figure 27. V015: analysed surface (Janbu corrected, FOS = 1.476).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.85	c' = 28.73	24.00°
Layer 2	Mohr-Coulomb	20.42	c' = 14.36	34.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Janbu corrected	1.479	1.476
Janbu simplified	1.446	1.442

The reference results come from design programs quoted in the soil nail manual; no independent value is published.

3.28 V016a — Stepped gabion wall, equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

4 m gabion wall of 1 m cubes with the steps facing outwards on a sloping two-layer profile, dry; mesh as an equivalent cohesion (59.7 kPa); the reference circle.

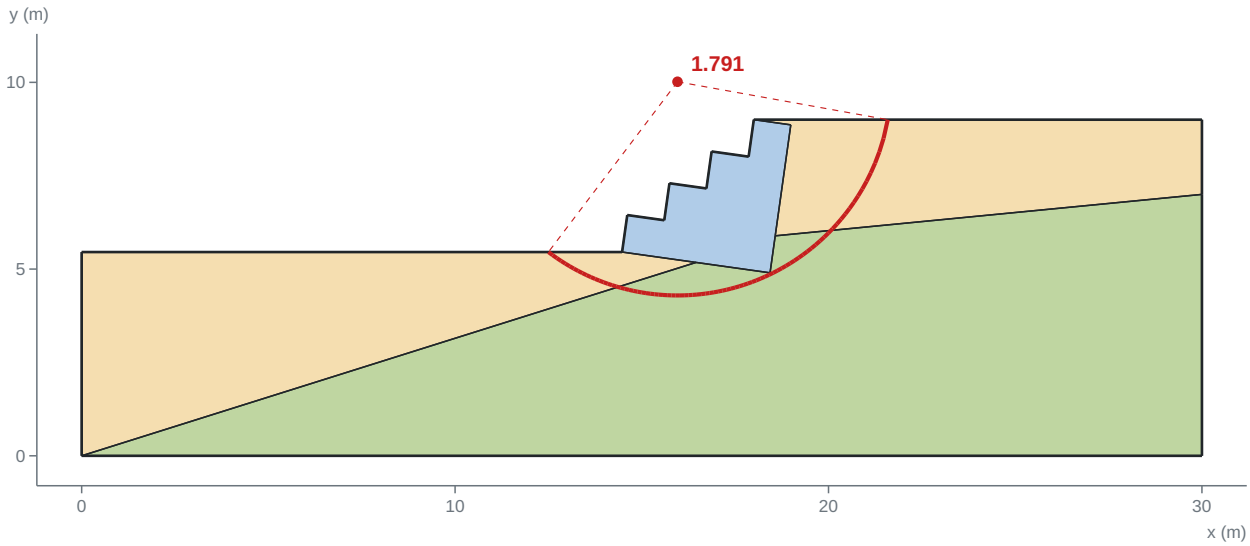


Figure 28. V016a: analysed circle (Bishop simplified, FOS = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 59.70	42.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.787–1.790	1.791
Spencer	1.791	1.797

The paper gives its results in figures only; the comparison is with the other programs.

3.29 V016b — Stepped gabion wall, mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V016a with the mesh as geosynthetic supports (T = 100 kN/m) at the course interfaces.

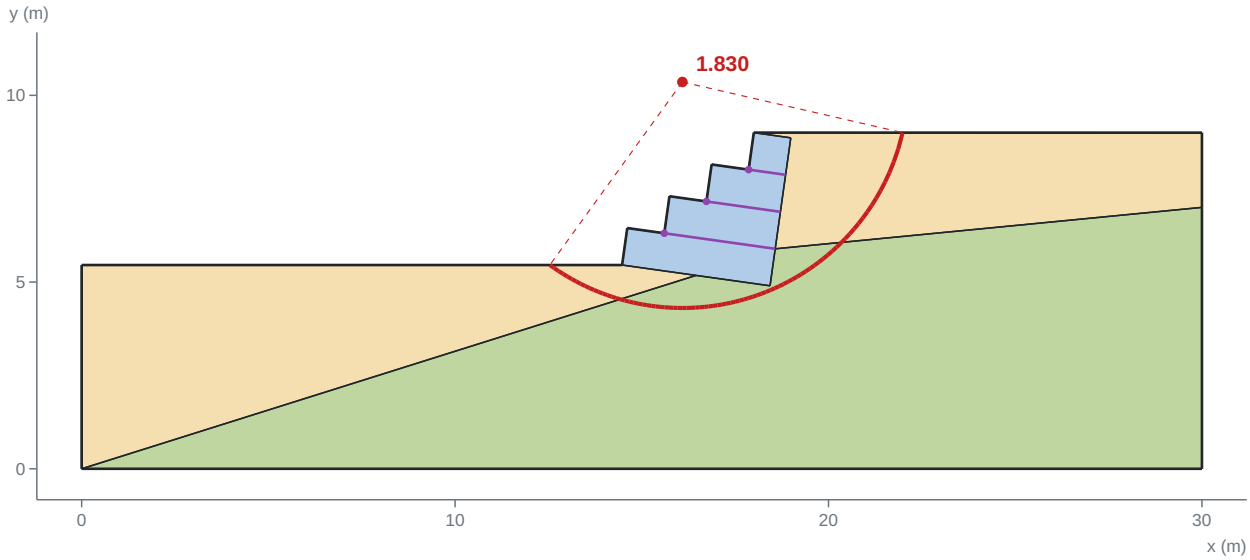


Figure 29. V016b: analysed circle (Bishop simplified, FOS = 1.830).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 0.00	42.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.830–1.835	1.830
Spencer	1.839	1.835

The paper gives its results in figures only; the comparison is with the other programs.

3.30 V016c — Gabion wall with weak joint layers

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V016 wall with 6 cm weak bands at the course interfaces (φ = 37.8°, c = 20.4 kPa from the joint tensile strength); the V016a reference circle, which passes below the wall and crosses no joint.

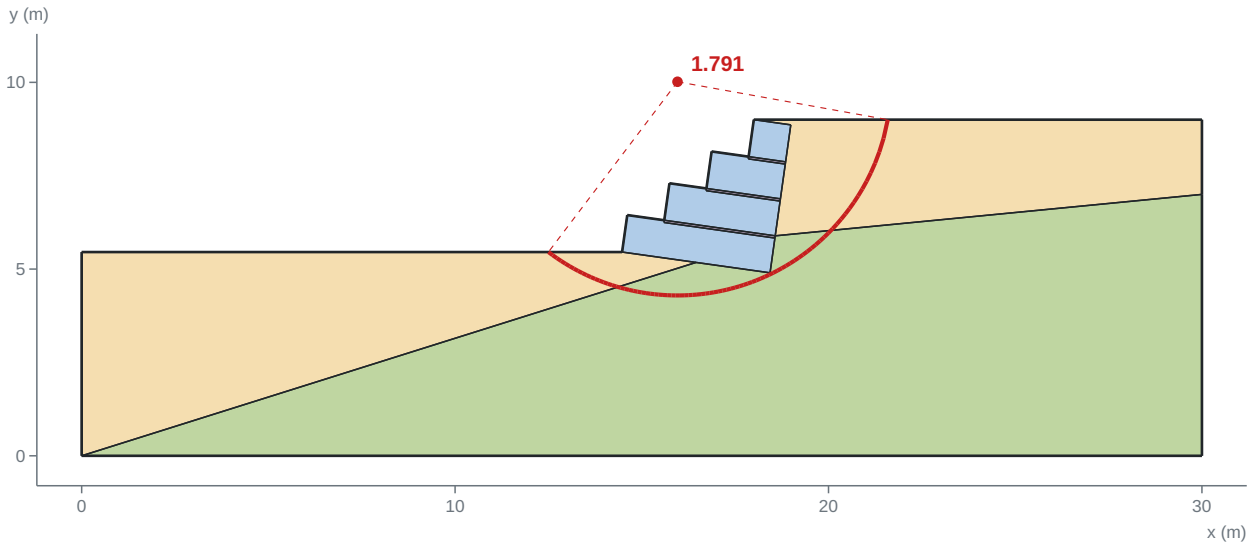


Figure 30. V016c: analysed circle (Bishop simplified, FOS = 1.791).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 5.00	30.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	25.00°
Gabion wall	Mohr-Coulomb	23.00	c' = 59.70	42.00°
Weak joint	Mohr-Coulomb	23.00	c' = 20.40	37.80°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.790	1.791

The paper gives its results in figures only; the comparison is with the other programs.

3.31 V017a — Ontario gabion wall (Cao et al. 2016), equivalent cohesion

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

6 m battered gabion wall retaining a 30° backfill with a 12 kPa crest surcharge and a water table in the retained slope; the gabion mesh as an equivalent cohesion (100 kPa); the reference circle. Courses from the base 4, 3, 2.5, 2, 1.5 and 1 m wide.

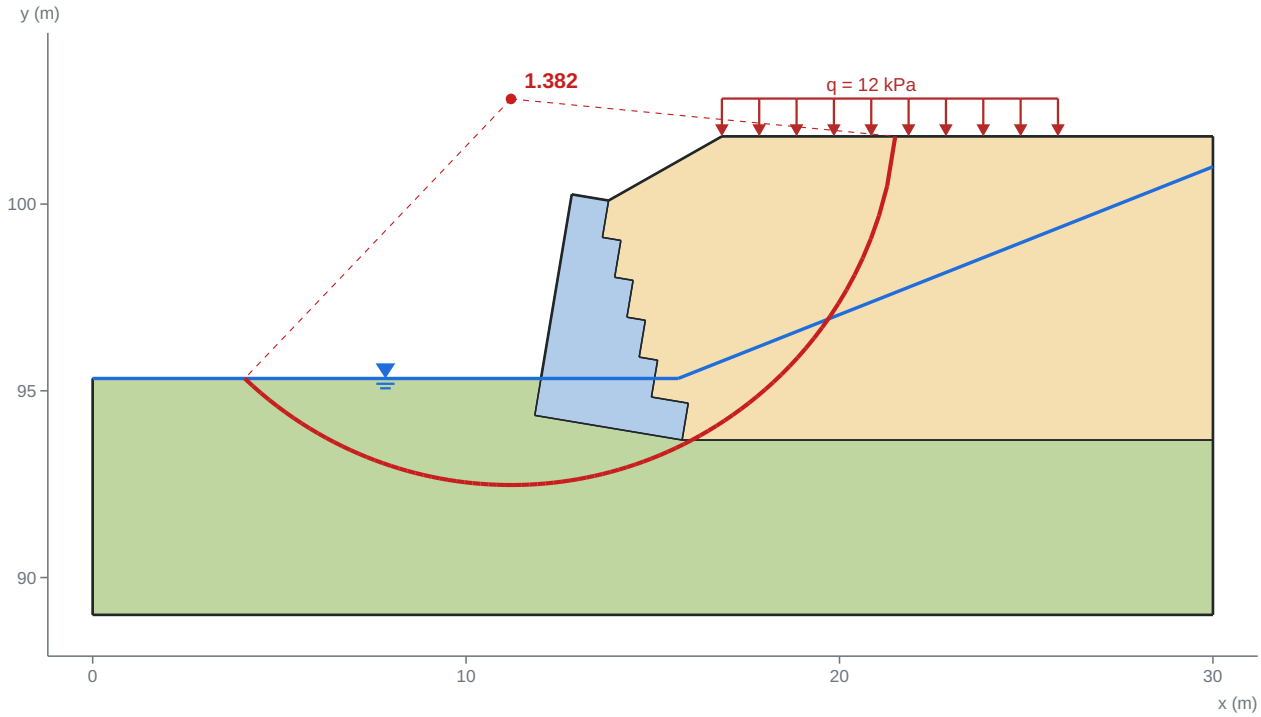


Figure 31. V017a: analysed circle (Bishop simplified, FOS = 1.382).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c' = 100.0	45.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.373	1.382
Spencer	1.386	1.398

The paper gives its results in figures only; the comparison is with the other programs.

3.32 V017b — Ontario gabion wall (Cao et al. 2016), mesh as geosynthetic supports

Cao et al. (2016); Grodecki (2017).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V017a with the steel mesh as geosynthetic supports at every course interface instead of the equivalent cohesion.

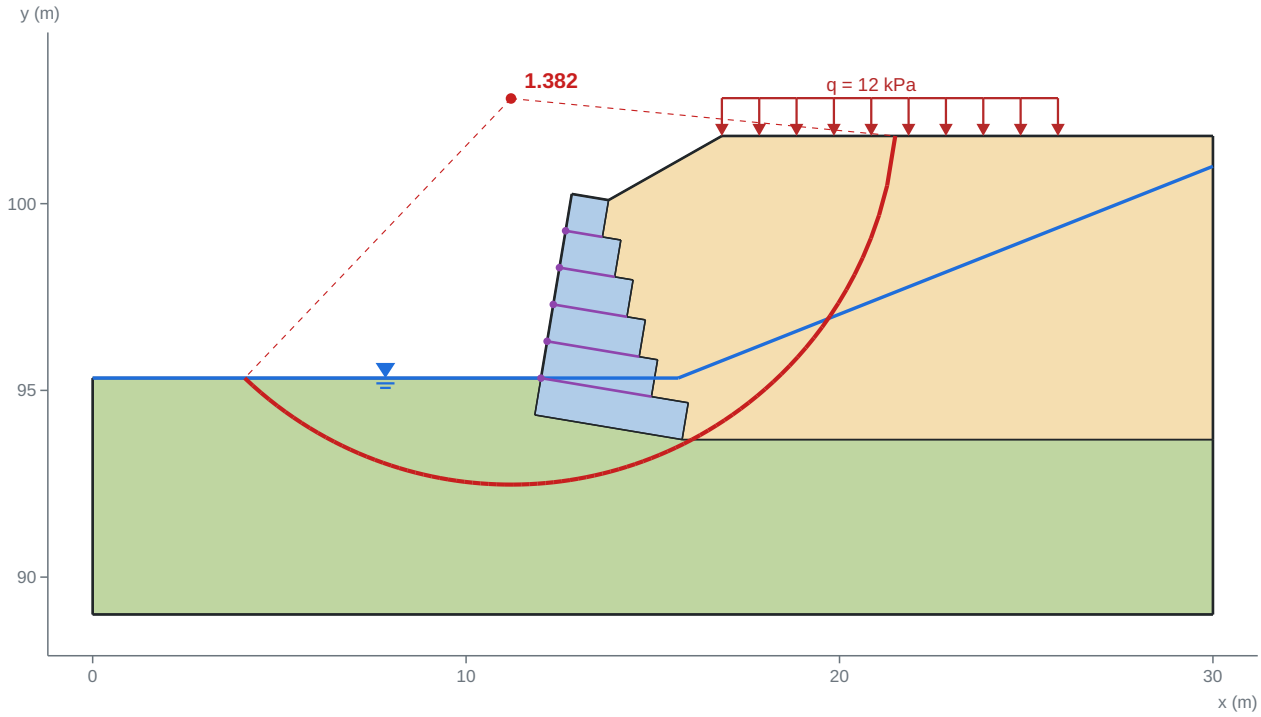


Figure 32. V017b: analysed circle (Bishop simplified, FOS = 1.382).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	21.00	c' = 0.00	32.00°
Soil #2	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Gabion wall	Mohr-Coulomb	20.00	c' = 0.00	45.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.378	1.382

The paper gives its results in figures only; the comparison is with the other programs.

3.33 V018 — Chen & Shao (1988) frictionless 60° slope, Prandtl mechanism

Chen, Z. & Shao, C. (1988). Evaluation of minimum factor of safety in slope stability analysis. Can. Geotech. J. 25(4), 735–748.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless, frictionless 10 m slope at 60° under the critical strip load (149.31 kPa) on the Prandtl surface (wedge, fan and passive wedge); theoretical FOS 1.0.

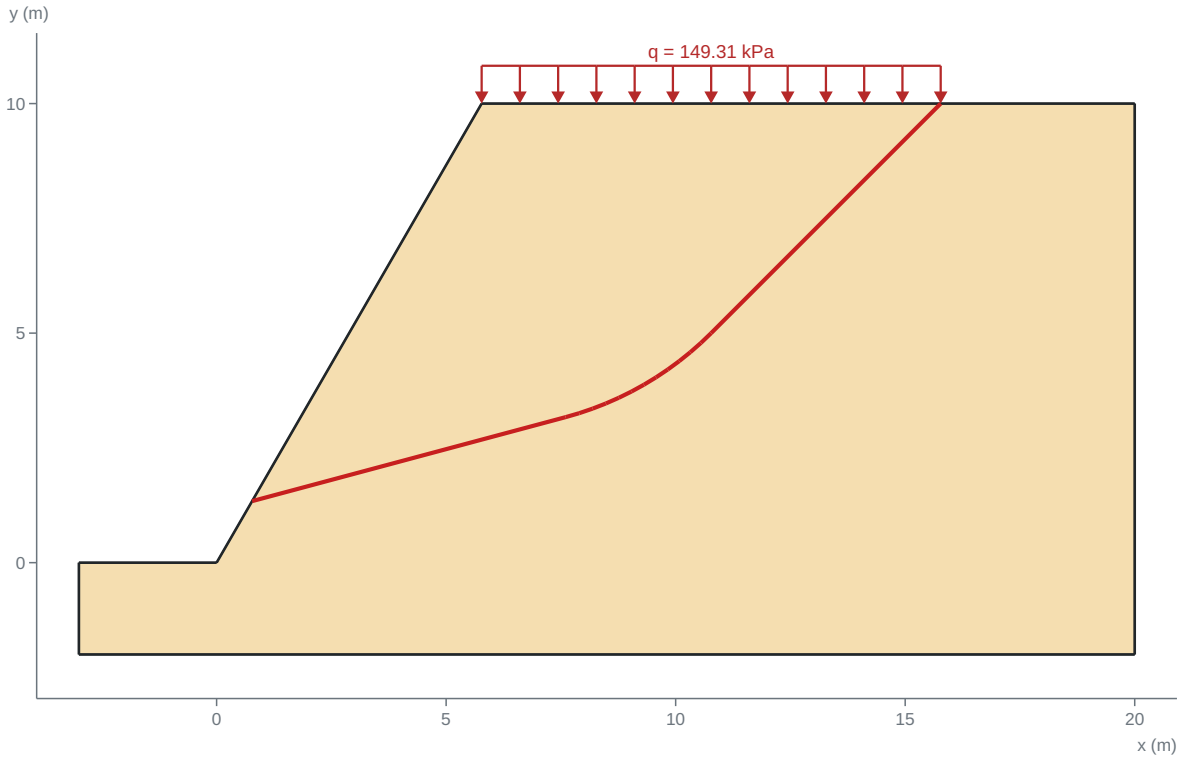


Figure 33. V018: analysed surface (Spencer, FOS = 1.052).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	0.00	c' = 49.00	0.00°

Published value	Method	Source	cSlope
1.05	Spencer	Chen & Shao (1988)	1.052

cSlope analysis	Other programs	cSlope
Spencer	1.051	1.052
Morgenstern-Price (specified f(x))	1.009–1.017	1.008

3.34 V019 — Cheng et al. 11 m homogeneous slope

Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.

Critical circle search · Units: SI (m, kPa, kN/m³)

11 m homogeneous slope (c = 10 kPa, φ = 30°, γ = 20 kN/m³); circular search.

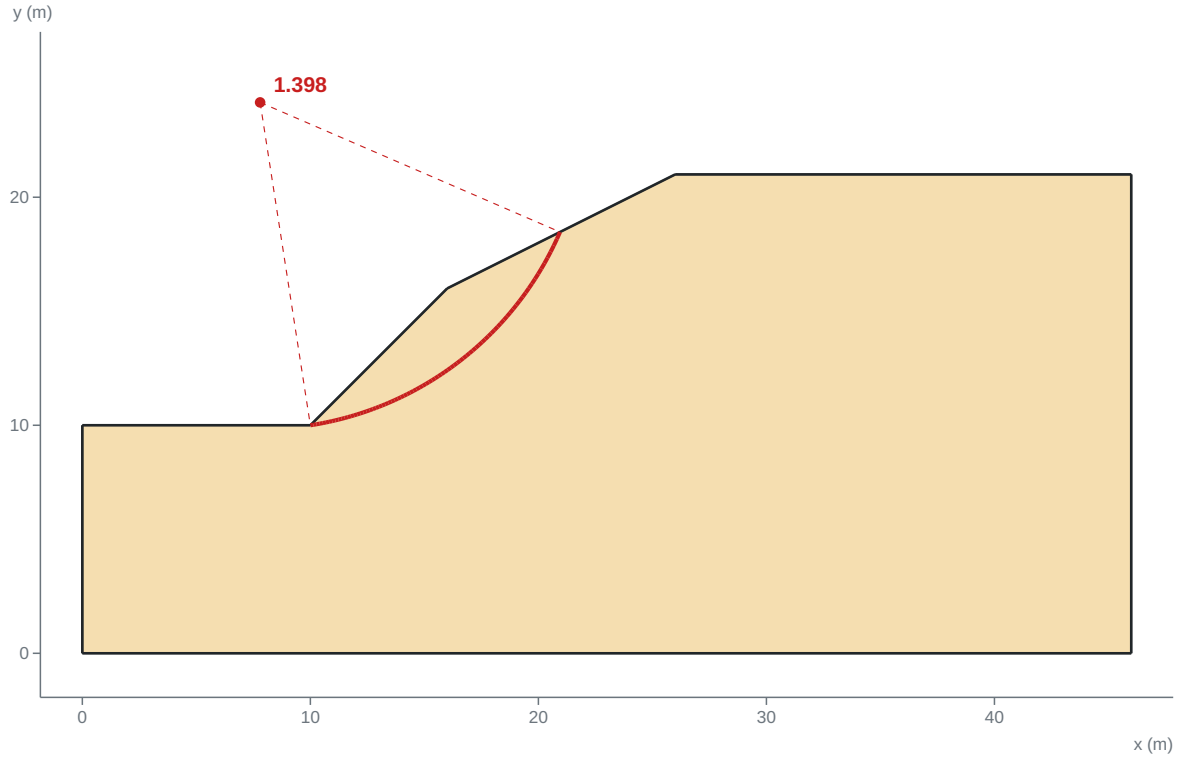


Figure 34. V019: critical circle (Spencer (40 slices, vertical pore-pressure head), FOS = 1.398).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c' = 10.00	30.00°

Published value	Method	Source	cSlope
1.383	Spencer	Cheng et al.	1.398

cSlope analysis	Other programs	cSlope
Spencer (40 slices, vertical pore-pressure head)	1.380–1.398	1.398
Spencer (vertical pore-pressure head)	–	1.398
Bishop simplified	1.401	1.401

3.35 V020 — Cheng et al. benched slope, global minimum

Cheng, Y.M., Lansivaara, T. & Wei, W.B. (2007). Two-dimensional slope stability analysis by limit equilibrium and strength reduction methods. Computers and Geotechnics 34(3), 137–150.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous benched slope; the global minimum on the mid-lower face by circular search.

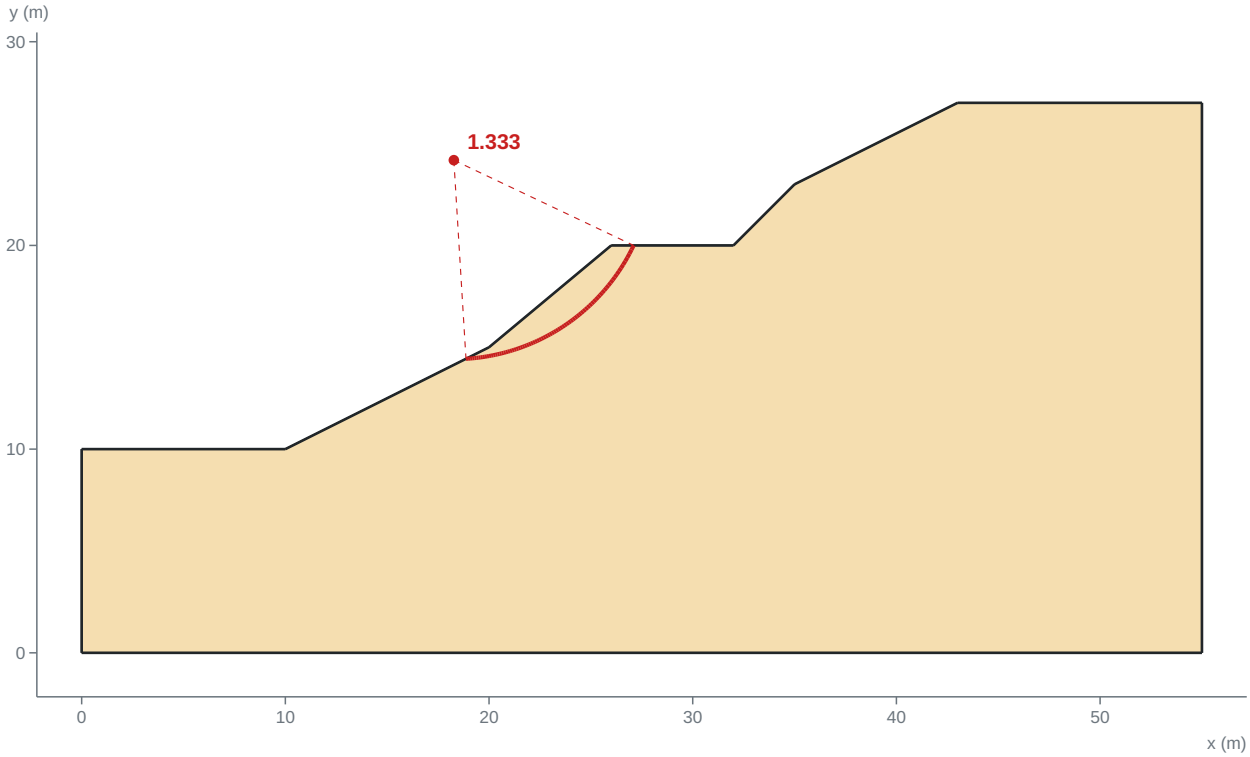


Figure 35. V020: critical circle (Spencer, FOS = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	20.00	c' = 5.00	30.00°

Published value	Method	Source	cSlope
1.33	Spencer	Cheng et al.	1.333

cSlope analysis	Other programs	cSlope
Spencer	1.336–1.338	1.333
Bishop simplified	1.342	1.337

3.36 V021a — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 1, the Congress Street cut as a commercial program models it (Su = 55, 43, 56 kPa under a strengthless sand cap, digitised piezometric line), on the critical circle tangent to the base of clay 2.

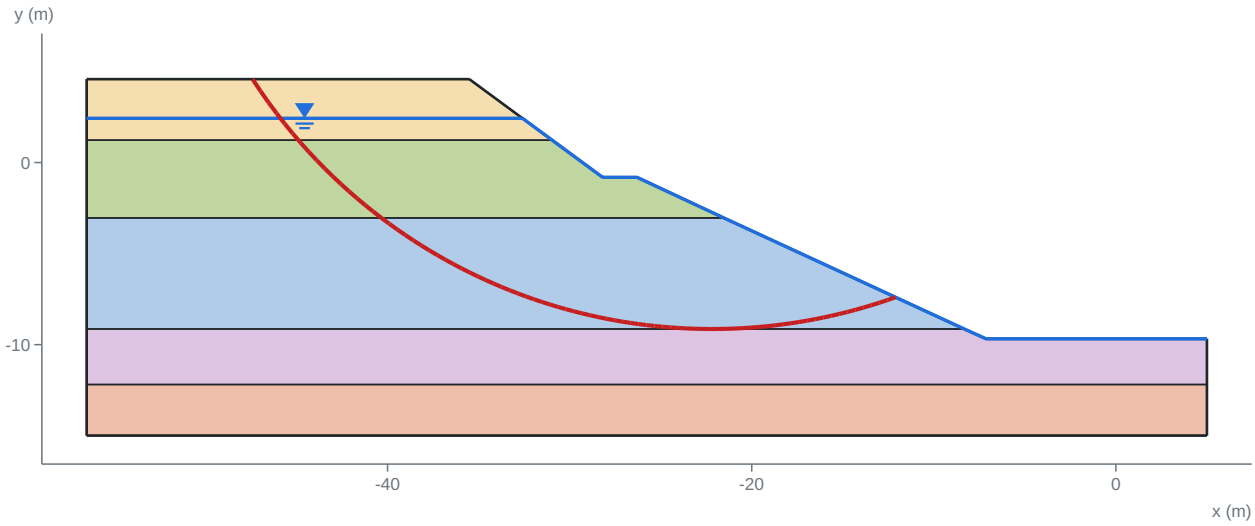


Figure 36. V021a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.128).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
1.1178	Bishop simplified	Chowdhury & Xu	1.128

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.132	1.128
Bishop simplified	–	1.128
Bishop simplified (60 slices)	1.128	1.128

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.246	0.239
β lognormal (Monte Carlo)	–	0.650	0.671

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.253	0.245

3.37 V021b — Chowdhury & Xu (1995) example 1, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V021a on the critical circle tangent to the base of clay 3.

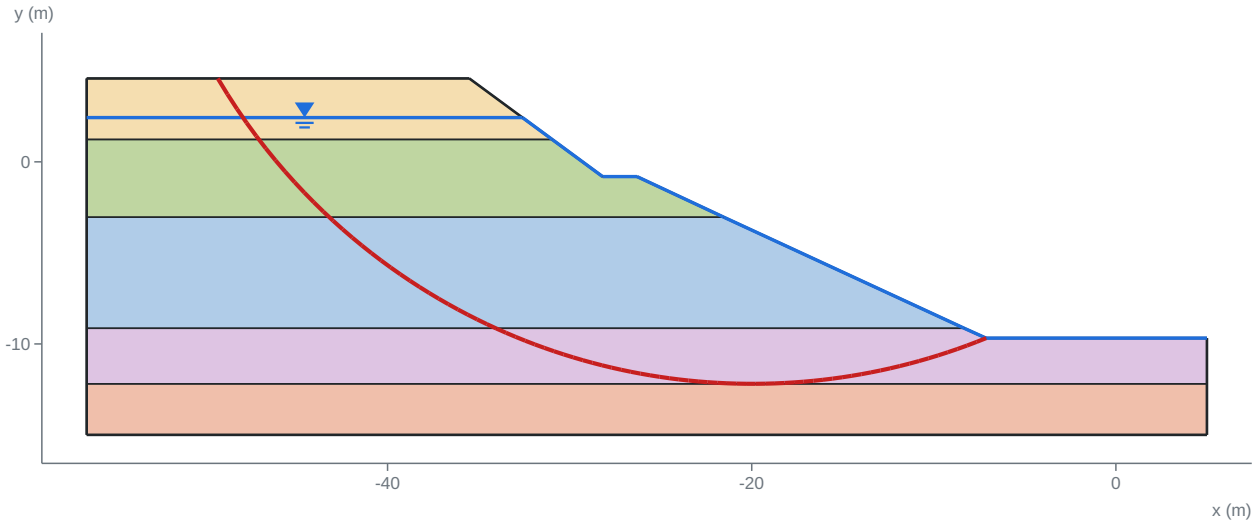


Figure 37. V021b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.110).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 55.00	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 43.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 56.00	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
1.1088	Bishop simplified	Chowdhury & Xu	1.110

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.116	1.110
Bishop simplified	–	1.110
Bishop simplified (60 slices)	1.109	1.110

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.279	0.282
β lognormal (Monte Carlo)	–	0.537	0.553

3.38 V021c — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2 (Su = 68.1, 39.3, 50.8 kPa), circle tangent to the base of clay 2.

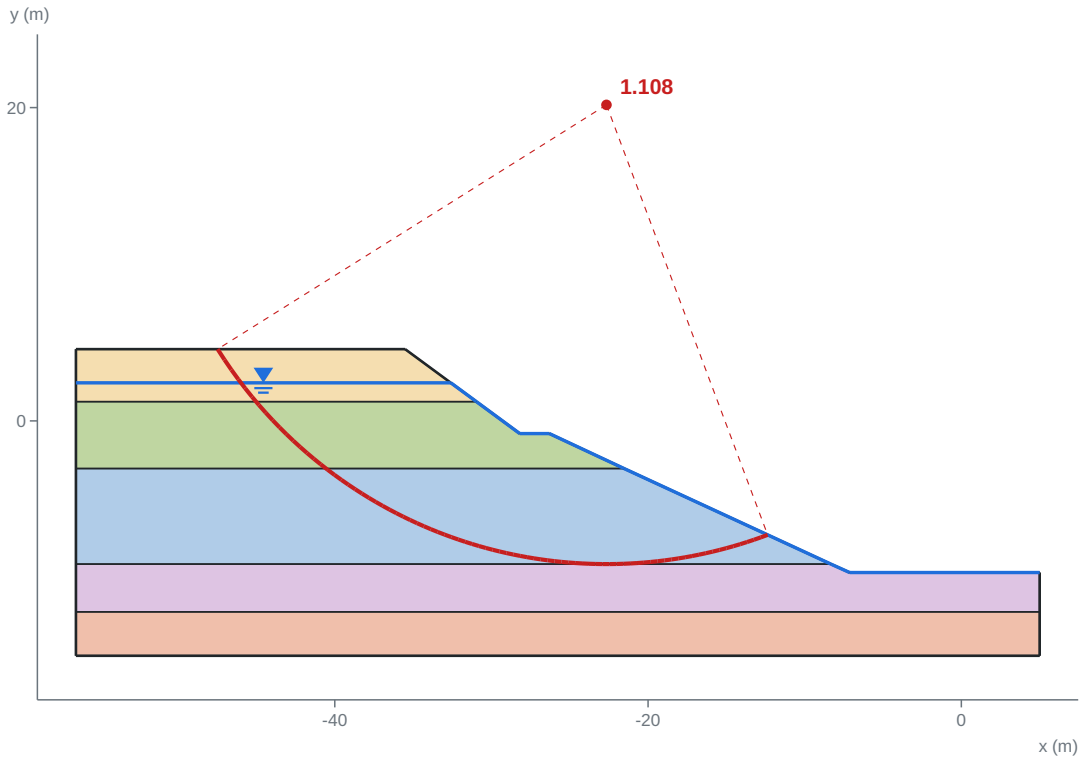


Figure 38. V021c: analysed circle (Morgenstern-Price (half-sine), FOS = 1.108).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
1.1096	Bishop simplified	Chowdhury & Xu	1.108

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.110	1.108
Bishop simplified	1.108	1.108

3.39 V021d — Chowdhury & Xu (1995) example 2, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 2, circle tangent to the base of clay 3.

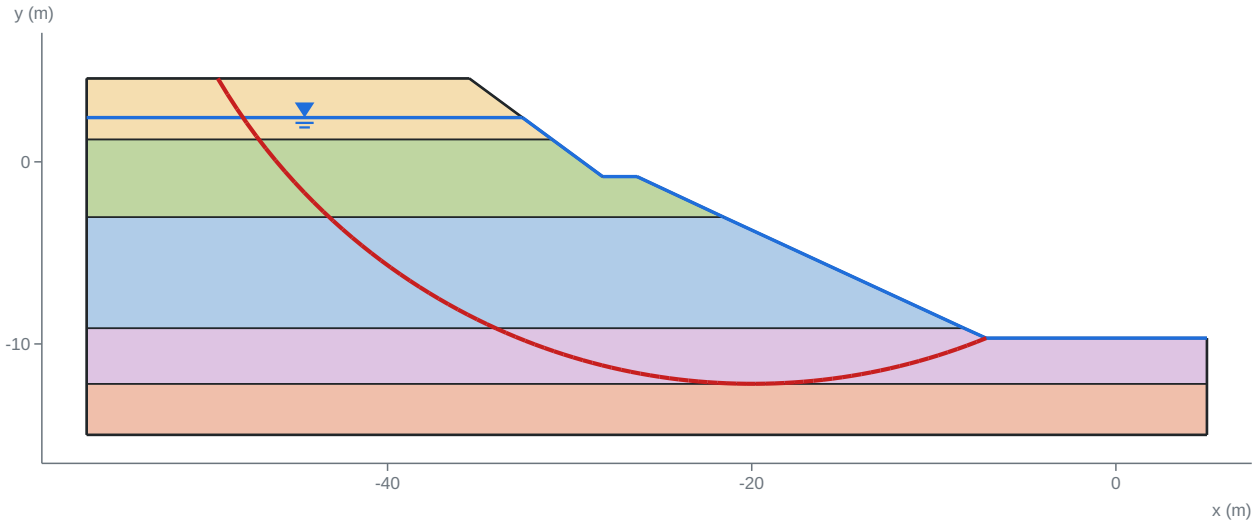


Figure 39. V021d: analysed circle (Morgenstern-Price (half-sine), FOS = 1.058).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 68.10	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 39.30	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 50.80	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
1.0639	Bishop simplified	Chowdhury & Xu	1.058

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.063	1.058
Bishop simplified	1.058	1.058

3.40 V021e — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3 (Su = 136, 80, 102 kPa), circle tangent to the base of clay 2.

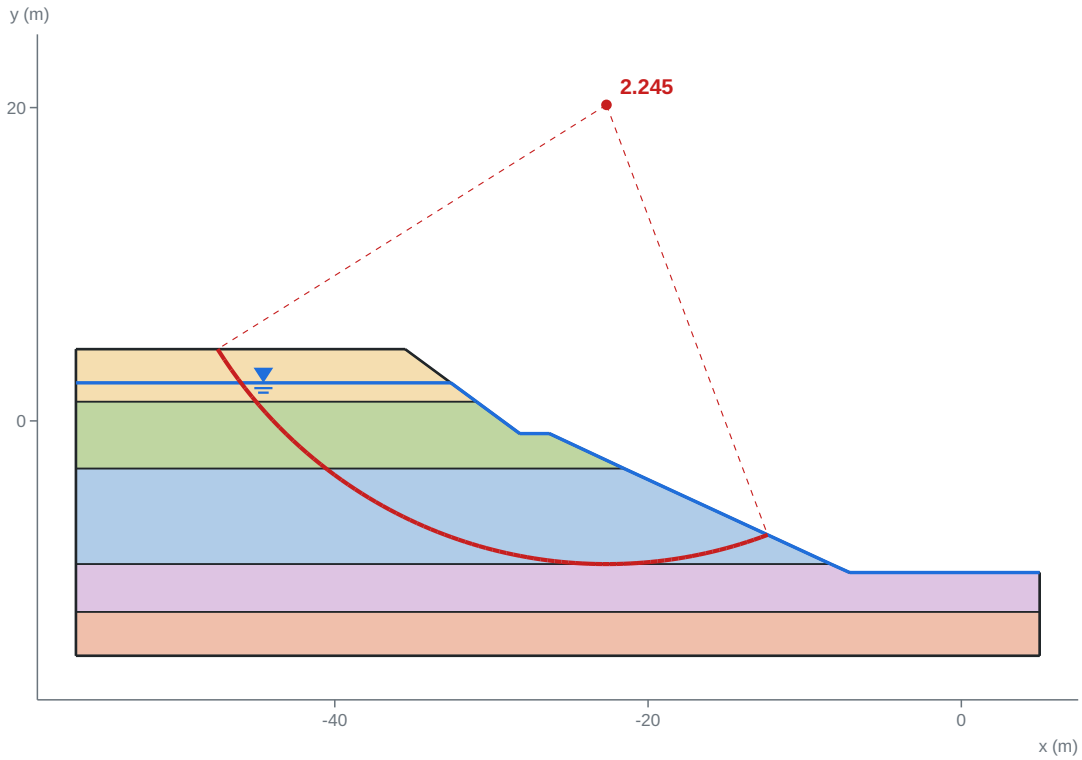


Figure 40. V021e: analysed circle (Morgenstern-Price (half-sine), FOS = 2.245).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
2.2343	Bishop simplified	Chowdhury & Xu	2.245

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	2.248	2.245
Bishop simplified	2.245	2.245

3.41 V021f — Chowdhury & Xu (1995) example 3, Congress Street cut, circle tangent to the base

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 3, circle tangent to the base of clay 3.

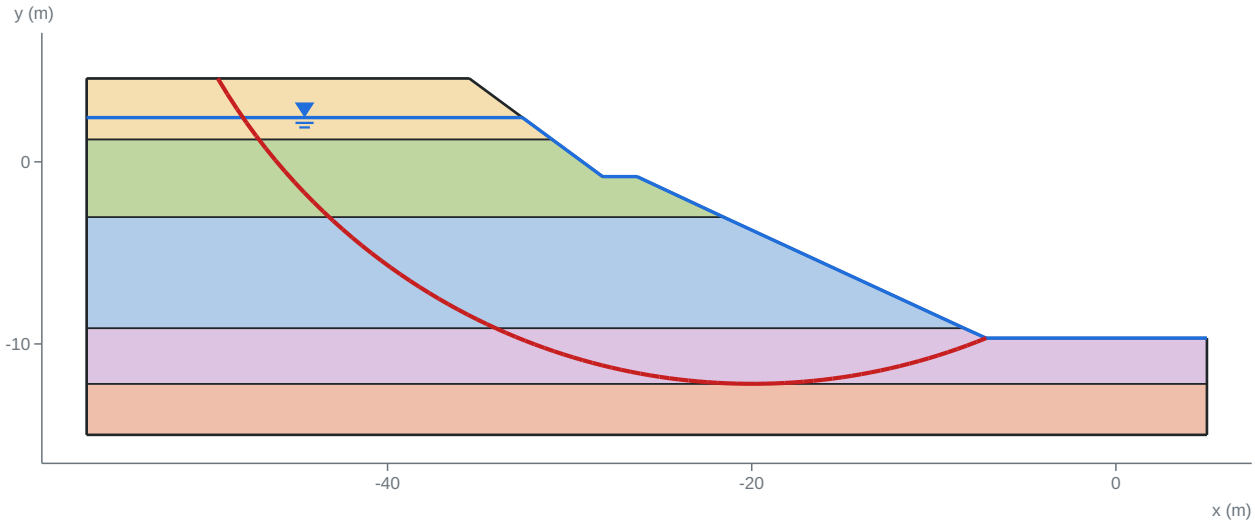


Figure 41. V021f: analysed circle (Morgenstern-Price (half-sine), FOS = 2.128).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Undrained (φ = 0)	21.00	Su = 136.0	0°
Clay 2	Undrained (φ = 0)	22.00	Su = 80.00	0°
Clay 3	Undrained (φ = 0)	22.00	Su = 102.0	0°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
2.1396	Bishop simplified	Chowdhury & Xu	2.128

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	2.138	2.128
Bishop simplified	2.128	2.128

3.42 V021g — Chowdhury & Xu (1995) example 4, Congress Street cut, circle tangent to the bas

Chowdhury & Xu (1995); Ireland (1954).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 4 (c-φ clays with the digitised piezometric line), circle tangent to the base of clay 2.

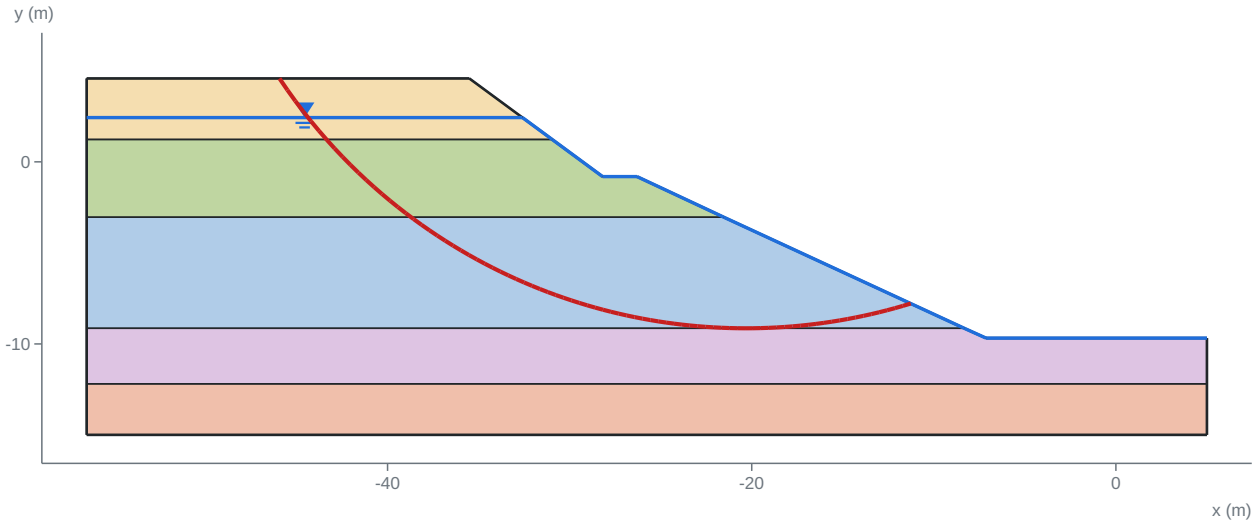


Figure 42. V021g: analysed circle (Morgenstern-Price (half-sine), FOS = 1.428).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Undrained (φ = 0)	22.00	Su = 0.00	0°
Clay 1	Mohr-Coulomb	17.00	c' = 55.00	5.00°
Clay 2	Mohr-Coulomb	22.00	c' = 43.00	7.00°
Clay 3	Mohr-Coulomb	22.00	c' = 56.00	8.00°
Bed	Mohr-Coulomb	22.00	c' = 200.0	35.00°

Published value	Method	Source	cSlope
1.4239	Bishop simplified	Chowdhury & Xu	1.428

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.422	1.428
Bishop simplified	–	1.428

3.43 V022a — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

Chowdhury & Xu example 5: embankment fill (c′ = 10 kPa, φ′ = 12°) on soft clay (Su = 40 kPa), the geometry as a commercial program's verification manual draws it; circle tangent to the fill–clay interface.

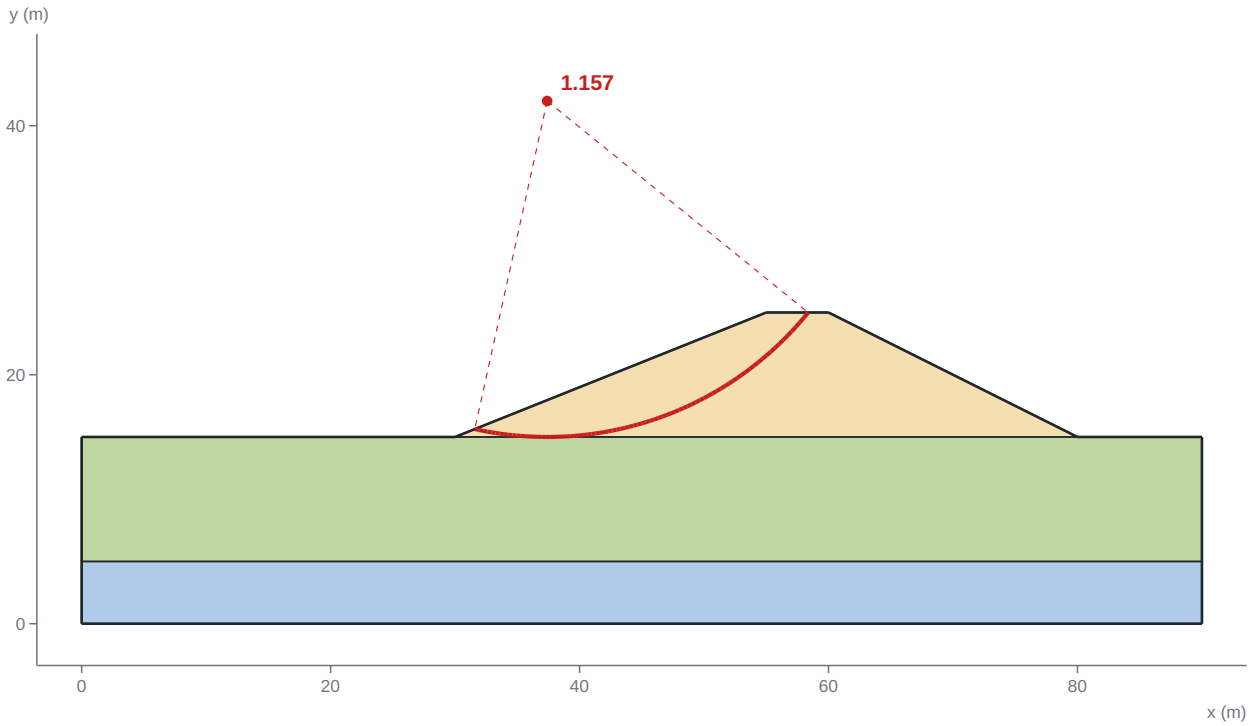


Figure 43. V022a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.157).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Published value	Method	Source	cSlope
1.1625	Bishop simplified	Chowdhury & Xu	1.159

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.158	1.157
Bishop simplified	–	1.159
Bishop simplified (60 slices)	1.160	1.159

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.206–0.212	0.206
β lognormal (Monte Carlo)	–	0.799–0.800	0.812
β lognormal (Taylor series)	–	0.787	0.788

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.210	0.208

3.44 V022b — Chowdhury & Xu (1995) example 5, embankment on soft clay, circle tangent to y =

Chowdhury & Xu (1995).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V022a with the circle tangent to the base of the soft clay.

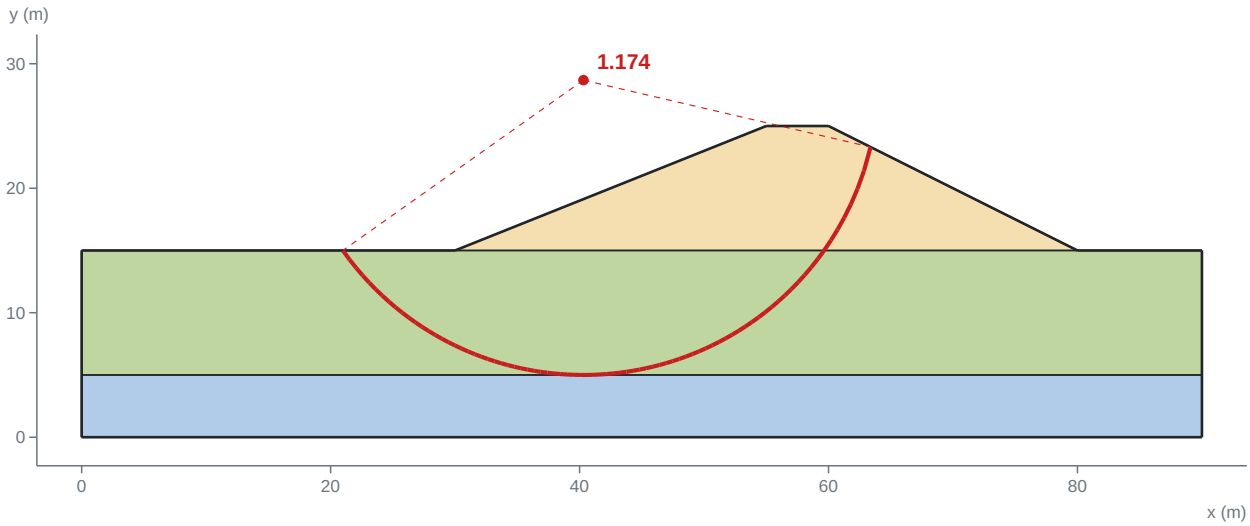


Figure 44. V022b: analysed circle (Morgenstern-Price (half-sine), FOS = 1.174).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 10.00	12.00°
Layer 2	Undrained (φ = 0)	18.00	Su = 40.00	0°
Hard layer	Undrained (φ = 0)	20.00	Su = 1000.0	0°

Published value	Method	Source	cSlope
1.1479	Bishop simplified	Chowdhury & Xu (1995)	1.177

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.178	1.174
Bishop simplified	1.185	1.177

Chowdhury & Xu's 1.1479 is about 2.5 % below the other programs on this family of circles.

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.199–0.209	0.212
β lognormal (Monte Carlo)	–	0.789–0.820	0.789
β lognormal (Taylor series)	–	0.798	0.802

Probabilistic analysis on the same surface: 3 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
Pf, share FOS < 1 (Monte Carlo)	–	0.214	0.216

3.45 V023 — LASH terminal trench, Duncan's Taylor series example

Duncan (2000), J. Geotech. Geoenviron. Eng. 126(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

Underwater trench in San Francisco Bay Mud: Su = 100 psf at el. −20 ft increasing 9.8 psf/ft, γ = 100 pcf, water at el. 0. Duncan's surface as a circle, Spencer; σγ = 3.3 pcf and σ of the Su gradient 1.2 psf/ft (a commercial program's Table 29.2). Converted to SI.

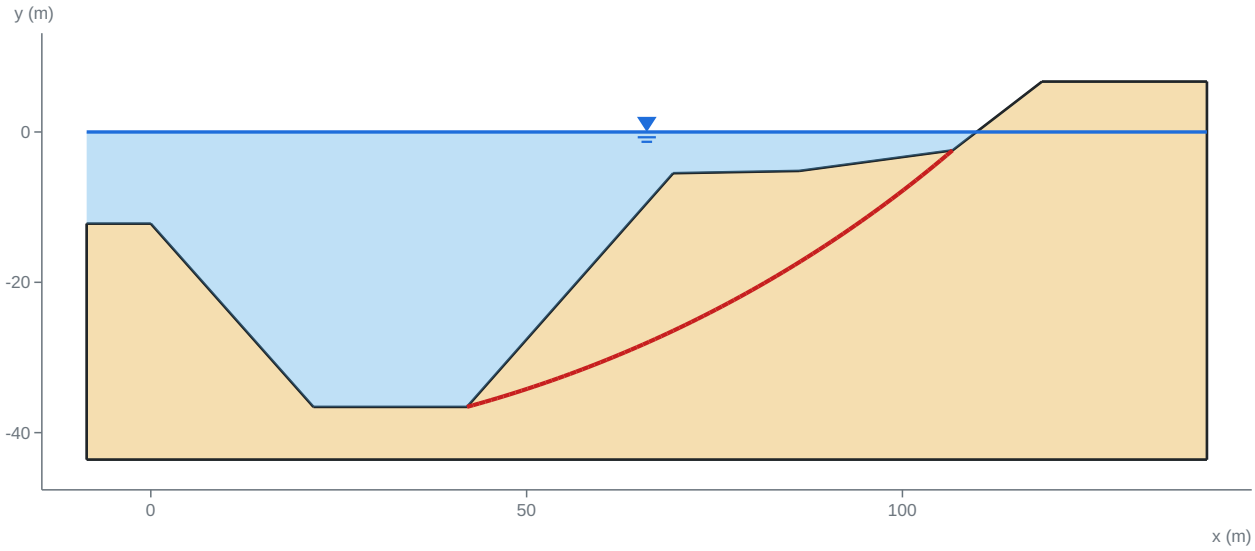


Figure 45. V023: analysed circle (Spencer, FOS = 1.138).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
San Francisco Bay Mud	Undrained (φ = 0)	15.71	Su = 4.79 at y = -6.10, +1.54 kPa/m below	0°

Published value	Method	Source	cSlope
1.17	not stated	Duncan (2000)	–

cSlope analysis	Other programs	cSlope
Spencer	1.145	1.138

Duncan (2000) prints two decimals; the other programs give 1.145 on the same input.

Probabilistic analysis on the same surface: 2 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Taylor series)	0.900 Duncan (2000)	–	0.892
Pf lognormal (Taylor series)	0.180 Duncan (2000)	–	0.186

3.46 V024 — Dam on a layered foundation with artesian uplift, piezometric line (D&W fig. 6.39)

Duncan & Wright, Soil Strength and Slope Stability, figs. 6.39–6.40.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a narrow clay core on 30 ft of clay over permeable sand; a commercial program's piezometric line from the underseepage; the circle tangent to el. 197 ft. Converted to SI.

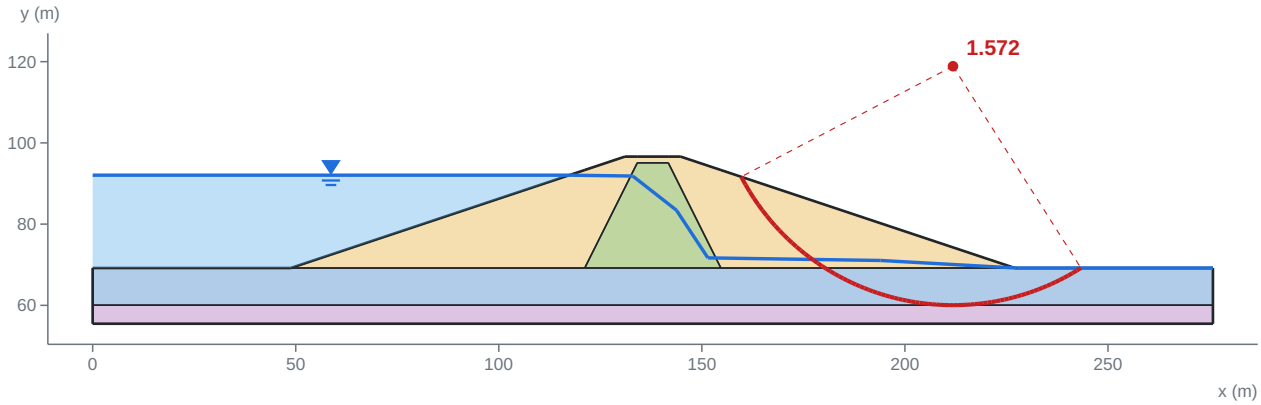


Figure 46. V024: analysed circle (Bishop simplified, FOS = 1.572).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Outer shell	Mohr-Coulomb	19.64	c' = 0.00	34.00°
Clay core	Mohr-Coulomb	19.16	c' = 4.79	26.00°
Foundation clay	Mohr-Coulomb	19.32	c' = 0.00	24.00°
Foundation sand	Mohr-Coulomb	19.95	c' = 0.00	32.00°

Published value	Method	Source	cSlope
1.57	Bishop simplified	Duncan & Wright	1.572

cSlope analysis	Other programs	cSlope
Bishop simplified	1.563–1.572	1.572
Spencer	1.557	1.562

3.47 V025 — Duncan & Wright fig. 6.38, piezometric line

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.38.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope with a nearly flat piezometric line; critical circle search. Converted to SI; the water line was traced from a figure.

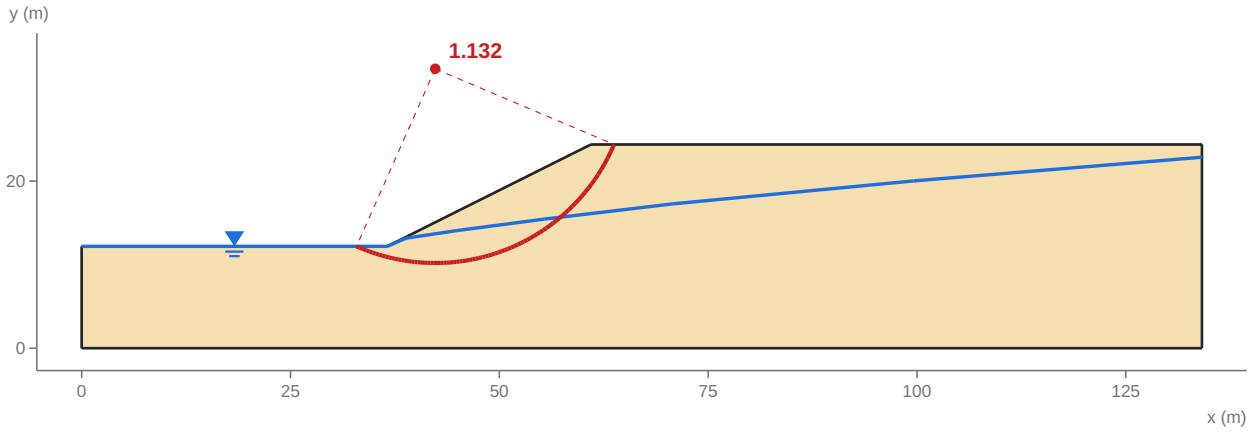


Figure 47. V025: critical circle (Bishop simplified, FOS = 1.132).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.64	c' = 9.58	20.00°

Published value	Method	Source	cSlope
1.141	not stated	Duncan & Wright	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.142	1.132
Spencer	1.142	1.132

3.48 V026 — Duncan & Wright fig. 7.12, embankment on clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.12.

Critical circle search · Units: imperial (ft, psf, pcf)

Sand embankment on a thick undrained clay foundation; critical circle search.

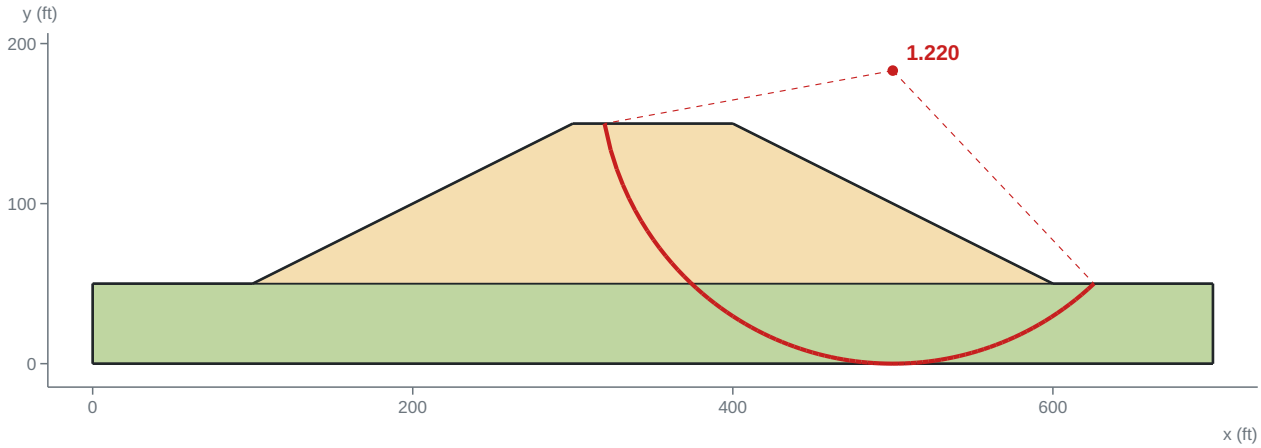


Figure 48. V026: critical circle (Bishop simplified, FOS = 1.220).

Material	Model	γ (pcf)	Strength (psf)	φ
Sand	Mohr-Coulomb	140.0	c' = 0.00	40.00°
Clay	Mohr-Coulomb	140.0	c' = 2500.0	0.00°

Published value	Method	Source	cSlope
1.22	Bishop simplified	Duncan & Wright	1.220
1.19	Spencer	Duncan & Wright	1.195

cSlope analysis	Other programs	cSlope
Bishop simplified	1.228	1.220
Spencer	–	1.195
Spencer, non-circular	1.190	1.170

3.49 V027 — Duncan & Wright fig. 14.3, undrained clay

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.3.

Critical circle search · Units: imperial (ft, psf, pcf)

Homogeneous undrained clay slope (φ = 0); critical circle search.

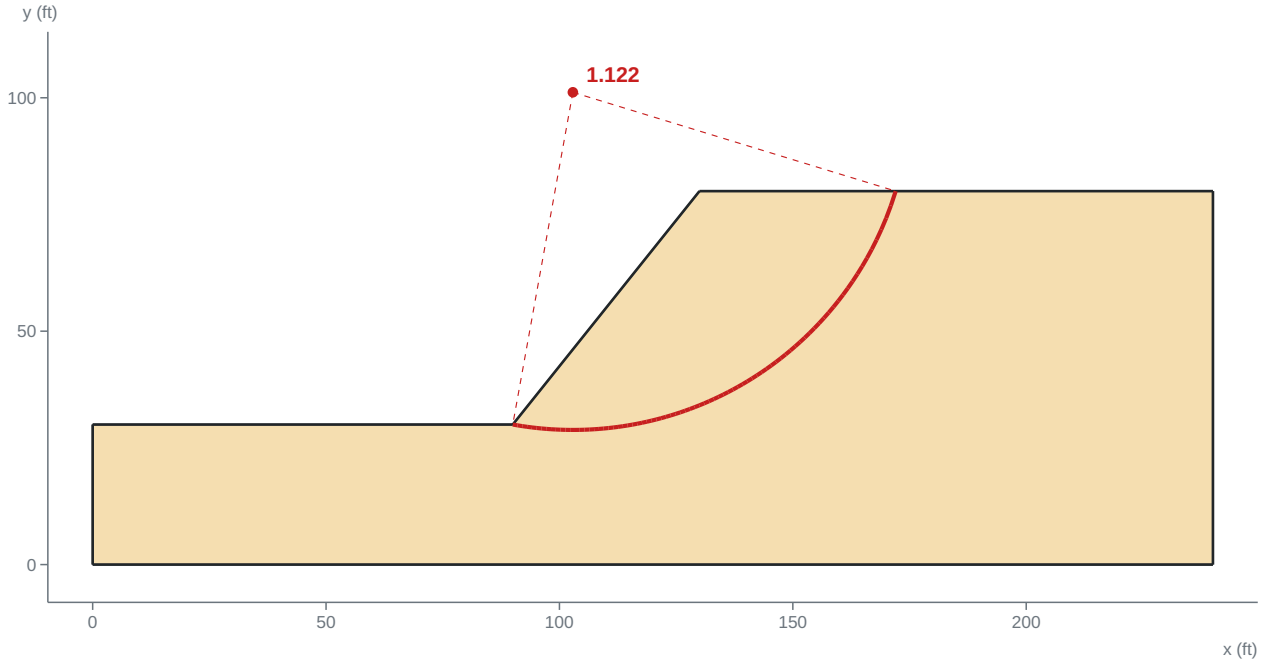


Figure 49. V027: critical circle (Bishop simplified, FOS = 1.122).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Published value	Method	Source	cSlope
1.124	Spencer	Duncan & Wright	1.132

cSlope analysis	Other programs	cSlope
Bishop simplified	1.126	1.122
Spencer	–	1.132

3.50 V028 — Duncan & Wright fig. 14.3, undrained clay on a 46.5 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V027 φ = 0 clay slope with a 46.5 ft thick foundation; critical-circle search (the minimum of the toe and base-tangent circles of Tables 78.4–78.5).

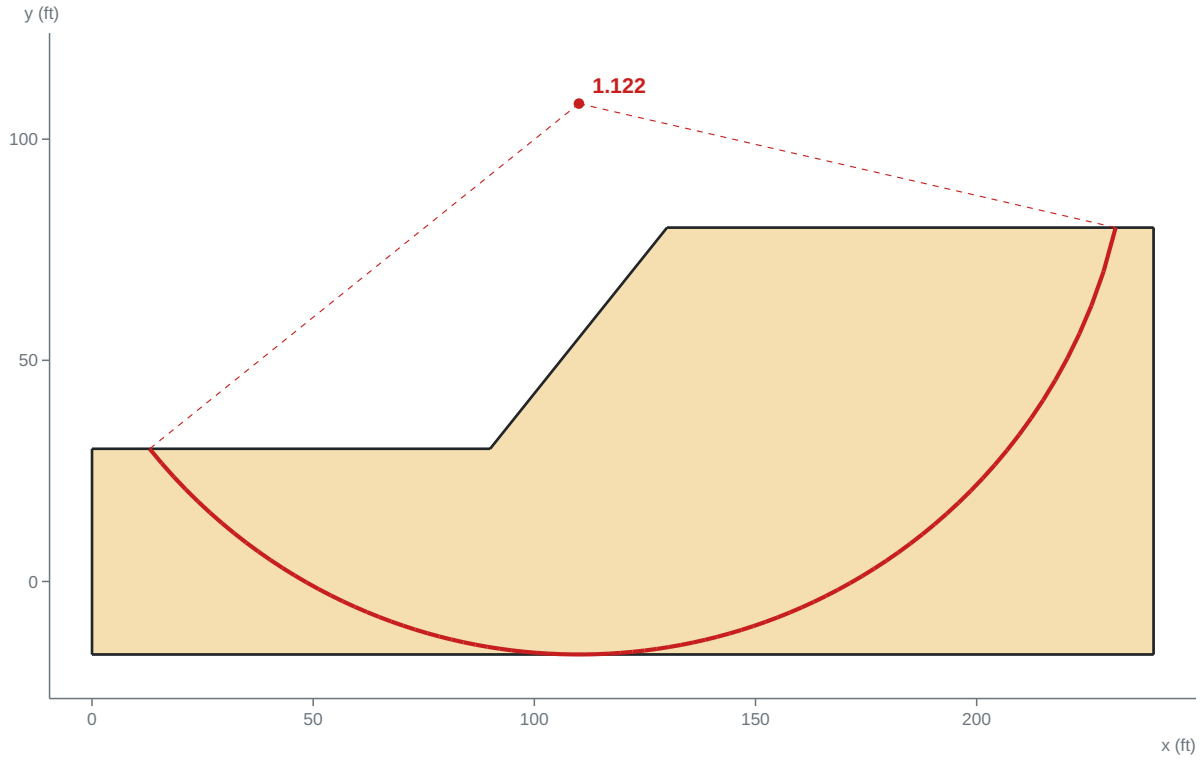


Figure 50. V028: critical circle (Bishop simplified, FOS = 1.122).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Published value	Method	Source	cSlope
1.124	Spencer	Duncan & Wright	1.122

cSlope analysis	Other programs	cSlope
Bishop simplified	1.126	1.122
Spencer	–	1.122

3.51 V029 — Duncan & Wright fig. 14.3, undrained clay on a 60 ft foundation

Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

As V028 with a 60 ft thick foundation (Tables 78.6–78.7).

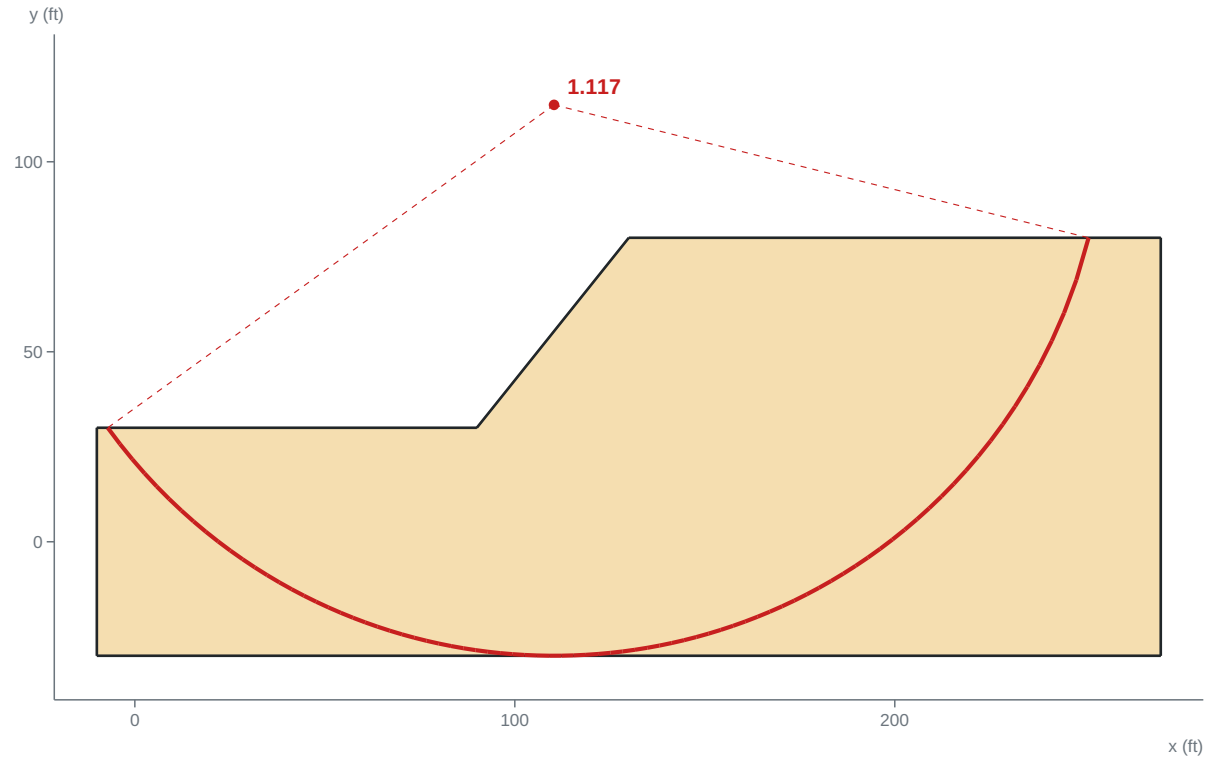


Figure 51. V029: critical circle (Bishop simplified, FOS = 1.117).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay	Mohr-Coulomb	100.0	c' = 1000.0	0.00°

Published value	Method	Source	cSlope
1.119	Spencer	Duncan & Wright	1.117

cSlope analysis	Other programs	cSlope
Bishop simplified	1.125	1.117
Spencer	–	1.117

3.52 V030 — Duncan & Wright fig. 14.4

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.4.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless fill on undrained clay; critical circle search.

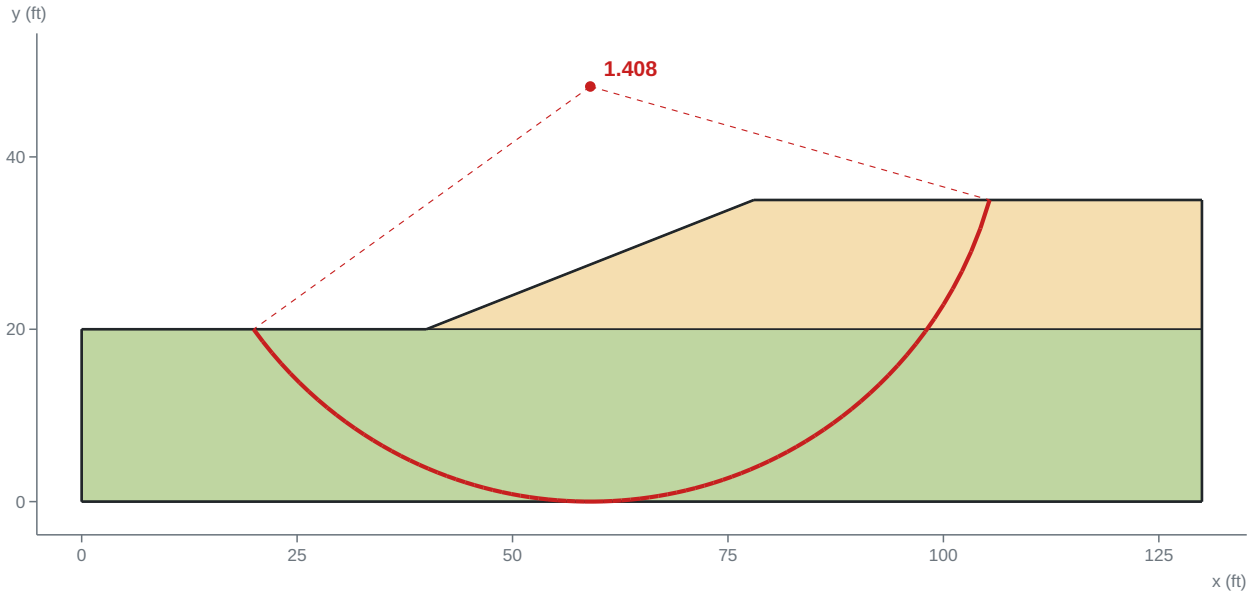


Figure 52. V030: critical circle (Bishop simplified, FOS = 1.408).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	120.0	c' = 0.00	30.00°
Clay	Mohr-Coulomb	120.0	c' = 450.0	0.00°

Published value	Method	Source	cSlope
1.40	Spencer	Duncan & Wright	1.399

cSlope analysis	Other programs	cSlope
Bishop simplified	1.412	1.408
Spencer	–	1.399

3.53 V031a — Duncan & Wright fig. 14.5, R = 87 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment over alternating cohesive and frictional layers; circle of radius 87 ft.

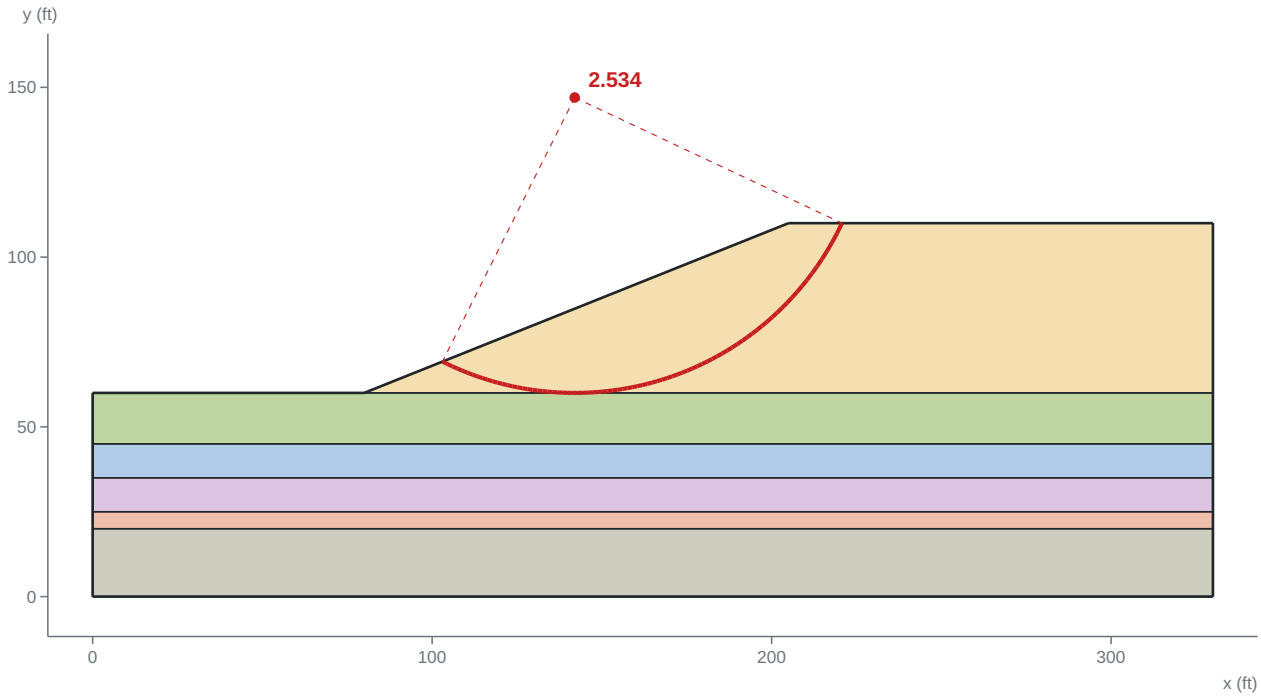


Figure 53. V031a: analysed circle (Bishop simplified, FOS = 2.534).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c' = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c' = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c' = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c' = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c' = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c' = 600.0	0.00°

Published value	Method	Source	cSlope
2.56	Spencer	Duncan & Wright	2.531

cSlope analysis	Other programs	cSlope
Bishop simplified	2.549	2.534
Spencer	2.545	2.531

3.54 V031b — Duncan & Wright fig. 14.5, R = 102 ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.5.

Fixed circle · Units: imperial (ft, psf, pcf)

As V031a with a deeper circle of radius 102 ft, reaching the weak clay layers.

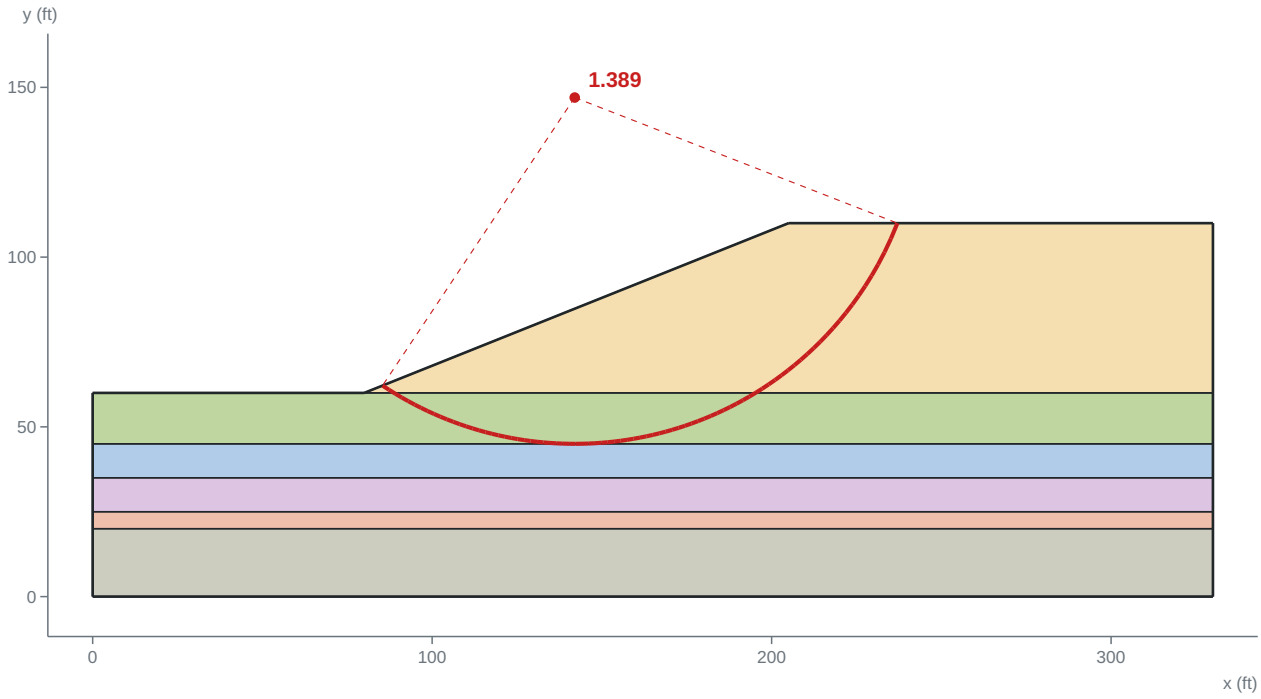


Figure 54. V031b: analysed circle (Bishop simplified, FOS = 1.389).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	120.0	c' = 1.00	35.00°
L60	Mohr-Coulomb	110.0	c' = 950.0	0.00°
L45	Mohr-Coulomb	122.0	c' = 1.00	32.00°
L35	Mohr-Coulomb	98.00	c' = 500.0	0.00°
L25	Mohr-Coulomb	131.0	c' = 1.00	37.00°
L20	Mohr-Coulomb	103.0	c' = 600.0	0.00°

Published value	Method	Source	cSlope
1.35	Spencer	Duncan & Wright	1.352

cSlope analysis	Other programs	cSlope
Bishop simplified	1.398	1.389
Spencer	1.359	1.352

3.55 V032a — Duncan & Wright fig. 14.7, deep circle

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7.

Critical circle search · Units: imperial (ft, psf, pcf)

Cohesionless embankment on an undrained clay foundation. Entry is limited to the crest and exit to the toe flat to find the published deep circle.

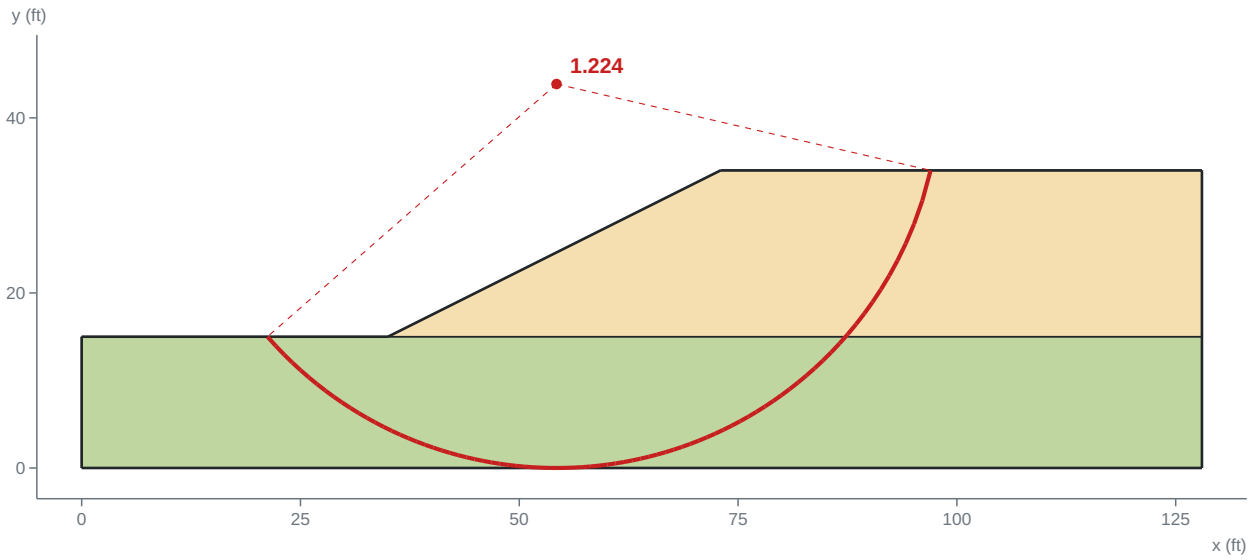


Figure 55. V032a: critical circle (Bishop simplified, FOS = 1.224).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c' = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c' = 500.0	0.00°

Published value	Method	Source	cSlope
1.21	Spencer	Duncan & Wright	1.205

cSlope analysis	Other programs	cSlope
Bishop simplified	1.230	1.224
Spencer	1.209	1.205

3.56 V032b — Duncan & Wright fig. 14.7, infinite-slope mechanism

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.7. Verification manual of a commercial slope stability program.

Critical circle search · Units: imperial (ft, psf, pcf)

The V032a embankment without search limits: the critical surface is the shallow infinite-slope sliver in the cohesionless fill (tan 30° / tan 26.6° = 1.155).

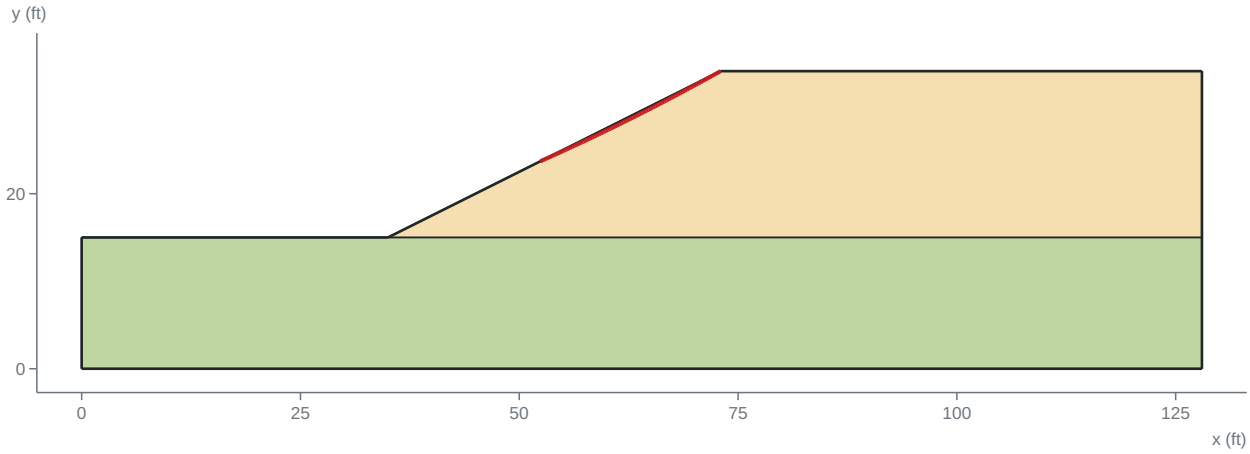


Figure 56. V032b: critical circle (Bishop simplified, FOS = 1.156).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	124.0	c' = 0.00	30.00°
Foundation	Mohr-Coulomb	98.00	c' = 500.0	0.00°

Published value	Method	Source	cSlope
1.15	Spencer	Duncan & Wright	1.156

cSlope analysis	Other programs	cSlope
Bishop simplified	1.155	1.156
Spencer	1.155	1.156

3.57 V033 — Duncan & Wright fig. 14.20a, water table

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20a.

Critical circle search · Units: SI (m, kPa, kN/m³)

Cohesive-frictional fill over sand with a piezometric line; critical circle search. Converted to SI.

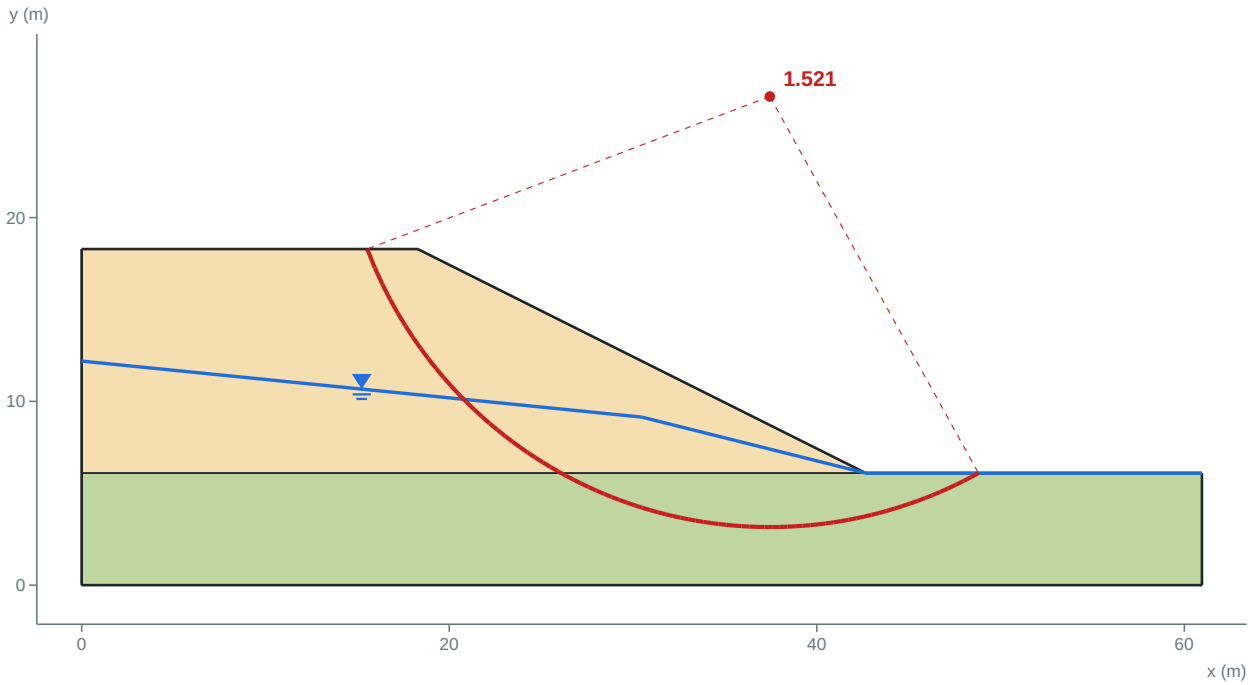


Figure 57. V033: critical circle (Bishop simplified, FOS = 1.521).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	19.64	c' = 28.73	25.00°
Sand	Mohr-Coulomb	20.74	c' = 0.00	30.00°

Published value	Method	Source	cSlope
1.529	Spencer	Duncan & Wright	1.533

cSlope analysis	Other programs	cSlope
Bishop simplified	1.533	1.521
Spencer	–	1.533
Spencer, non-circular	1.534	1.522

3.58 V034a — Duncan & Wright fig. 14.20b, Su = 200 + 15·depth

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

Granular fill over clay with undrained strength increasing linearly with depth; critical circle search.

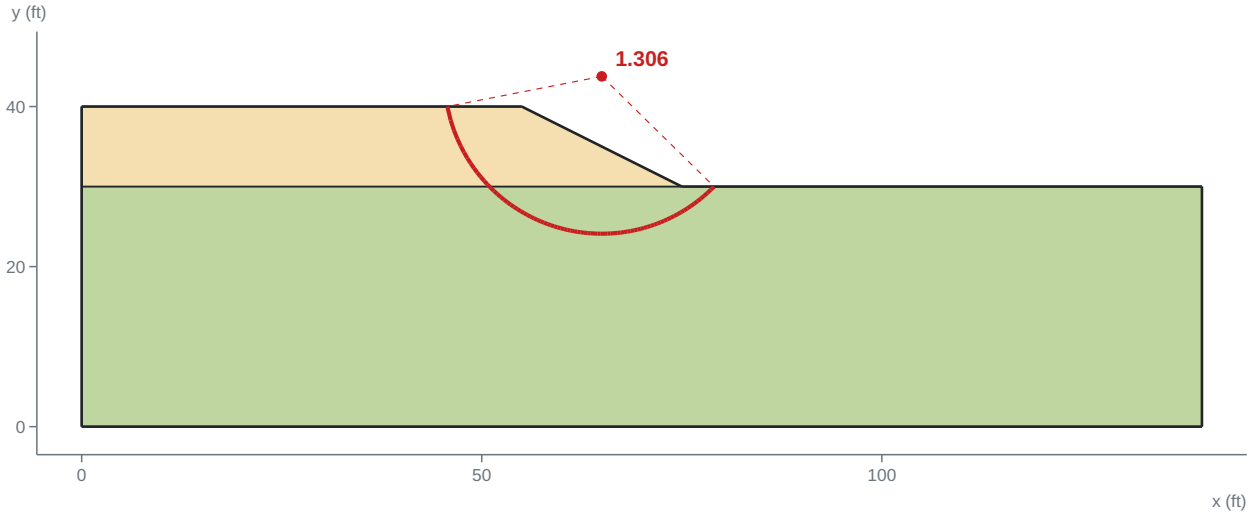


Figure 58. V034a: critical circle (Bishop simplified, FOS = 1.306).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c' = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 200.0 at y = 30.00, +15.00 psf/ft below	0°

Published value	Method	Source	cSlope
1.276	Spencer	Duncan & Wright	1.276

cSlope analysis	Other programs	cSlope
Bishop simplified	1.313	1.306
Spencer	–	1.276
Spencer, non-circular	1.262	1.260

3.59 V034b — Duncan & Wright fig. 14.20b, Su = 300 psf

Duncan & Wright, Soil Strength and Slope Stability, fig. 14.20b.

Critical circle search · Units: imperial (ft, psf, pcf)

As V034a with a constant undrained strength of 300 psf.

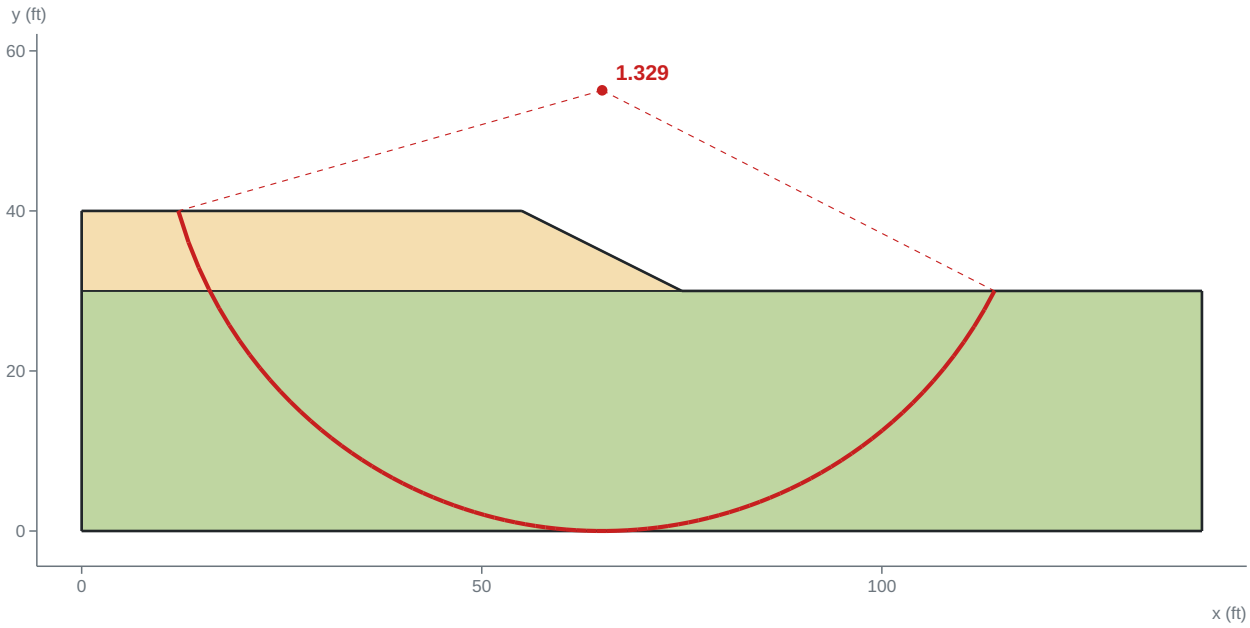


Figure 59. V034b: critical circle (Bishop simplified, FOS = 1.329).

Material	Model	γ (pcf)	Strength (psf)	φ
Fill	Mohr-Coulomb	123.0	c' = 0.00	36.00°
Clay	Undrained (φ = 0)	97.00	Su = 300.0	0°

Published value	Method	Source	cSlope
1.328	Spencer	Duncan & Wright	1.328

cSlope analysis	Other programs	cSlope
Bishop simplified	1.335	1.329
Spencer	–	1.328
Spencer, non-circular	1.182	1.168

3.60 V035a — Duncan & Wright tieback, active

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous slope with a 9000 lb/ft tieback applied horizontally at mid-height as an active force, on the reference printed GLE circle. Converted to SI.

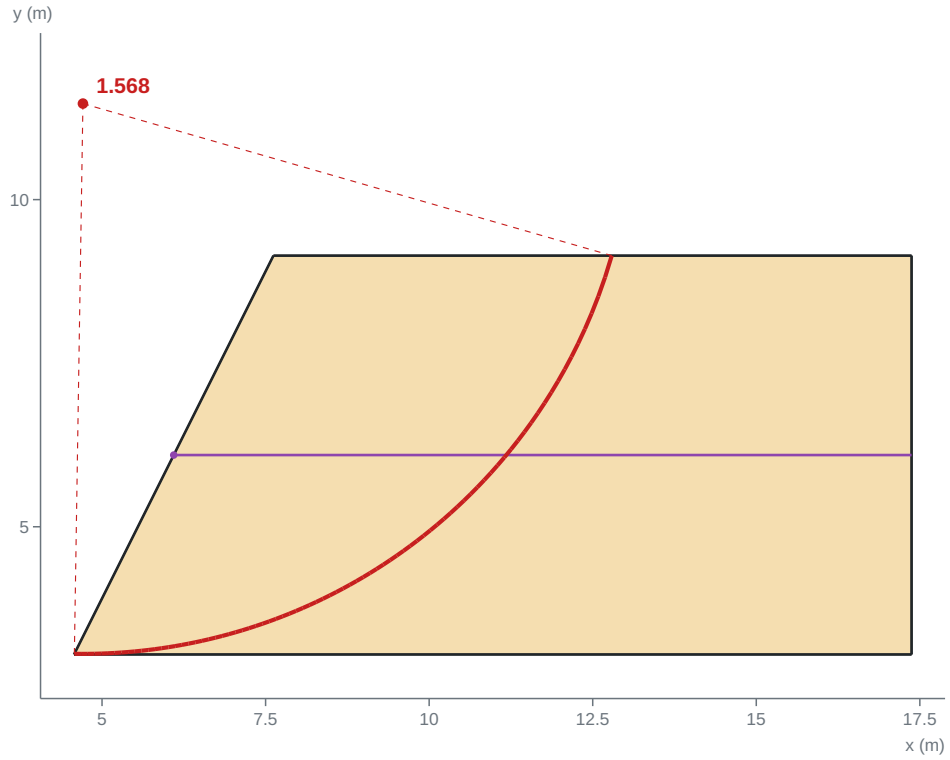


Figure 60. V035a: analysed circle (Morgenstern-Price (half-sine), FOS = 1.568).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Published value	Method	Source	cSlope
1.51	not stated	Duncan & Wright	–

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.575	1.568

Duncan & Wright's 1.51 is about 4 % below the other programs on this surface.

3.61 V035b — Duncan & Wright tieback, passive

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.34.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V035a with the tieback applied as a passive force (divided by F), on the reference printed Bishop circle.

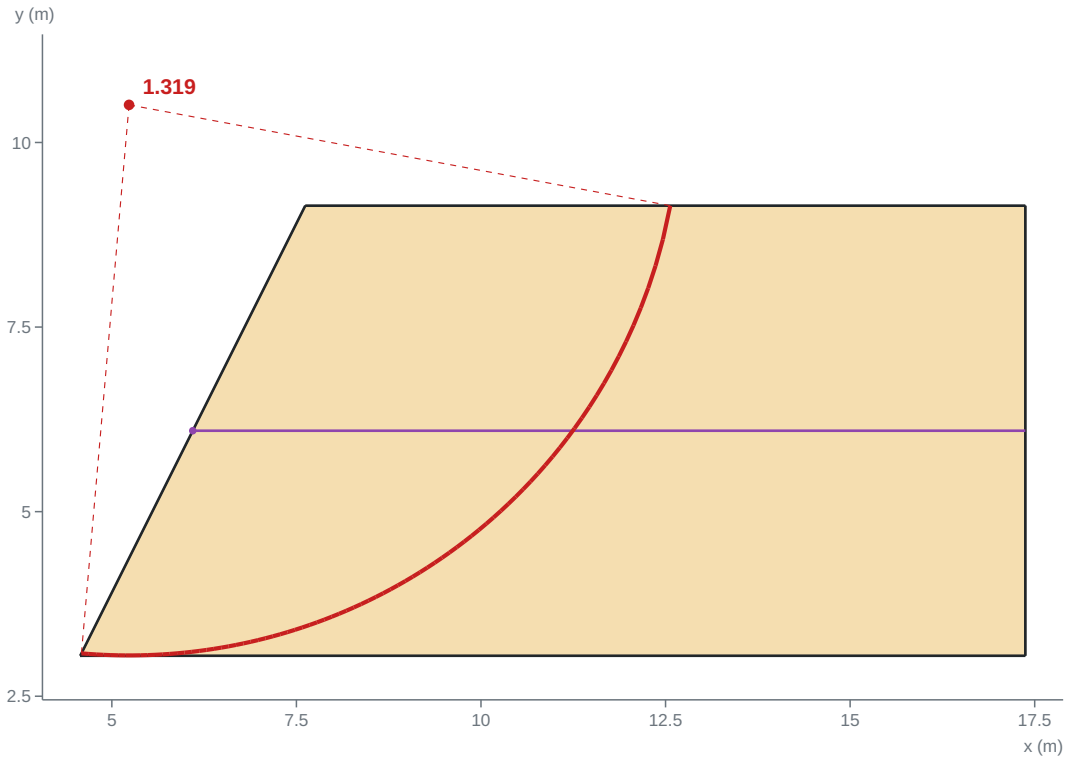


Figure 61. V035b: analysed circle (Bishop simplified, FOS = 1.319).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
saturated clay	Undrained (φ = 0)	15.39	Su = 16.76	0°

Published value	Method	Source	cSlope
1.32	Bishop simplified	Duncan & Wright	1.319

cSlope analysis	Other programs	cSlope
Bishop simplified	1.324	1.319

3.62 V036a — Embankment on an undrained foundation, cz = 0 (D&W fig. 15.9)

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment (c′ = 0, φ′ = 35°, γ = 125 pcf) on a 20 ft undrained foundation, cu = 300 psf, constant; circular search. Converted to SI.

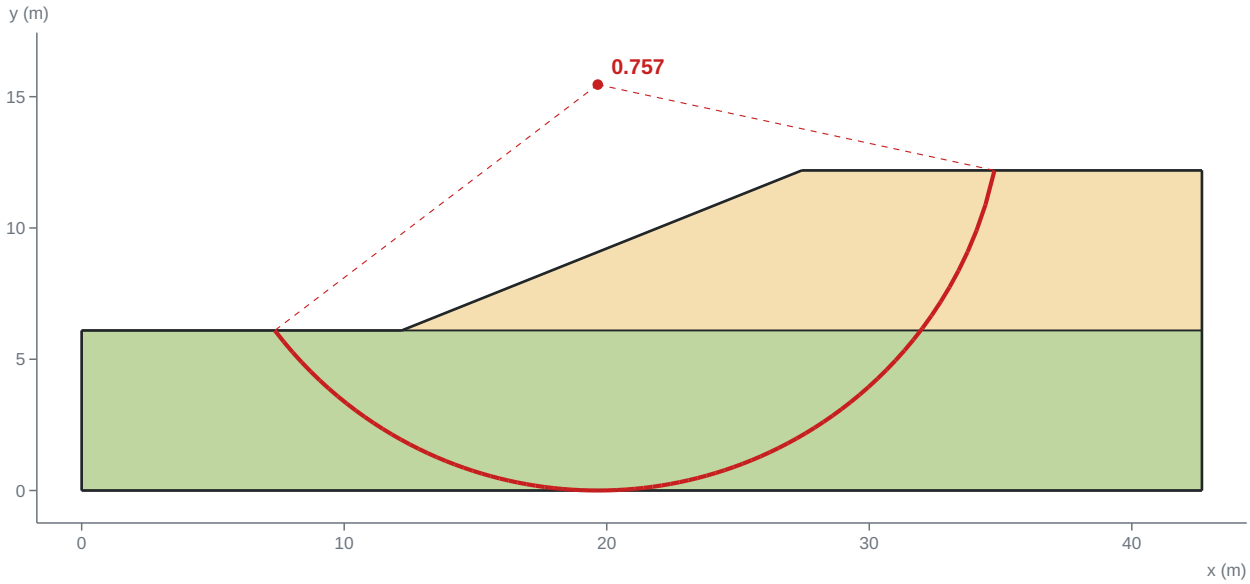


Figure 62. V036a: critical circle (Bishop simplified, FOS = 0.757).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36	0°

Published value	Method	Source	cSlope
0.75	Spencer	Duncan & Wright	0.752

cSlope analysis	Other programs	cSlope
Bishop simplified	0.761	0.757
Spencer	0.756	0.752
Spencer, non-circular	0.740	0.736

3.63 V036b — Embankment on an undrained foundation, cz = 5 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 5 psf/ft with depth.

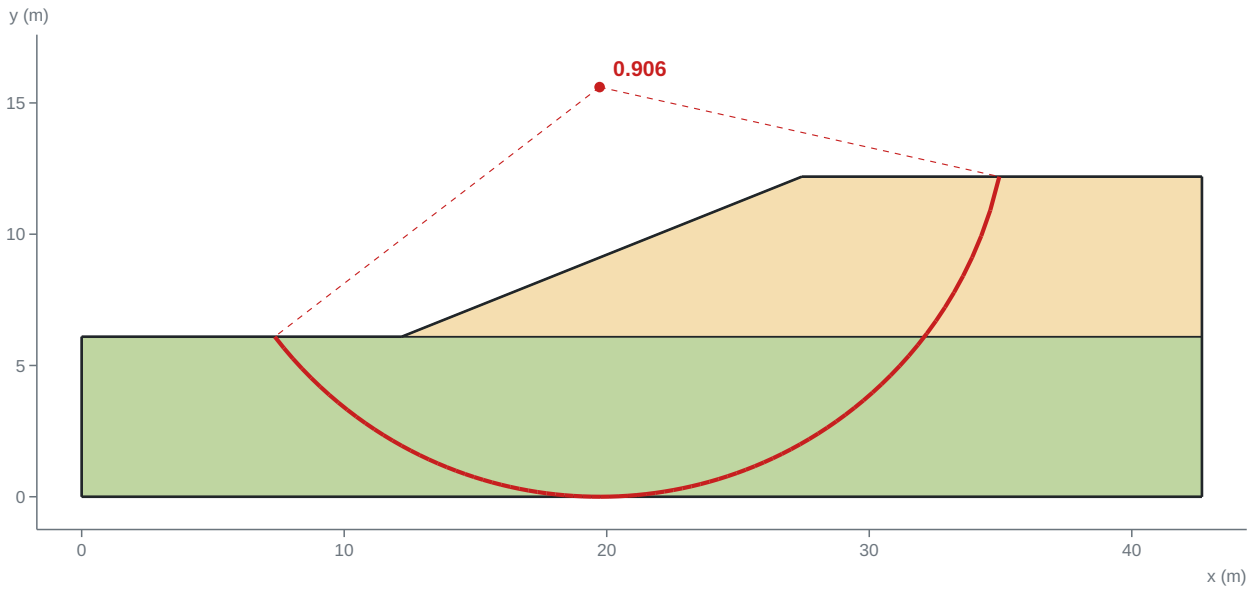


Figure 63. V036b: critical circle (Bishop simplified, FOS = 0.906).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +0.79 kPa/m below	0°

Published value	Method	Source	cSlope
0.90	Spencer	Duncan & Wright	0.898

cSlope analysis	Other programs	cSlope
Bishop simplified	0.909	0.906
Spencer	0.898	0.898

3.64 V036c — Embankment on an undrained foundation, cz = 10 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 10 psf/ft with depth.

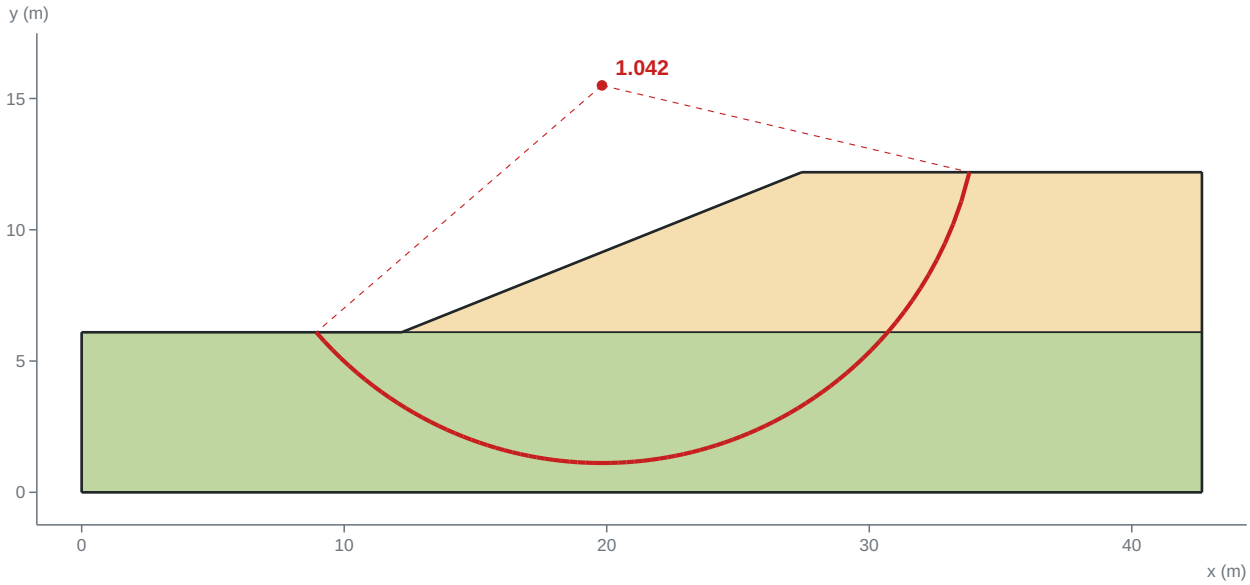


Figure 64. V036c: critical circle (Bishop simplified, FOS = 1.042).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +1.57 kPa/m below	0°

Published value	Method	Source	cSlope
1.03	Spencer	Duncan & Wright	1.028

cSlope analysis	Other programs	cSlope
Bishop simplified	1.045	1.042
Spencer	1.032	1.028
Spencer, non-circular	1.018	1.017

3.65 V036d — Embankment on an undrained foundation, cz = 15 psf/ft

Duncan & Wright, Soil Strength and Slope Stability, fig. 15.9.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V036a with cu increasing 15 psf/ft with depth.

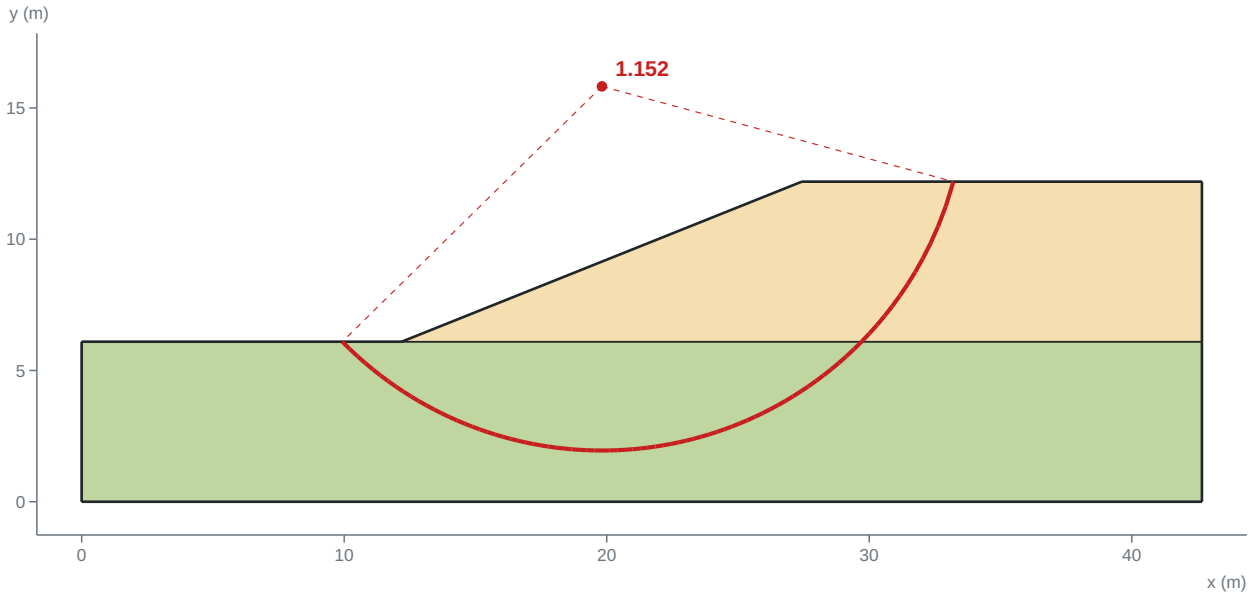


Figure 65. V036d: critical circle (Bishop simplified, FOS = 1.152).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation	Undrained (φ = 0)	15.71	Su = 14.36 at y = 6.10, +2.36 kPa/m below	0°

Published value	Method	Source	cSlope
1.13	Spencer	Duncan & Wright	1.132

cSlope analysis	Other programs	cSlope
Bishop simplified	1.154	1.152
Spencer	1.134	1.132
Spencer, non-circular	1.116	1.119

3.66 V037 — Geogrid-reinforced fill on rock (D&W fig. 7.28)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.28.

Critical circle search · Units: SI (m, kPa, kN/m³)

24 ft fill slope (φ′ = 37°) on a strong rock foundation with five 800 lb/ft geogrids, 20 ft long, every 4 ft; circular search for surfaces through the toe. Converted to SI.

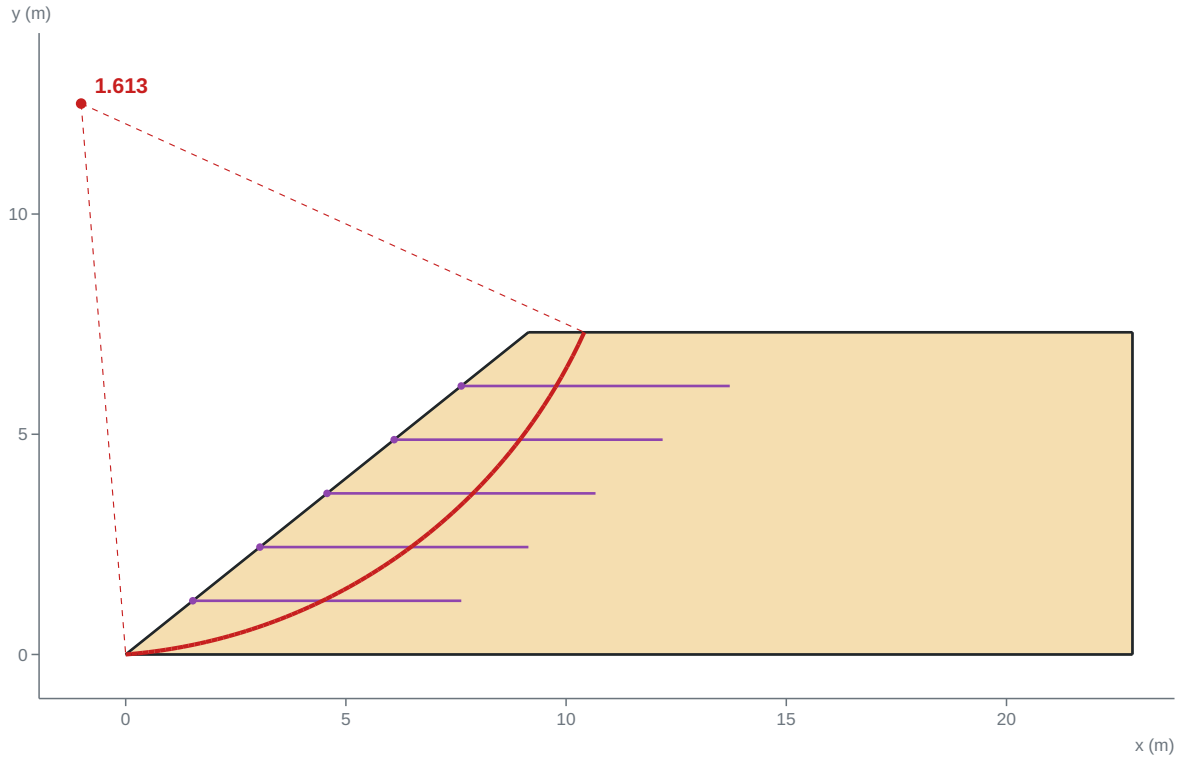


Figure 66. V037: critical circle (Bishop simplified, FOS = 1.613).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
fill	Mohr-Coulomb	20.42	c' = 0.00	37.00°

Published value	Method	Source	cSlope
1.62	Spencer	Duncan, Wright & Brandon (2014)	1.607

cSlope analysis	Other programs	cSlope
Bishop simplified	1.629	1.613
Spencer	1.620	1.607

3.67 V038 — Homogeneous dam with a piezometric line (D&W fig. 7.19)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.19.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment (c′ = 100 psf, φ′ = 30°, γ = 100 pcf) with the pool at el. 40 ft and a commercial program's piezometric line; critical-circle search. Converted to SI.

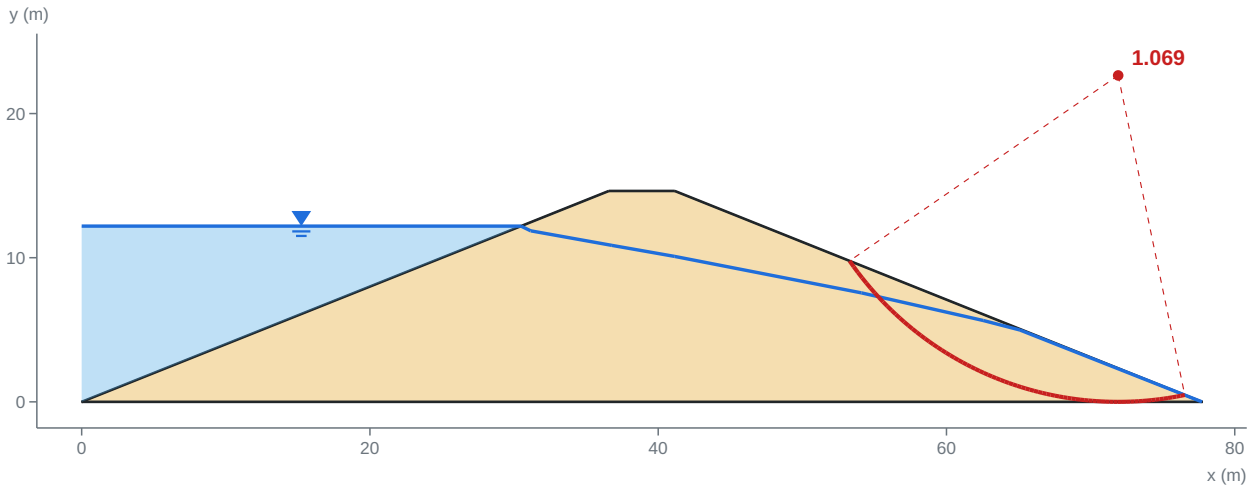


Figure 67. V038: critical circle (Bishop simplified, FOS = 1.069).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.71	c' = 4.79	30.00°

Published value	Method	Source	cSlope
1.16	not stated	Duncan & Wright	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.070–1.090	1.069
Spencer	1.100	1.077

Duncan & Wright's 1.16 is for γ = 125 pcf and their printed phreatic line, not this program input (100 pcf, a 9-point line); on their input cSlope gives 1.157 (V038b).

3.68 V038b — Homogeneous dam with a piezometric line, Duncan & Wright's input

Duncan, J.M., Wright, S.G. & Brandon, T.L. (2014). Soil Strength and Slope Stability, 2nd ed., Wiley; §7.7, Figs. 7.30–7.36, Table 7.8.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V038 dam with the unit weight Duncan, Wright & Brandon's slice calculations use (125 pcf) and their printed phreatic line (30 points, Fig. 7.31); critical-circle search. Converted to SI.

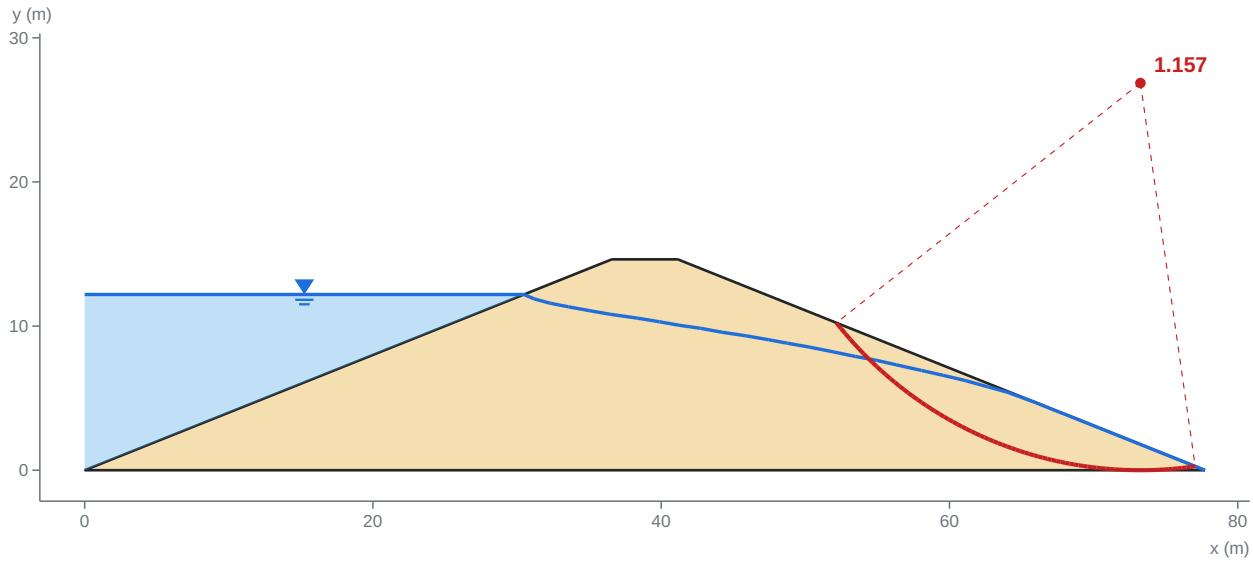


Figure 68. V038b: critical circle (Spencer, FOS = 1.157).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	19.64	c' = 4.79	30.00°

Published value	Method	Source	cSlope
1.16	Spencer	Duncan & Wright	1.157
1.19	Spencer, finite-element pore pressures	Duncan, Wright & Brandon (2014)	1.157

cSlope analysis	Other programs	cSlope
Spencer	–	1.157

The book contradicts itself on the unit weight: its figures say 100 pcf, its hand calculations and slice weights use 125 pcf. With 125 pcf and the book's line cSlope reproduces the 1.16 that all three of the book's programs give (Table 7.8); with 100 pcf it gives 1.03.

3.69 V039 — James Bay dyke (Duncan & Wright fig. 7.16)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

Granular fill embankment with a wide berm (c′ = 0, φ′ = 30°) on a 4 m clay crust (Su = 41), 8 m marine clay (34.5) and 6.5 m lacustrine clay (31.2); circular search. The critical circle is deep and tangent to the base of the lacustrine clay.

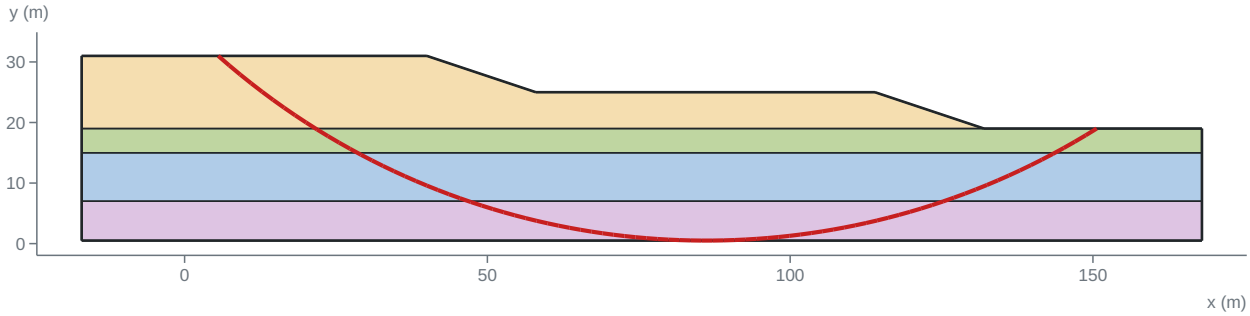


Figure 69. V039: critical circle (Spencer, FOS = 1.447).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Clay crust	Undrained (φ = 0)	20.00	Su = 41.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Published value	Method	Source	cSlope
1.45	Spencer	Duncan & Wright	1.447
1.16	Spencer, non-circular	Duncan & Wright	1.162

cSlope analysis	Other programs	cSlope
Spencer	–	1.447
Spencer, non-circular	1.167	1.162

3.70 V040 — James Bay dyke, reference model (deterministic)

El-Ramly et al. (2002); Duncan & Wright fig. 7.16.

Critical circle search · Units: SI (m, kPa, kN/m³)

The James Bay dyke as a commercial program models it (crest 36 m, berm at 30 m, clay crust Su = 43 kPa, till below y = 5.5); critical-circle search, deterministic.

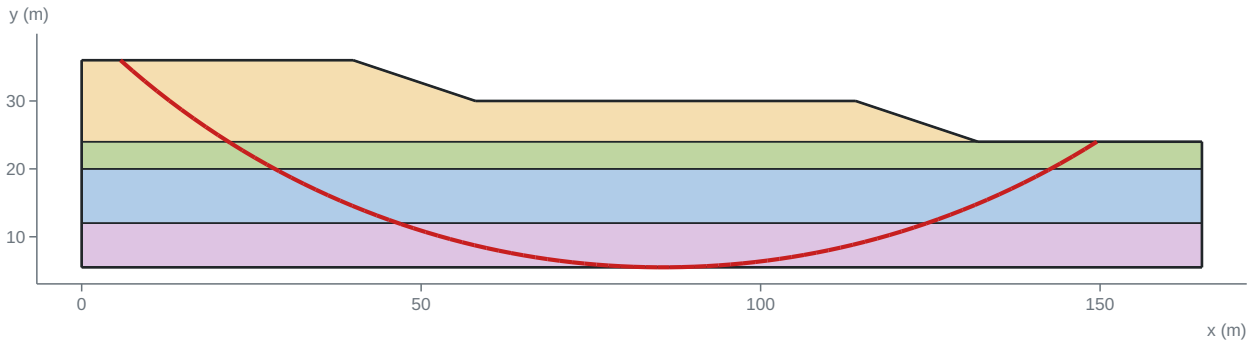


Figure 70. V040: critical circle (Bishop simplified, FOS = 1.459).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.00	c' = 0.00	30.00°
Clay crust	Undrained (φ = 0)	18.80	Su = 43.00	0°
Marine clay	Undrained (φ = 0)	18.80	Su = 34.50	0°
Lacustrine clay	Undrained (φ = 0)	20.30	Su = 31.20	0°

Published value	Method	Source	cSlope
1.46	Bishop simplified	El-Ramly et al. (2002)	1.459

cSlope analysis	Other programs	cSlope
Bishop simplified	1.461	1.459

3.71 V041a — Submerged slope, pool 30 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

Fully submerged homogeneous slope (c = 100 psf, φ = 20°, γ = 128 pcf) with the pool 30 ft above the crest: the extra water weight and pore pressure cancel. Converted to SI.

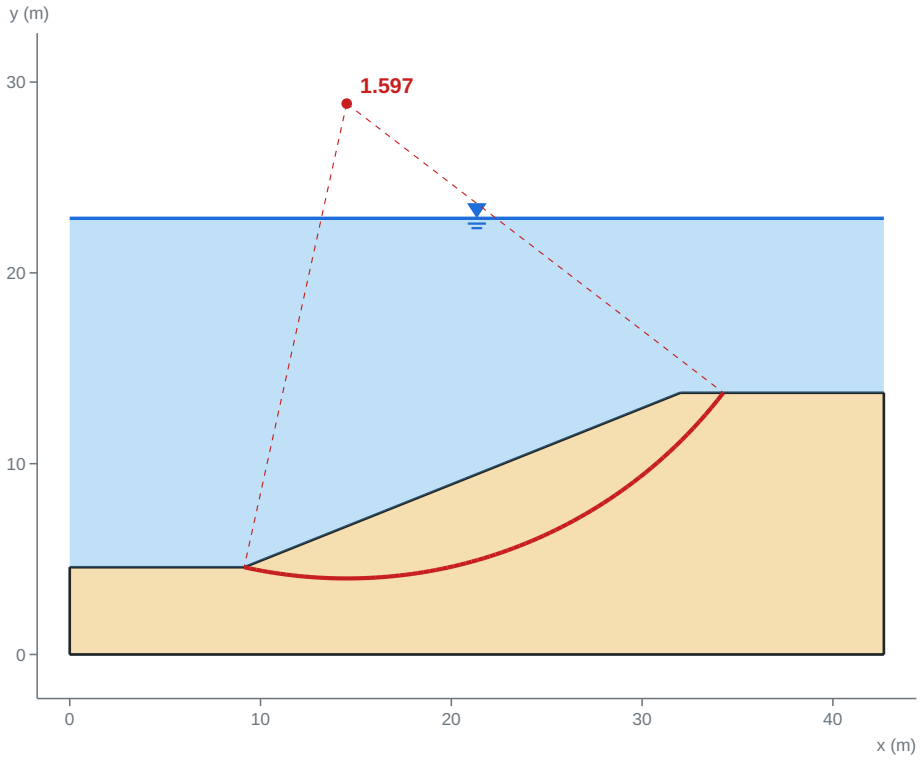


Figure 71. V041a: critical circle (Bishop simplified, FOS = 1.597).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c' = 4.79	20.00°

Published value	Method	Source	cSlope
1.60	Spencer	Duncan & Wright	1.594

cSlope analysis	Other programs	cSlope
Bishop simplified	1.603	1.597
Spencer	1.599	1.594
Spencer, non-circular	1.592	1.591

3.72 V041b — Submerged slope, pool 60 ft above the crest (D&W fig. 6.27)

Duncan & Wright, Soil Strength and Slope Stability, fig. 6.27.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V041a with the pool 60 ft above the crest: the same factor of safety.

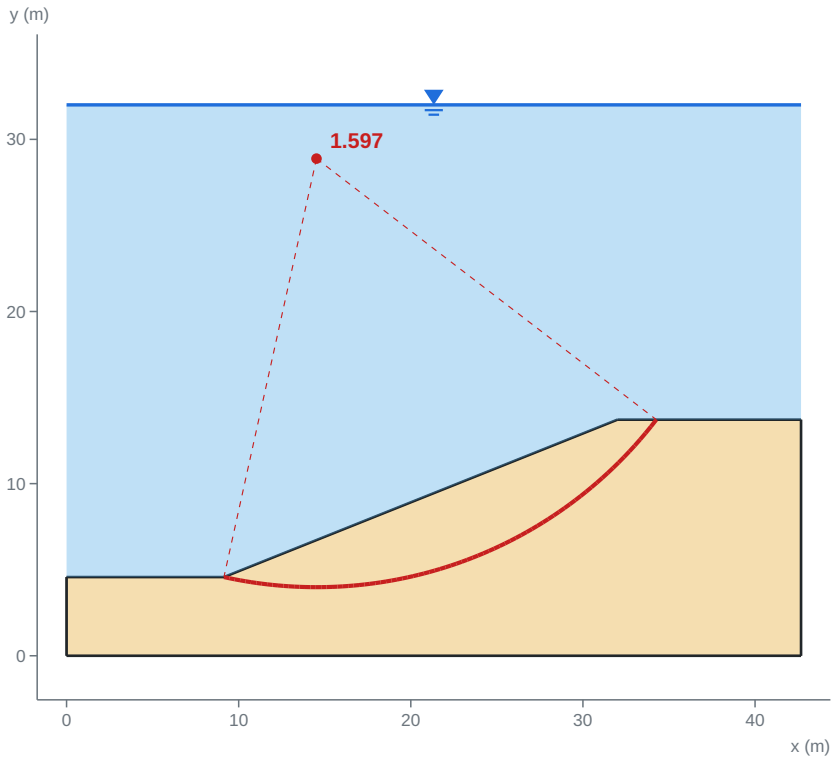


Figure 72. V041b: critical circle (Bishop simplified, FOS = 1.597).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.11	c' = 4.79	20.00°

Published value	Method	Source	cSlope
1.60	Spencer	Duncan & Wright	1.594

cSlope analysis	Other programs	cSlope
Bishop simplified	1.603	1.597
Spencer	1.599	1.594
Spencer, non-circular	1.590	1.589

3.73 V042 — Thick-core dam with a piezometric line (D&W fig. 7.24)

Duncan & Wright, Soil Strength and Slope Stability, fig. 7.24.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric dam with a thick clay core (φ′ = 20°) in a shell (φ′ = 38°), pond at el. 315 ft, a commercial program's piezometric line; the deep circle. Converted to SI.

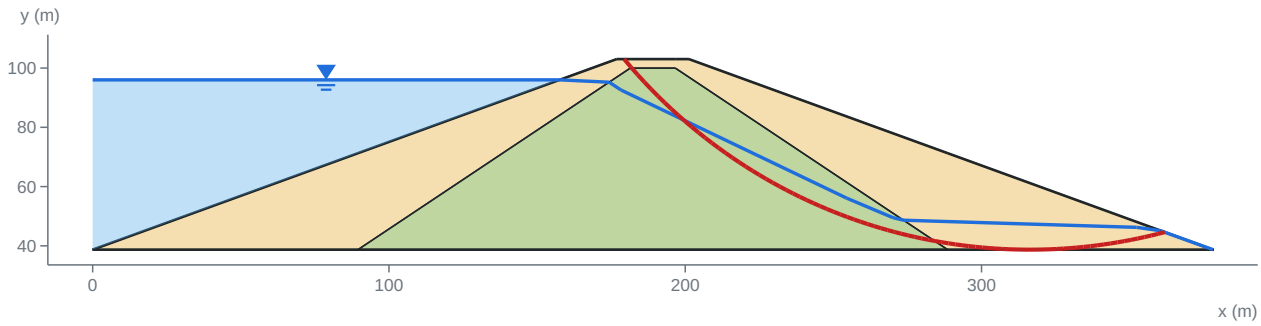


Figure 73. V042: analysed circle (Bishop simplified, FOS = 1.582).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Shell	Mohr-Coulomb	21.99	c' = 0.00	38.00°
Core	Mohr-Coulomb	18.85	c' = 0.00	20.00°

Published value	Method	Source	cSlope
1.67	not stated	Duncan & Wright	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.584	1.582
Spencer	1.648	1.649

3.74 V043 — Syncrude tailings dyke (El-Ramly et al.), composite circle

El-Ramly, H., Morgenstern, N.R. & Cruden, D.M. (2003). Probabilistic stability analysis of a tailings dyke on presheared clay-shale. Can. Geotech. J. 40(1), 192–208.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Cohesionless tailings sand, glacio-fluvial sands and tills (φ = 34° here; El-Ramly et al. give the sandy till 35.7°) on a presheared clay-shale (φ = 7.5°) with a water table; the reference circle (327.5, 394, R = 124) reaches below the model base, so the surface follows the base there (composite).

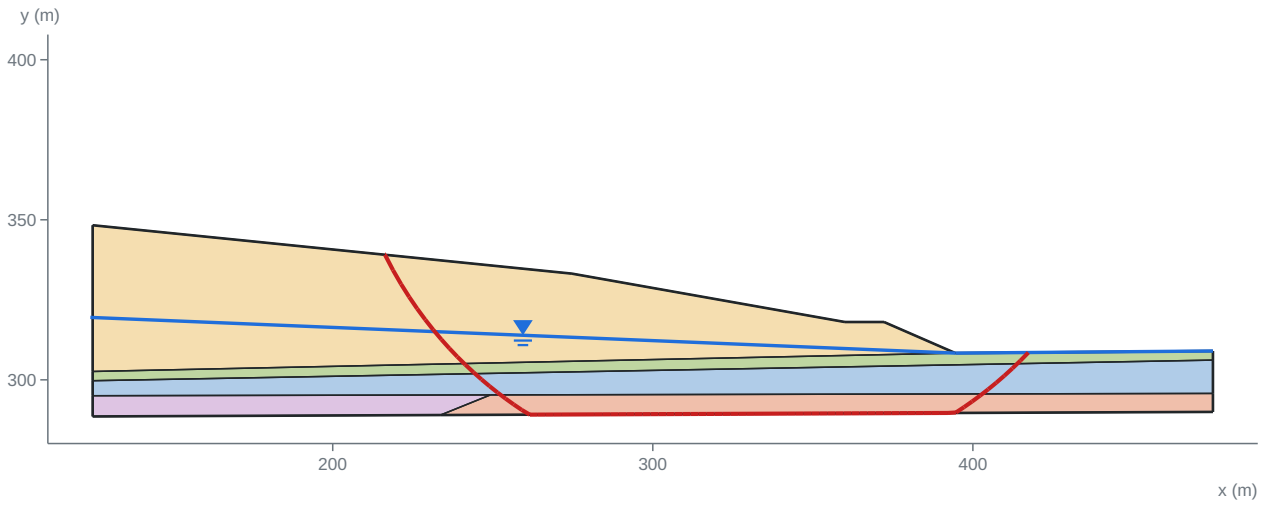


Figure 74. V043: analysed surface (Bishop simplified, FOS = 1.320).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Tailing sand (TS)	Mohr-Coulomb	20.00	c' = 0.00	34.00°
Glacio-fluvial sand (Pf4)	Mohr-Coulomb	17.00	c' = 0.00	34.00°
Sandy till (Pgs)	Mohr-Coulomb	17.00	c' = 0.00	34.00°
Clayey till (Pgc)	Mohr-Coulomb	17.00	c' = 0.00	7.50°
Disturbed clay-shale (Kca)	Mohr-Coulomb	17.00	c' = 0.00	7.50°

Published value	Method	Source	cSlope
1.31	Bishop simplified	El-Ramly et al.	1.320

cSlope analysis	Other programs	cSlope
Bishop simplified	1.320	1.320

3.75 V044 — Fredlund & Krahn (1977), dry

Fredlund, D.G. & Krahn, J. (1977). Comparison of slope stability methods of analysis. Can. Geotech. J. 14(3).

Fixed circle · Units: imperial (ft, psf, pcf)

Homogeneous 2:1 slope, 40 ft high, on a firm base; the classic comparison problem of Fredlund & Krahn analysed with the published circle and no pore pressure.

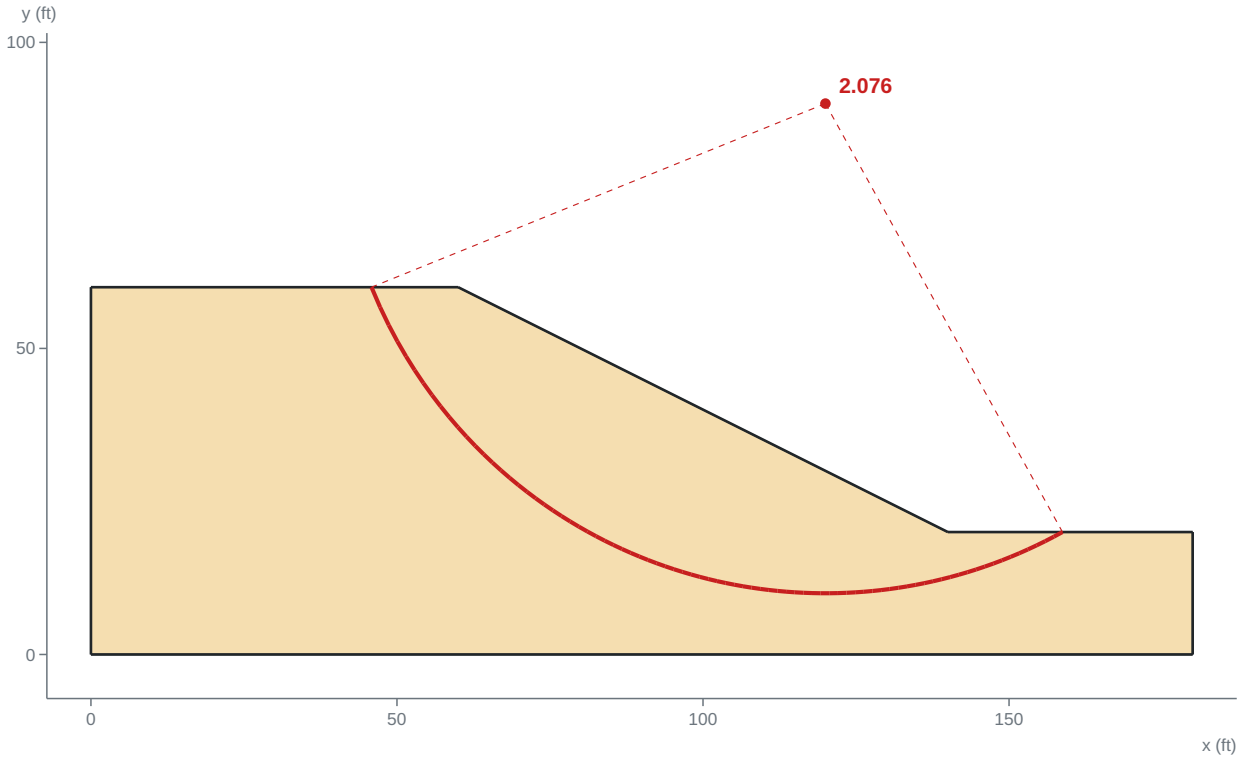


Figure 75. V044: analysed circle (Bishop simplified, FOS = 2.076).

Material	Model	γ (pcf)	Strength (psf)	φ
Soil	Mohr-Coulomb	120.0	c' = 600.0	20.00°

Published value	Method	Source	cSlope
2.08	Bishop simplified	Fredlund & Krahn	2.076
2.076	Morgenstern-Price (half-sine)	Fredlund & Krahn	2.072
2.073	Spencer	Fredlund & Krahn	2.072

cSlope analysis	Other programs	cSlope
Bishop simplified	–	2.076
Morgenstern-Price (half-sine)	–	2.072
Spencer	–	2.072

3.76 V045 — Fredlund & Krahn (1977), with water table

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Fixed circle · Units: SI (m, kPa, kN/m³)

The Fredlund & Krahn slope with a piezometric line, analysed on the published circle. Converted to SI.

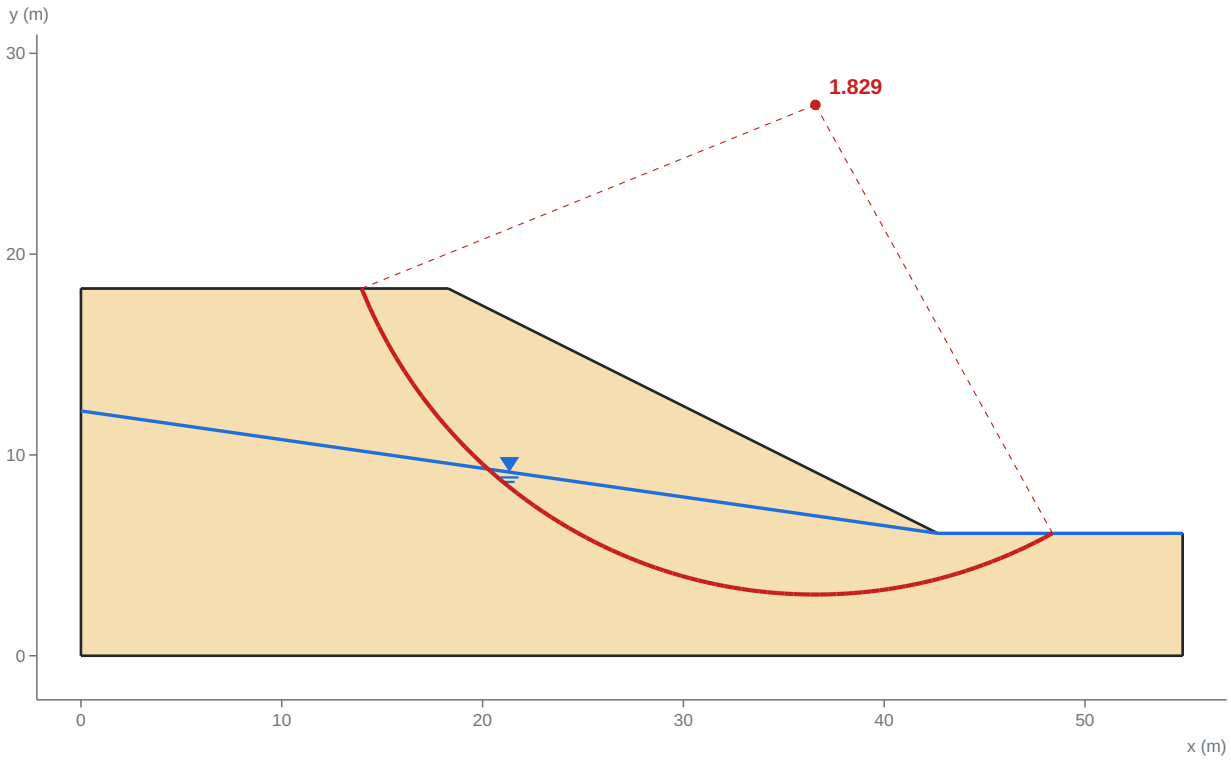


Figure 76. V045: analysed circle (Bishop simplified, FOS = 1.829).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	18.85	c' = 28.73	20.00°

Published value	Method	Source	cSlope
1.834	Bishop simplified	Fredlund & Krahn	1.829
1.832	Morgenstern-Price (half-sine)	Fredlund & Krahn	1.827
1.83	Spencer	Fredlund & Krahn	1.827

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.829
Morgenstern-Price (half-sine)	–	1.827
Spencer	–	1.827

3.77 V046a — Fredlund & Krahn slope on a weak seam, composite surface

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Specified non-circular surface · Units: imperial (ft, psf, pcf)

The Fredlund & Krahn slope with a 1 ft weak seam (c′ = 0, φ′ = 10°) on bedrock at el. 15 (their case 2); the published circle (120, 90, R = 80) reaches below the seam, so the surface follows the arc down to the bedrock, runs along the seam and climbs back on the arc. Dry.

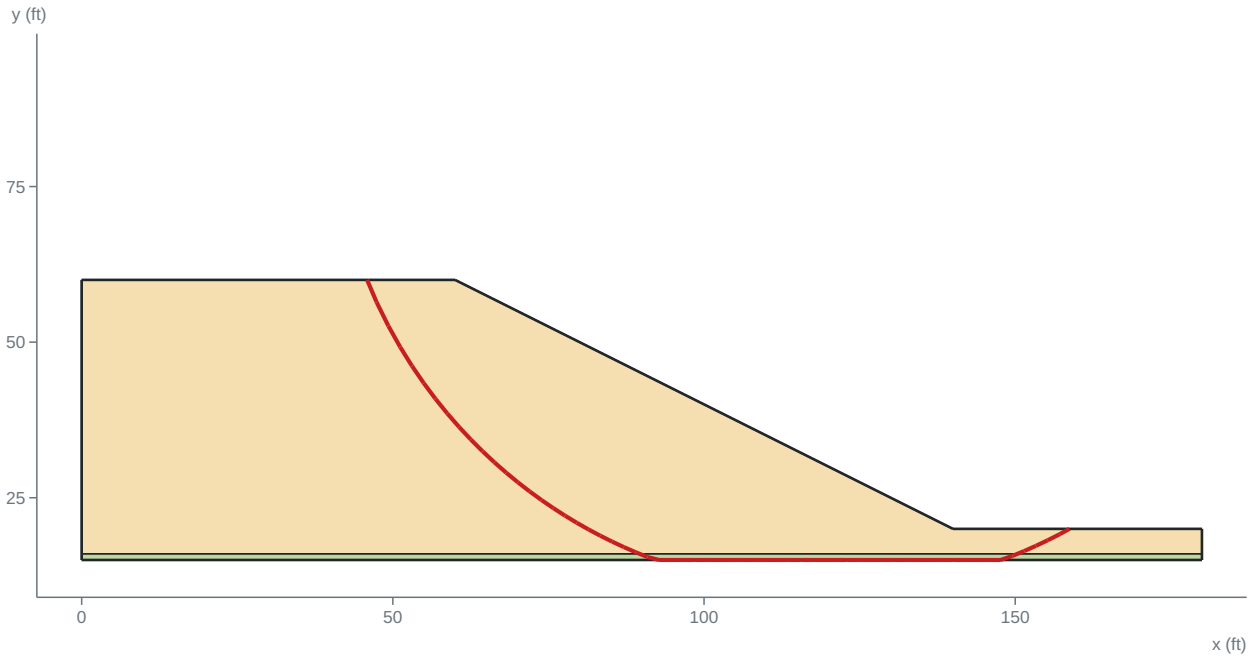


Figure 77. V046a: analysed surface (Bishop simplified, FOS = 1.380).

Material	Model	γ (pcf)	Strength (psf)	φ
Upper soil	Mohr-Coulomb	120.0	c' = 600.0	20.00°
Weak seam	Mohr-Coulomb	120.0	c' = 0.00	10.00°

Published value	Method	Source	cSlope
1.37	Morgenstern-Price (half-sine)	Fredlund & Krahn	1.370
1.377	Bishop simplified	Fredlund & Krahn	1.380
1.373	Spencer	Fredlund & Krahn	1.379

cSlope analysis	Other programs	cSlope
Bishop simplified	1.382	1.380
Spencer	1.382	1.379
Morgenstern-Price (half-sine)	–	1.370

3.78 V046b — Fredlund & Krahn slope on a weak seam, with a piezometric line

Fredlund, D.G. & Krahn, J. (1977). Can. Geotech. J. 14(3).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

As V046a with the piezometric line of V045 (from (0, 40) ft to the toe) for both materials: Fredlund & Krahn's case 6. Converted to SI.

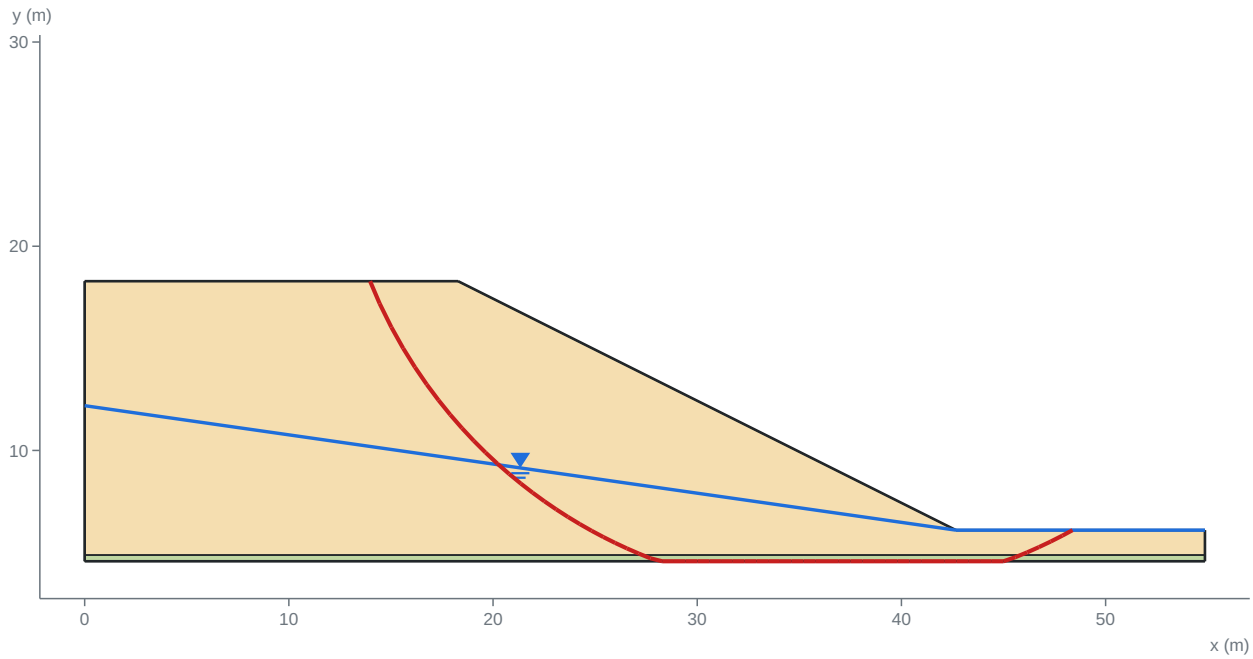


Figure 78. V046b: analysed surface (Bishop simplified, FOS = 1.241).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper soil	Mohr-Coulomb	18.85	c' = 28.73	20.00°
Weak seam	Mohr-Coulomb	18.85	c' = 0.00	10.00°

Published value	Method	Source	cSlope
1.248	Bishop simplified	Fredlund & Krahn	1.241
1.245	Spencer	Fredlund & Krahn	1.242

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.241
Spencer	–	1.242
Morgenstern-Price (half-sine)	–	1.235

3.79 V047a — ACADS 1(a), homogeneous slope, dry

Giam & Donald (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(a): homogeneous 2:1 slope, 10 m high (c′ = 3 kPa, φ′ = 19.6°, γ = 20 kN/m³), dry; critical-circle search.

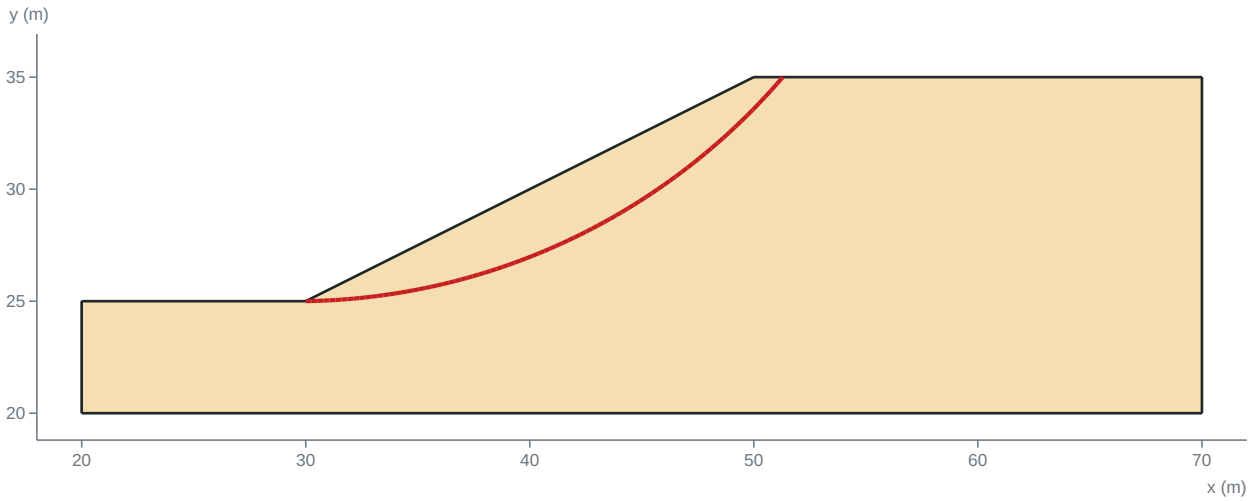


Figure 79. V047a: critical circle (Bishop simplified, FOS = 0.985).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 3.00	19.60°

Published value	Method	Source	cSlope
1.00	Spencer	ACADS	0.984

cSlope analysis	Other programs	cSlope
Bishop simplified	0.987–0.992	0.985
Spencer	0.986	0.984
Janbu corrected	0.990	0.986
Morgenstern-Price (half-sine)	0.991	0.984

3.80 V047b — ACADS tension crack, reference crack depth 3.6 m (results correspond to a dry c

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 1(b) as a commercial program models it: c′ = 32 kPa, φ′ = 10°, a 3.6 m tension crack (dry); critical-circle search.

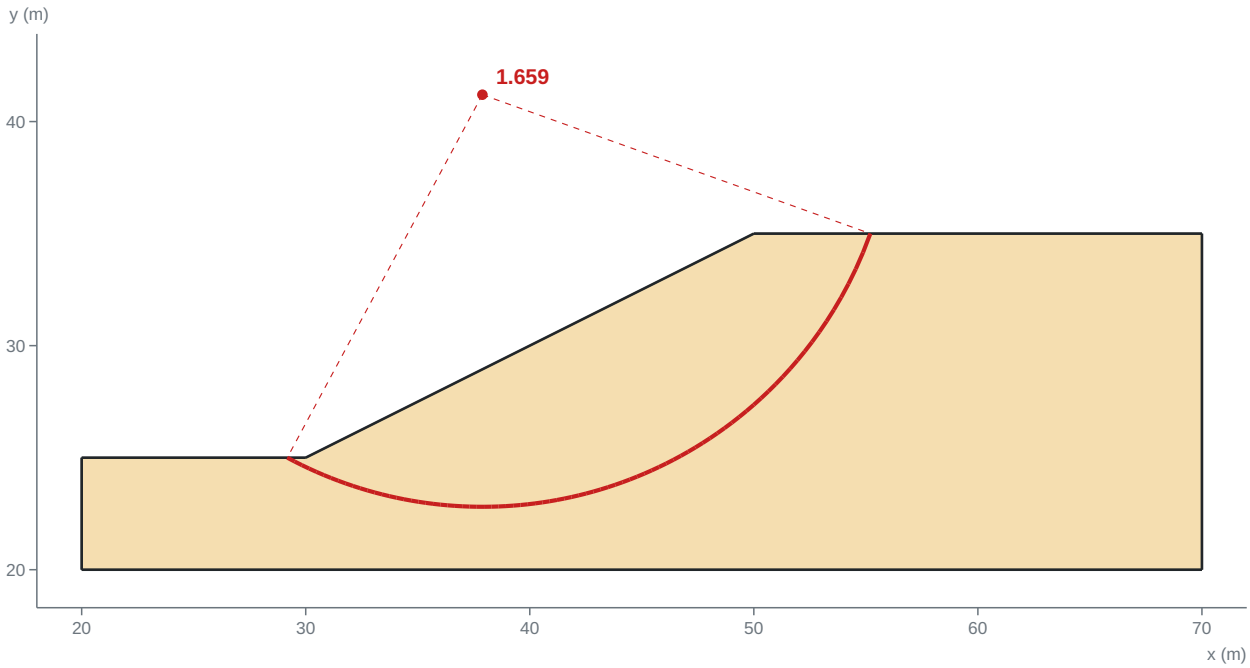


Figure 80. V047b: critical circle (Bishop simplified, FOS = 1.659).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 32.00	10.00°

Published value	Method	Source	cSlope
1.65–1.70	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.664	1.659
Morgenstern-Price (half-sine)	1.660	1.656

3.81 V048 — ACADS 1(b), water-filled tension crack

Giam & Donald (1989), ACADS problem 1(b).

Critical circle search · Units: SI (m, kPa, kN/m³)

The ACADS 1(a) slope (10 m high, 2:1) with c' = 32 kPa, φ' = 10° and a water-filled tension crack 4 m deep, the crack zone of a commercial program's Fig. 2.1 (35 → 31 m; the formula 2c/(γ√Ka) gives 3.814 m); critical circle search.

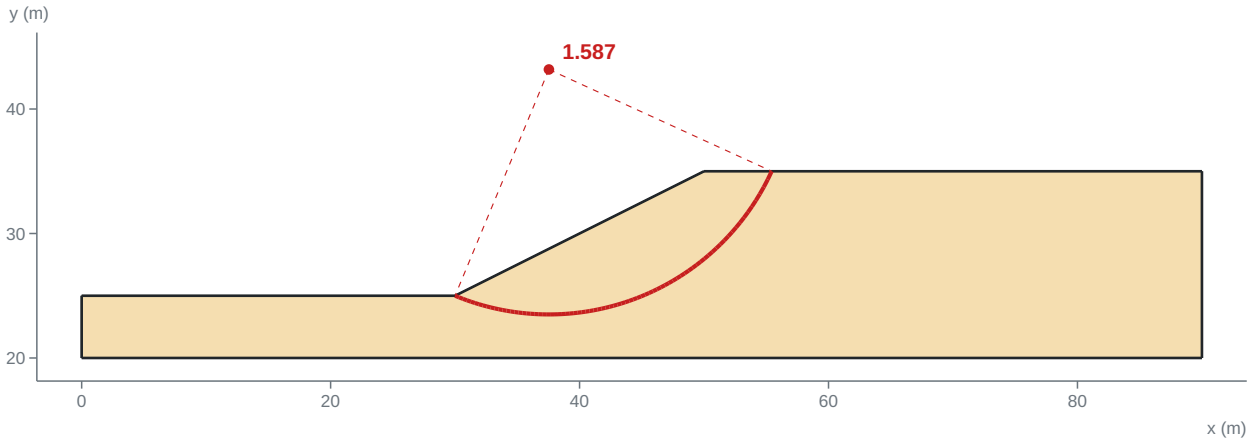


Figure 81. V048: critical circle (Bishop simplified, FOS = 1.587).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	20.00	c' = 32.00	10.00°

Published value	Method	Source	cSlope
1.65	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.596	1.587
Spencer	1.592	1.582
Janbu corrected	1.489	1.475

The ACADS reference 1.65 corresponds to a dry tension crack; with the crack filled with water the programs give about 1.59–1.60.

3.82 V049 — ACADS 1(c), three-layer slope

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

Non-homogeneous three-layer slope (ACADS 1c); critical-circle search.

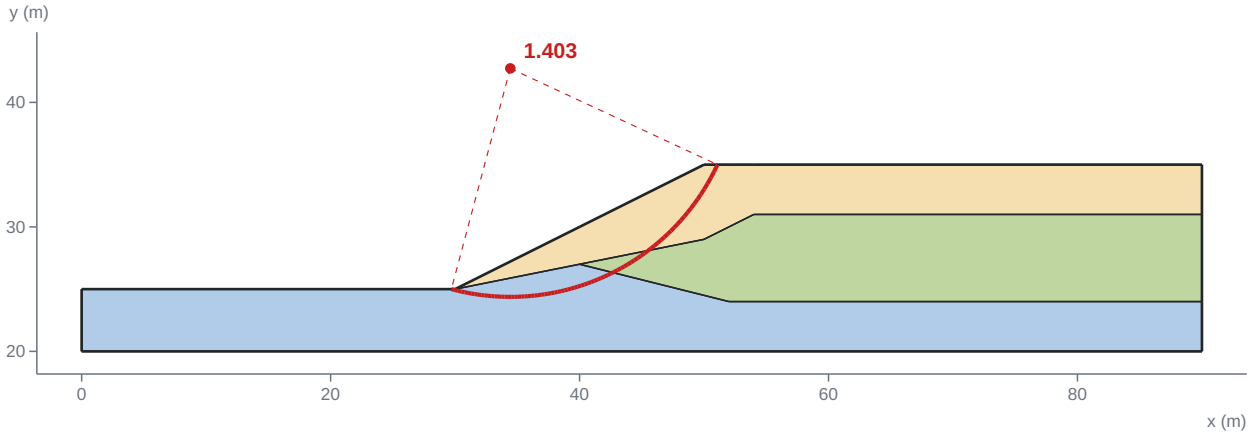


Figure 82. V049: critical circle (Bishop simplified, FOS = 1.403).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil #1	Mohr-Coulomb	19.50	c' = 0.00	38.00°
Soil #2	Mohr-Coulomb	19.50	c' = 5.30	23.00°
Soil #3	Mohr-Coulomb	19.50	c' = 7.20	20.00°

Published value	Method	Source	cSlope
1.39	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.403–1.405	1.403
Spencer	1.375	1.372
Janbu corrected	1.357	1.355
Morgenstern-Price (half-sine)	1.382	1.372

3.83 V050a — ACADS 3(b), weak layer, specified non-circular surface

Giam & Donald (1989), ACADS problem 3(b).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Slope with a 0.5 m weak layer (c' = 0, φ' = 10°) and a water table at its base, analysed on the published four-point non-circular surface that runs along the weak layer.

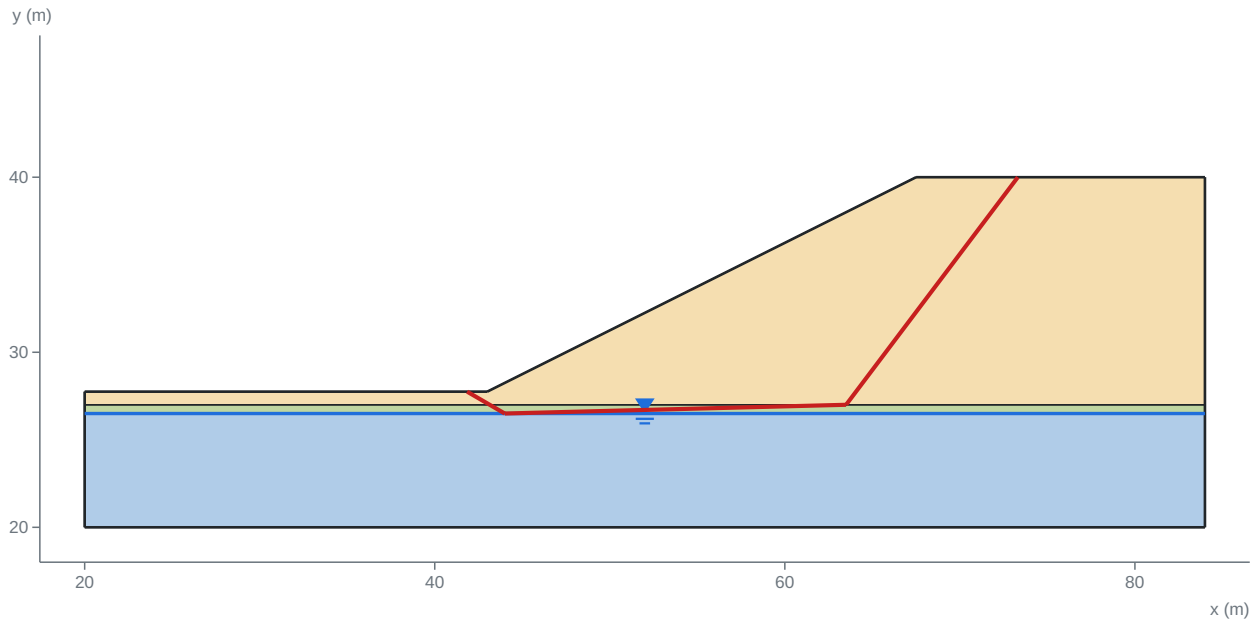


Figure 83. V050a: analysed surface (Janbu corrected, FOS = 1.294).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c' = 28.50	20.00°

Published value	Method	Source	cSlope
1.34	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Janbu corrected	1.294	1.294
Spencer	1.277	1.276
Morgenstern-Price (half-sine)	1.261	1.260

The ACADS referee value comes from the 1989 survey; programs today give 1.26–1.29 on the specified surface.

3.84 V050b — ACADS 3(a), weak layer, non-circular search

Giam & Donald (1989), ACADS problem 3(a).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

As V050a, with a circular search followed by non-circular optimisation; the critical surface slides along the weak layer with a back scarp to the crest.

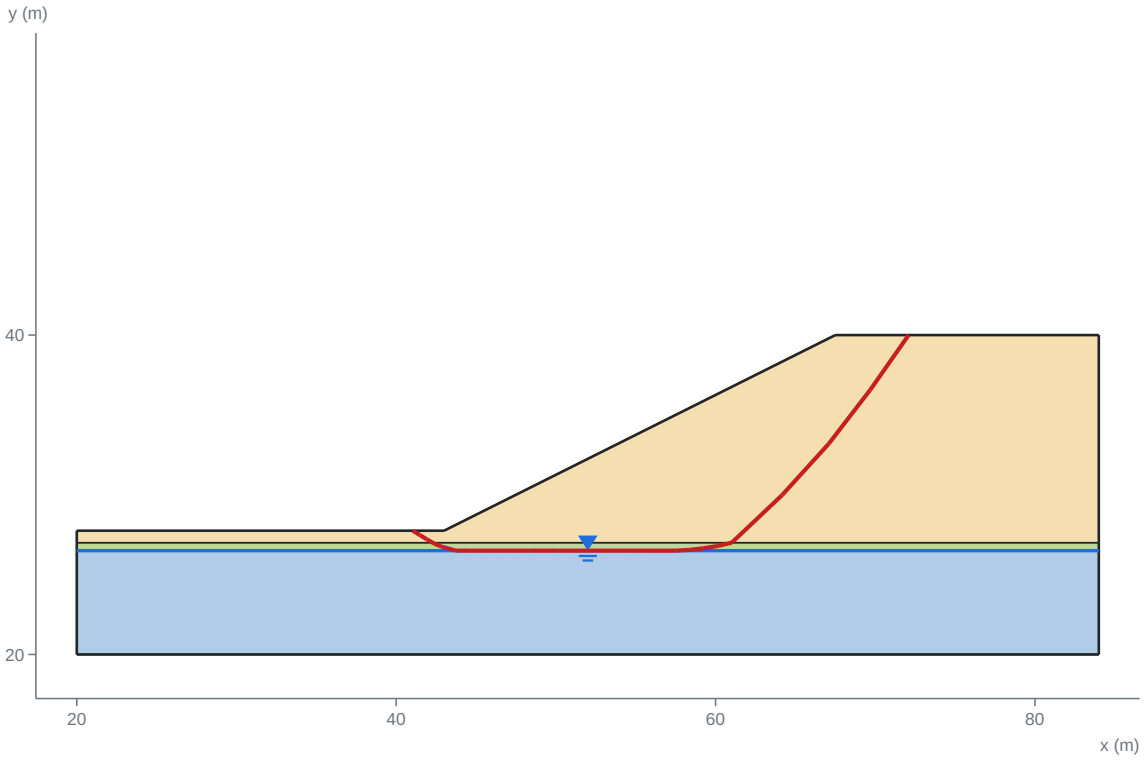


Figure 84. V050b: critical surface (Spencer, non-circular, FOS = 1.225).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°
Soil 1 below	Mohr-Coulomb	18.84	c' = 28.50	20.00°

Published value	Method	Source	cSlope
1.24–1.27	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.246	1.225

ACADS accepted 1.24–1.27; cSlope's search finds a surface along the weak layer at 1.228 (Spencer), unchanged at 50, 100 and 200 slices.

3.85 V051 — ACADS 4, weak layer, water table and surcharges

Giam & Donald (1989), ACADS slope stability programs review.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Weak-layer slope with a piezometric line and two surcharge strips (20 kPa on the bench, 20–40 kPa ramp on the crest); non-circular search. The published spread is wide: a search-difficulty benchmark.

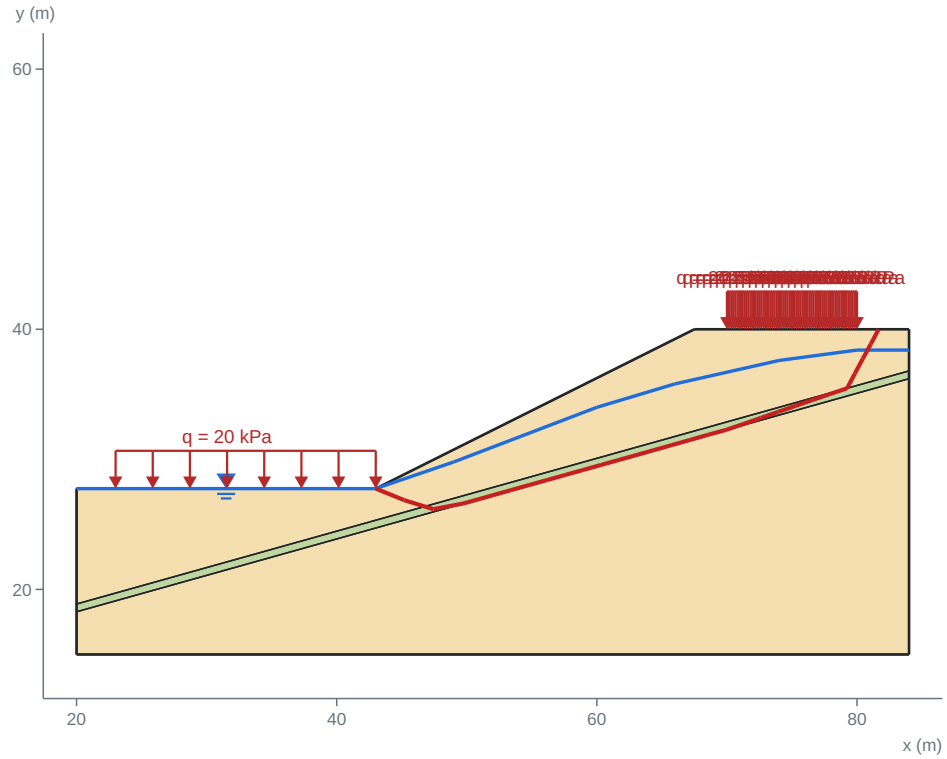


Figure 85. V051: critical surface (Spencer, non-circular, FOS = 0.714).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.84	c' = 28.50	20.00°
Weak Layer	Mohr-Coulomb	18.84	c' = 0.00	10.00°

Published value	Method	Source	cSlope
0.78	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Spencer, non-circular	0.707	0.714

The ACADS referee value 0.78 comes from the 1989 survey, where the participating programs averaged 0.81; optimised non-circular searches today give about 0.68–0.71.

3.86 V052a — Talbingo Dam, end of construction

Giam & Donald (1989), ACADS slope stability programs review.

Critical circle search · Units: SI (m, kPa, kN/m³)

ACADS 2(a): Talbingo Dam, four zones, critical-circle search. The minimum is a shallow slide parallel to the steeper upstream face.

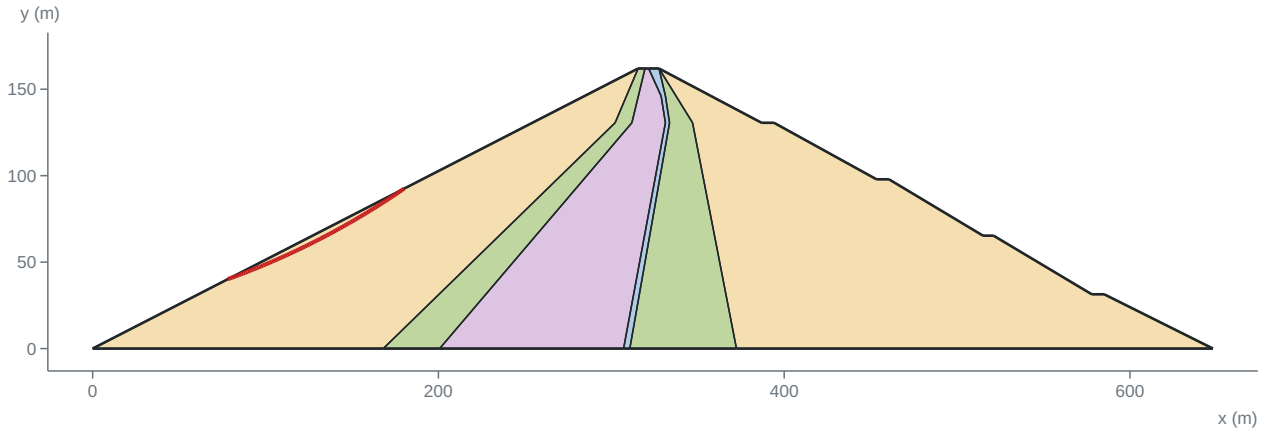


Figure 86. V052a: critical circle (Bishop simplified, FOS = 1.964).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Published value	Method	Source	cSlope
1.95	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.948–1.951	1.964
Spencer	1.948	1.964
Janbu corrected	1.949	1.974

3.87 V052b — Talbingo Dam, specified circle

Giam & Donald (1989), ACADS slope stability programs review.

Fixed circle · Units: SI (m, kPa, kN/m³)

ACADS 2(b): Talbingo Dam on the circle (100.3, 291.0, R = 278.8).

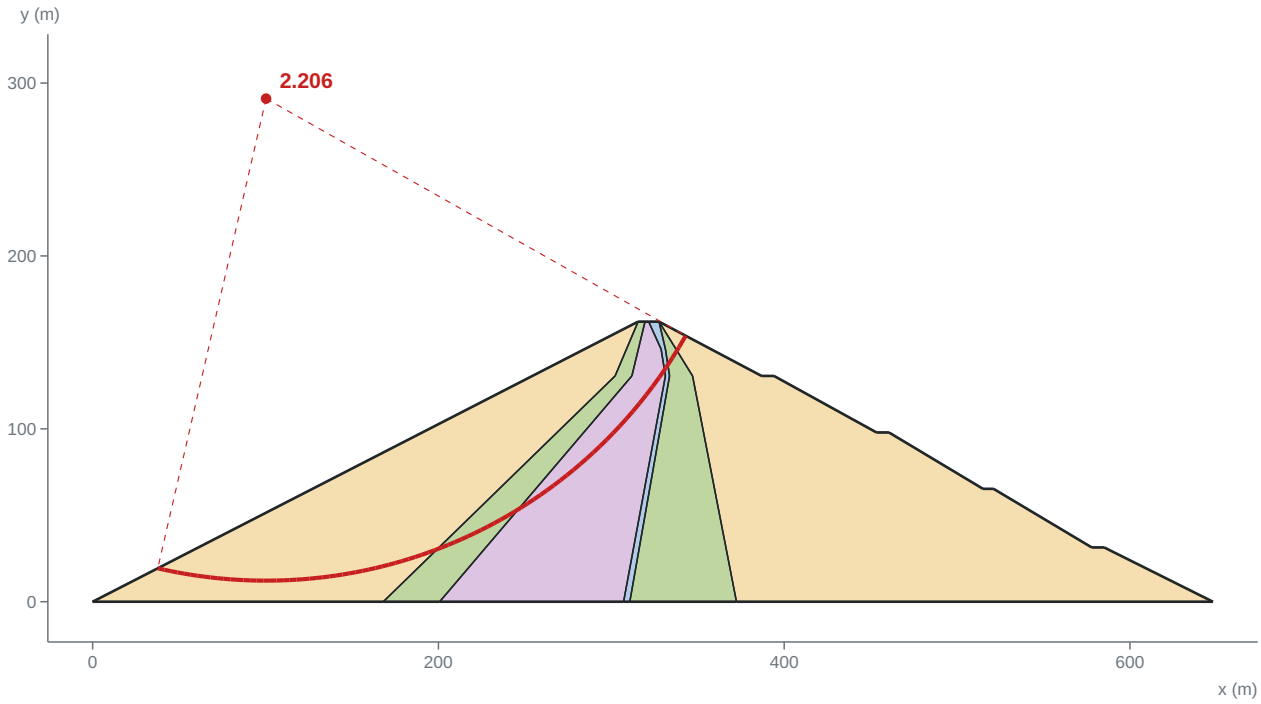


Figure 87. V052b: analysed circle (Bishop simplified, FOS = 2.206).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Rockfill	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Transitions	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Filter	Mohr-Coulomb	20.40	c' = 0.00	45.00°
Core	Mohr-Coulomb	18.10	c' = 85.00	23.00°

Published value	Method	Source	cSlope
2.29	not stated	Giam & Donald (1989), ACADS	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.208	2.206
Spencer	2.292	2.290
Janbu corrected	2.073	2.073
Morgenstern-Price (half-sine)	2.299	2.299

3.88 V053 — Greco (1996) ex. 5 / Chen & Shao weak-layer slope, reference geometry

Greco (1996) example 5; Chen & Shao (1988).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Greco's weak-seam slope (V054) with a commercial program's geometry and water table; circular search with non-circular optimisation, vertical pore-pressure head.

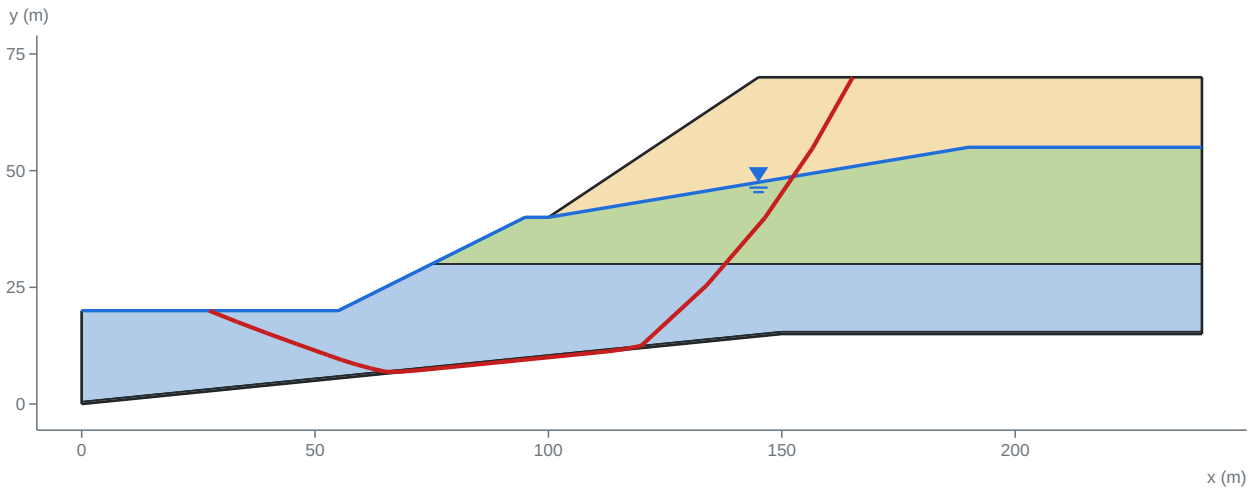


Figure 88. V053: critical surface (Spencer, non-circular, FOS = 1.006).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c' = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c' = 19.80	30.00°
Layer 4	Mohr-Coulomb	21.50	c' = 9.80	16.00°

Published value	Method	Source	cSlope
1.009–1.025	Spencer, non-circular (test problem 3)	Chen & Shao (1988)	1.006
0.97	Spencer, non-circular (global minimum along the weak layer)	Greco (1996)	1.006
1.08	Spencer, local minimum (nearly circular surface from the toe)	Greco (1996)	1.091

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.010	1.006
Spencer	–	1.091

Chen & Shao give 1.009–1.025 with Spencer's assumption for a surface sliding along the weak seam (their test problem 3, Tables 5–6). Greco's Monte Carlo search finds a global minimum of 0.97 along the weak layer, entering the level ground far from the toe, and a local minimum of 1.08 on a nearly circular surface from the toe; the programs' block searches and cSlope give 1.01 on a commercial program's version of the geometry, 4 % above Greco's global minimum.

3.89 V054 — Greco (1996) example 5 / Chen & Shao (1988)

Greco (1996), example 5; Chen & Shao (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Four layers including a weak seam, with a water table; critical circle search, vertical pore-pressure head.

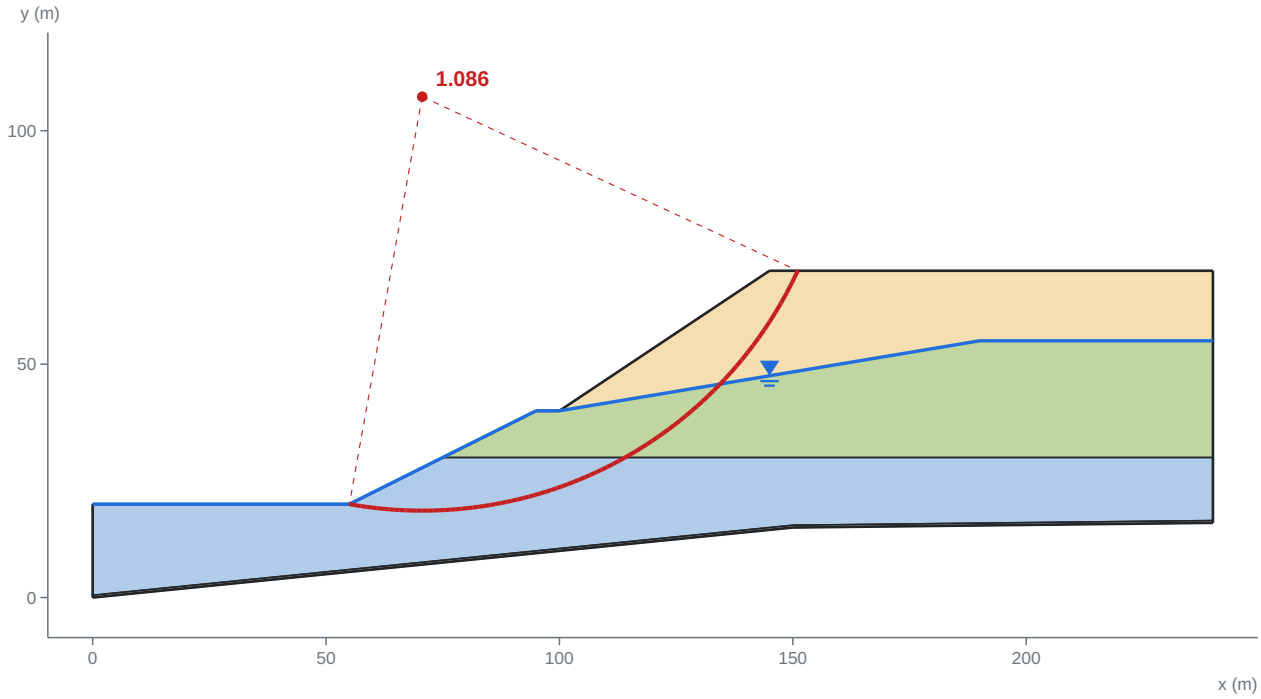


Figure 89. V054: critical circle (Bishop simplified, FOS = 1.086).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	20.00	c' = 9.80	35.00°
Layer 2	Mohr-Coulomb	19.00	c' = 58.80	25.00°
Layer 3	Mohr-Coulomb	21.50	c' = 19.80	30.00°
Seam	Mohr-Coulomb	21.50	c' = 9.80	16.00°

Published value	Method	Source	cSlope
1.08	Spencer, local minimum (nearly circular surface from the toe)	Greco (1996)	1.091

cSlope analysis	Other programs	cSlope
Bishop simplified	1.087	1.086
Spencer	1.093	1.091

Greco's 1.08 is the local minimum of his search, a nearly circular surface from the toe; his global minimum, 0.97 along the weak layer, is compared in V053.

3.90 V055 — Griffiths & Lane example 2, slope with a foundation layer

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The 2:1 slope (c/γH = 0.05, φ = 20°) on a foundation of the same soil; unconstrained circular search finds the toe circle. Converted to SI.

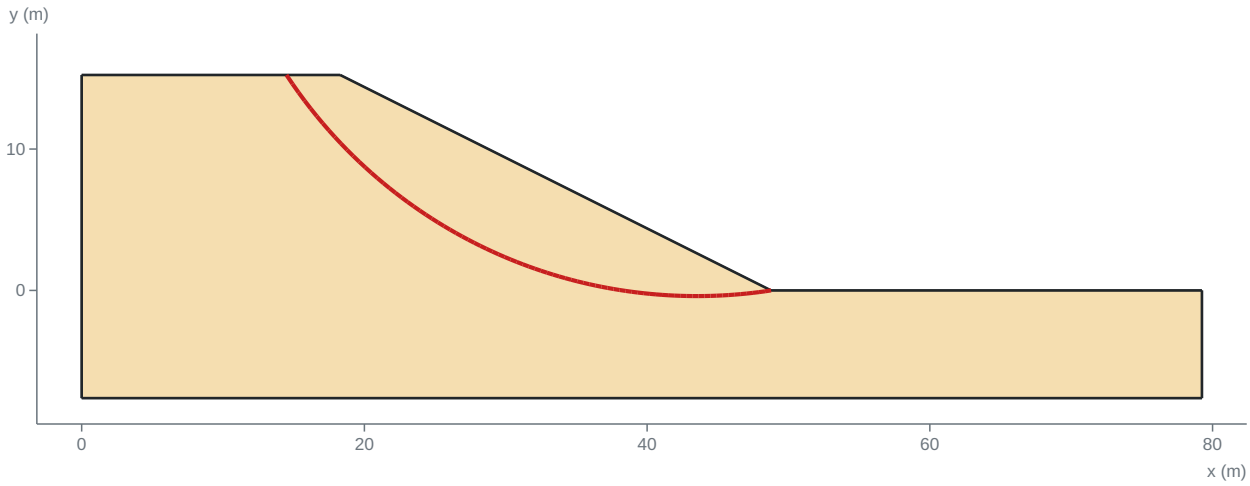


Figure 90. V055: critical circle (Spencer, FOS = 1.366).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Mohr-Coulomb	19.64	c' = 14.96	20.00°

Published value	Method	Source	cSlope
1.40	finite elements	Griffiths & Lane	1.366

cSlope analysis	Other programs	cSlope
Spencer	1.366	1.366

3.91 V056 — Griffiths & Lane example 3, thin weak layer, cu2/cu1 = 0.2

Griffiths & Lane (1999), Géotechnique 49(3).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Undrained clay slope with a thin weak layer at a fifth of the strength, on the three-line wedge through the layer (the paper's Janbu mechanism). Converted to SI.

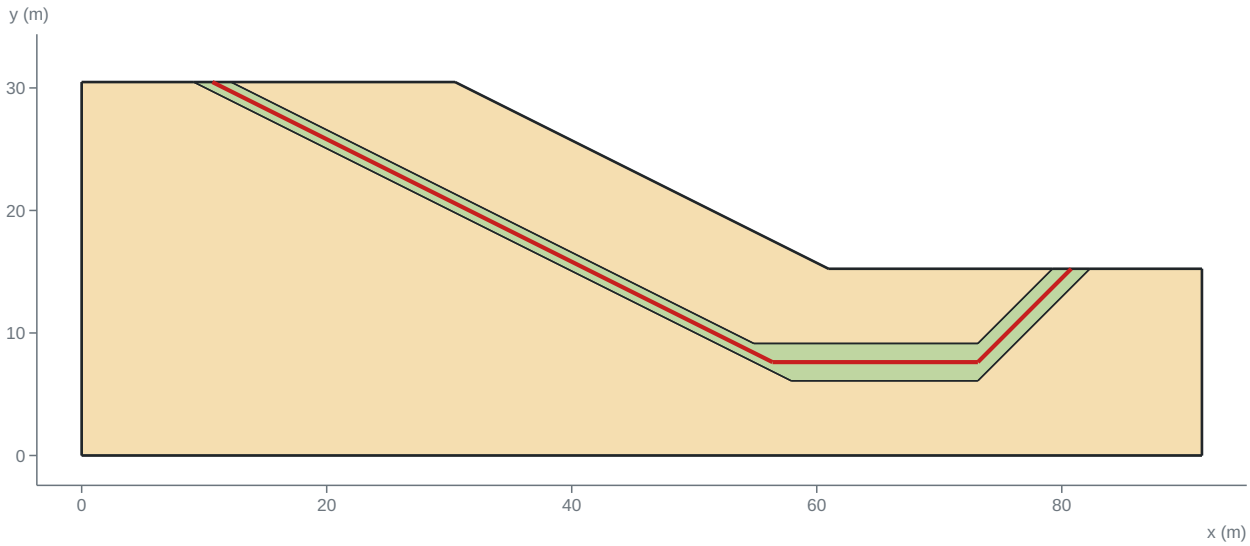


Figure 91. V056: analysed surface (Janbu corrected, FOS = 0.499).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Surrounding clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Thin weak layer	Undrained (φ = 0)	19.64	Su = 14.96	0°

Published value	Method	Source	cSlope
0.485	Janbu wedge	Griffiths & Lane	0.499

cSlope analysis	Other programs	cSlope
Janbu corrected	–	0.499

3.92 V057a — Griffiths & Lane example 4, cu2/cu1 = 1 (deep base circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Undrained clay slope over a foundation of the same strength; the deep base circle. Converted to SI.

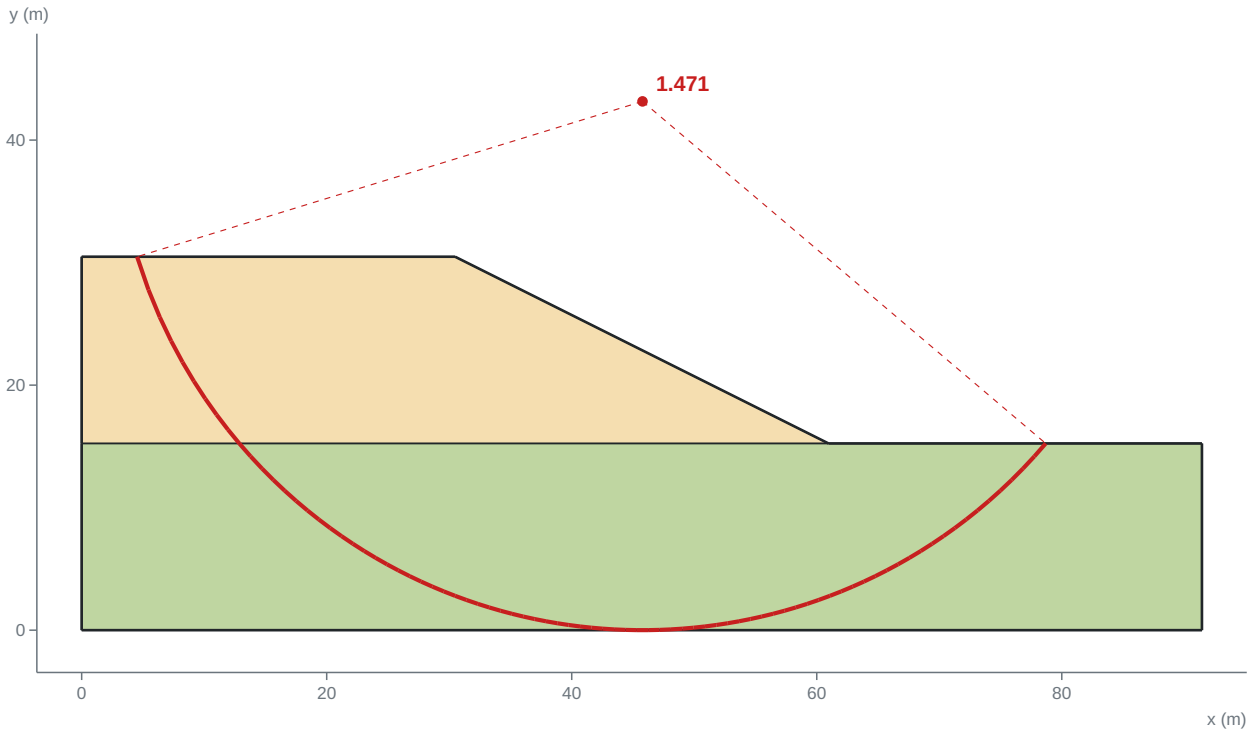


Figure 92. V057a: critical circle (Spencer, FOS = 1.471).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 74.81	0°

Published value	Method	Source	cSlope
1.46	limit-equilibrium chart	Griffiths & Lane	1.471
1.47	stability chart (Taylor)	Griffiths & Lane (1999)	–
1.45	finite elements	Griffiths & Lane (1999)	–

cSlope analysis	Other programs	cSlope
Spencer	1.468	1.471

3.93 V057b — Griffiths & Lane example 4, cu2/cu1 = 2 (toe circle)

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

The same slope over a foundation twice as strong; the shallow toe circle. Converted to SI.

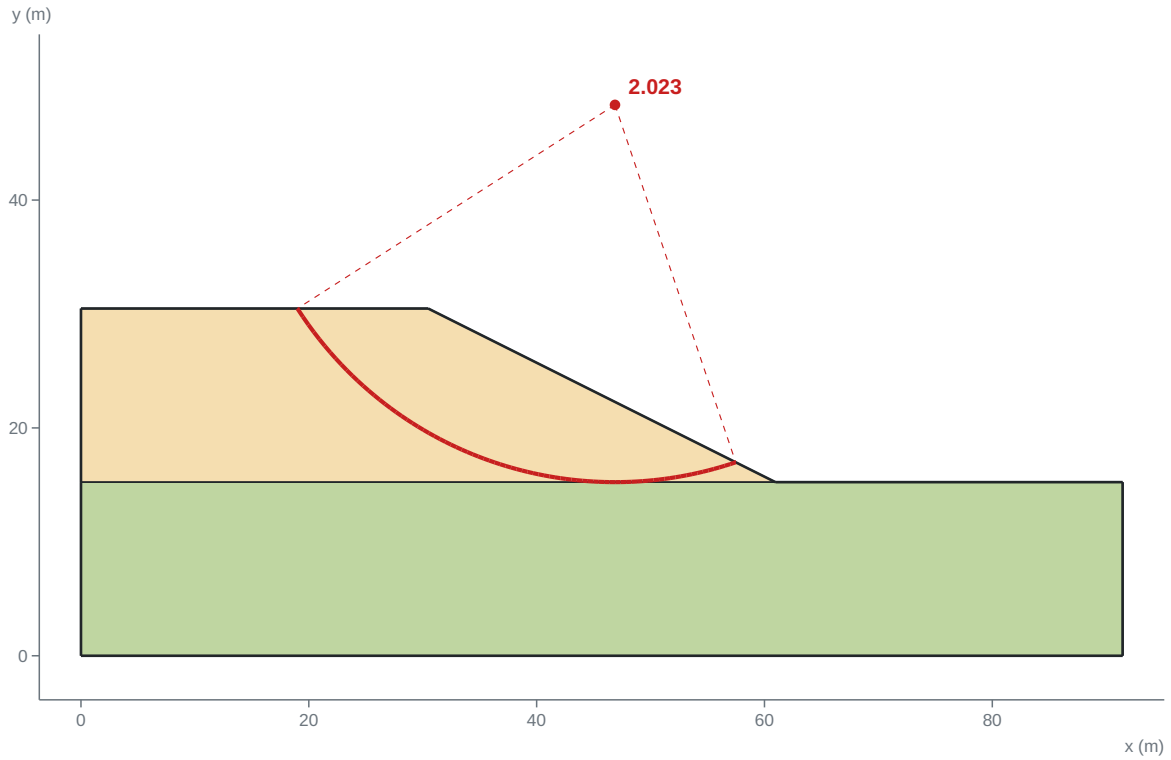


Figure 93. V057b: critical circle (Spencer, FOS = 2.023).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Slope clay	Undrained (φ = 0)	19.64	Su = 74.81	0°
Foundation clay	Undrained (φ = 0)	19.64	Su = 149.6	0°

Published value	Method	Source	cSlope
2.04	limit-equilibrium chart	Griffiths & Lane	2.023
2.10	stability chart (Taylor)	Griffiths & Lane (1999)	–
2.03	finite elements	Griffiths & Lane (1999)	–

cSlope analysis	Other programs	cSlope
Spencer	2.022	2.023

3.94 V058 — Griffiths & Lane example 6, earth dam with full reservoir

Griffiths & Lane (1999), Géotechnique 49(3).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two-sided earth dam with the reservoir full and a piezometric line; circular search. Converted to SI.

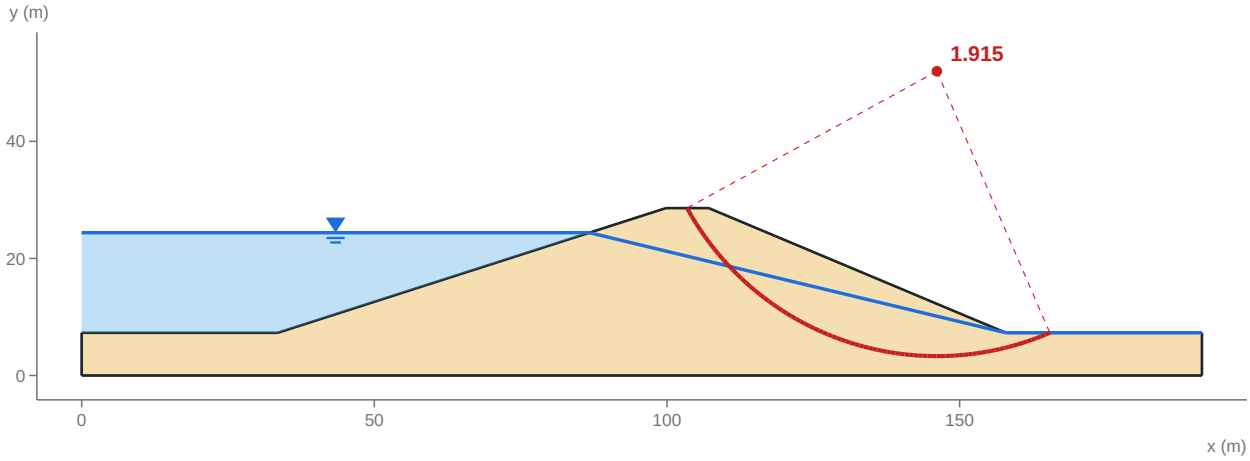


Figure 94. V058: critical circle (Spencer, FOS = 1.915).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.20	c' = 13.80	37.00°

Published value	Method	Source	cSlope
1.90	finite elements	Griffiths & Lane	1.915

cSlope analysis	Other programs	cSlope
Spencer	1.915	1.915

3.95 V059a — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.4

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 84 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

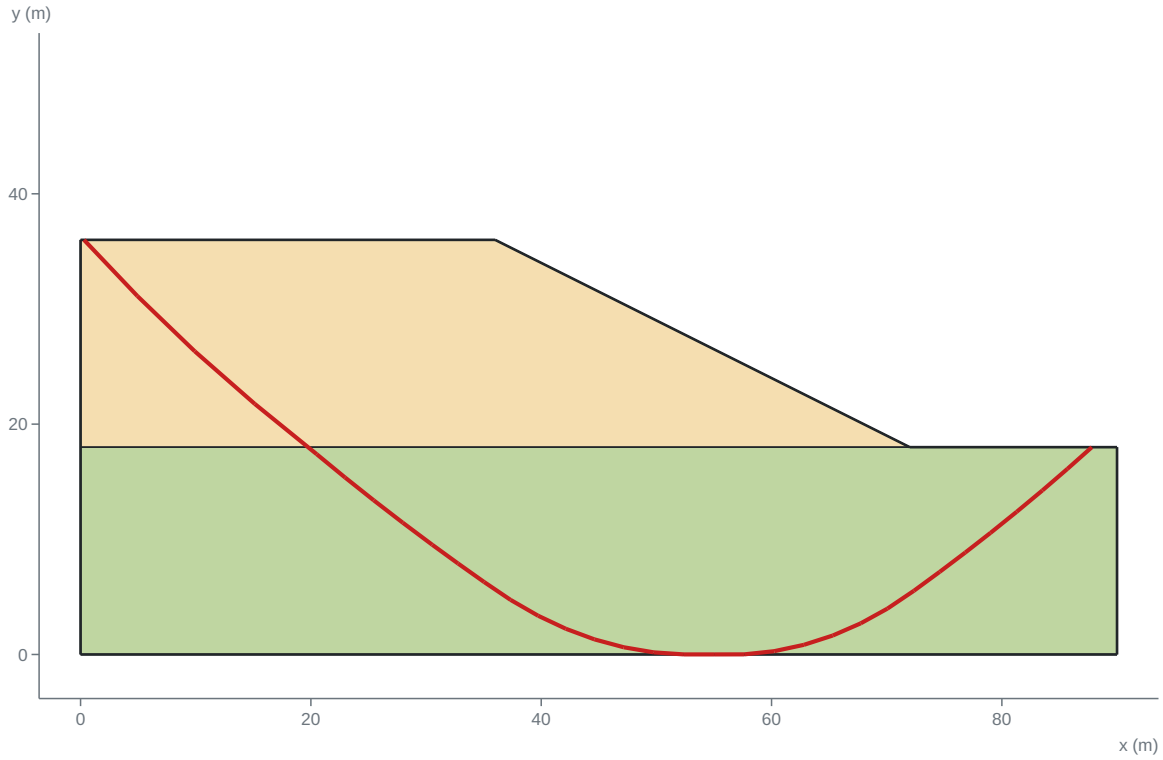


Figure 95. V059a: critical surface (Spencer, non-circular, FOS = 1.214).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 84.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.215	1.214

The paper gives its results in figures only; the comparison is with the other programs.

3.96 V059b — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.5

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 90 kPa), γ = 20, firm base; optimised non-circular search. The deep mechanism governs.

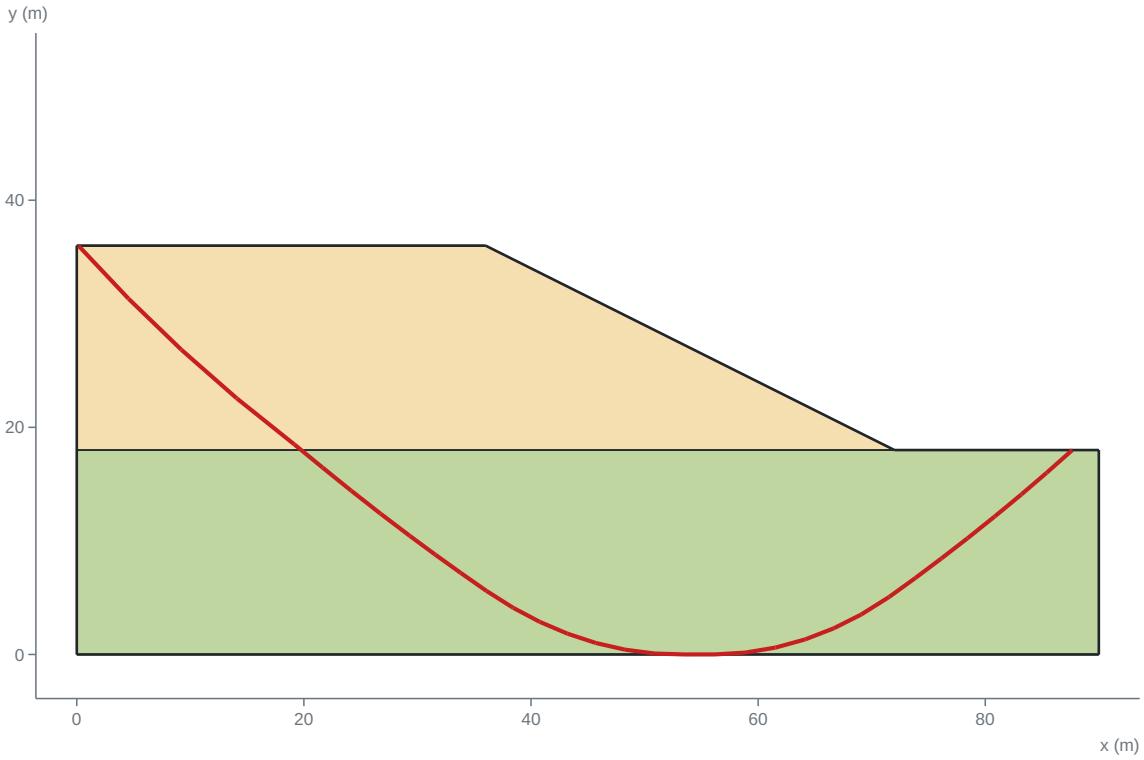


Figure 96. V059b: critical surface (Spencer, non-circular, FOS = 1.289).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 90.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.290	1.289

The paper gives its results in figures only; the comparison is with the other programs.

3.97 V059c — Guo & Griffiths two-layer undrained slope, cu2/cu1 = 1.6

Guo & Griffiths (2020), Can. Geotech. J. 57.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

18 m embankment at 1V:2H (Su = 60 kPa) on an 18 m undrained foundation (Su = 96 kPa), γ = 20, firm base; optimised non-circular search. The shallow mechanism governs.

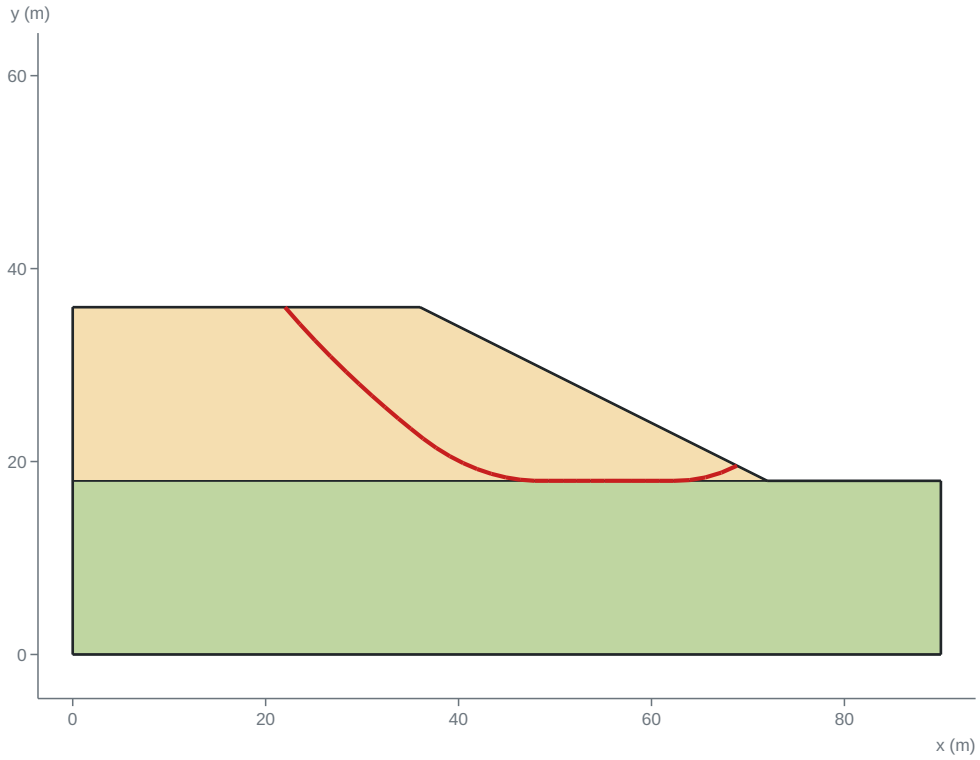


Figure 97. V059c: critical surface (Spencer, non-circular, FOS = 1.312).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Undrained (φ = 0)	20.00	Su = 60.00	0°
Foundation	Undrained (φ = 0)	20.00	Su = 96.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.315	1.312

The paper gives its results in figures only; the comparison is with the other programs.

3.98 V060a — Cannon Dam end of construction (Hassan & Wolff), critical circle

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Critical circle search · Units: SI (m, kPa, kN/m³)

Two clay fills with a sand-filter strip under the crest and a spoil-covered downstream toe; Bishop search for the minimum factor of safety at the mean strengths. The reliability index on the minimum-β surface is V060b.

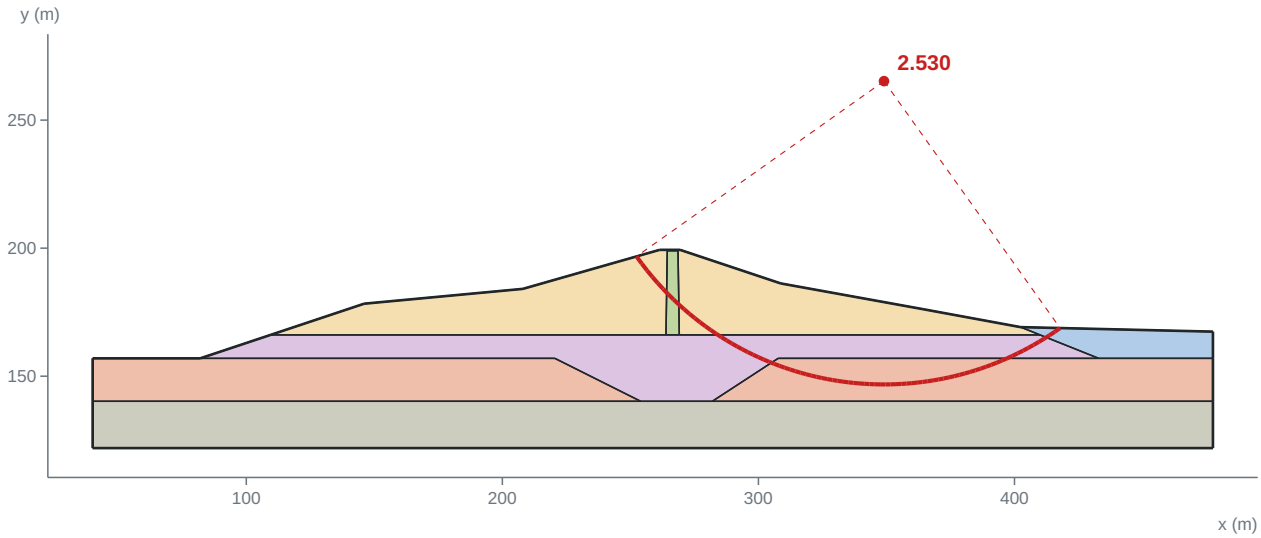


Figure 98. V060a: critical circle (Bishop simplified, FOS = 2.530).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c' = 1000.0	45.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	2.529	2.530

Hassan & Wolff give 2.753 on a fixed circle (V062a), not a search result.

3.99 V060b — Cannon Dam end of construction (Hassan & Wolff), reliability on the minimum-β c

Hassan & Wolff (1999), J. Geotech. Geoenviron. Eng. 125(4).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V060a dam on the circle of minimum reliability index: Phase I fill φ = 8.5° ± 8.5°, c = 117.8 ± 58.9 kPa; Phase II fill c = 143.6 ± 79 kPa, φ = 15° ± 9°, treated as independent normal variables (another program's Taylor series has no correlation term); Bishop.

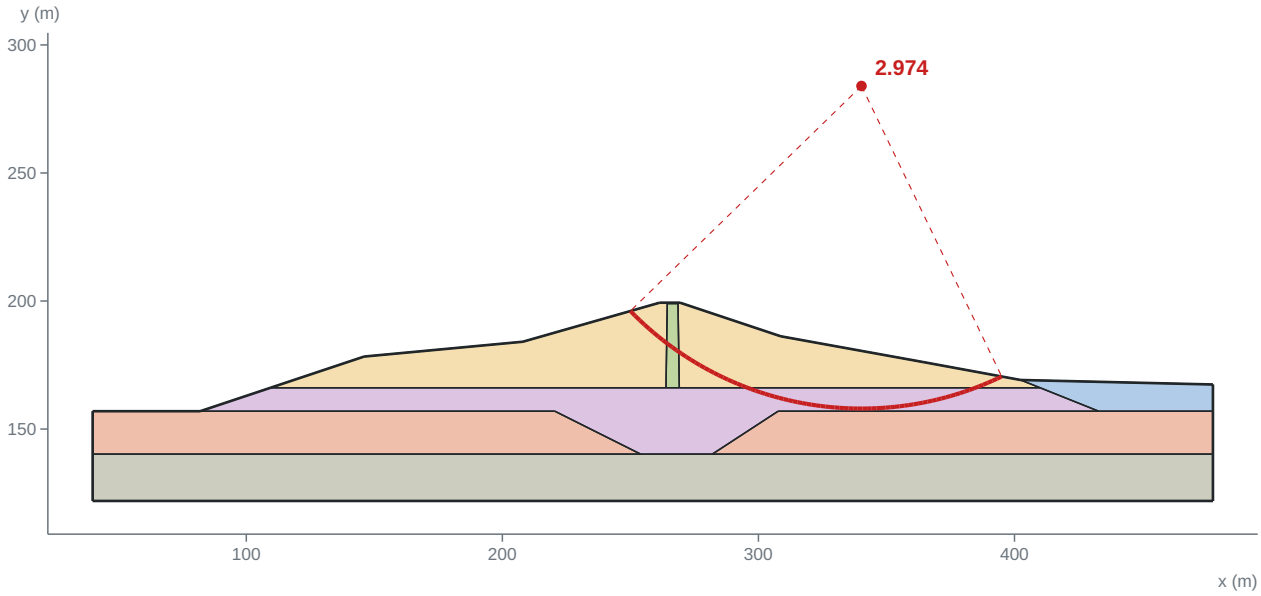


Figure 99. V060b: analysed circle (Bishop simplified, FOS = 2.974).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II clay fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Phase I clay fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	23.00	c' = 1000.0	45.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	2.973	2.974

The source gives reliability indices on fixed circles; no factor of safety is published for the minimum-β circle.

Probabilistic analysis on the same surface: 4 normal variables (independent); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Taylor series)	–	3.353	3.354

3.100 V061 — Clarence Cannon Dam (Wolff & Harr), specified surface

Wolff & Harr (1987).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Phase I and II fills with a chimney drain, a horizontal drain blanket (el. 548.2–552.8 ft) and a cutoff trench to rock, flat water table at el. 557 ft; the printed non-circular surface. Converted to SI.

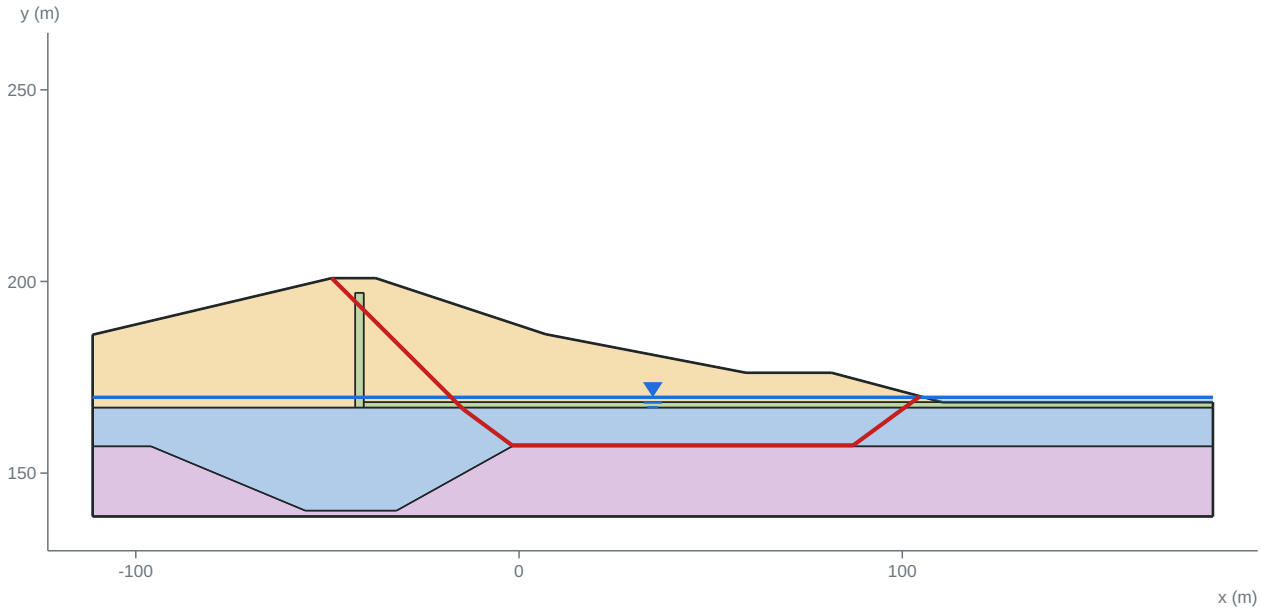


Figure 100. V061: analysed surface (Morgenstern-Price (half-sine), FOS = 2.348).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	23.56	c' = 138.9	14.80°
Sand drain	Mohr-Coulomb	18.85	c' = 0.00	30.00°
Phase I fill	Mohr-Coulomb	23.56	c' = 106.8	6.34°
Foundation sand	Mohr-Coulomb	18.85	c' = 0.00	30.00°

Published value	Method	Source	cSlope
2.36	Morgenstern-Price (half-sine)	Wolff & Harr	2.348

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	2.343	2.348
Spencer	2.383	2.388

3.101 V062a — Clarence Cannon Dam, surface 7A

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7A (centre 320.175, 313.336; tangent elevation 145.347), Bishop.

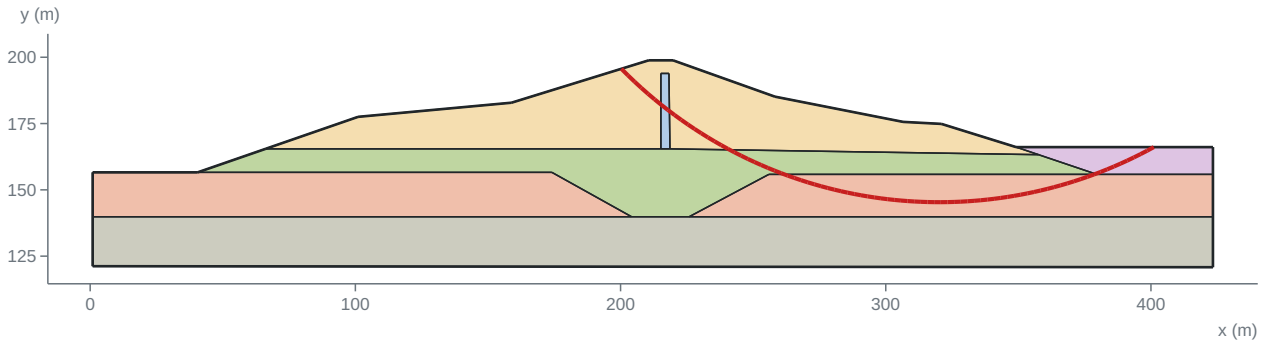


Figure 101. V062a: analysed circle (Bishop simplified, FOS = 2.549).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.753	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.551–2.560	2.549

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	10.953	11.034

3.102 V062b — Clarence Cannon Dam, surface 7B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7B (centre 295.235, 277.589; tangent elevation 156.298), Bishop.

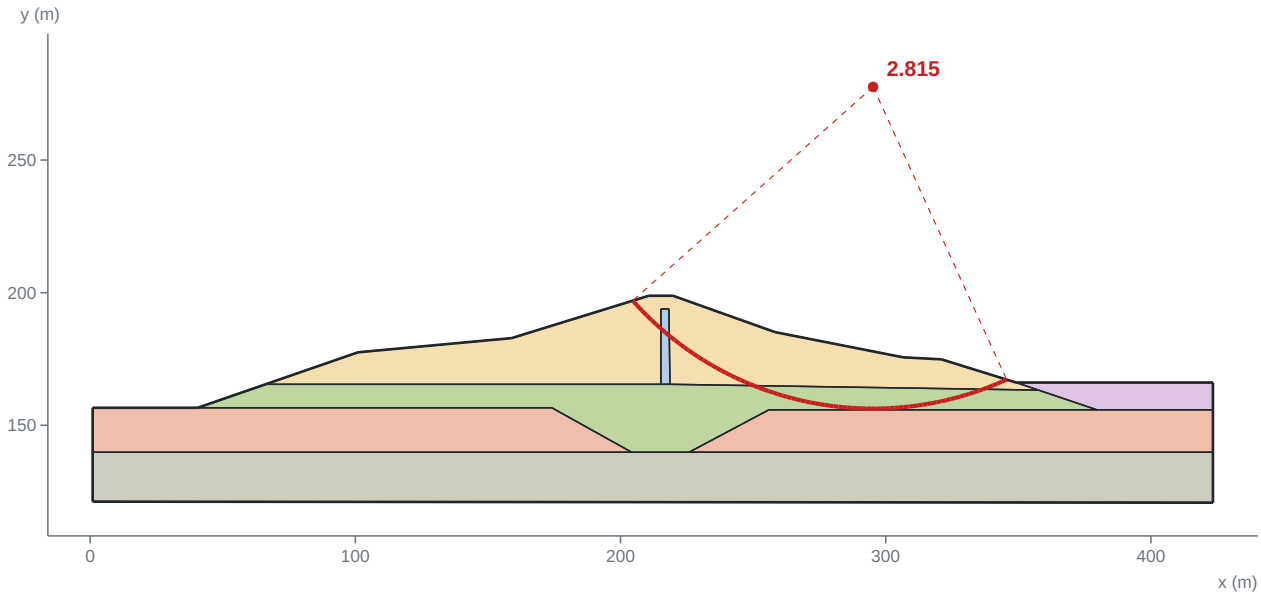


Figure 102. V062b: analysed circle (Bishop simplified, FOS = 2.815).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.352	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.806–2.820	2.815

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.351	4.402

3.103 V062c — Clarence Cannon Dam, surface 7C

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7C (centre 303.683, 310.206; tangent elevation 155.928), Bishop.

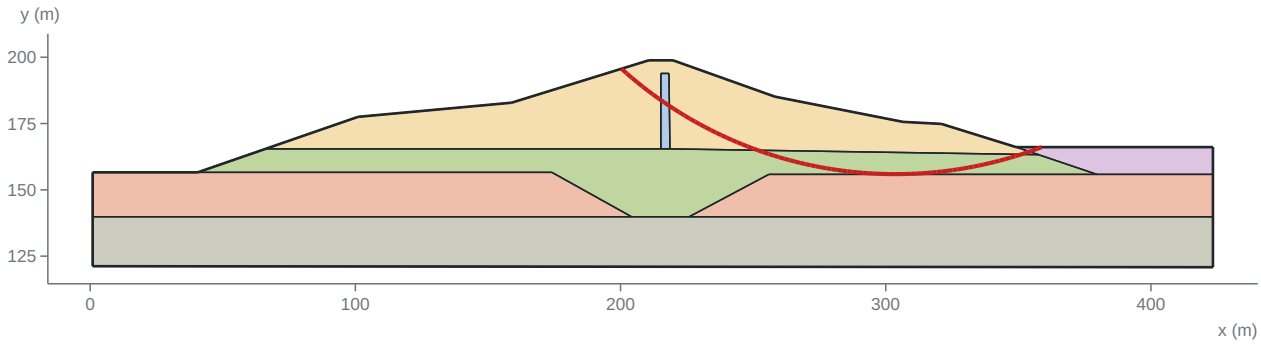


Figure 103. V062c: analysed circle (Bishop simplified, FOS = 2.779).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.523	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.771–2.777	2.779

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.263	4.325

3.104 V062d — Clarence Cannon Dam, surface 7D

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7D (centre 322.152, 310.206; tangent elevation 145.36), Bishop.

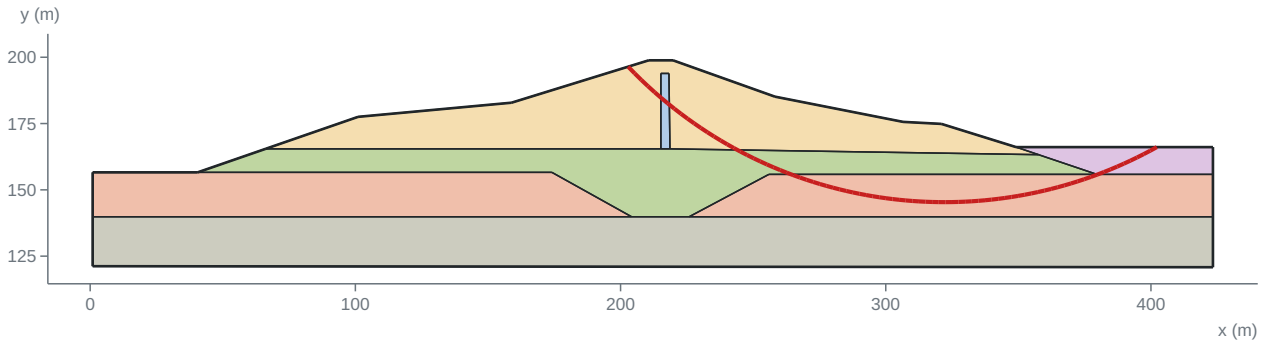


Figure 104. V062d: analysed circle (Bishop simplified, FOS = 2.577).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.457	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.583–2.589	2.577

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	11.092	11.076

3.105 V062e — Clarence Cannon Dam, surface 7E

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 7E (centre 327.693, 324.018; tangent elevation 147.163), Bishop.

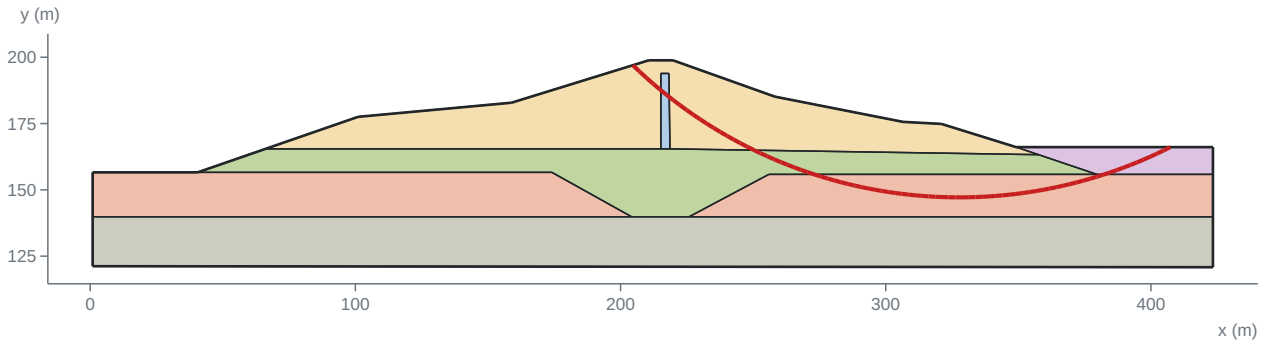


Figure 105. V062e: analysed circle (Bishop simplified, FOS = 2.690).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.602	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.692–2.703	2.690

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	10.281	10.329

3.106 V062f — Clarence Cannon Dam, surface 8B

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8B (centre 304.239, 307.858; tangent elevation 155.311), Bishop.

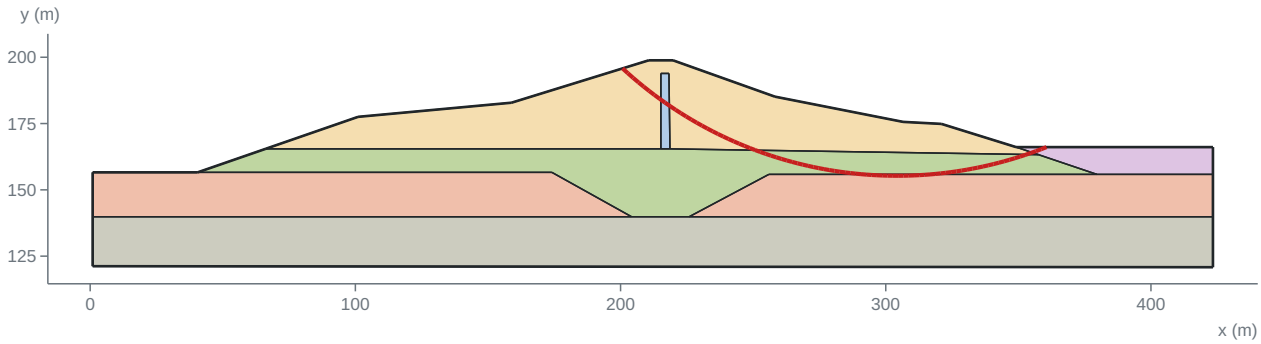


Figure 106. V062f: analysed circle (Bishop simplified, FOS = 2.673).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
2.995	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.672–2.673	2.673

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	4.858	4.933

3.107 V062g — Clarence Cannon Dam, surface 8F

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8F (centre 313.114, 304.784; tangent elevation 160.368), Bishop.

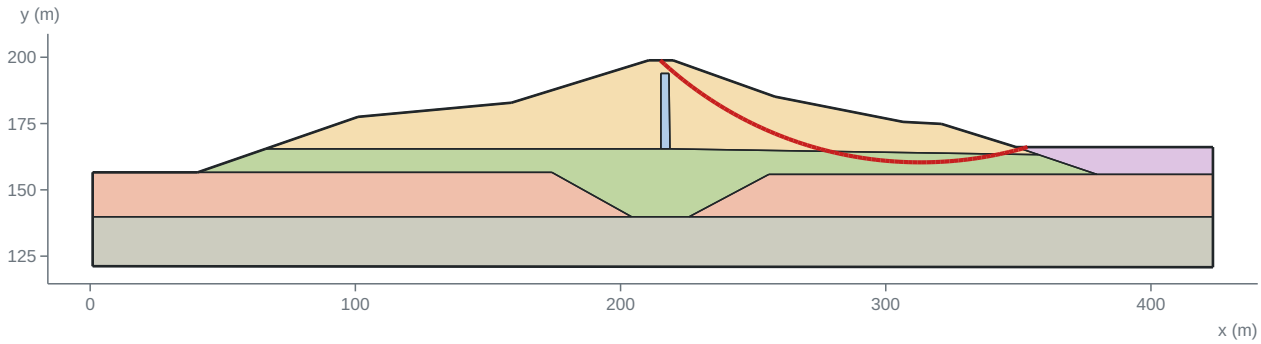


Figure 107. V062g: analysed circle (Bishop simplified, FOS = 3.592).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
3.916	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	3.586–3.598	3.592

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	5.485	5.528

3.108 V062h — Clarence Cannon Dam, surface 8G

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8G (centre 314.856, 305.604; tangent elevation 165.756), Bishop.

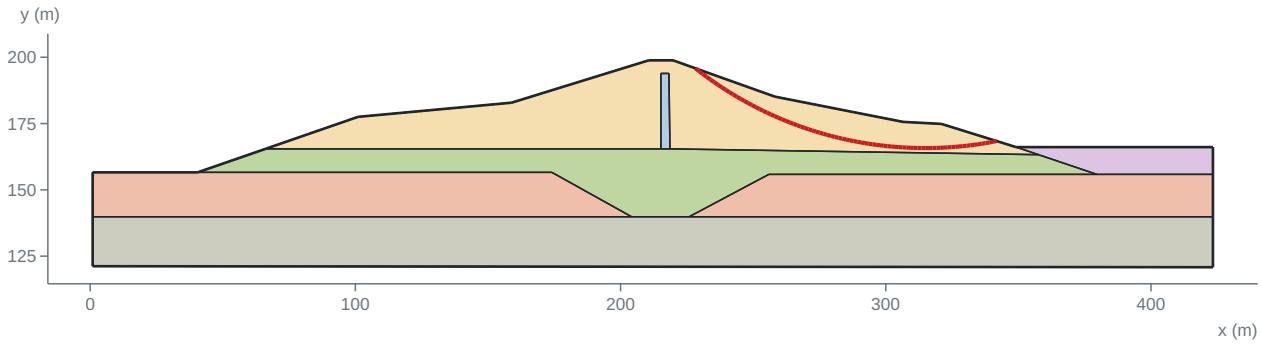


Figure 108. V062h: analysed circle (Bishop simplified, FOS = 6.063).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
10.576	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	6.069–6.074	6.063

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	5.563	5.490

3.109 V062i — Clarence Cannon Dam, surface 8H

Hassan & Wolff (1999).

Fixed circle · Units: SI (m, kPa, kN/m³)

Six-zone embankment dam on sand over limestone, analysed on the published circle 8H (centre 316.198, 305.604; tangent elevation 170.537), Bishop.

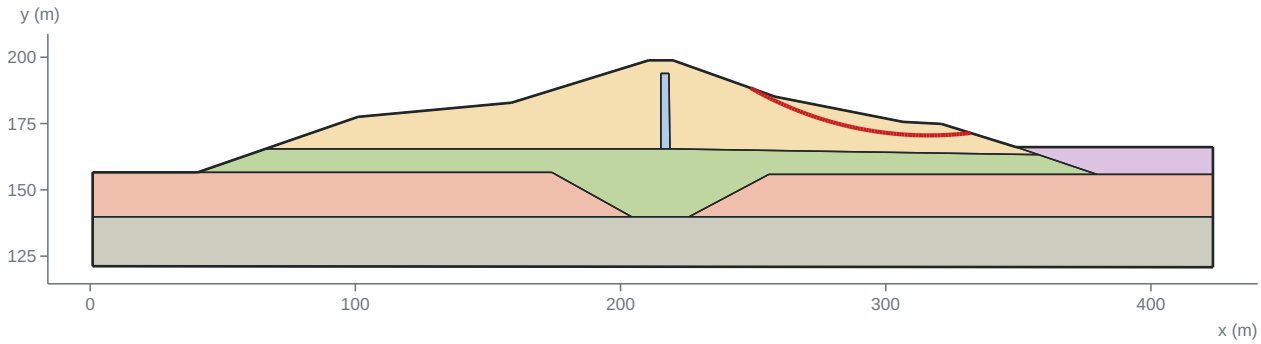


Figure 109. V062i: analysed circle (Bishop simplified, FOS = 11.567).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Phase II fill	Mohr-Coulomb	22.00	c' = 143.6	15.00°
Phase I fill	Mohr-Coulomb	22.00	c' = 117.8	8.50°
Sand filter	Mohr-Coulomb	22.00	c' = 0.00	35.00°
Spoil fill	Mohr-Coulomb	25.00	c' = 5.00	35.00°
Foundation sand	Mohr-Coulomb	20.00	c' = 5.00	18.00°
Limestone	Mohr-Coulomb	20.00	c' = 100.0	35.00°

Published value	Method	Source	cSlope
6.293	not stated	Hassan & Wolff (1999)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	11.583	11.567

Hassan & Wolff's own factors of safety differ from all programs by 3–75 %, while the programs agree with each other within 0.5 %.

Probabilistic analysis on the same surface: 4 normal variables (ρ = -0.55, ρ = 0.1 between pairs); Taylor series (mean ± σ) and Monte Carlo with 10000 samples (seed 1). Differences in β and Pf are absolute.

Quantity	Published	Other programs	cSlope
β lognormal (Monte Carlo)	–	6.394	6.431

3.110 V063 — Huang & Jia earth dam, dry

Huang, M. & Jia, C.-Q. (2009). Strength reduction FEM in stability analysis of soil slopes subjected to transient unsaturated seepage. Computers and Geotechnics 36(1–2), 93–101.

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous earth dam (c′ = 13.8 kPa, φ′ = 37°, γ = 18.2 kN/m³) before the reservoir is filled; circular search.

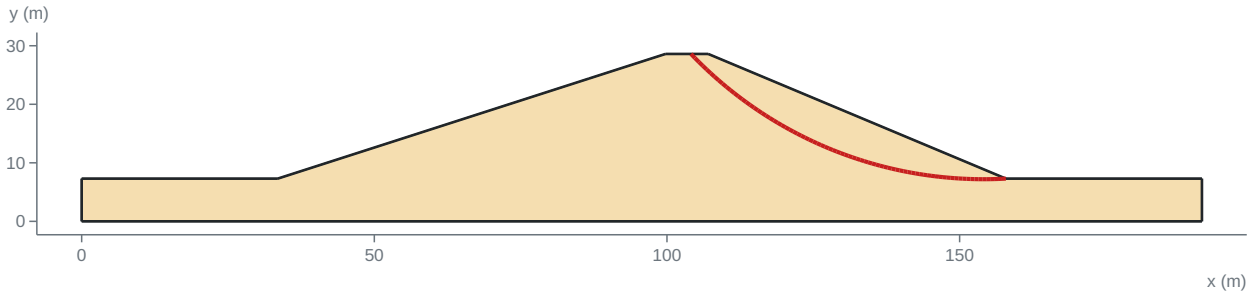


Figure 110. V063: critical circle (Spencer, FOS = 2.451).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	18.20	c' = 13.80	37.00°

Published value	Method	Source	cSlope
2.43	Bishop simplified	Huang & Jia	2.453
2.50	finite-element strength reduction	Huang & Jia (2009)	–

cSlope analysis	Other programs	cSlope
Spencer	2.455	2.451
Bishop simplified	–	2.453

Huang & Jia's limit-equilibrium value 2.43 is by Bishop's method (their p. 95 and fig. 3a); their strength-reduction finite-element value is 2.50.

3.111 V064 — Leshchinsky & Han MSE wall, five tiers

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Five 1.8 m tiers (Ta = 10.1 kN/m).

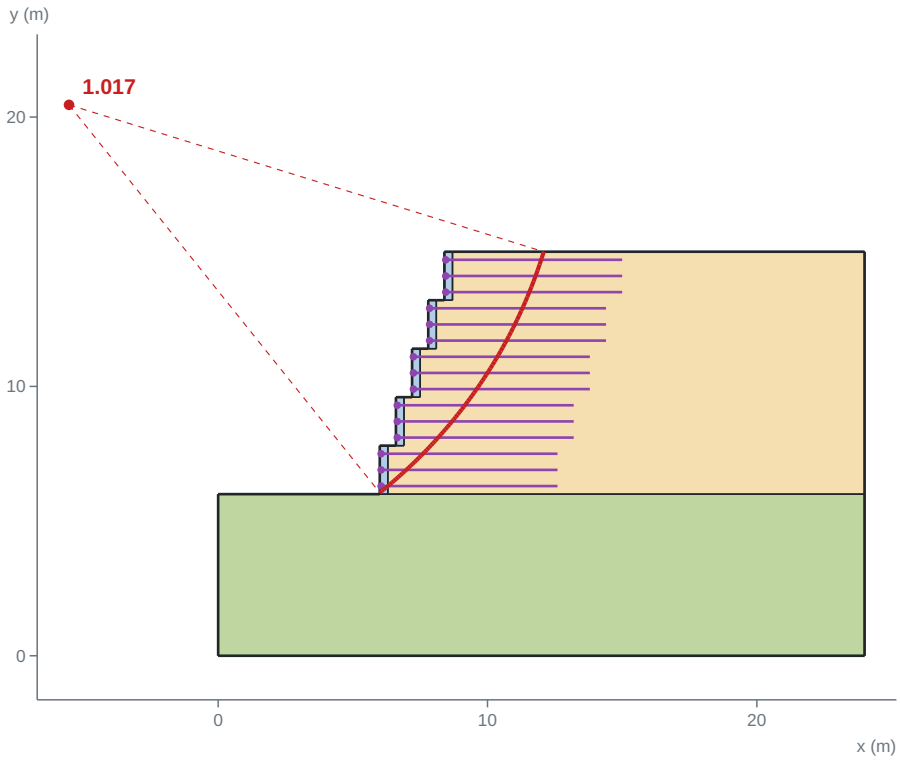


Figure 111. V064: analysed circle (Bishop simplified, FOS = 1.017).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	1.017

cSlope analysis	Other programs	cSlope
Bishop simplified	1.020	1.017

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.112 V065 — Leshchinsky & Han MSE wall, weak foundation

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a on a c = 0, φ = 18° foundation.

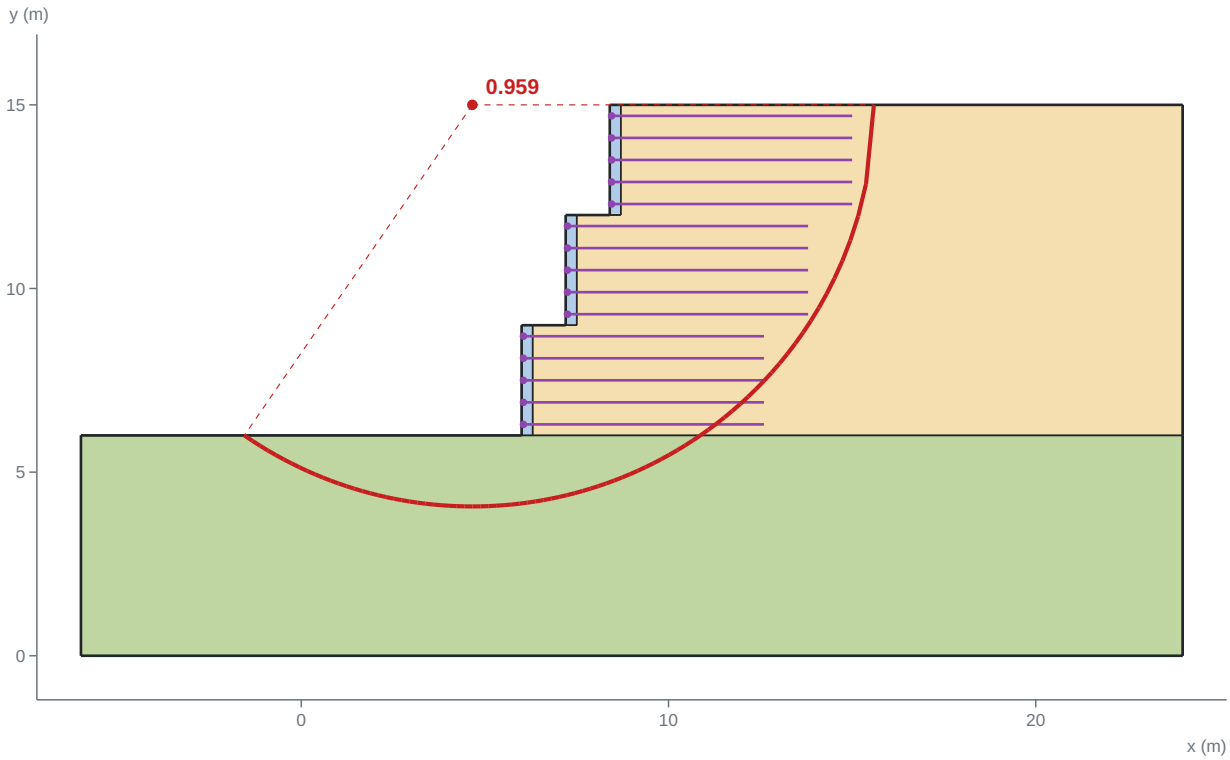


Figure 112. V065: analysed circle (Spencer, FOS = 0.959).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 0.00	18.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	0.977
0.86	finite differences	Leshchinsky & Han (2004)	–
0.98	Bishop simplified, deep circle (read from fig. 7a)	Leshchinsky & Han (2004)	0.977
0.92	Spencer, three-part wedge	Leshchinsky & Han (2004)	0.959

cSlope analysis	Other programs	cSlope
Spencer	0.964	0.959
Bishop simplified	–	0.977

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.113 V066a — Leshchinsky & Han three-tier MSE wall, baseline

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

Three 3 m tiers offset 1.2 m, block facing, geotextiles every 0.6 m (L = 6.3 m, Ta = 10 kN/m, applied horizontally, passive), a commercial program's critical circle.

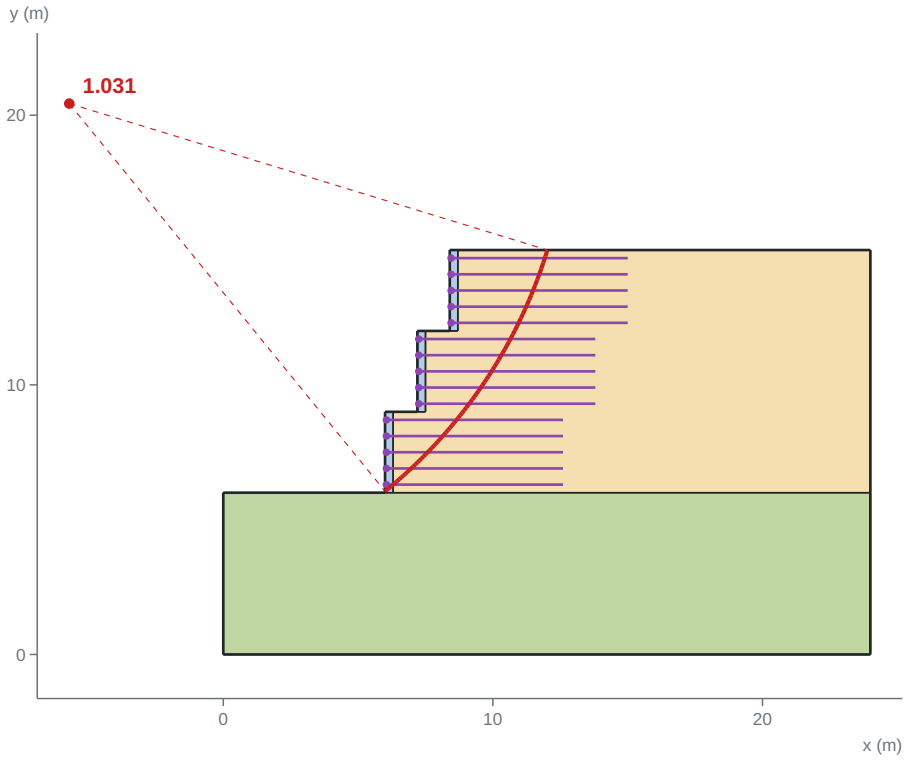


Figure 113. V066a: analysed circle (Bishop simplified, FOS = 1.031).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	1.031
0.99	finite differences	Leshchinsky & Han (2004)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.040	1.031

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.114 V066b — Leshchinsky & Han MSE wall, L = 4.2 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with 4.2 m geotextiles (Ta = 11.4 kN/m).

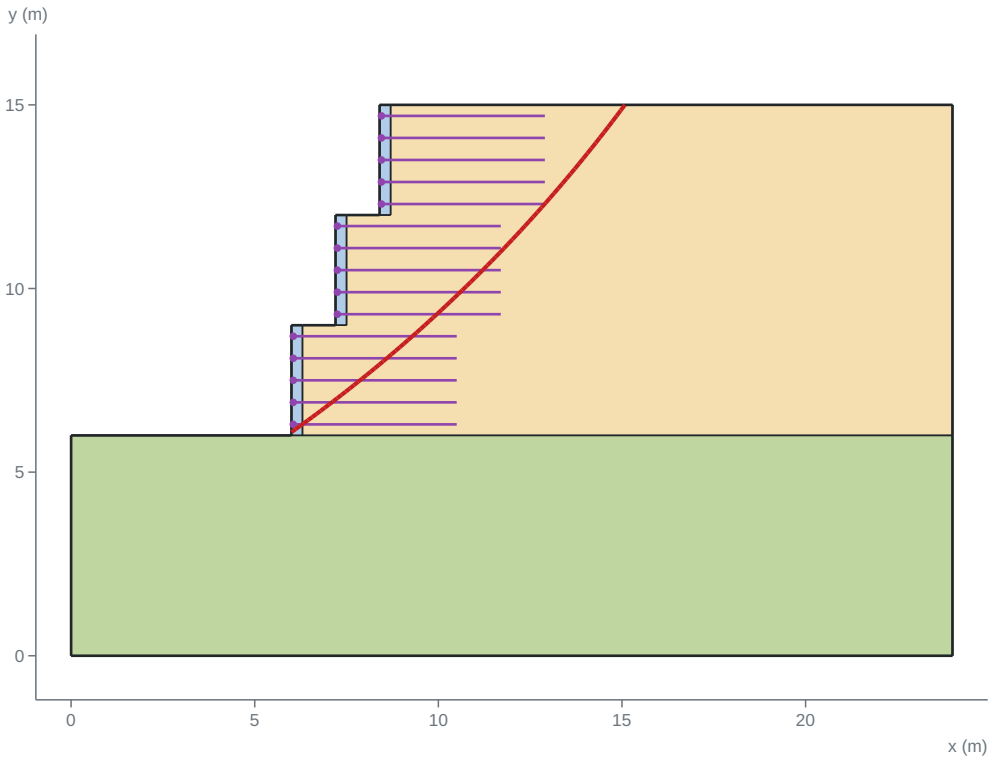


Figure 114. V066b: analysed circle (Spencer, FOS = 1.011).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	1.013
0.98	finite differences	Leshchinsky & Han (2004)	–

cSlope analysis	Other programs	cSlope
Spencer	1.011	1.011
Bishop simplified	–	1.013

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.115 V066c — Leshchinsky & Han MSE wall, two reinforcement types

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with two reinforcement strengths (7.5 and 11.0 kN/m).

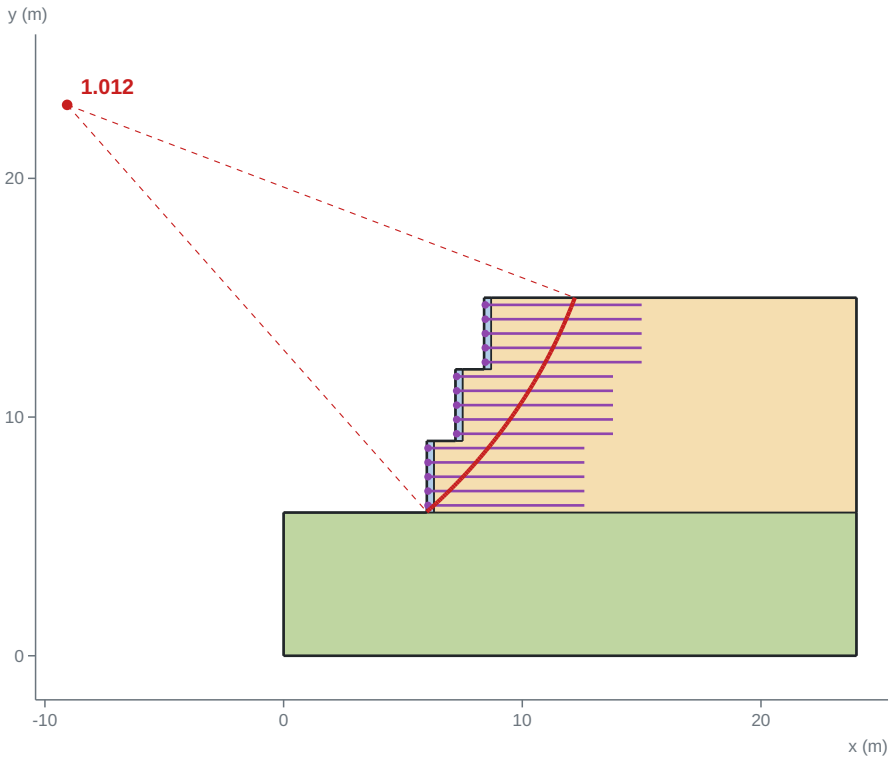


Figure 115. V066c: analysed circle (Bishop simplified, FOS = 1.012).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified	Leshchinsky & Han	1.012
1.01	finite differences	Leshchinsky & Han (2004)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.012

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.116 V066d — Leshchinsky & Han MSE wall, water hw = 3 m

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a with the water table 3 m above the toe: pore pressure in the foundation only (drained fill); Ta = 9.25 kN/m. The pond in front of the wall is left out of this comparison: another program's water loads apply no pressure to the vertical facing, whereas cSlope does.

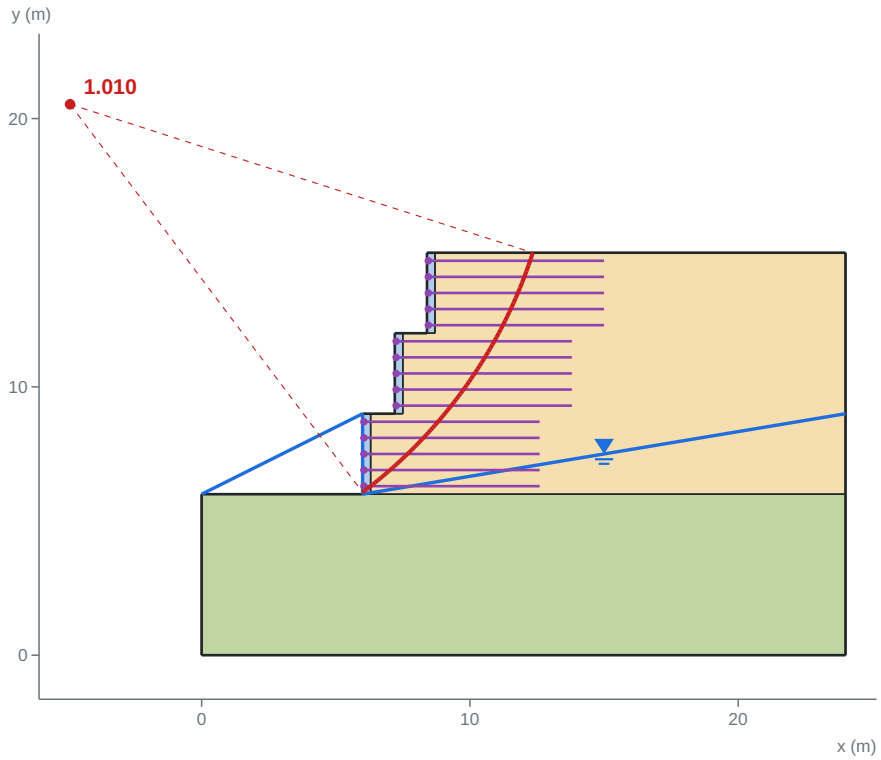


Figure 116. V066d: analysed circle (Bishop simplified, FOS = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	1.010
1.01	finite differences	Leshchinsky & Han (2004)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.010	1.010

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value.

3.117 V066e — Leshchinsky & Han MSE wall, surcharge

Leshchinsky & Han (2004), J. Geotech. Geoenviron. Eng. 130(12).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V066a (Ta = 10 kN/m) with a 20 kPa surcharge on the top tier (x = 8.7 to 24).

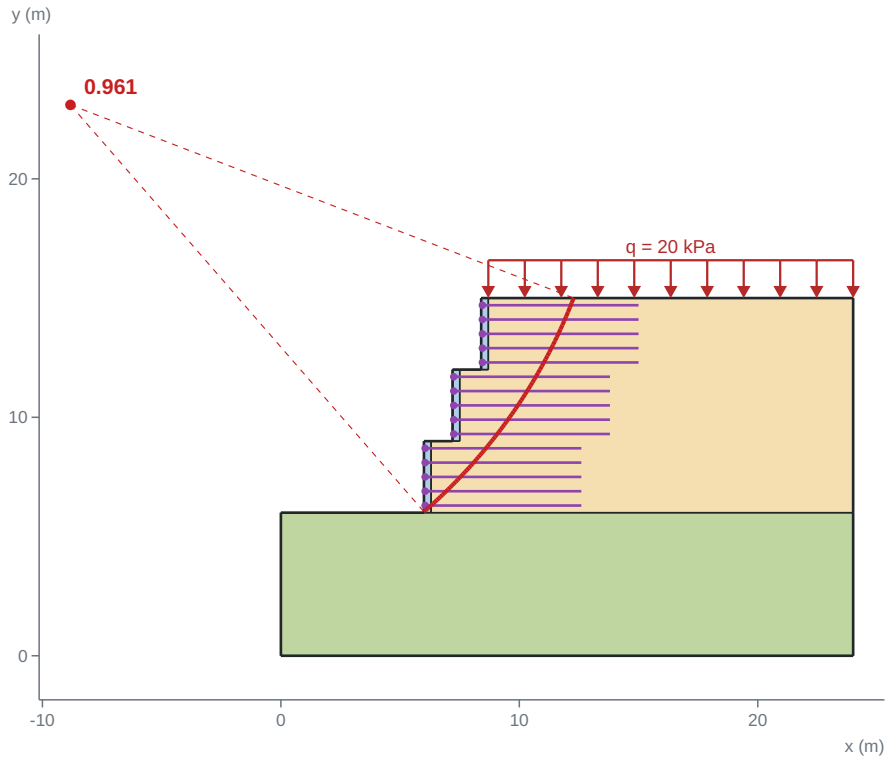


Figure 117. V066e: analysed circle (Bishop simplified, FOS = 0.961).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Reinforced and retained fill	Mohr-Coulomb	18.00	c' = 0.00	34.00°
Foundation soil	Mohr-Coulomb	18.00	c' = 10.00	34.00°
Blocks	Mohr-Coulomb	18.00	c' = 2.50	34.00°

Published value	Method	Source	cSlope
1.00	Bishop simplified, circular	Leshchinsky & Han (2004)	0.961
1.02	finite differences	Leshchinsky & Han (2004)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	0.958	0.961

Leshchinsky & Han give 1.00 with Bishop's method for every variant of the wall (the reinforcement was designed for that value), and a finite-difference value. Here the reinforcement keeps the baseline strength of 10 kN/m, as in the program compared with, not the 11.6 kN/m designed for the surcharge, so the published 1.00 is not directly comparable.

3.118 V067 — Low (1989), layered undrained clay

Low (1989).

Critical circle search · Units: SI (m, kPa, kN/m³)

Embankment on three undrained clay layers of different strength; critical circle search.

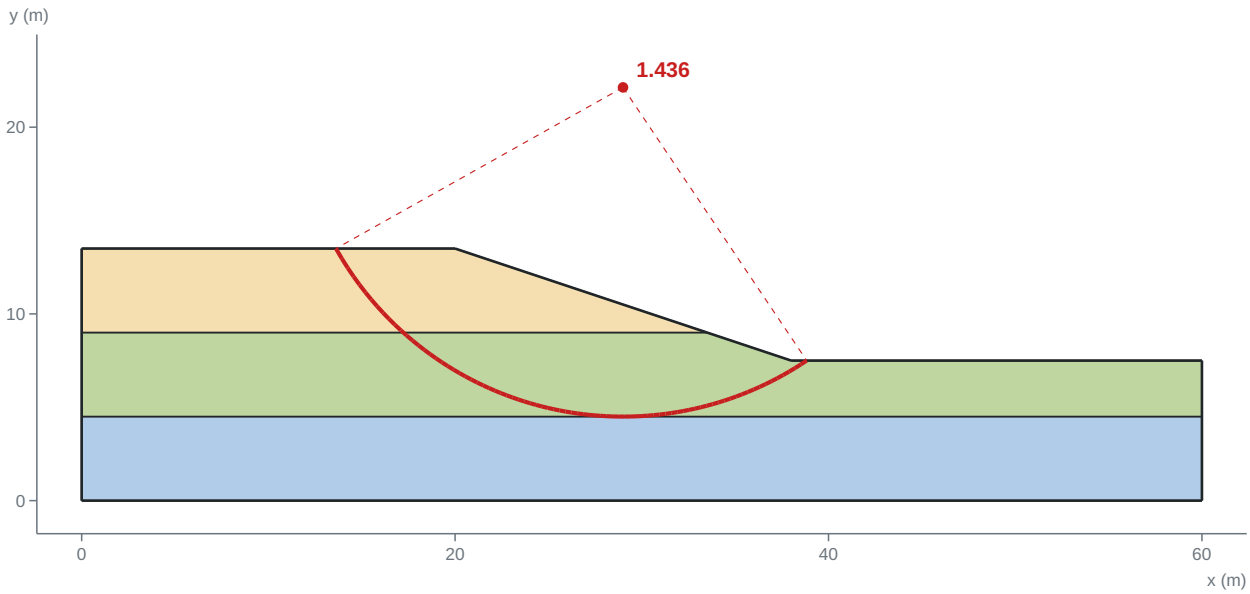


Figure 118. V067: critical circle (Bishop simplified, FOS = 1.436).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
A	Undrained (φ = 0)	18.00	Su = 30.00	0°
B	Undrained (φ = 0)	18.00	Su = 20.00	0°
C	Undrained (φ = 0)	18.00	Su = 150.0	0°

Published value	Method	Source	cSlope
1.44	Ordinary	Low (1989)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.439	1.436

3.119 V068 — Low (1989), stiff crust over soft clays, published circle

Low, B.K. (1989). Stability analysis of embankments on soft ground. J. Geotech. Eng. 115(2), 211–227; example 2, fig. 11.

Fixed circle · Units: SI (m, kPa, kN/m³)

Stiff upper slope on two undrained clays, the lowest with Su increasing linearly with depth (S = f(datum)), on the critical circle a commercial program prints with its Bishop value (centre (18.001, 16.000), R 15.556).

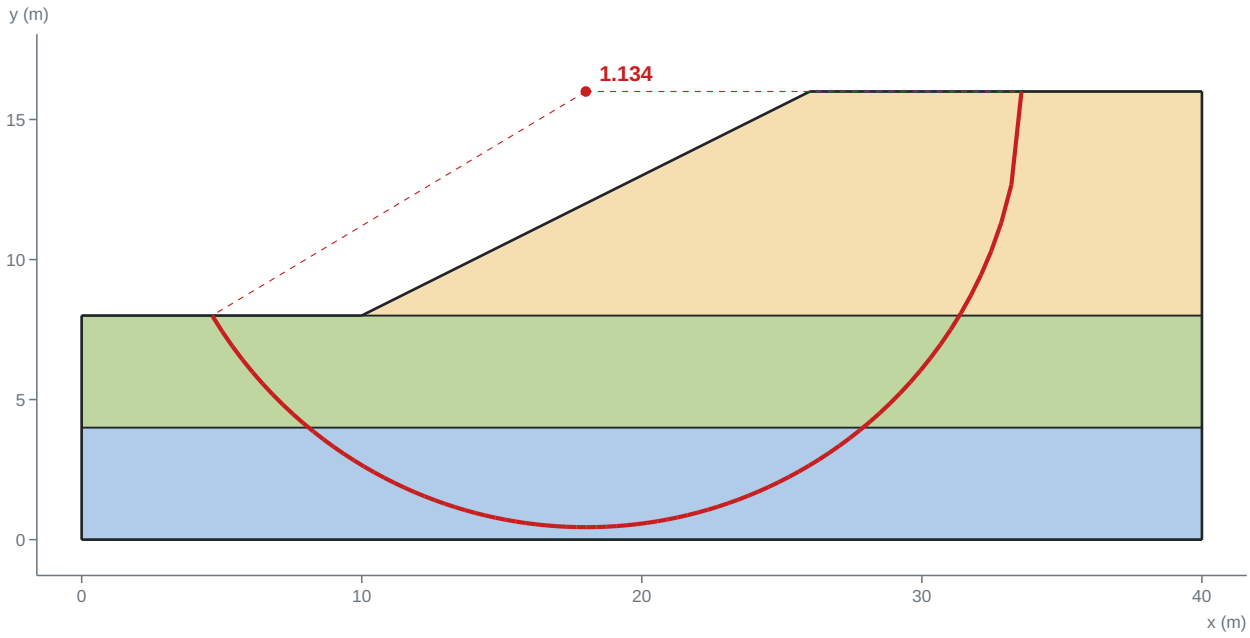


Figure 119. V068: analysed circle (Bishop simplified, FOS = 1.134).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper	Mohr-Coulomb	20.00	c' = 95.00	15.00°
Middle	Mohr-Coulomb	20.00	c' = 15.00	0.00°
Lower	S = f(datum)	20.00	Su = 15.00 at y = 4.00, 3.75 kPa/m below	0°

Published value	Method	Source	cSlope
1.14	Bishop simplified	Low	1.134
1.36	Ordinary	Low (1989)	–
1.31	Ordinary and Bishop with φ = 0 in the bank	Low (1989)	–
1.38	hand calculation (Low's method)	Low (1989)	–
1.17	Spencer, non-circular (random search)	Kim et al. (2002)	–
1.13–1.27	limit analysis, lower and upper bound	Kim et al. (2002)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.134
Spencer, non-circular	–	–

The comparison program's 1.19 is not reproduced on the circle it prints with that value: cSlope gives 1.13 there, close to Low's Bishop value 1.14, which is the check. Low himself calls that Bishop value misleading, an artefact of negative effective normal stresses in the stiff bank with φ = 15°: with φ = 0 in the bank both Ordinary and Bishop give 1.31, and his own hand method 1.38. Kim et al. (2002) give 1.17 with Spencer's method on a non-circular surface and limit-analysis bounds of 1.13–1.27.

3.120 V069a — Morgenstern (1963) complete drawdown, 3:1

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2).

Critical circle search · Units: SI (m, kPa, kN/m³)

3:1 slope after complete drawdown: piezometric line on the slope face, base at the toe (D = 1). Pore pressure from the vertical head. Converted to SI.

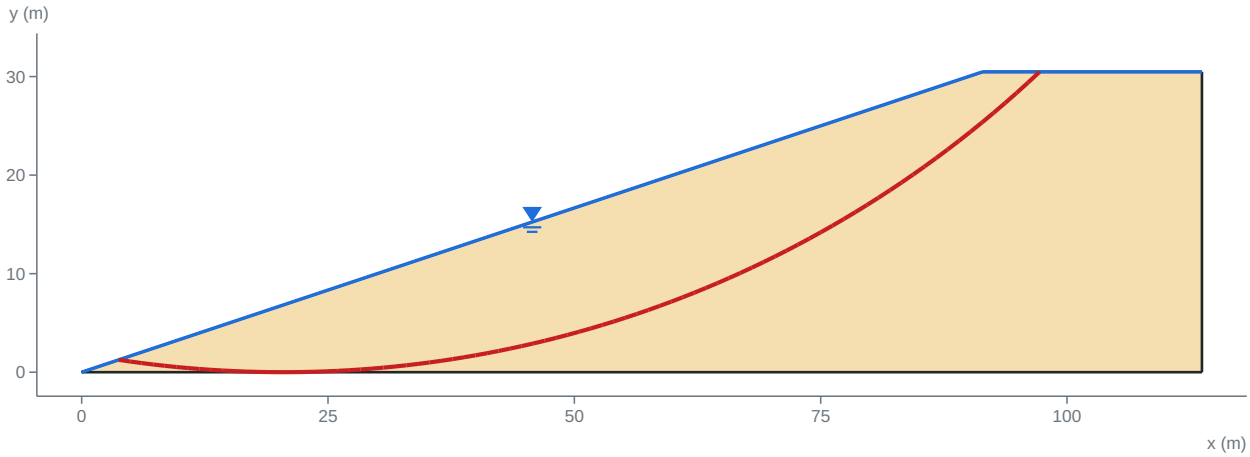


Figure 120. V069a: critical circle (Bishop simplified, FOS = 1.200).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Published value	Method	Source	cSlope
1.20	stability chart	Morgenstern	1.200

cSlope analysis	Other programs	cSlope
Bishop simplified	–	1.200
Spencer	–	1.205

3.121 V069b — Morgenstern complete drawdown, the reference circle

Morgenstern, N. (1963). Stability charts for earth slopes during rapid drawdown. Géotechnique 13(2). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V069a slope after complete drawdown on the reference critical circle (Fig. 100.2); pore pressure from the vertical head. Converted to SI.

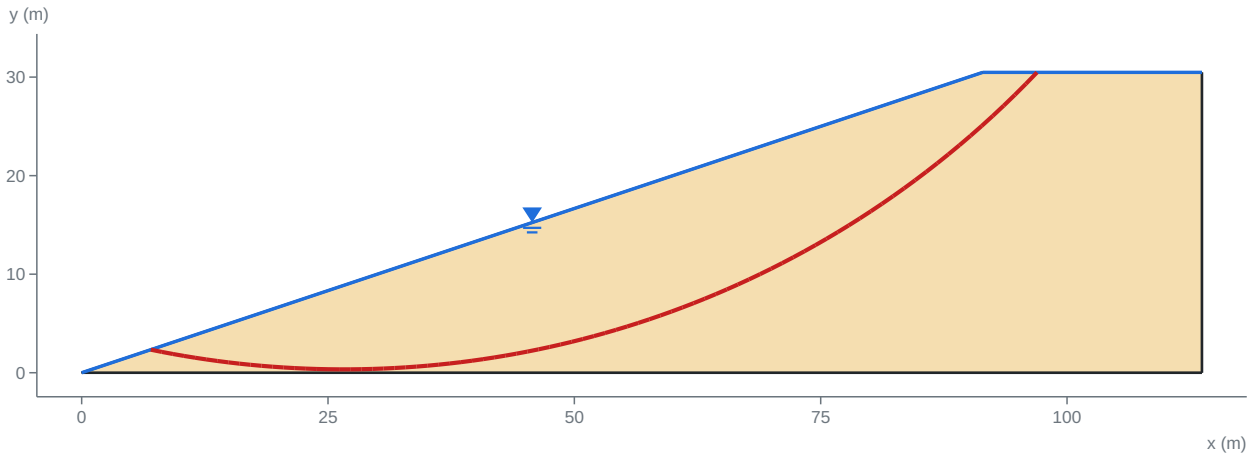


Figure 121. V069b: analysed circle (Bishop simplified, FOS = 1.209).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Published value	Method	Source	cSlope
1.20	stability chart	Morgenstern (1963)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.212	1.209

Morgenstern's 1.2 is read from a stability chart.

3.122 V070 — Partial drawdown, $\bar{B} = 1$ (Morgenstern chart)

Morgenstern (1963).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous embankment after a partial drawdown from 100 to 50 ft with $\bar{B} = 1$: the piezometric line follows the face above the pool and stays at the pool below it, which loads the submerged face. Converted to SI.

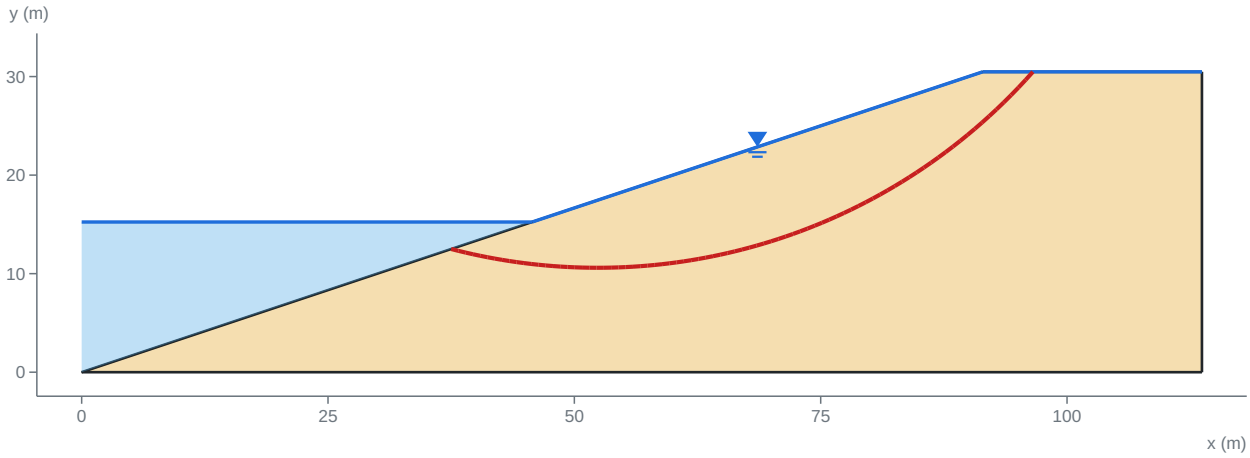


Figure 122. V070: critical circle (Bishop simplified, FOS = 1.415).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Morgenstern soil	Mohr-Coulomb	19.60	c' = 14.94	30.00°

Published value	Method	Source	cSlope
1.41	stability chart	Morgenstern	1.415
1.52	stability chart, circles tangent to the base	Morgenstern (1963)	–
1.48	stability chart, circles tangent to H/2	Morgenstern (1963)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.417	1.415
Spencer	1.422	1.421

Morgenstern's 1.41 is the lowest of three chart readings for circles tangent to the base, to H/2 and to H/4 (1.52, 1.48 and 1.41, the last interpolated), with his remark that a slightly lower value could perhaps be found; a free search may therefore come in below it.

3.123 V071a — Pockoski & Duncan (2000) slope 1

Pockoski & Duncan (2000), slope 1.

Fixed circle · Units: SI (m, kPa, kN/m³)

2:1 slope with a water table, on the circle published by a commercial program; pore pressure from the vertical head below the piezometric line. Converted to SI.

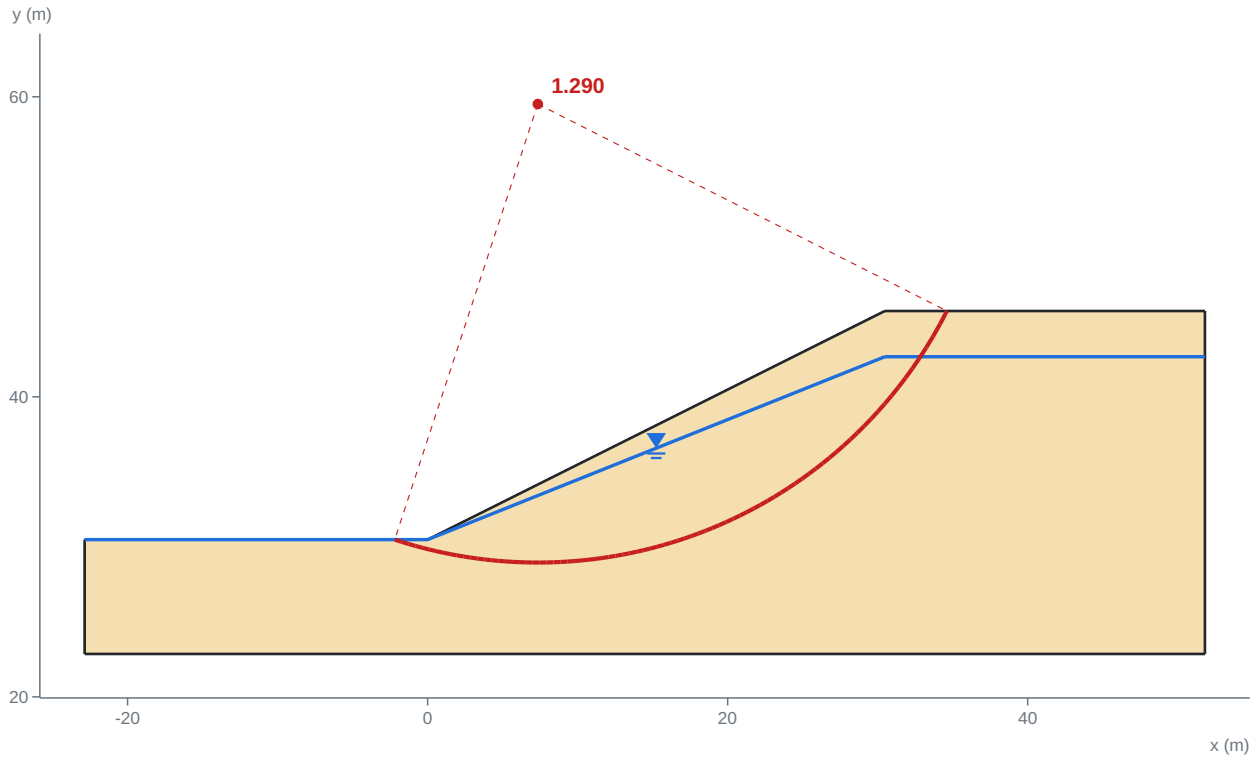


Figure 123. V071a: analysed circle (Bishop simplified, FOS = 1.290).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.293	1.290
Spencer	1.300	1.296

Pockoski & Duncan compare computer programs and publish no independent value.

3.124 V071b — Pockoski & Duncan slope 1, circular search

Pockoski & Duncan (2000), slope 1. Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V071a slope (Pockoski & Duncan slope 1) with a circular search instead of the given circle; pore pressure from the vertical head. Converted to SI.

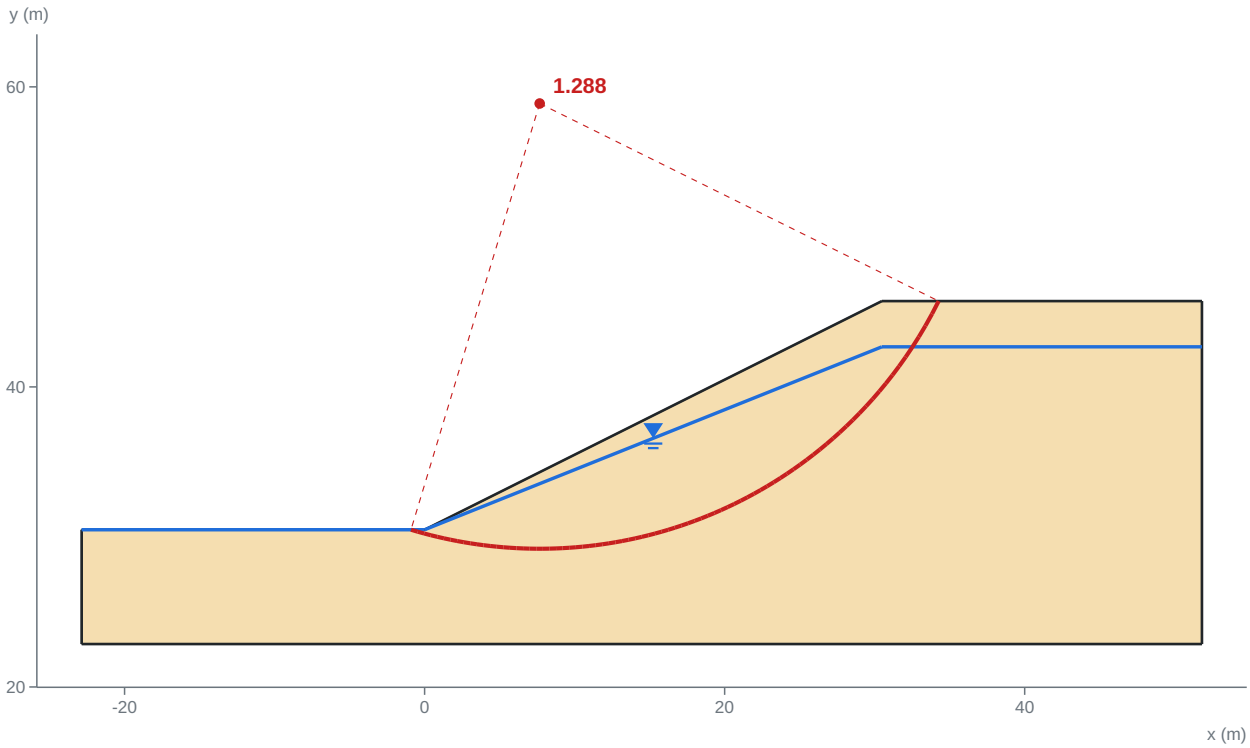


Figure 124. V071b: critical circle (Bishop simplified, FOS = 1.288).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.293	1.288
Spencer	1.300	1.294
Janbu simplified	1.151	1.147

Pockoski & Duncan compare computer programs and publish no independent value.

3.125 V072a — Pockoski & Duncan test slope 2, dry tension crack

Pockoski & Duncan (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

The V071a slope with a dry 5.5 ft tension crack, on the reference critical circle. Converted to SI.

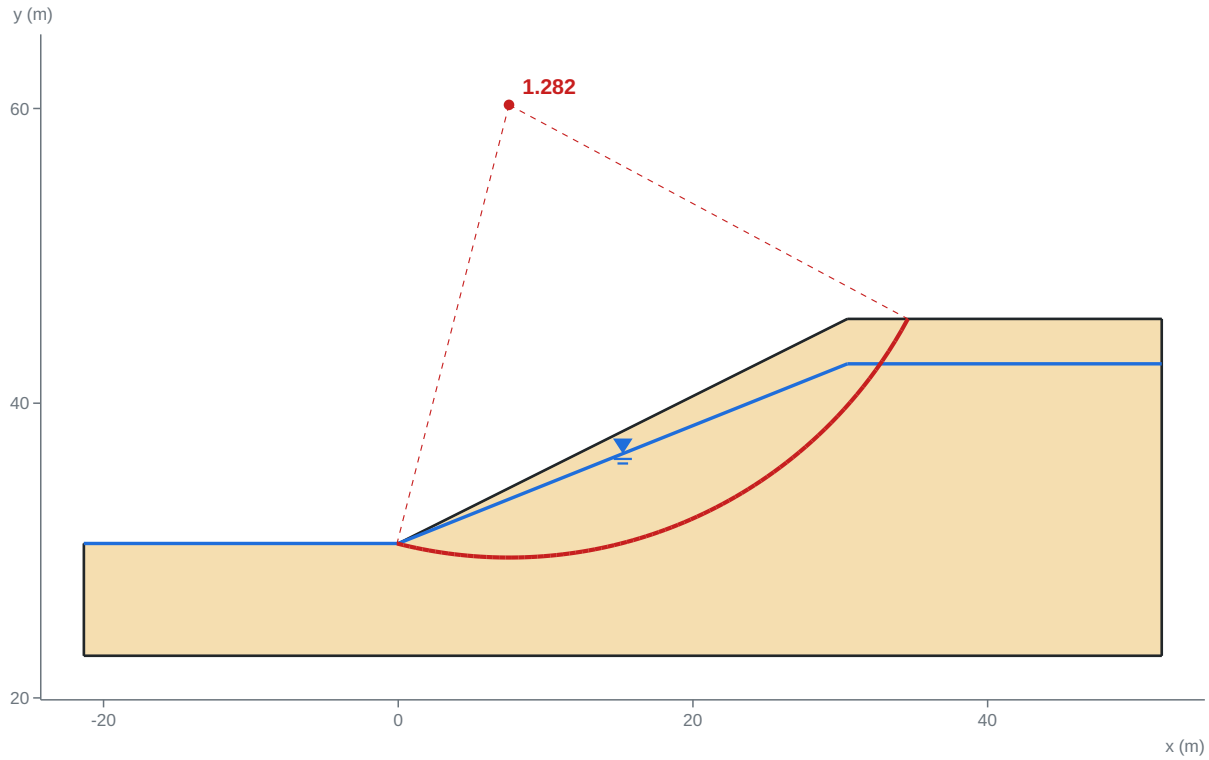


Figure 125. V072a: analysed circle (Bishop simplified, FOS = 1.282).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.285	1.282
Spencer	1.290	1.287

Pockoski & Duncan compare computer programs and publish no independent value.

3.126 V072b — Pockoski & Duncan slope 2, circular search

Pockoski & Duncan (2000). Verification manual of a commercial slope stability program.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V072a slope (dry 5.5 ft tension crack) with a circular search. Converted to SI.

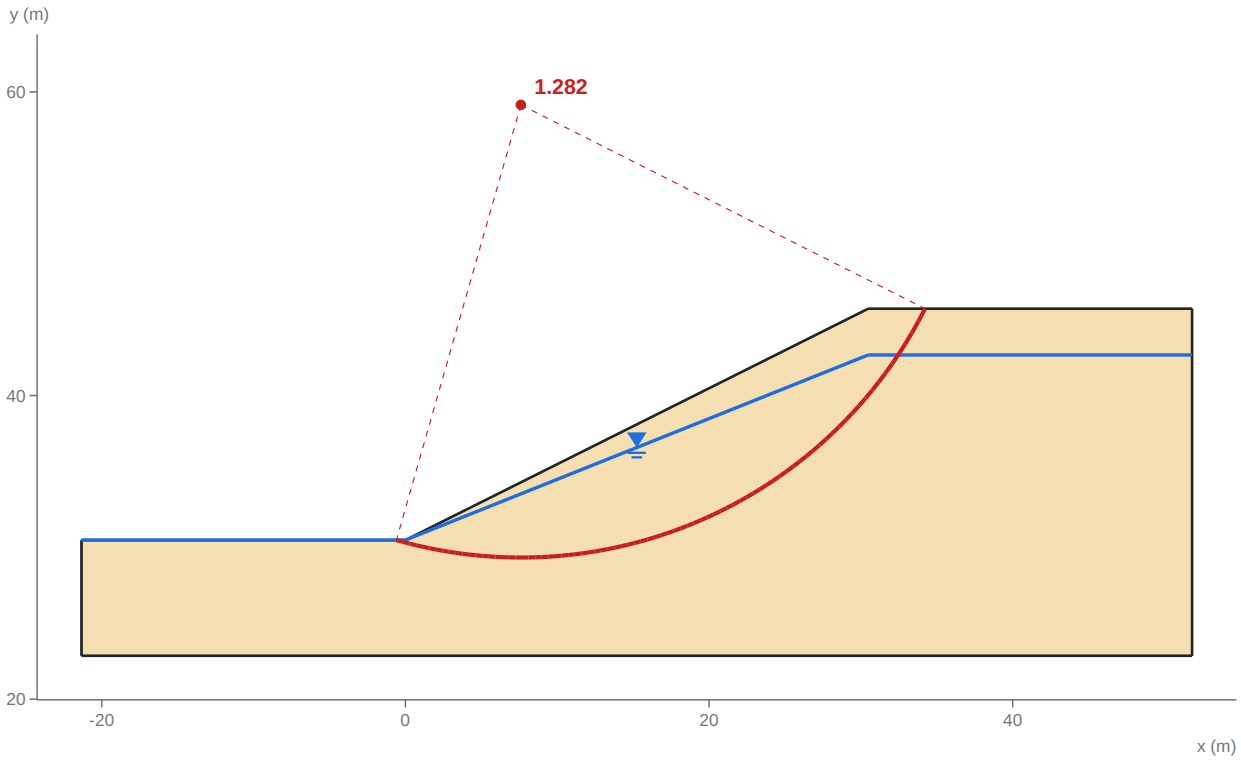


Figure 126. V072b: critical circle (Bishop simplified, FOS = 1.282).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	18.85	c' = 14.36	30.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.285	1.282
Spencer	1.290	1.287
Janbu simplified	1.141	1.135

Pockoski & Duncan compare computer programs and publish no independent value.

3.127 V073 — Pockoski & Duncan test slope 3, clay seam and dry crack

Pockoski & Duncan (2000).

Critical circle search · Units: SI (m, kPa, kN/m³)

Sandy clay over a 5 ft highly plastic clay seam on the model base, water table and a dry 6 ft tension crack; circular search. Converted to SI.

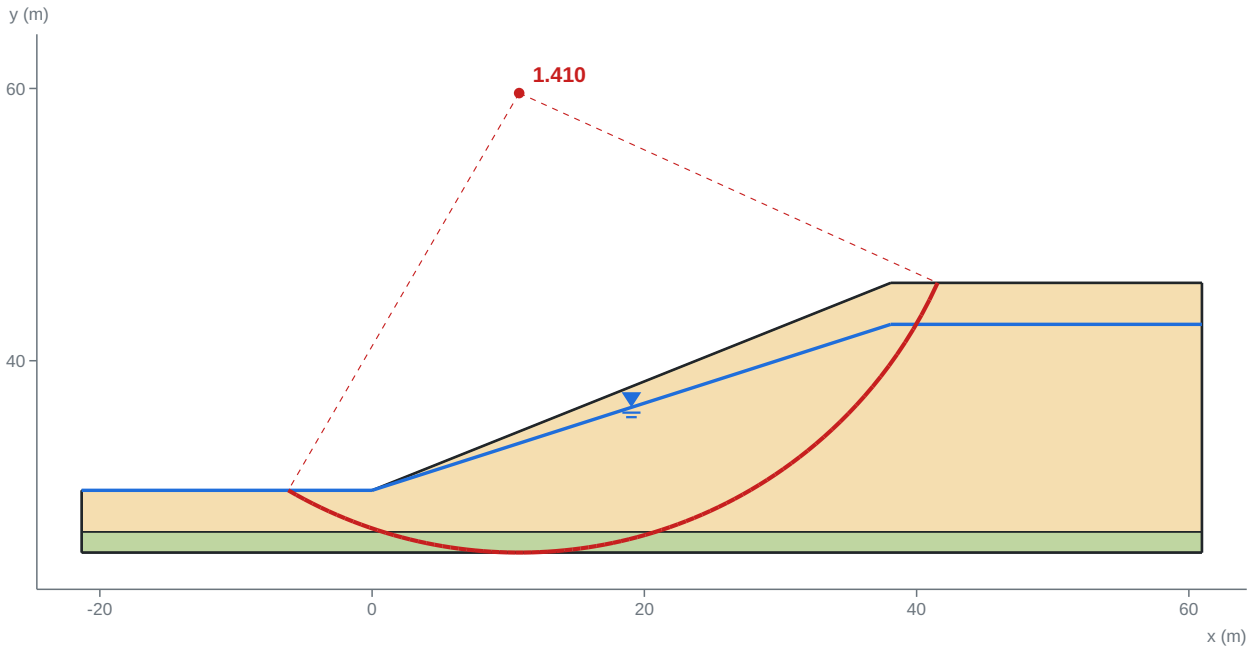


Figure 127. V073: critical circle (Bishop simplified, FOS = 1.410).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Mohr-Coulomb	20.42	c' = 14.36	35.00°
Plastic clay seam	Mohr-Coulomb	20.42	c' = 0.00	25.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.417	1.410
Spencer	1.422	1.415

Pockoski & Duncan compare computer programs and publish no independent value.

3.128 V074 — Soil-nailed wall with tension crack and surcharges

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

25 ft nailed wall in undrained clay (800 psf) with crest surcharges, a dry 7 ft crack and five passive nail rows at 15°; the reference circle truncated by the crack. Converted to SI.

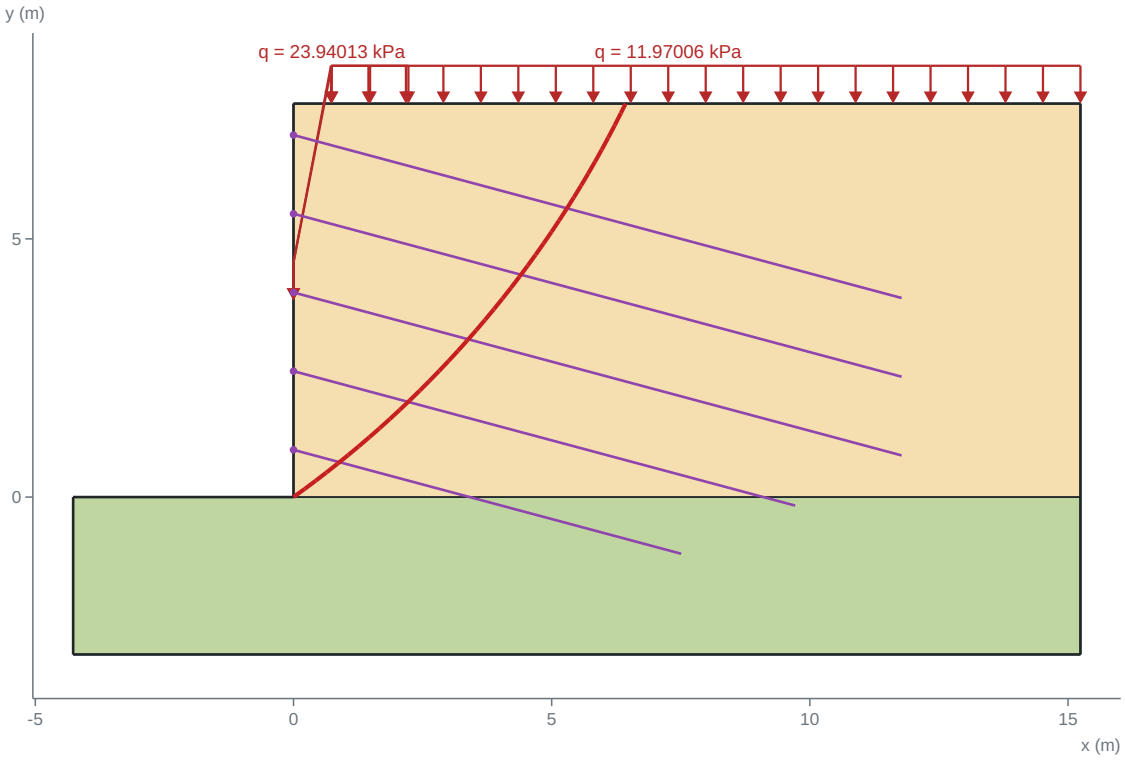


Figure 128. V074: analysed surface (Spencer, FOS = 1.010).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sandy clay	Undrained (φ = 0)	18.85	Su = 38.30	0°
Firm soil	Mohr-Coulomb	20.42	c' = 239.4	35.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer	1.009	1.010
Janbu corrected	1.073	1.075

Pockoski & Duncan compare computer programs and publish no independent value.

3.129 V075 — Tieback wall in sand, drawdown water table

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Single-row tieback wall in sand (φ′ = 30°; the tieback from 12 ft up the wall, 15° down, 37 ft long of which 22 ft bonded) with a water table drawn down to the wall toe (digitised from the reference figure) and pore pressure from the vertical head; the printed surface from the wall toe. Converted to SI.

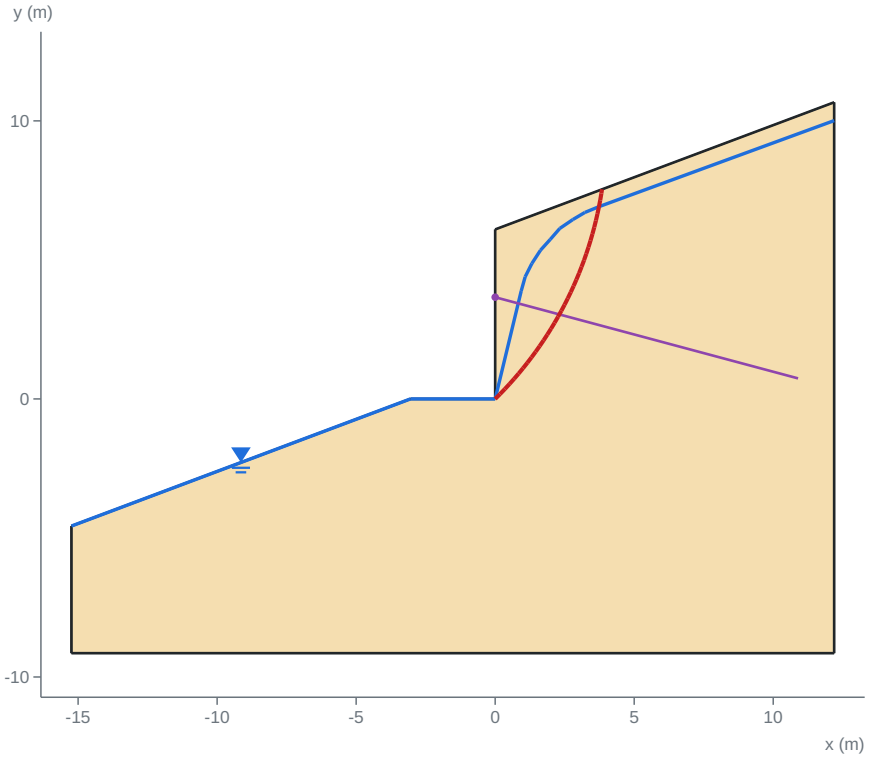


Figure 129. V075: analysed surface (Spencer, FOS = 0.578).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	18.85	c' = 0.00	30.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer	0.570	0.578

Pockoski & Duncan compare computer programs and publish no independent value.

3.130 V076 — Tied-back wall in layered soil

Pockoski & Duncan (2000), Comparison of computer programs for analysis of reinforced slopes. Virginia Tech.

Fixed circle · Units: SI (m, kPa, kN/m³)

44 ft tied-back excavation in eight layers, water table at grade in front of the wall and at el. 102.5 behind it (a vertical step at the wall), three tieback rows at 20° (88 ft, 40 ft bond). a commercial program's critical circle. Converted to SI.

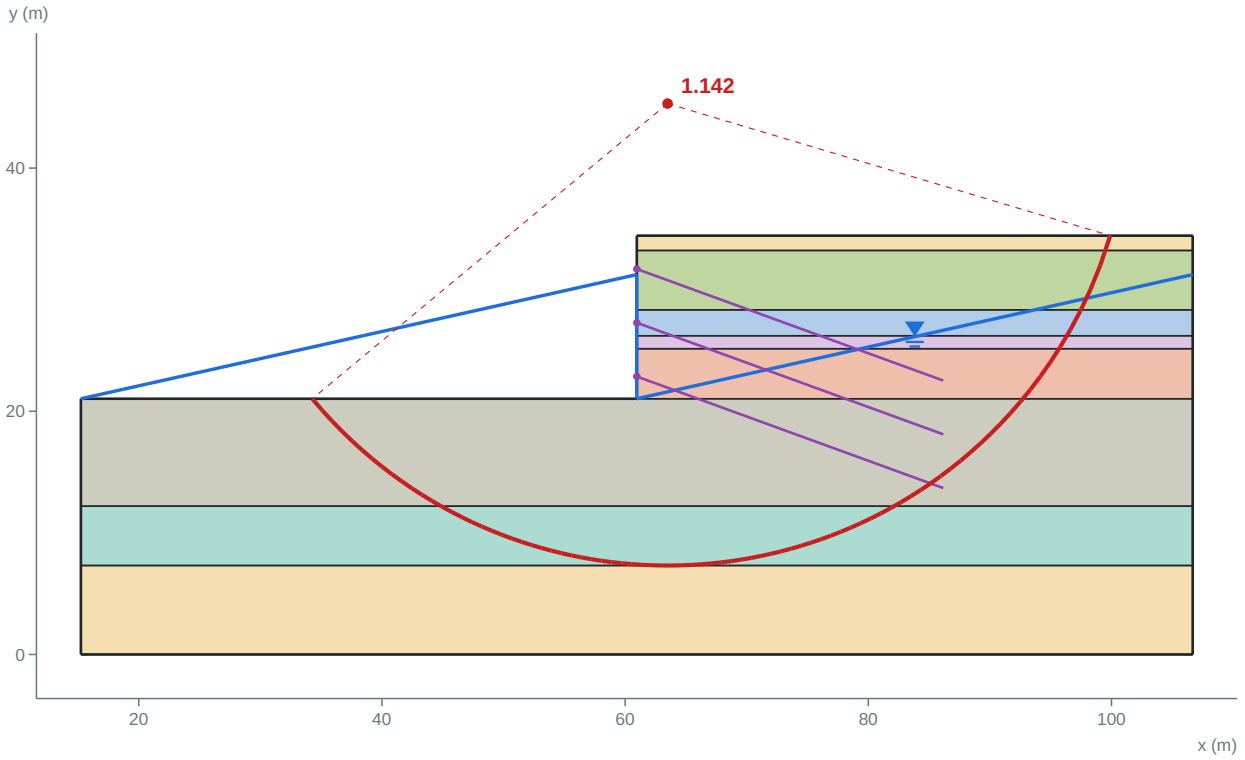


Figure 130. V076: analysed circle (Bishop simplified, FOS = 1.142).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Granular Fill	Mohr-Coulomb	18.91	c' = 0.00	30.00°
Cohesive Fill	Mohr-Coulomb	18.02	c' = 0.00	30.00°
Organic Silt	Undrained (φ = 0)	17.31	Su = 43.09	0°
OC Crust	Undrained (φ = 0)	18.50	Su = 119.0	0°
Upper Marine Clay	Undrained (φ = 0)	18.50	Su = 79.96	0°
Middle Marine Clay	Undrained (φ = 0)	18.50	Su = 45.97	0°
Lower Marine Clay	Undrained (φ = 0)	18.50	Su = 51.95	0°
Glaciomarine Deposits	Undrained (φ = 0)	23.11	Su = 71.82	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.147	1.142
Spencer	1.145	1.140

Pockoski & Duncan compare computer programs and publish no independent value.

3.131 V077 — Prandtl bearing capacity on level ground

Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Weightless (γ = 1e-6 kN/m³) c = 20 kPa soil (φ = 0) loaded by a strip load of 102.83 kPa = c·(2 + π) over 10 m, the exact bearing capacity. Analysed on the Prandtl mechanism of Fig. 26.2: active wedge under the load, a 90° fan arc of radius 7.071 m about the load's edge (5, 10), lowest point 2.929, and the passive wedge.

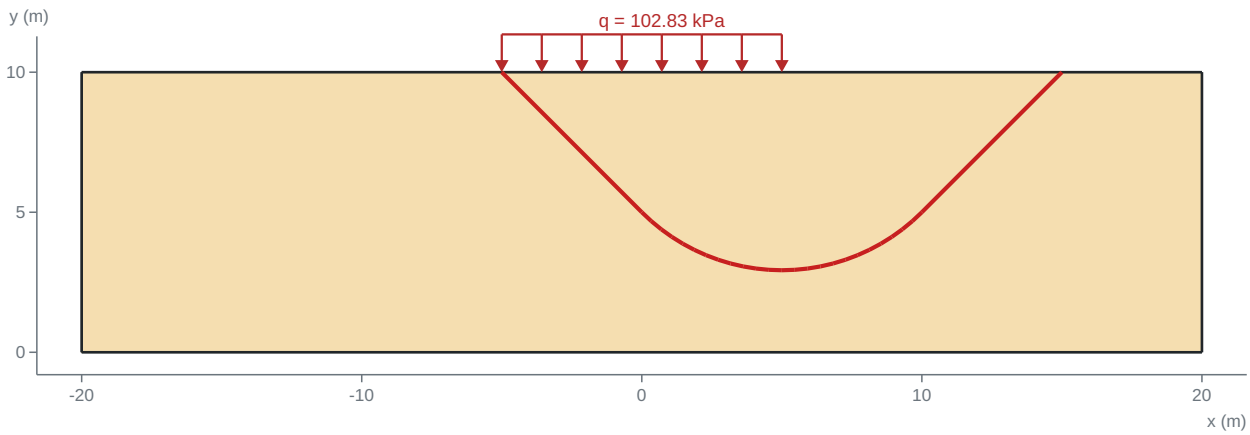


Figure 131. V077: analysed surface (Spencer, FOS = 0.943).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil	Undrained (φ = 0)	0.00	Su = 20.00	0°

Published value	Method	Source	cSlope
1.00	plasticity theory	Prandtl (1920)	–

cSlope analysis	Other programs	cSlope
Spencer	0.940	0.943

The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 6 % less on this mechanism.

3.132 V078 — Prandtl bearing capacity, fully specified Prandtl surface (reference values)

Prandtl (1920). Über die Härte plastischer Körper. Nachr. Ges. Wiss. Göttingen, Math.-phys. Kl., 74–85.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Prandtl bearing capacity as a commercial program specifies it: weightless c = 20 kPa soil, 102.83 kPa over 10 m, the fully specified Prandtl surface (active wedge, 90° fan about the load edge, passive wedge).

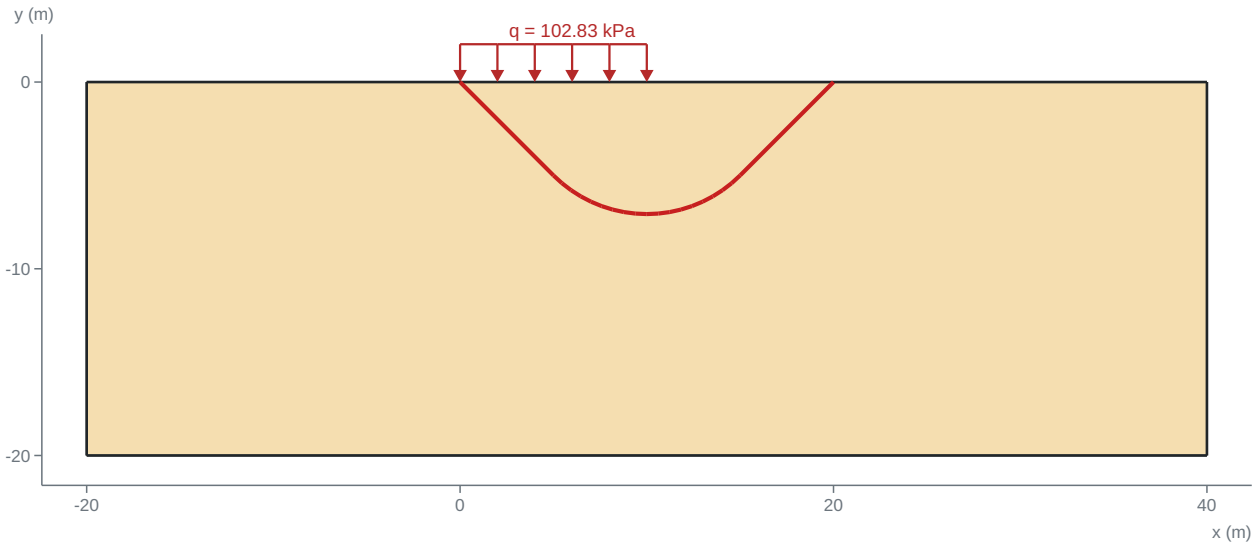


Figure 132. V078: analysed surface (Morgenstern-Price (half-sine), FOS = 0.958).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Undrained (φ = 0)	0.00	Su = 20.00	0°

Published value	Method	Source	cSlope
1.00	plasticity theory	Prandtl (1920)	–

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	0.960	0.958

The theoretical factor of safety is 1.0; limit equilibrium with slices gives about 4 % less on this mechanism.

3.133 V079 — Priest (1993) rigid block on a plane, water-filled crack

Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Priest's example 8.1 as two commercial programs model it: a 30 m face with the crest at x = 17.0 (60.46°, Priest's is 60° with the crest at 17.32, see V080), a 30° plane from the toe, a vertical 15 m tension crack holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack, which gives Priest's linear pressure along the plane.

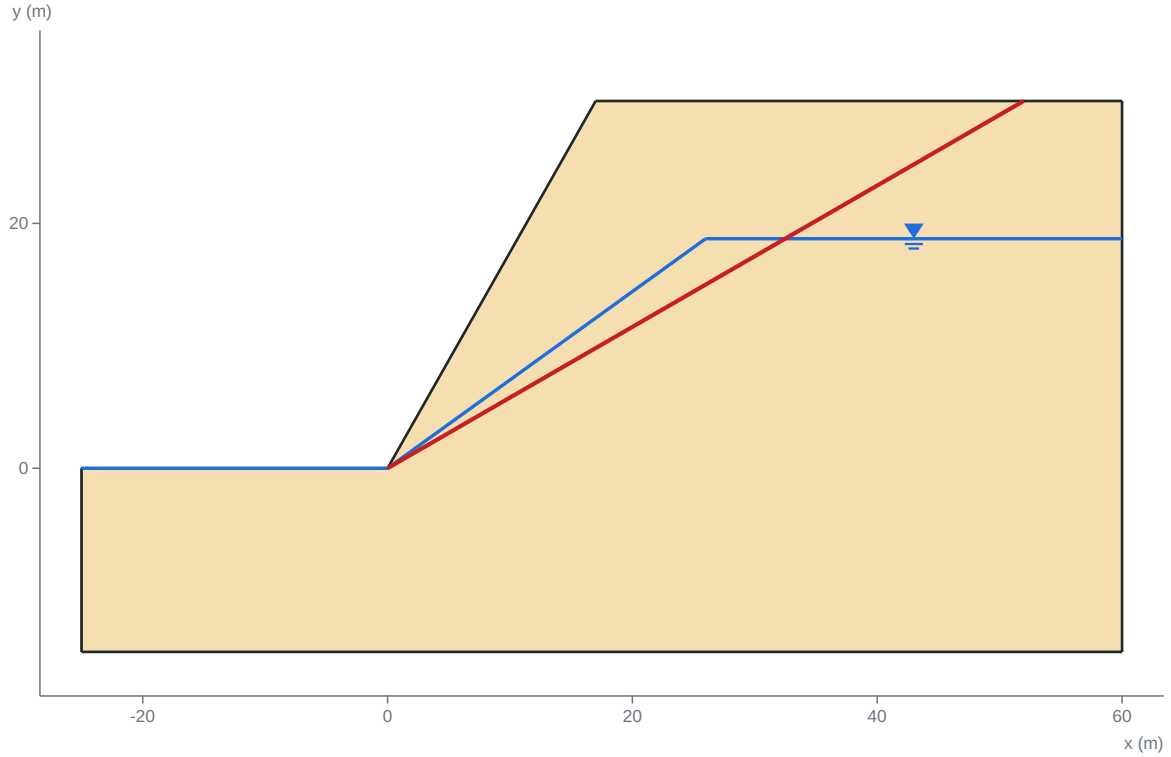


Figure 133. V079: analysed surface (Janbu simplified, FOS = 1.048).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c' = 20.00	30.00°

Published value	Method	Source	cSlope
1.049	rigid block, force equilibrium	Priest (1993)	–

cSlope analysis	Other programs	cSlope
Janbu simplified	1.049	1.048
Spencer	1.049	1.048

3.134 V080 — Priest (1993) rigid block, reference geometry (60° face)

Priest, S.D. (1993). Discontinuity Analysis for Rock Engineering. Chapman & Hall, London; example 8.1, pp. 224–225.

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

Priest's example 8.1 with his geometry: a 60° face 30 m high (crest at x = 17.321), a 30° plane from the toe, a vertical 15 m tension crack 8.66 m behind the crest holding 3.75 m of water, and a water table rising from the toe to 3.75 m above the plane at the crack.

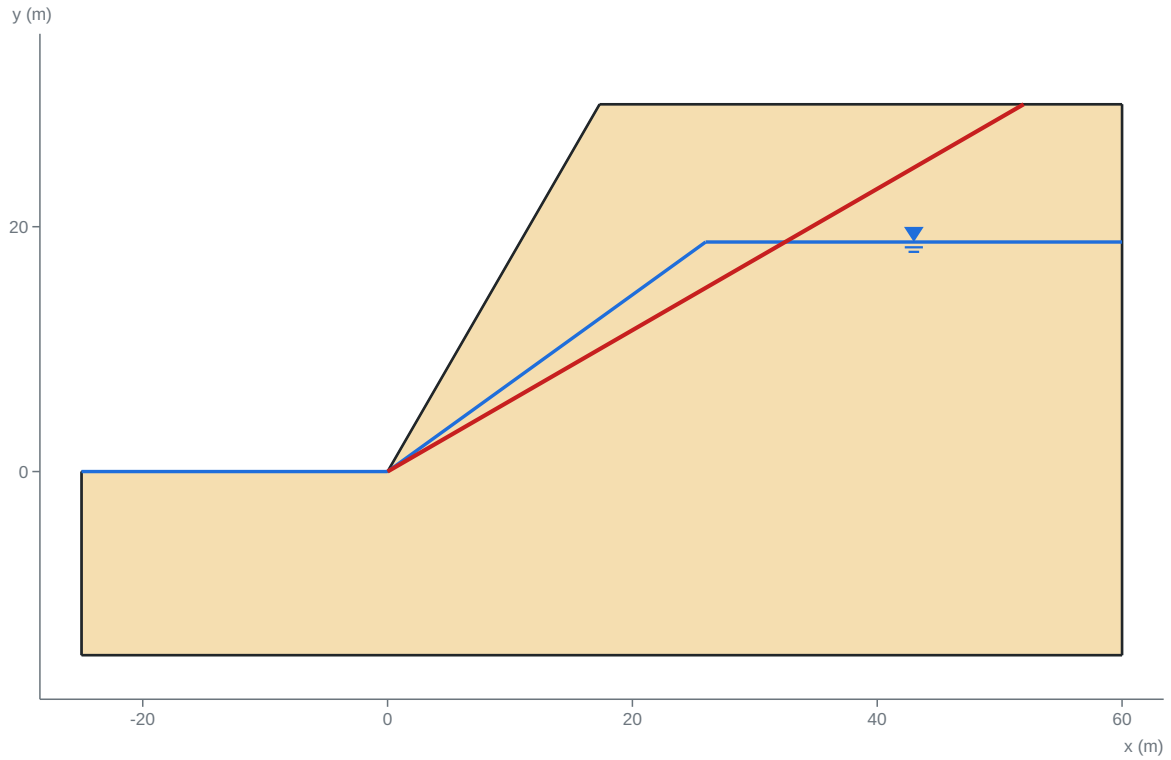


Figure 134. V080: analysed surface (Morgenstern-Price (half-sine), FOS = 1.049).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Block	Mohr-Coulomb	25.00	c' = 20.00	30.00°

Published value	Method	Source	cSlope
1.049	Spencer	Priest	1.049
1.049	rigid block, force equilibrium	Priest (1993)	–

cSlope analysis	Other programs	cSlope
Morgenstern-Price (half-sine)	1.049	1.049
Spencer	–	1.049

3.135 V081a — Sharma (1996) reinforcement example, unreinforced

Sharma (1996).

Fixed circle · Units: SI (m, kPa, kN/m³)

45° cohesionless sand slope with a 40 kPa surcharge behind the crest, on the published toe circle.

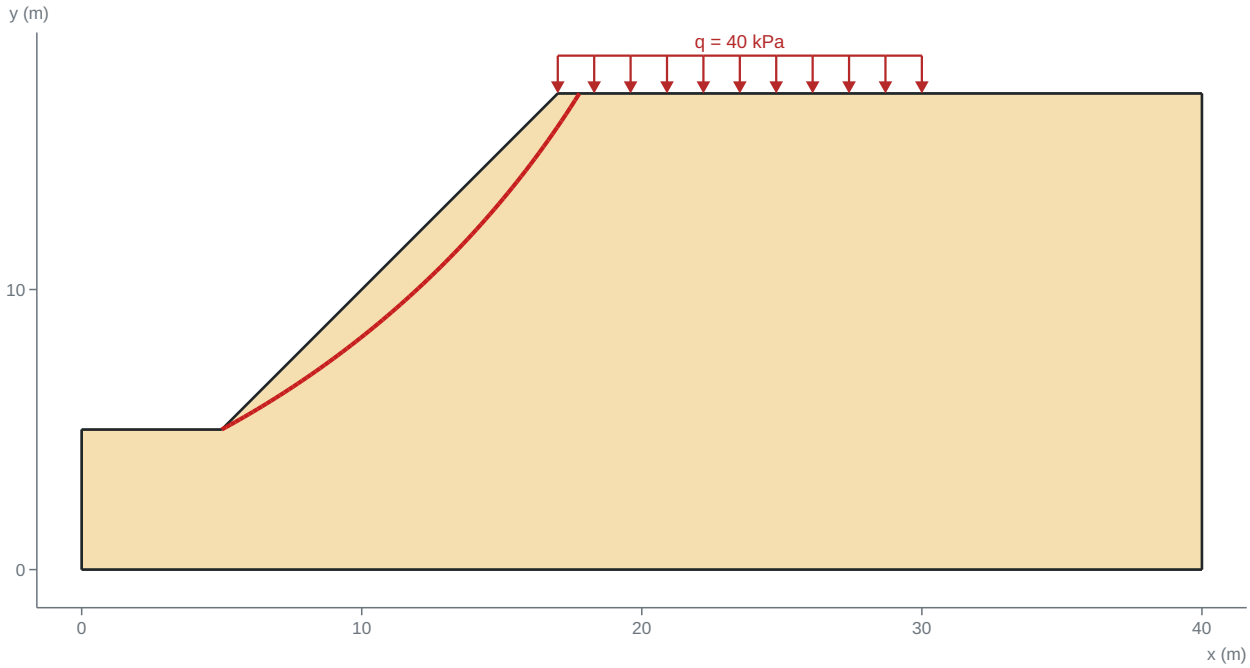


Figure 135. V081a: analysed circle (Bishop simplified, FOS = 0.764).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Sand	Mohr-Coulomb	20.00	c' = 0.00	36.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	0.764	0.764

The source compares programs and publishes no independent value.

3.136 V081b — Sharma (1996) slope with a reinforced zone (φ = 54.93°)

Sharma (1996). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V081a sand slope (φ′ = 36°, 40 kPa surcharge) with a 7.6 m reinforced zone of φ = 54.93° (tan φ = 1.96 tan 36°) behind the face, on the reference critical toe circle (8.707, 18.307, R = 13.813).

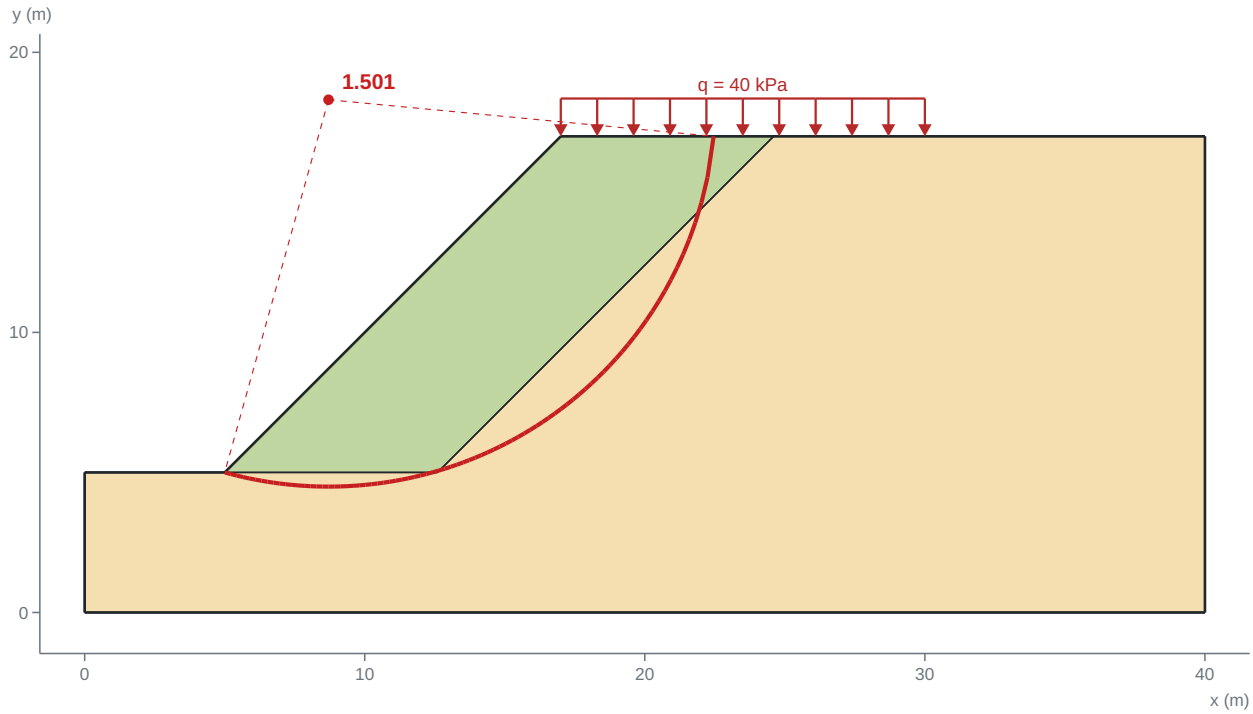


Figure 136. V081b: analysed circle (Bishop simplified, FOS = 1.501).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	20.00	c' = 0.00	36.00°
Reinforced zone	Mohr-Coulomb	20.00	c' = 0.00	54.93°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.504	1.501

The source compares programs and publishes no independent value.

3.137 V082a — Sharma (1996) slope, 11 ft dry tension crack instead of the zero-strength cap

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V082c slope with the zero-strength cap replaced by an 11 ft dry tension crack (Table 27.4, analysis 3), on the V082c circle; phreatic correction. Converted to SI.

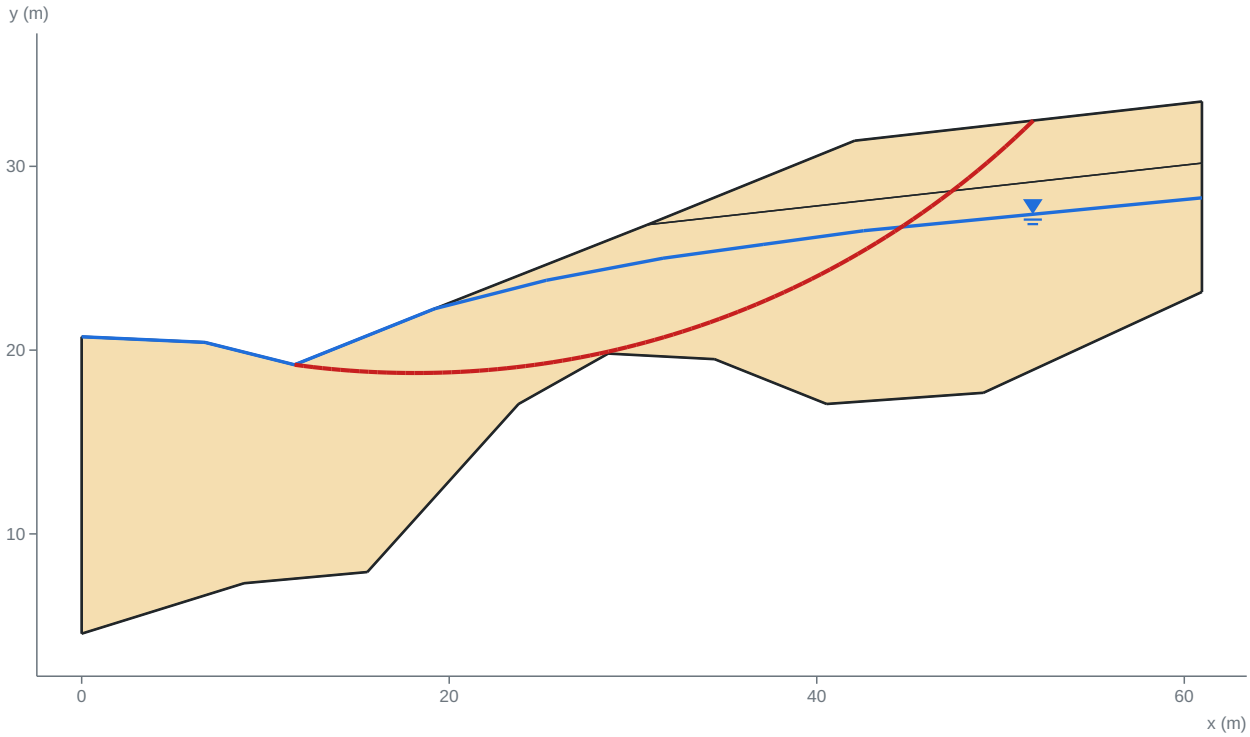


Figure 137. V082a: analysed circle (Bishop simplified, FOS = 1.534).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.532	1.534
Spencer	1.532	1.533
Morgenstern-Price (half-sine)	1.532	1.533
Janbu corrected	1.544	1.546

The source compares programs and publishes no independent value.

3.138 V082b — Sharma (1996) slope, 11 ft tension crack with 6 ft of water

Sharma (1996); Malkawi et al. (2001). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V082a with 6 ft of water in the 11 ft crack (Table 27.4, analysis 4).

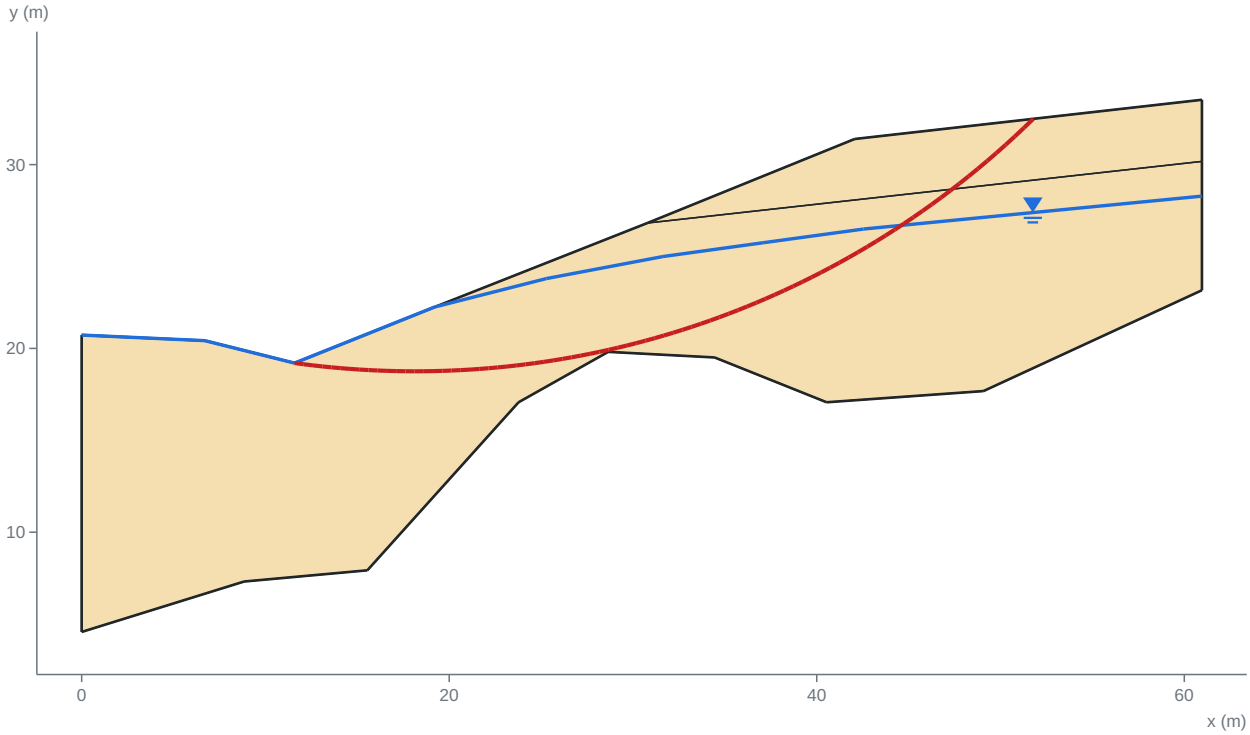


Figure 138. V082b: analysed circle (Bishop simplified, FOS = 1.513).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.511	1.513
Spencer	1.510	1.512
Morgenstern-Price (half-sine)	1.510	1.512
Janbu corrected	1.520	1.522

The source compares programs and publishes no independent value.

3.139 V082c — Sharma (1996) slope over undulating bedrock, zero-strength cap

Sharma (1996); Malkawi et al. (2001).

Fixed circle · Units: SI (m, kPa, kN/m³)

Two-material slope over undulating bedrock with a zero-strength cap layer and a water table (phreatic correction); soil 1 has moist/saturated unit weights 116.4/124.2 pcf. Specified circle (59.52, 219.21, R = 157.68). Converted to SI.

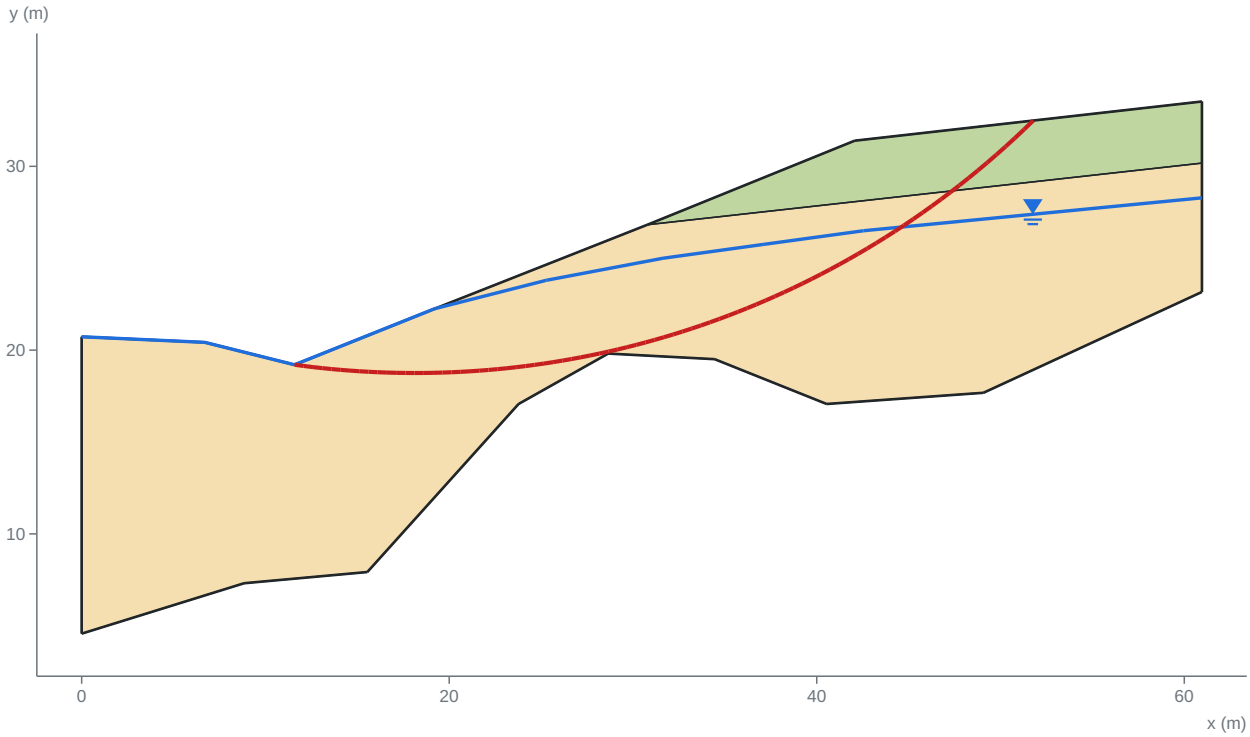


Figure 139. V082c: analysed circle (Bishop simplified, FOS = 1.396).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.396	1.396
Spencer	1.402	1.402
Janbu corrected	1.391	1.392
Morgenstern-Price (half-sine)	1.398	1.398

The source compares programs and publishes no independent value.

3.140 V082d — Sharma (1996) slope over undulating bedrock, circular search with exit limits

Sharma (1996); Malkawi et al. (2001).

Critical circle search · Units: SI (m, kPa, kN/m³)

The V082c slope (zero-strength cap, water table with phreatic correction) with a circular search whose circles must exit between x = 38 and 70 ft at the toe and 120 and 180 ft at the crest (Table 27.3). Converted to SI.

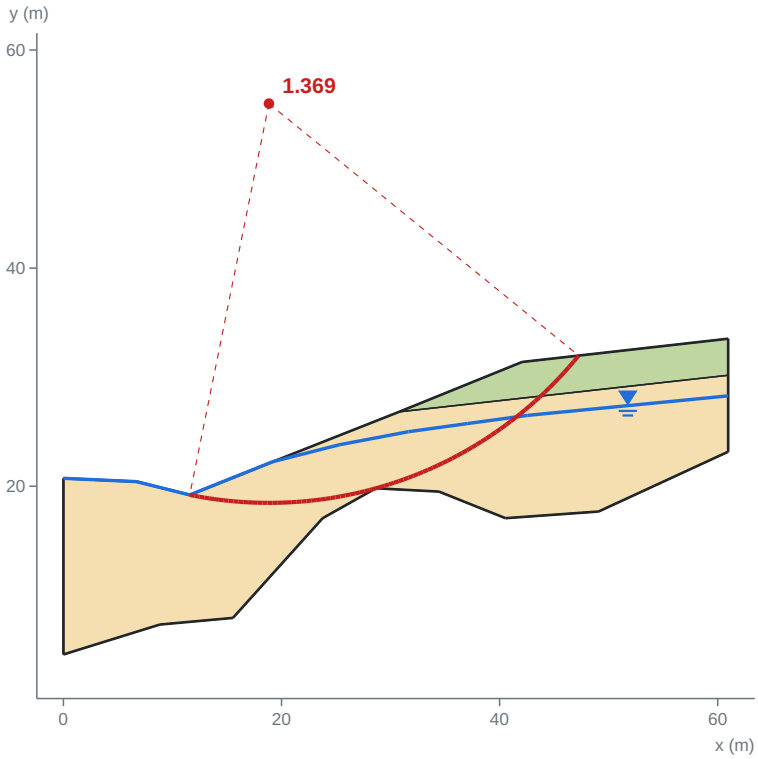


Figure 140. V082d: critical circle (Bishop simplified, FOS = 1.369).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Mohr-Coulomb	18.28	c' = 23.94	14.00°
Soil 2 (zero strength)	Undrained (φ = 0)	18.28	Su = 0.00	0°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.376	1.369
Spencer	1.382	1.376
Morgenstern-Price (half-sine)	1.378	1.371
Janbu corrected	1.345	1.343

The source compares programs and publishes no independent value.

3.141 V083 — Soil-nailed wall in clay (Amherst test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

6.1 m vertical cut in undrained clay (Su = 25 kPa) with two nail rows at 20° (tensile 118 kN, plate 86 kN, bond 15 kN/m, spacing 1.5 m; axial, passive) and the facing weight as a line load; the critical plane printed by a commercial program, from the toe (0, 0) to the crest (6.279, 6.1), 44.2°.

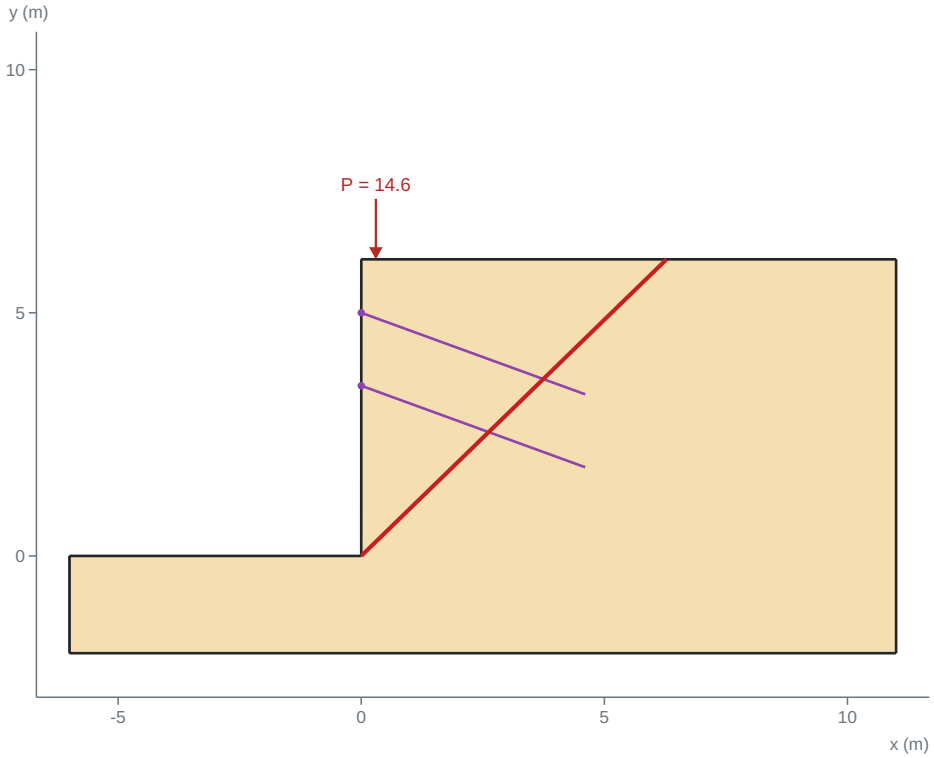


Figure 141. V083: analysed surface (Janbu corrected, FOS = 0.885).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Amherst Clay	Undrained (φ = 0)	18.90	Su = 25.00	0°

Published value	Method	Source	cSlope
0.887	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	0.890	0.885

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.142 V084a — Soil-nailed wall in sand (Clouterre test wall)

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

7 m cut in sand (c′ = 3 kPa, φ′ = 38°) with seven passive nail rows at 10° (tensile 15 kN, plate 59 kN, bond 7.5 kN/m, i.e. the tensile capacity develops over 2 m; spacing 1.15 m) and the facing weight; 55° plane through the toe. V084b … V084f are the other planes of Table 48.3.

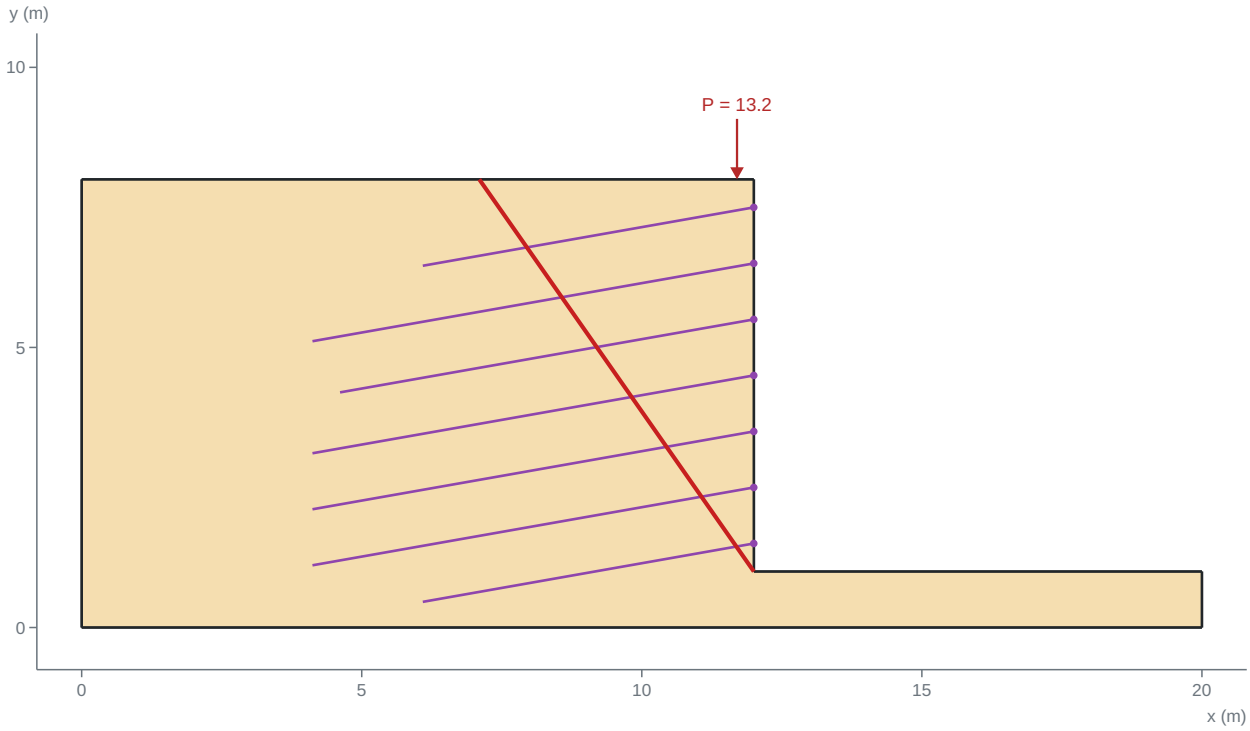


Figure 142. V084a: analysed surface (Janbu corrected, FOS = 0.988).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
0.989	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	0.989	0.988

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.143 V084b — Soil-nailed wall in sand (Clouterre test wall), 45° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 45° plane through the toe.

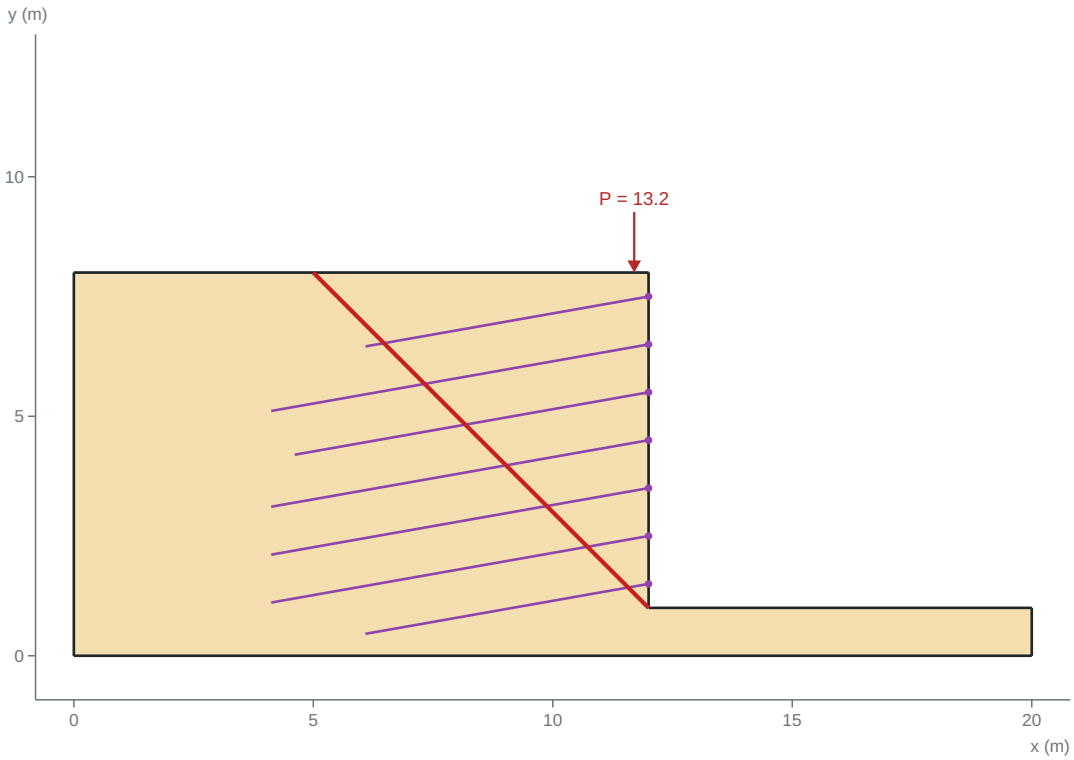


Figure 143. V084b: analysed surface (Janbu corrected, FOS = 1.124).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
1.176	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	1.123	1.124

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 5 %.

3.144 V084c — Soil-nailed wall in sand (Clouterre test wall), 50° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 50° plane through the toe.

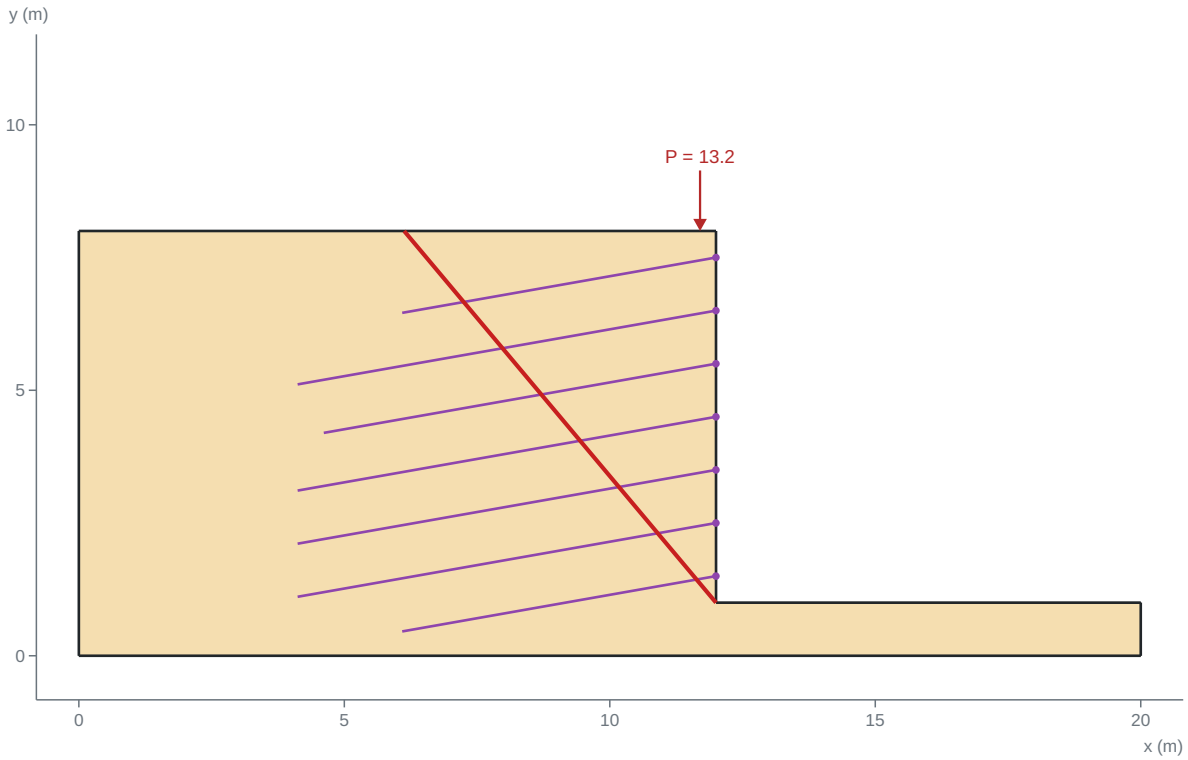


Figure 144. V084c: analysed surface (Janbu corrected, FOS = 1.043).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
1.07	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	1.043	1.043

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.145 V084d — Soil-nailed wall in sand (Clouterre test wall), 60° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 60° plane through the toe.

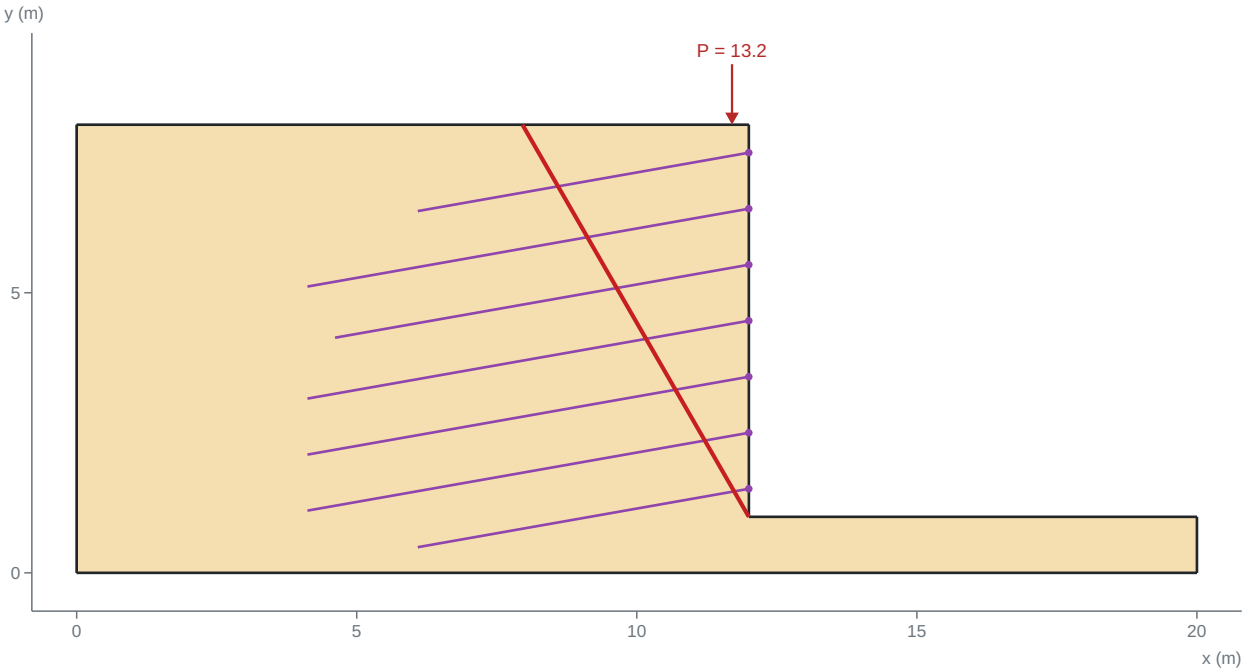


Figure 145. V084d: analysed surface (Janbu corrected, FOS = 0.944).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
0.929	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	0.945	0.944

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.146 V084e — Soil-nailed wall in sand (Clouterre test wall), 65° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 65° plane through the toe.

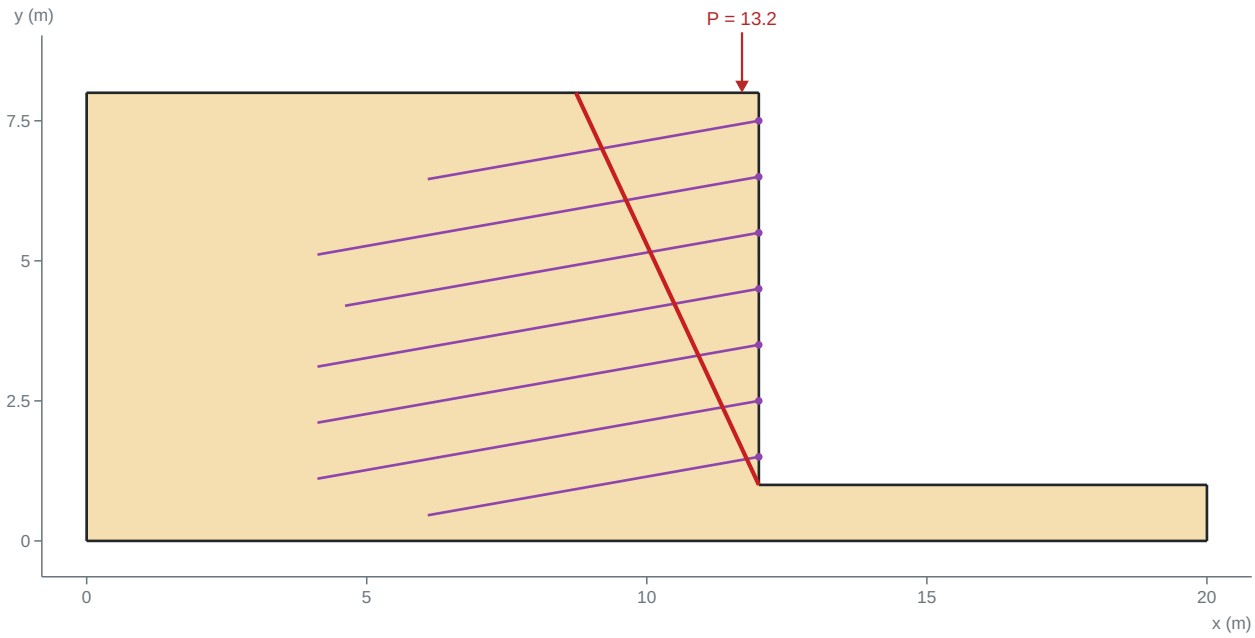


Figure 146. V084e: analysed surface (Janbu corrected, FOS = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
0.893	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	0.922	0.920

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.147 V084f — Soil-nailed wall in sand (Clouterre test wall), 70° plane

Sheahan & Ho (2003).

Specified non-circular surface · Units: SI (m, kPa, kN/m³)

The V084a wall on the 70° plane through the toe.

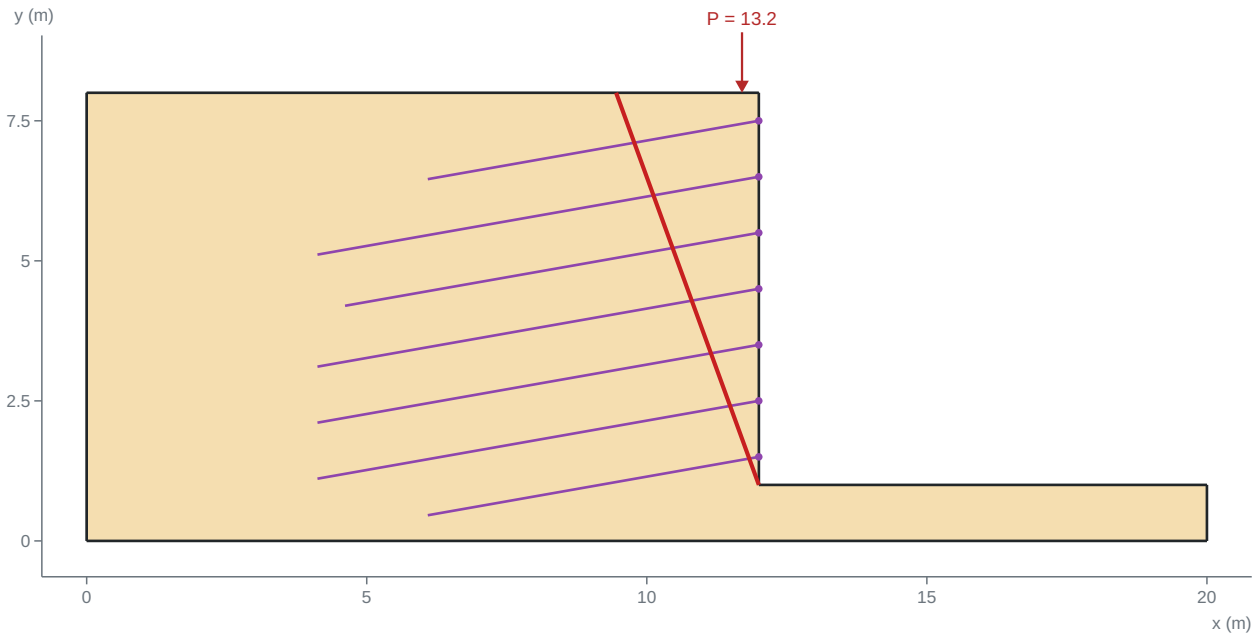


Figure 147. V084f: analysed surface (Janbu corrected, FOS = 0.920).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fontainebleau Sand	Mohr-Coulomb	20.00	c' = 3.00	38.00°

Published value	Method	Source	cSlope
0.887	force-equilibrium trial wedge	Sheahan & Ho (2003)	–

cSlope analysis	Other programs	cSlope
Janbu corrected	0.923	0.920

Sheahan & Ho use a force-equilibrium trial wedge; Janbu's method on the same planes differs by up to 4 %.

3.148 V085 — Bradwell excavated slope, stiff-fissured London clay

Skempton, A.W. & La Rochelle, P. (1965). The Bradwell slip: a short-term failure in London clay. Géotechnique 15(3), 221–242; Duncan, Wright & Brandon (2014), Soil Strength and Slope Stability, 2nd ed., §7.7.3.

Critical circle search · Units: imperial (ft, psf, pcf)

Excavation in London clay (six sublayers with Su increasing with depth, S = f(datum)) under Marsh clay and a clay fill cracked to its full depth (11.4 ft); search for the deep failure, entering on the crest and leaving at the excavation toe.

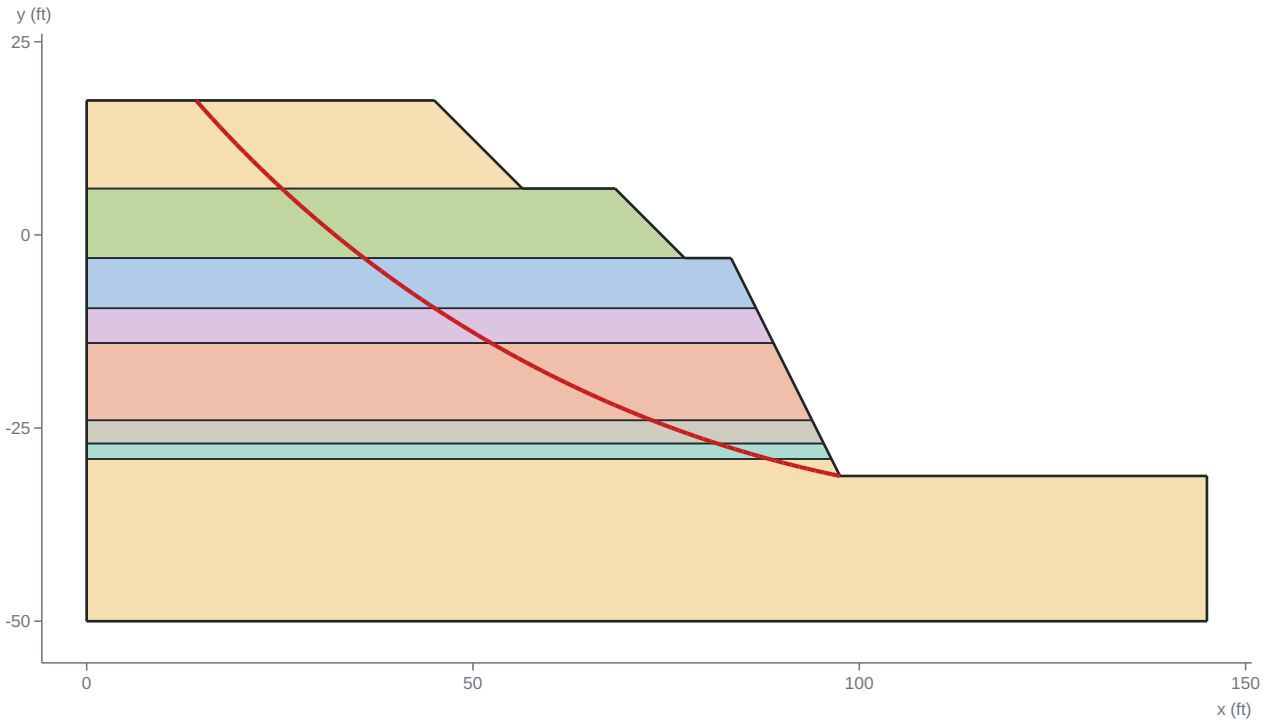


Figure 148. V085: critical circle (Bishop simplified, FOS = 1.759).

Material	Model	γ (pcf)	Strength (psf)	φ
Clay fill	Mohr-Coulomb	110.0	c' = 1.00	35.00°
Marsh clay	Mohr-Coulomb	105.0	c' = 300.0	0.00°
Brown London clay 1	S = f(datum)	120.0	Su = 750.0 at y = -3.00, 90.00 psf/ft below	0°
Brown London clay 2	S = f(datum)	120.0	Su = 1335.0 at y = -9.50, 82.00 psf/ft below	0°
Brown London clay 3	S = f(datum)	120.0	Su = 1704.0 at y = -14.00, 53.00 psf/ft below	0°
Brown London clay 4	S = f(datum)	120.0	Su = 2234.0 at y = -24.00, 47.00 psf/ft below	0°
Blue London clay 1	S = f(datum)	120.0	Su = 2375.0 at y = -27.00, 47.00 psf/ft below	0°
Blue London clay 2	S = f(datum)	120.0	Su = 2469.0 at y = -29.00, 39.00 psf/ft below	0°

Published value	Method	Source	cSlope
1.76	Bishop simplified	Duncan & Wright	1.759
1.63	Janbu simplified	Duncan & Wright	1.627

cSlope analysis	Other programs	cSlope
Bishop simplified	1.762	1.759
Spencer	1.758	1.759
Janbu corrected	1.736	1.735
Janbu simplified	1.628	1.627

3.149 V086a — Tandjiria embankment, clay fill, geosynthetic T = 169 kN/m

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V086b clay-fill embankment with a 169 kN/m geosynthetic at its base (axial, active, fully anchored) and the water-filled 2 m crack, on the critical circle.

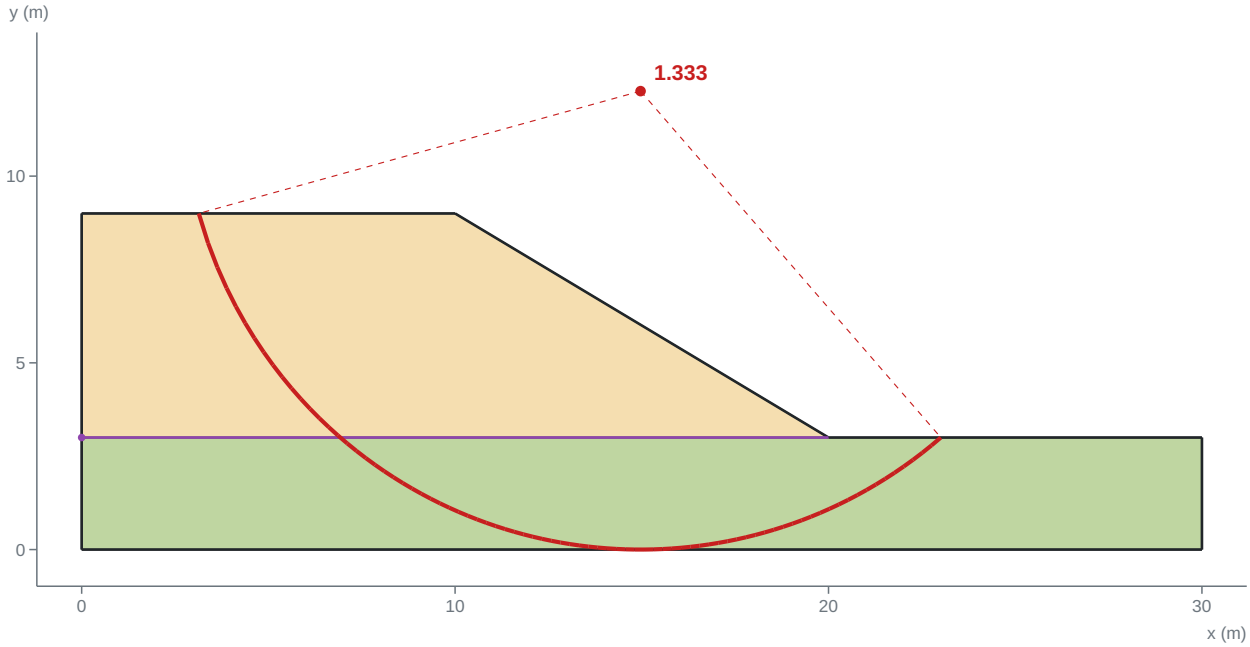


Figure 149. V086a: analysed circle (Spencer, FOS = 1.333).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft Clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Published value	Method	Source	cSlope
1.35	generalized method of slices, at T = 170.5 kN/m	Tandjiria, Low & Teh (2002)	–

cSlope analysis	Other programs	cSlope
Spencer	1.332–1.350	1.333

3.150 V086b — Tandjiria embankment, clay fill, water-filled crack

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Critical circle search · Units: SI (m, kPa, kN/m³)

Half-embankment (9 m crest, 10 m wide crest, 1V:1.67H face) of clay fill (Su = 20 kPa) on soft clay (Su = 20 kPa), γ = 19.4; 2 m water-filled tension crack, unreinforced; circular search. Also run with the automatic (Rankine) crack depth.

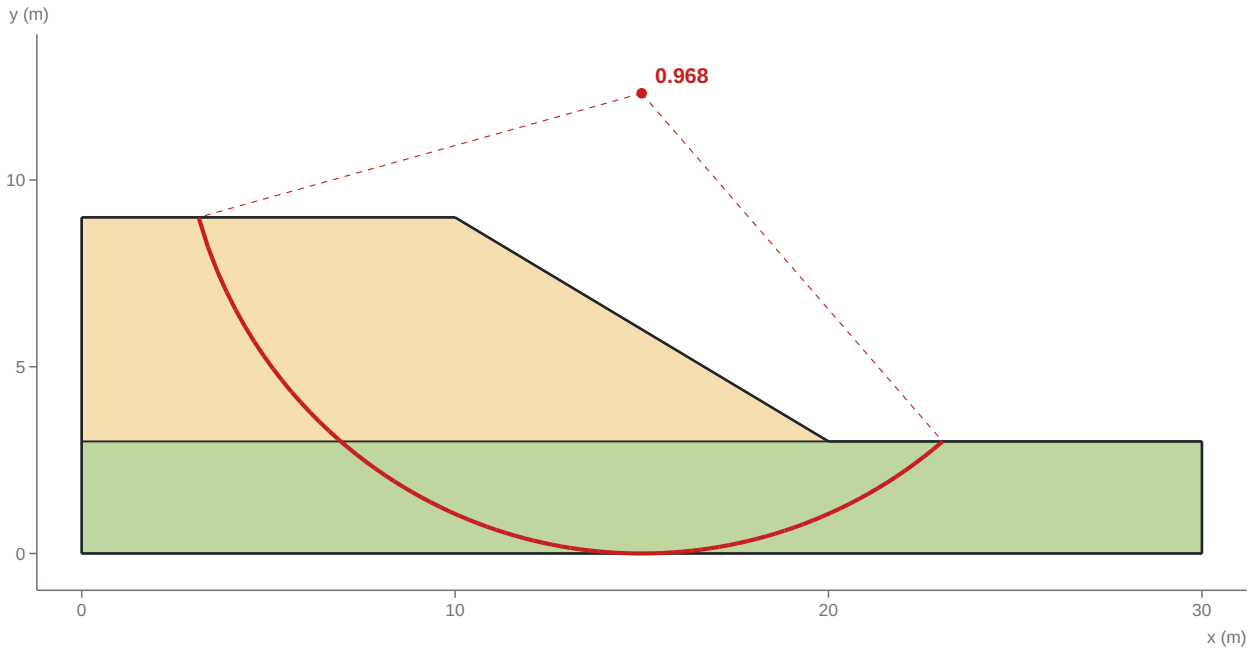


Figure 150. V086b: critical circle (Spencer (2 m tension crack), FOS = 0.968).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Published value	Method	Source	cSlope
0.981	Spencer	Tandjiria, Low & Teh (2002)	0.968

cSlope analysis	Other programs	cSlope
Spencer (2 m tension crack)	0.975	0.968
Spencer (tension crack of automatic depth)	0.975	0.966
Morgenstern-Price (half-sine)	0.975	0.968

3.151 V086c — Tandjiria embankment, sand fill, dry crack

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086 embankment as a sand fill (φ′ = 37°, γ = 17) on soft clay (Su = 20 kPa, γ = 20) with a dry 2 m tension crack, unreinforced; circular search.

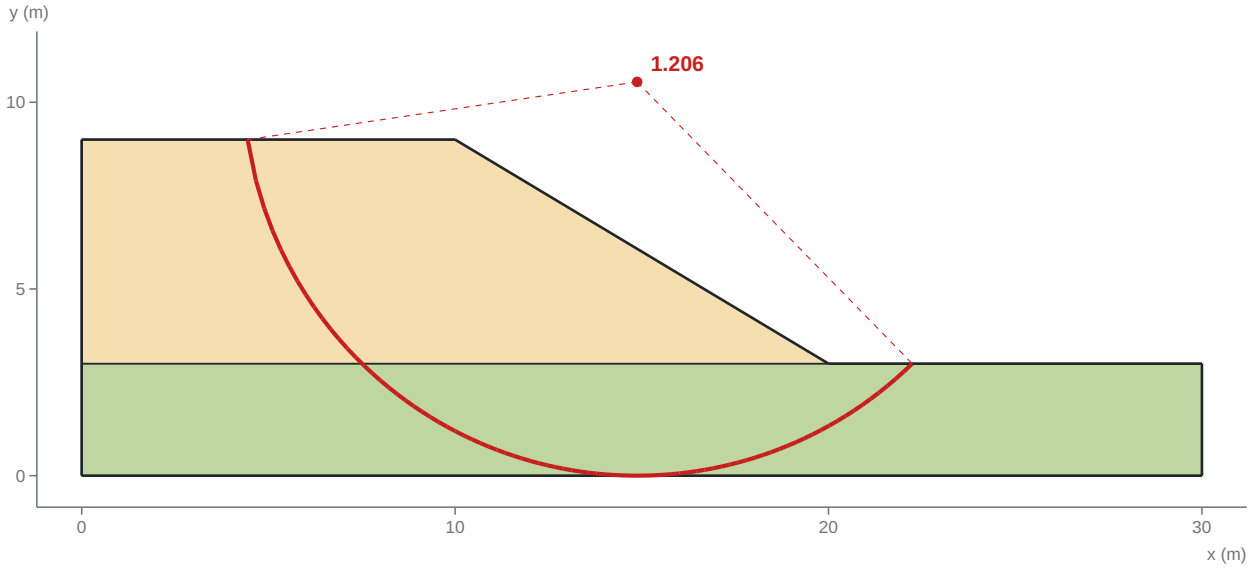


Figure 151. V086c: critical circle (Spencer, FOS = 1.206).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	17.00	c' = 0.00	37.00°
Soft clay	Undrained (φ = 0)	20.00	Su = 20.00	0°

Published value	Method	Source	cSlope
1.219	Spencer	Tandjiria, Low & Teh (2002)	1.206

cSlope analysis	Other programs	cSlope
Spencer	1.209	1.206
Morgenstern-Price (half-sine)	1.218	1.215

3.152 V086d — Tandjiria embankment, clay fill, the paper's geometry

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 1, Table 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086b clay-fill embankment as the paper gives it: a 30° face (p. 431, Fig. 4) and a water-filled tension crack 2c/γ = 2.06 m deep (Table 2 note), unreinforced; circular search.

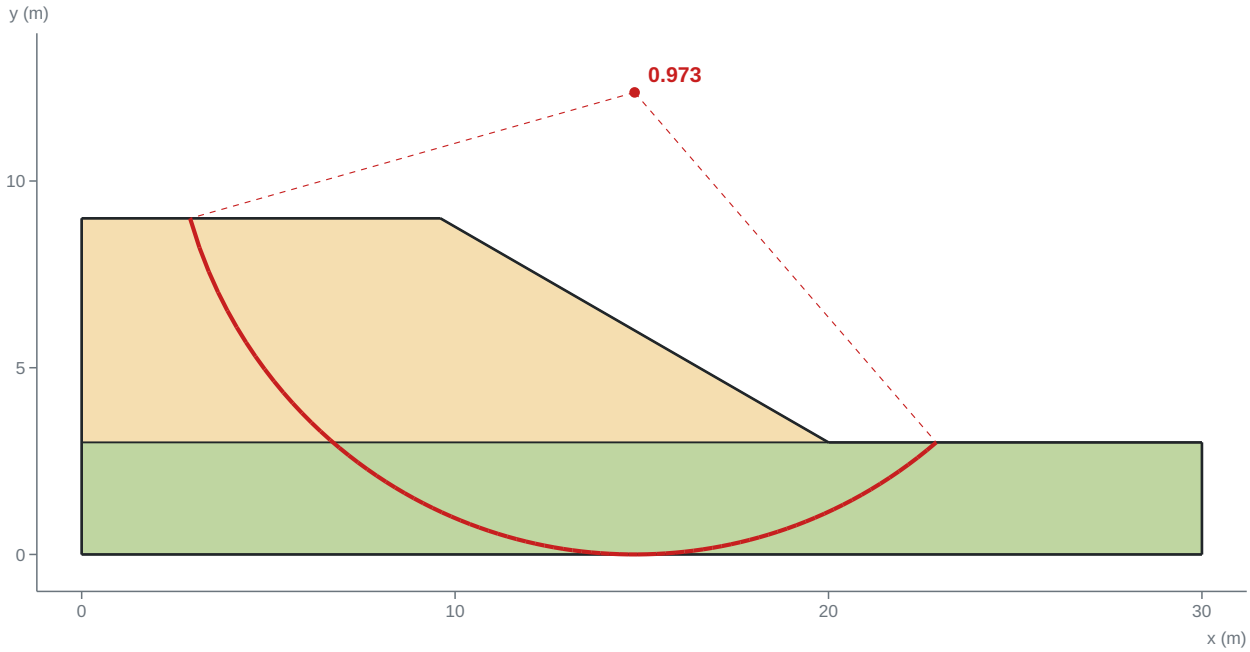


Figure 152. V086d: critical circle (Spencer, FOS = 0.973).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Undrained (φ = 0)	19.40	Su = 20.00	0°
Soft clay	Undrained (φ = 0)	19.40	Su = 20.00	0°

Published value	Method	Source	cSlope
0.981	Spencer	Tandjiria, Low & Teh	0.973

cSlope analysis	Other programs	cSlope
Spencer	–	0.973

The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed and is taken as the fill's (19.4).

3.153 V086e — Tandjiria embankment, sand fill, the paper's geometry

Tandjiria, V., Low, B.K. & Teh, C.I. (2002). Effect of reinforcement force distribution on stability of embankments. Geotextiles and Geomembranes 20, 423–443; problem 2.

Critical circle search · Units: SI (m, kPa, kN/m³)

The V086c sand-fill embankment as the paper gives it: a 30° face and no tension crack (the paper mentions one only for the clay fill), unreinforced; circular search.

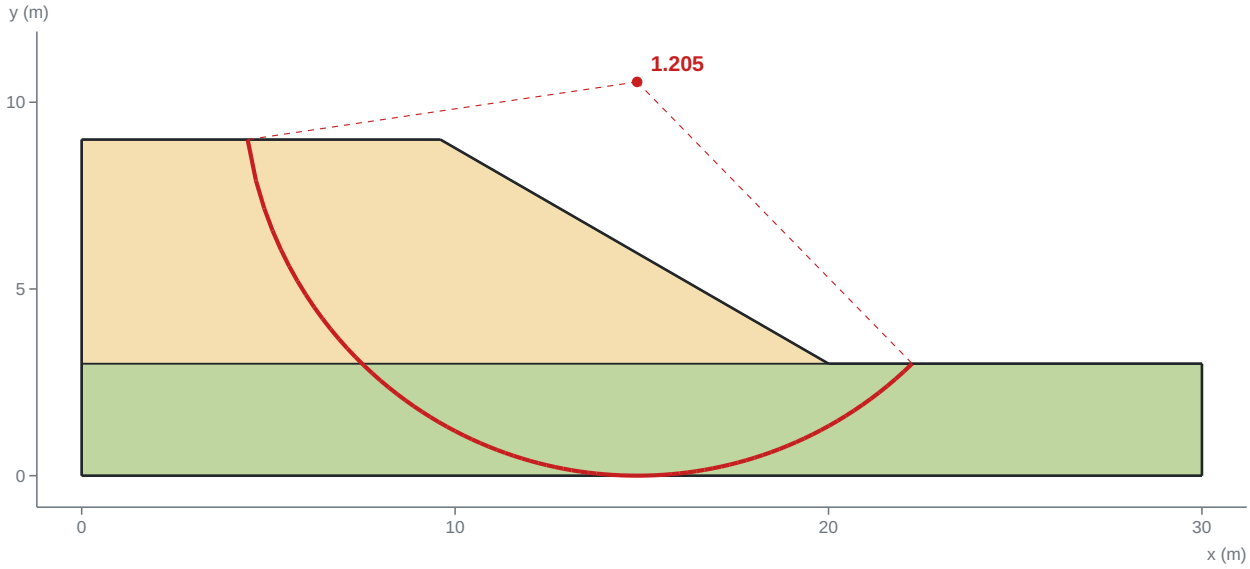


Figure 153. V086e: critical circle (Spencer, FOS = 1.205).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Fill	Mohr-Coulomb	17.00	c' = 0.00	37.00°
Soft clay	Undrained (φ = 0)	20.00	Su = 20.00	0°

Published value	Method	Source	cSlope
1.219	Spencer	Tandjiria, Low & Teh	1.205

cSlope analysis	Other programs	cSlope
Spencer	–	1.205

The paper uses the generalized method of slices without naming its side-force function; Spencer's method is cSlope's nearest. The soft clay's unit weight is not printed; 20 kN/m³ is assumed (17 gives 0.1 % less).

3.154 V087 — USACE dam, chart-check properties (EM 1110-2-1902 fig. 4-3)

USACE EM 1110-2-1902 (2003), fig. 4-3.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family with single unit weights (embankment c = 200 psf, φ = 25°, γ = 115; sand 0/35/130; clay 0/27/115), slightly flatter faces and ponded water at el. 20 on both faces. the reference circle (−135, 169, R = 169.0), ending at the crest (−15, 50). Converted to SI.

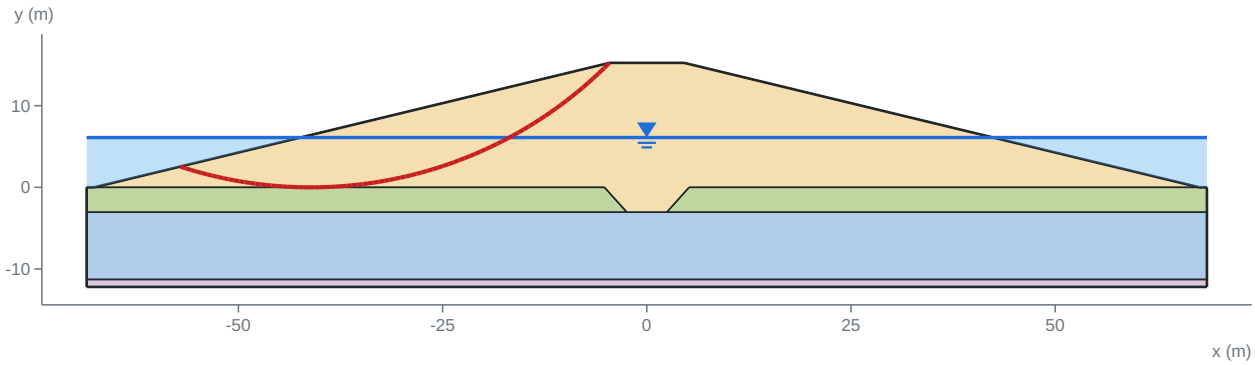


Figure 154. V087: analysed circle (Janbu corrected, FOS = 2.248).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 9.58	25.00°
Sand	Mohr-Coulomb	20.42	c' = 0.00	35.00°
Foundation clay	Mohr-Coulomb	18.07	c' = 0.00	27.00°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Published value	Method	Source	cSlope
2.30	not stated	USACE EM 1110-2-1902	–

cSlope analysis	Other programs	cSlope
Janbu corrected	2.290	2.248

3.155 V088 — USACE dam, end of construction (EM 1110-2-1902 fig. 4-1)

USACE EM 1110-2-1902 (2003), fig. 4-1.

Fixed circle · Units: SI (m, kPa, kN/m³)

Symmetric 50 ft embankment (crest ±12 ft, toes ±206.18 ft) with c = 1000 psf, φ = 5°, a core trench through a 10 ft sand blanket over foundation clay (Su = 3000 psf) and rock; groundwater at the sand top, 7 ft dry crest crack; moist/saturated unit weights. Specified circle (102, 163, R = 163). Converted to SI.

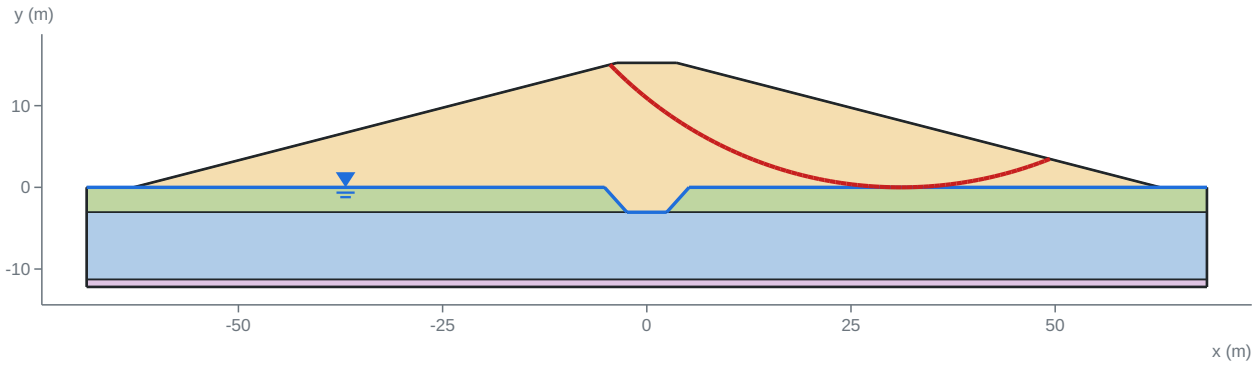


Figure 155. V088: analysed circle (Spencer, FOS = 2.442).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 47.88	5.00°
Sand	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation clay	Undrained (φ = 0)	17.28	Su = 143.6	0°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Published value	Method	Source	cSlope
2.44	Spencer	USACE EM 1110-2-1902	2.442

cSlope analysis	Other programs	cSlope
Spencer	2.445	2.442
Bishop simplified	2.447	2.442
Janbu corrected	2.430	2.444
Morgenstern-Price (half-sine)	2.447	2.442

3.156 V089 — USACE dam, upstream low pool (EM 1110-2-1902 fig. 4-2)

USACE EM 1110-2-1902 (2003), fig. 4-2.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V088 dam family, 45 ft high (crest ±15 ft, toes ±200 ft), with drained strengths (embankment c = 100 psf, φ = 25°; sand 0/35; clay 0/28; rock 0/45; moist/saturated unit weights) and the pool at el. 20 standing on the upstream face. USACE's circle (−102, 163, R = 173), tangent to the clay top. Converted to SI.

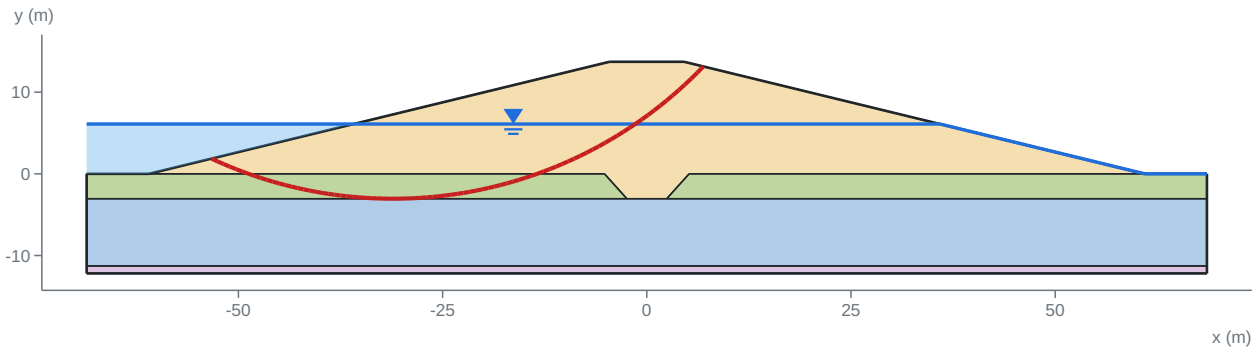


Figure 156. V089: analysed circle (Bishop simplified, FOS = 2.698).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	18.07	c' = 4.79	25.00°
Sand	Mohr-Coulomb	19.64	c' = 0.00	35.00°
Foundation clay	Mohr-Coulomb	17.28	c' = 0.00	28.00°
Rock	Mohr-Coulomb	25.13	c' = 0.00	45.00°

Published value	Method	Source	cSlope
2.71	Bishop simplified	USACE EM 1110-2-1902	2.698

cSlope analysis	Other programs	cSlope
Bishop simplified	2.716	2.698
Spencer	2.736	2.721
Janbu corrected	2.650	2.648
Morgenstern-Price (half-sine)	2.744	2.727

3.157 V090 — USACE EM 1110-2-1902, example F-5

US Army Corps of Engineers (2003). Slope Stability, EM 1110-2-1902, Appendix F.

Fixed circle · Units: imperial (ft, psf, pcf)

Embankment (c = 1780 psf, φ = 5°) on a foundation (c = 1600 psf, φ = 2°), analysed with the circle given in the Corps of Engineers example.

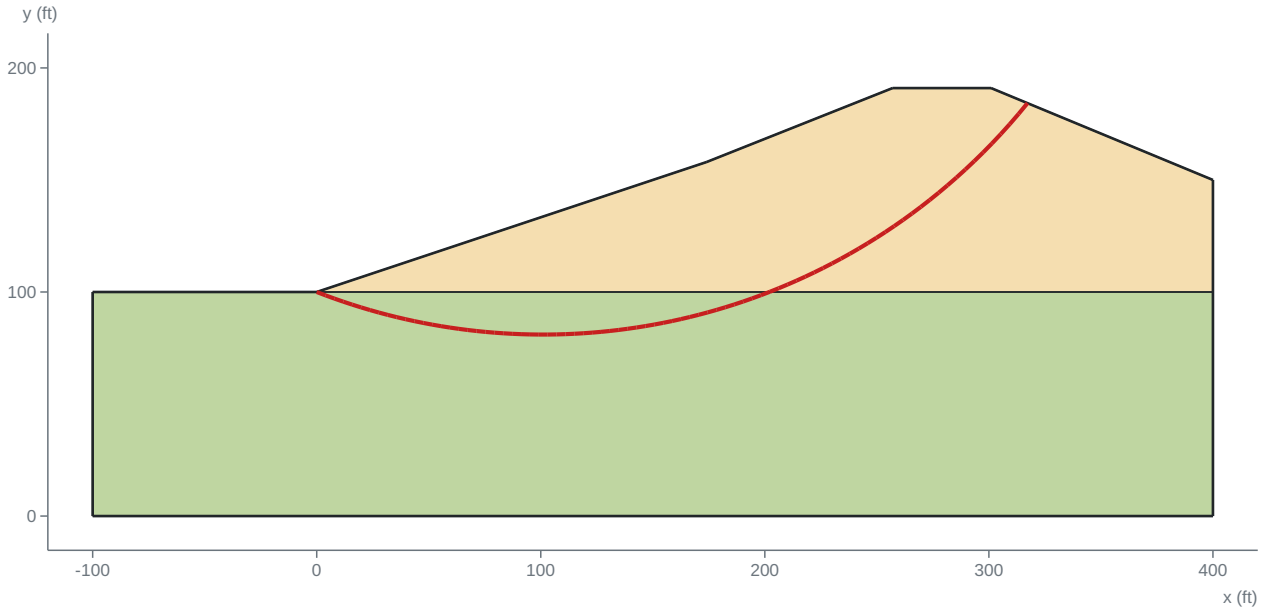


Figure 157. V090: analysed circle (Bishop simplified, FOS = 1.320).

Material	Model	γ (pcf)	Strength (psf)	φ
Embankment	Mohr-Coulomb	135.0	c' = 1780.0	5.00°
Foundation	Mohr-Coulomb	127.0	c' = 1600.0	2.00°

Published value	Method	Source	cSlope
1.33	not stated	USACE EM 1110-2-1902	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.332	1.320
Spencer	1.328	1.317
Janbu corrected	1.345	1.341
Morgenstern-Price (half-sine)	1.327	1.316

3.158 V091 — USACE example E-2 (fig. E-10), ponded water

USACE EM 1110-2-1902 (2003), appendix E, example E-2, fig. E-10.

Fixed circle · Units: SI (m, kPa, kN/m³)

Undrained three-layer slope with 8 ft of water ponded against its toe (pool el. 0), on a commercial program's circle (48.262, 28, R = 47.683 ft; Fig. 68.2). Converted to SI.

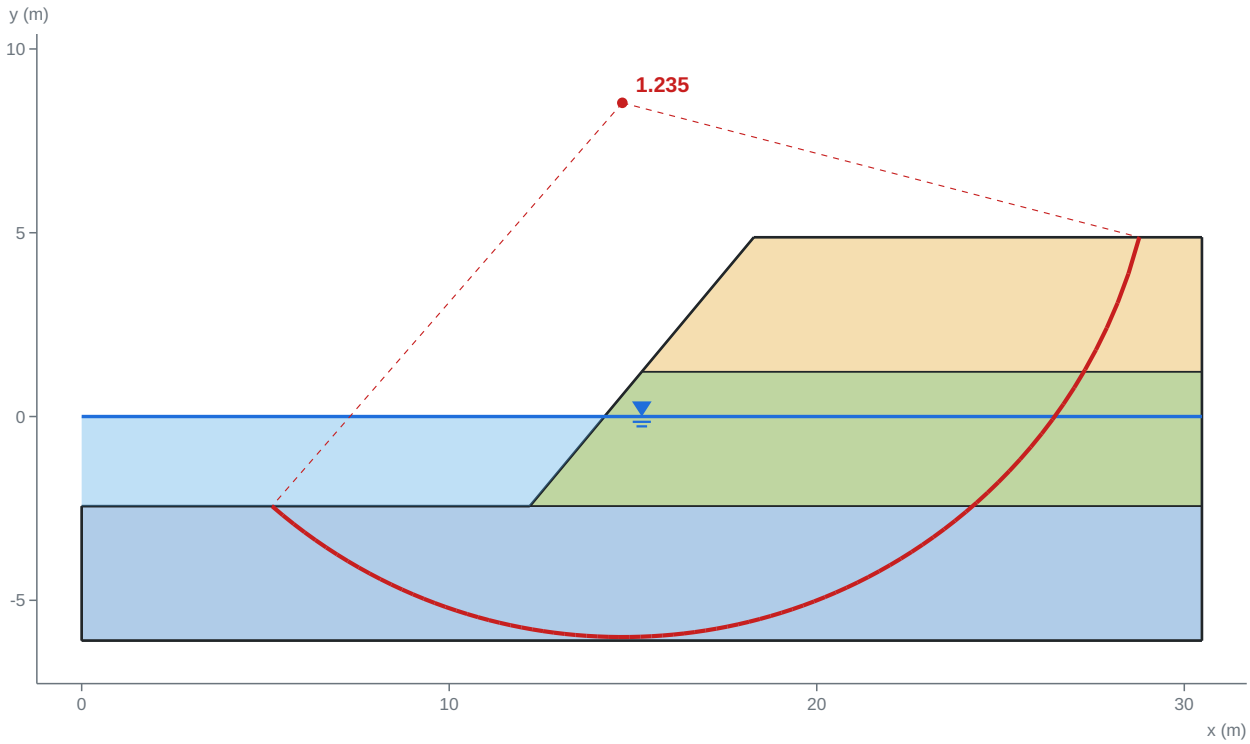


Figure 158. V091: analysed circle (Bishop simplified, FOS = 1.235).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil 1	Undrained (φ = 0)	18.85	Su = 28.73	0°
Soil 2	Undrained (φ = 0)	15.71	Su = 19.15	0°
Soil 3	Undrained (φ = 0)	16.49	Su = 23.94	0°

Published value	Method	Source	cSlope
1.24	stability chart (example E-2)	USACE EM 1110-2-1902	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.241	1.235
Spencer	1.241	1.235
Morgenstern-Price (half-sine)	1.244	1.235
Janbu corrected	1.385	1.380

USACE's chart solution for this circle (example E-2, p. E-20) gives 1.24; the programs and cSlope give 1.235–1.244 on it. The manual's shallower circle tangent to el. −8 ft (example E-1) gives 1.14 and is the more critical of the two.

3.159 V092 — USACE example F-6, steady-seepage dam

USACE EM 1110-2-1902, example F-6.

Fixed circle · Units: SI (m, kPa, kN/m³)

112 ft embankment on a granular foundation under steady seepage: piezometric line from the pool through the chimney drain to the tailwater, which stands on the downstream toe (free water). The upstream face rises from (30.2, 100) ft to the crest (60.8, 112). USACE's circle. Converted to SI.

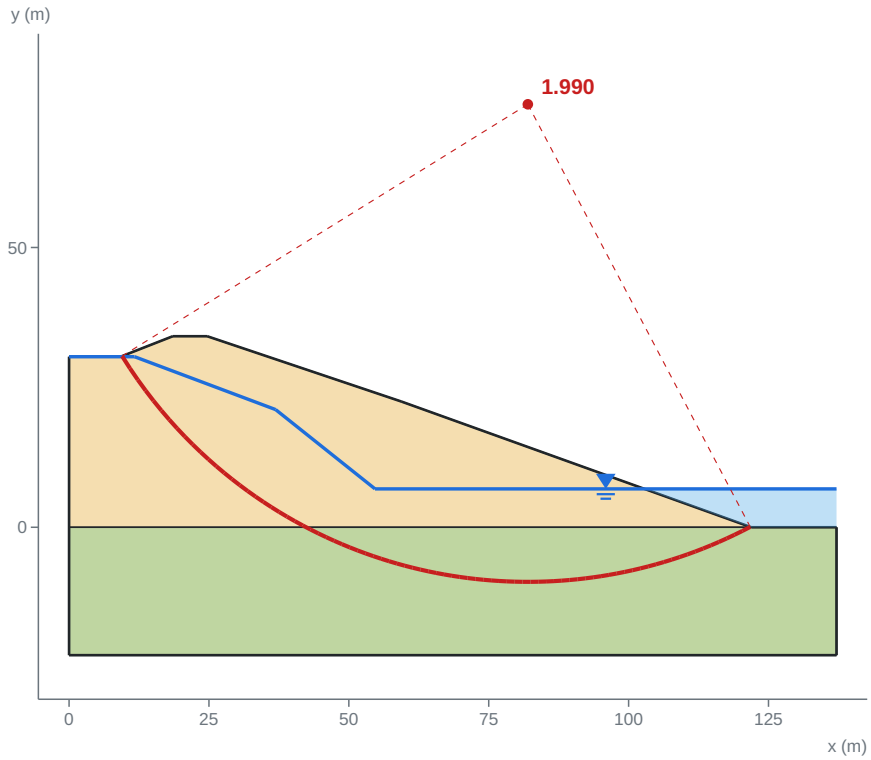


Figure 159. V092: analysed circle (Bishop simplified, FOS = 1.990).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Embankment	Mohr-Coulomb	20.42	c' = 0.00	34.00°
Foundation	Mohr-Coulomb	19.64	c' = 0.00	35.00°

Published value	Method	Source	cSlope
2.01	Simplified Bishop, nine-slice hand calculation on the manual's own piezometric line (example F-6)	USACE EM 1110-2-1902	–
2.07	Modified Swedish (θ = 19.7°) on the manual's own piezometric line (example F-6)	USACE EM 1110-2-1902	–

cSlope analysis	Other programs	cSlope
Bishop simplified	2.011	1.990
Spencer	2.026	2.005
Janbu corrected	1.830	1.824
Morgenstern-Price (half-sine)	2.027	2.005

USACE's values are for the manual's own piezometric line, which its slice table (fig. F-17a) places lower than the line of the program version analysed here: the kink at about el. 53–59 ft instead of 69 ft, and the tailwater level reached about 18 ft further upstream. On the manual's line cSlope gives Bishop 2.05–2.06 and Spencer 2.06–2.08, depending on where the table's slices are placed: within 0.5 % of USACE's Modified Swedish value 2.07, and 2–2.6 % above its nine-slice Bishop hand calculation. The

manual's values are shown for reference and not compared with cSlope's analyses on the program's line.

3.160 V093 — Yamagami & Ueta (1988)

Yamagami & Ueta (1988).

Critical circle search · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 5 m high; critical circle search.

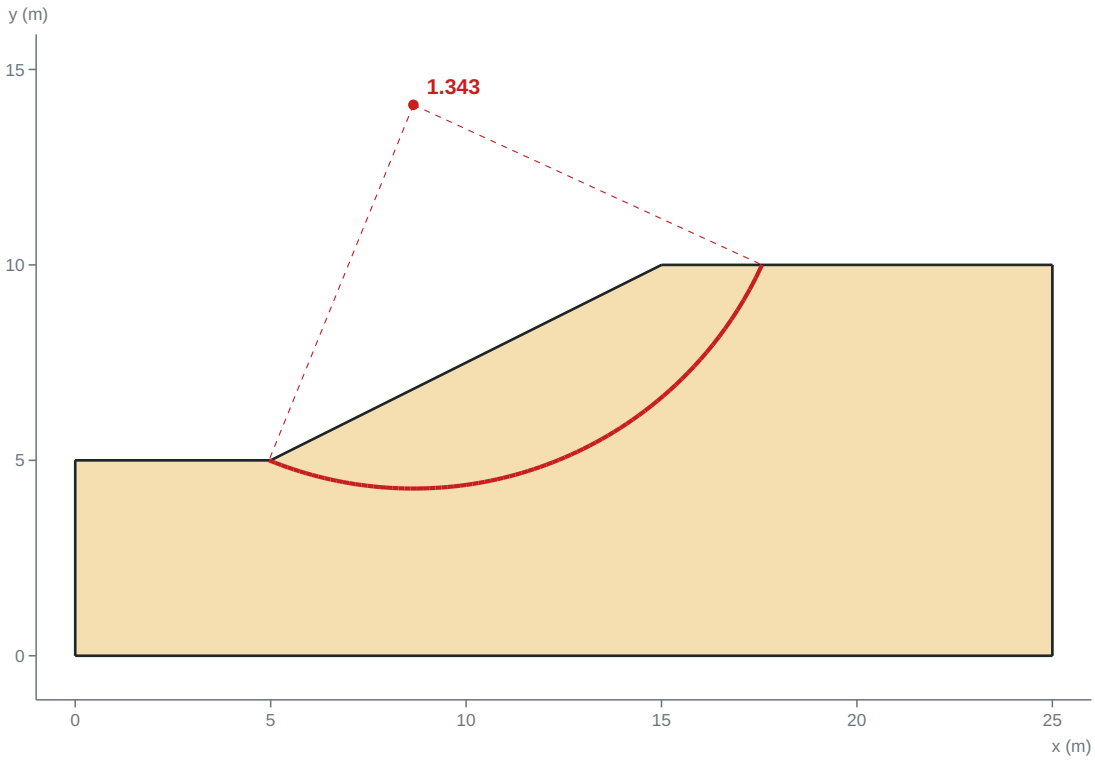


Figure 160. V093: critical circle (Bishop simplified, FOS = 1.343).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	17.64	c' = 9.80	10.00°

Published value	Method	Source	cSlope
1.348	Bishop simplified	Yamagami & Ueta (1988)	1.343
1.282	Ordinary	Yamagami & Ueta (1988)	–
1.185	Janbu simplified, non-circular	Yamagami & Ueta (1988)	1.176
1.339	Spencer, non-circular	Yamagami & Ueta (1988)	1.322
1.33	Spencer, non-circular	Greco (1996)	1.322

cSlope analysis	Other programs	cSlope
Bishop simplified	1.344	1.343
Spencer, non-circular	1.325	1.322
Janbu simplified, non-circular	1.178	1.176

3.161 V094 — Yamagami & Ueta (1988) four-layer slope, non-circular

Greco (1996) example 4; Yamagami & Ueta (1988).

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

Four-layer slope without water whose critical surface is non-circular; circular search followed by non-circular optimisation.

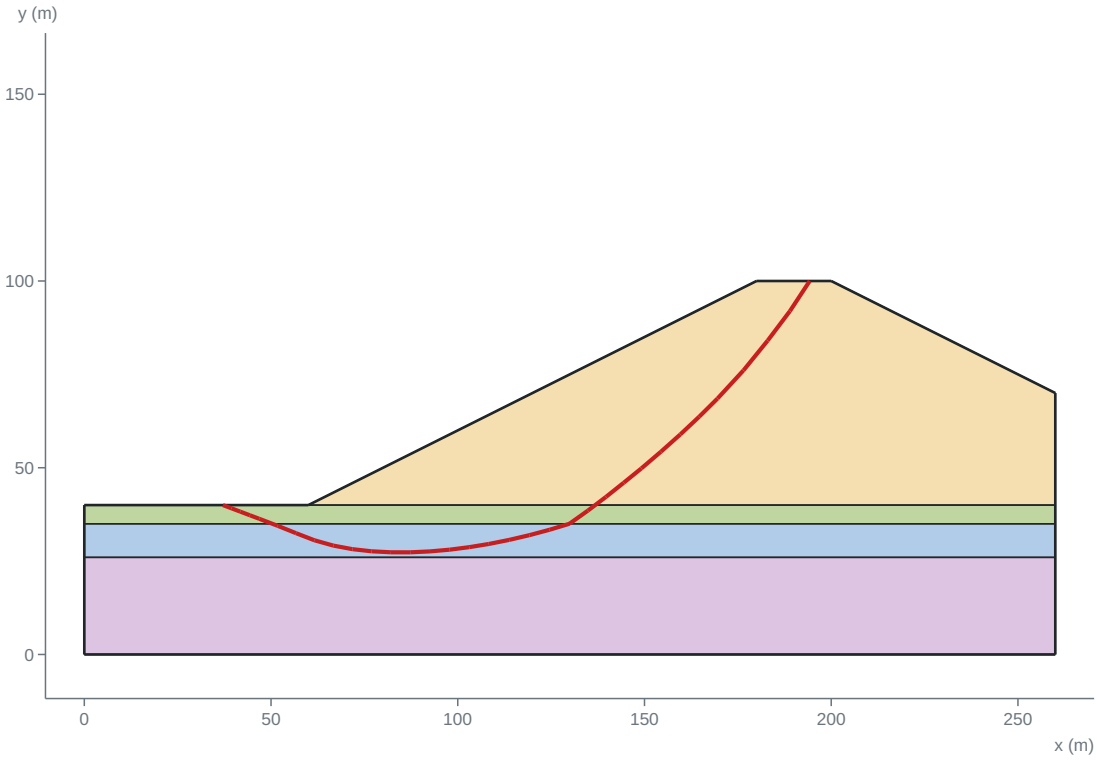


Figure 161. V094: critical surface (Spencer, non-circular, FOS = 1.400).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Upper layer	Mohr-Coulomb	20.38	c' = 49.00	29.00°
Layer 2	Mohr-Coulomb	17.64	c' = 0.00	30.00°
Layer 3	Mohr-Coulomb	20.38	c' = 7.84	20.00°
Bottom layer	Mohr-Coulomb	17.64	c' = 0.00	30.00°

Published value	Method	Source	cSlope
1.40–1.42	Spencer, non-circular	Greco (1996)	1.400

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.398	1.400
Morgenstern-Price (half-sine), non-circular	1.389	1.389

3.162 V095a — Yamagami micro-pile slope, no pile

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

Homogeneous 2:1 slope, 4 m high, without piles, on the published circle.

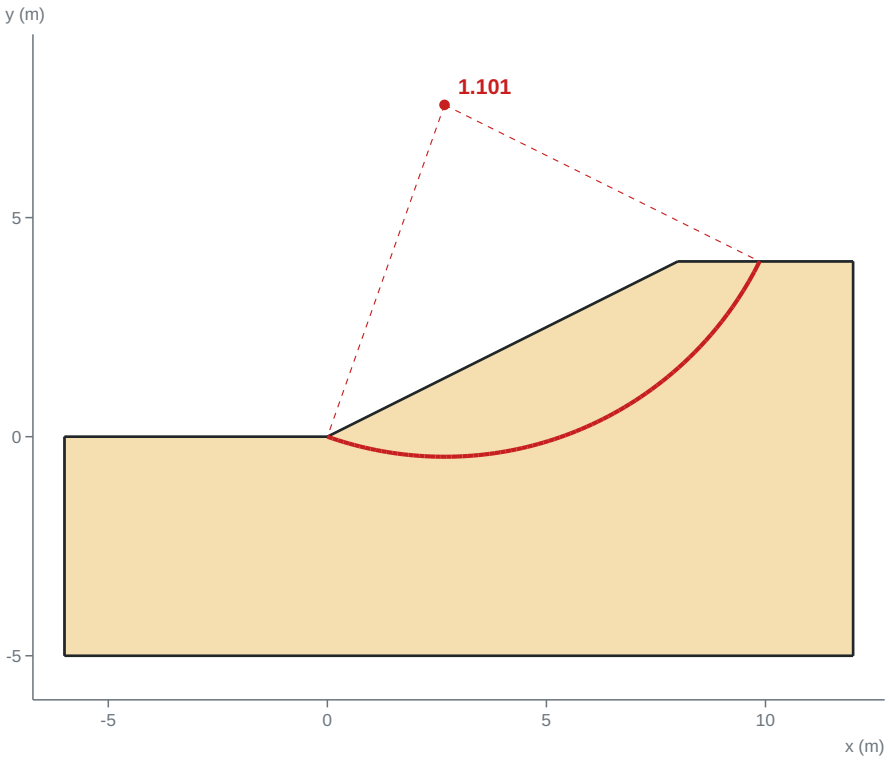


Figure 162. V095a: analysed circle (Bishop simplified, FOS = 1.101).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Soil	Mohr-Coulomb	15.68	c' = 4.90	10.00°

Published value	Method	Source	cSlope
1.10	not stated	Yamagami et al. (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.102	1.101

3.163 V095c — Homogeneous slope with a micro-pile row

Yamagami et al. (2000).

Fixed circle · Units: SI (m, kPa, kN/m³)

As V095b with a micro-pile row (1 m spacing) whose shear strength of 10.7 kN acts as a passive force tangent to the slip surface, on the reference with-pile circle (2.674, 7.376, R = 8.102).

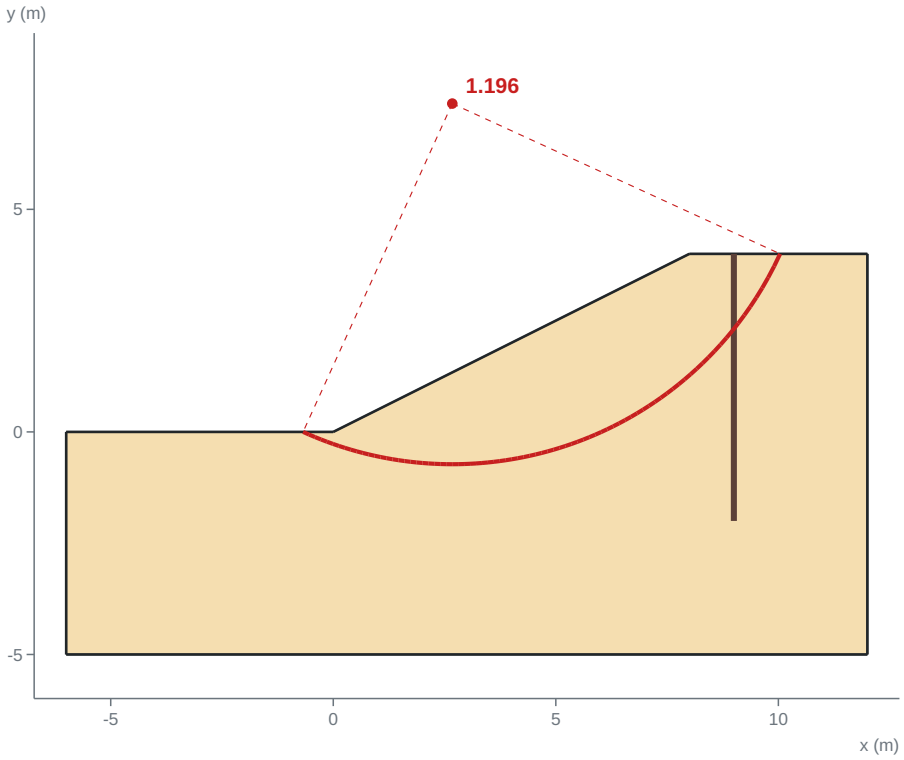


Figure 163. V095c: analysed circle (Bishop simplified, FOS = 1.196).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Material 1	Mohr-Coulomb	15.68	c' = 4.90	10.00°

Published value	Method	Source	cSlope
1.20	not stated	Yamagami et al. (2000)	–

cSlope analysis	Other programs	cSlope
Bishop simplified	1.193	1.196

3.164 V096a — Zhu & Lee heterogeneous slope, dry, deep circle

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Critical circle search · Units: SI (m, kPa, kN/m³)

Four horizontal layers under a two-bench slope with a dry tension crack 1.015 m deep at the crest (Fig. 52.1; the depth of a commercial program's surface end below the crest), dry; the circular search lands in the deep family (surface 3).

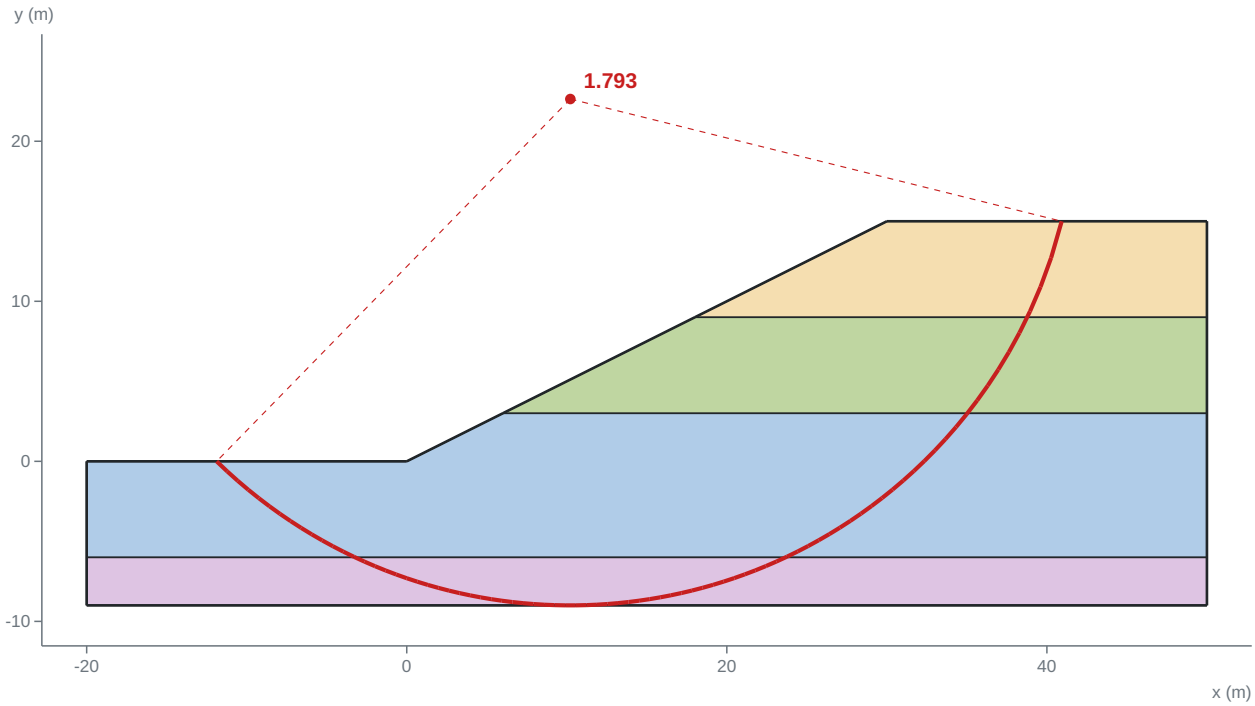


Figure 164. V096a: critical circle (Bishop simplified, FOS = 1.793).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c' = 10.00	12.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.804	1.793
Spencer	1.804	1.794
Morgenstern-Price (half-sine)	1.790	1.780

The circle and the 1.015 m crack follow a program's version of the problem. The paper gives its surface 3 only in a figure, and it enters the crest about 3 m nearer the slope than the program's circle; the paper's values for it are checked on the paper's own circle in V096d (dry) and V096e (with the water table).

3.165 V096b — Zhu & Lee heterogeneous slope with a water table, deep circle

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Critical circle search · Units: SI (m, kPa, kN/m³)

As V096a (dry 1.015 m crack) with the water table of Table 52.2.

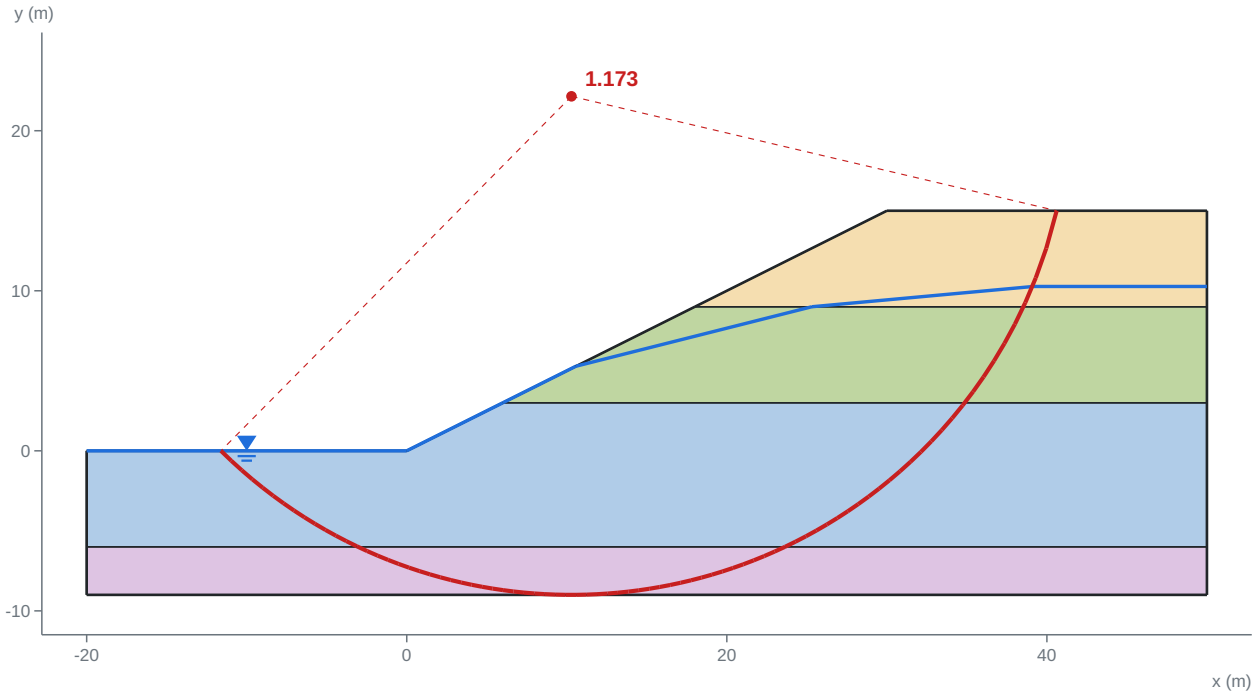


Figure 165. V096b: critical circle (Bishop simplified, FOS = 1.173).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1 (top)	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4 (bottom)	Mohr-Coulomb	18.00	c' = 10.00	12.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.176	1.173
Spencer	1.189	1.187
Morgenstern-Price (half-sine)	1.174	1.172

As V096a: the paper's values for its surface 3 with the water table are checked on the paper's own circle in V096e.

3.166 V096c — Zhu & Lee slope, surface 3 (the reference circle), dry, 1 m crack

Zhu & Lee (2002), Int. J. Numer. Anal. Meth. Geomech. 26(15). Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V096a slope with the dry 1.015 m crack on the reference surface 3 circle (10.911, 23.208, R = 32.106; Fig. 52.4).

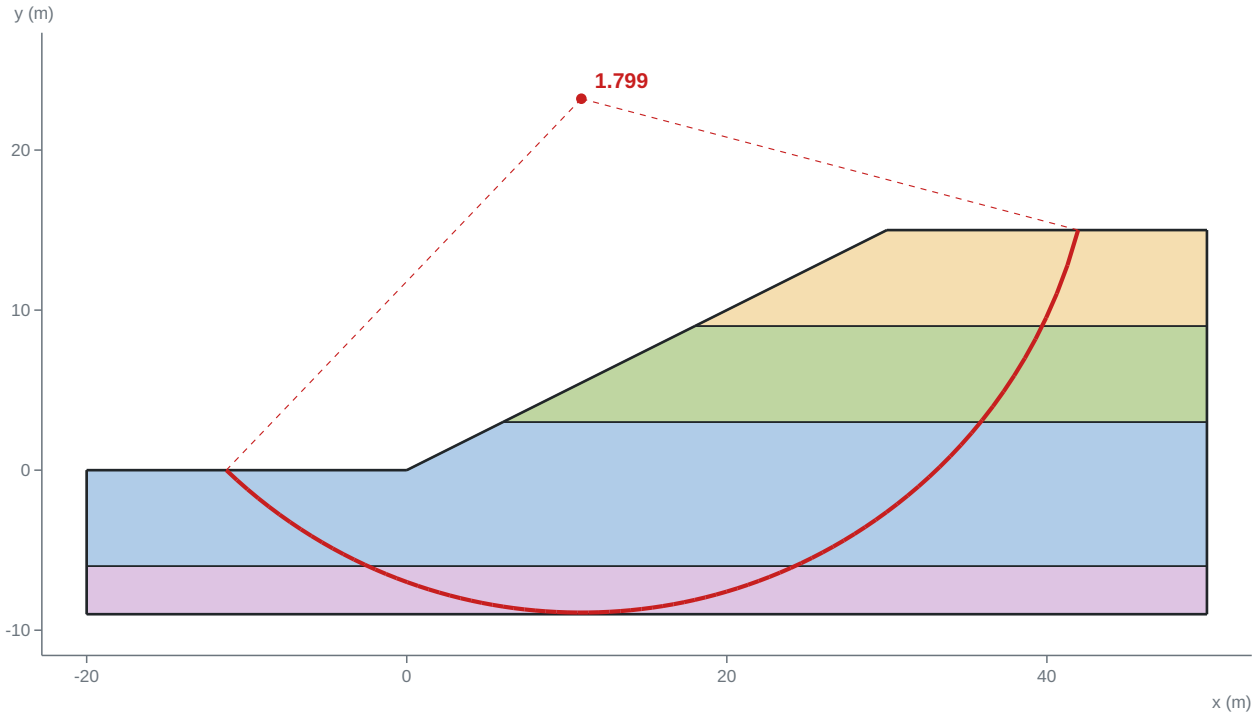


Figure 166. V096c: analysed circle (Spencer, FOS = 1.799).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c' = 10.00	12.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer	1.804	1.799
Bishop simplified	1.804	1.799

The program's reference circle enters the crest at x = 41.9 m, about 3 m further from the slope than the paper's surface 3 as drawn in its fig. 3; the paper's values are checked on the paper's circle in V096d and V096e.

3.167 V096d — Zhu & Lee slope, the paper's surface 3, dry

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Fixed circle · Units: SI (m, kPa, kN/m³)

The V096a slope, dry, no tension crack, on the paper's surface 3 digitised from its fig. 3: least-squares circle (9.7, 21.3, R = 29.8) through the drawn surface, leaving the lower bench at x = −11.1 and entering the crest at x = 38.8 (±0.3 m).

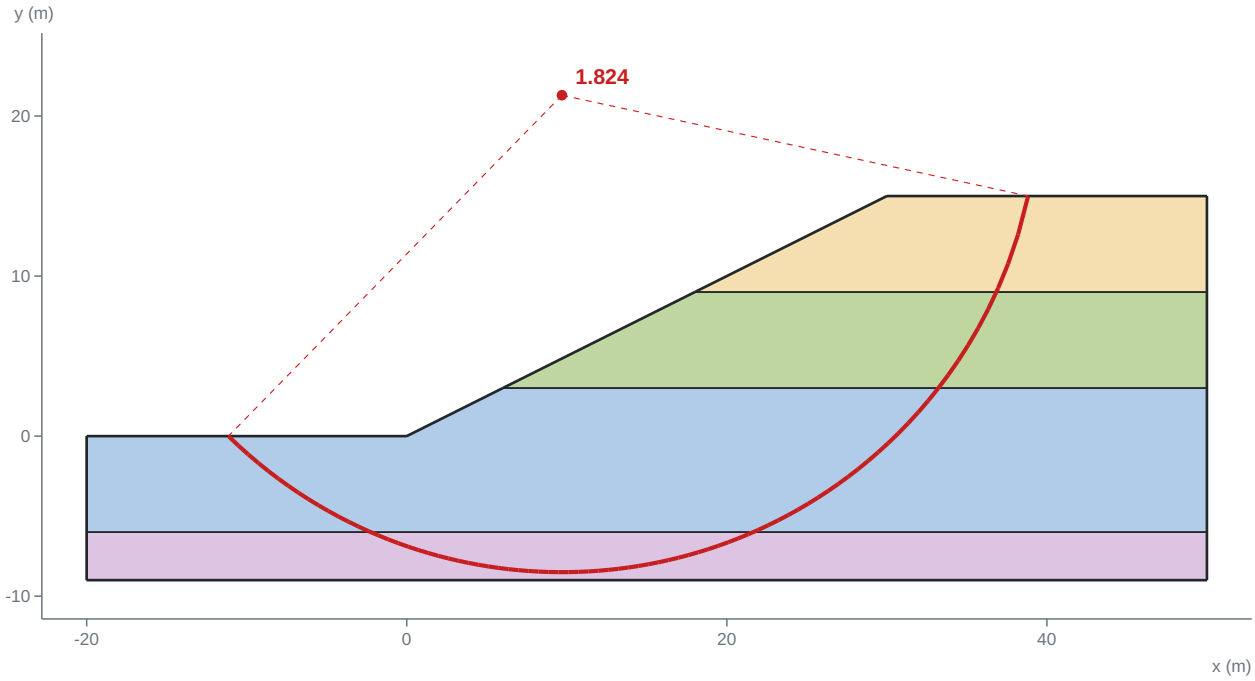


Figure 167. V096d: analysed circle (Spencer, FOS = 1.824).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c' = 10.00	12.00°

Published value	Method	Source	cSlope
1.836	Spencer	Zhu & Lee	1.824
1.823	Morgenstern-Price (half-sine)	Zhu & Lee	1.808
1.744	explicit solution (the paper's method)	Zhu & Lee (2002)	–
1.429	Bishop simplified	Zhu & Lee (2002) (disputed: see the note)	1.823
1.229	Ordinary	Zhu & Lee (2002)	–

cSlope analysis	Other programs	cSlope
Spencer	–	1.824
Morgenstern-Price (half-sine)	–	1.808
Bishop simplified	–	1.823

The paper gives surface 3 only in a figure. On the circle fitted to it cSlope comes within 1 % of the paper's Spencer and Morgenstern-Price values (1.836, 1.823), against 2 % on the program's circle of V096c; moving the fitted centre or radius by 0.3 m

changes the result by up to 1 %. The paper's simplified Bishop value 1.429 is 22 % below its own rigorous values for the same circle, where cSlope's Bishop gives 1.823 as expected on a circle; it is shown but not counted in the error statistics.

3.168 V096e — Zhu & Lee slope, the paper's surface 3, with the water table

Zhu, D.Y. & Lee, C.F. (2002). Explicit limit equilibrium solution for slope stability. Int. J. Numer. Anal. Meth. Geomech. 26(15), 1573–1590.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V096d with the water table of the paper's fig. 3 (the points of V096b, which match the drawn line within 0.3 m).

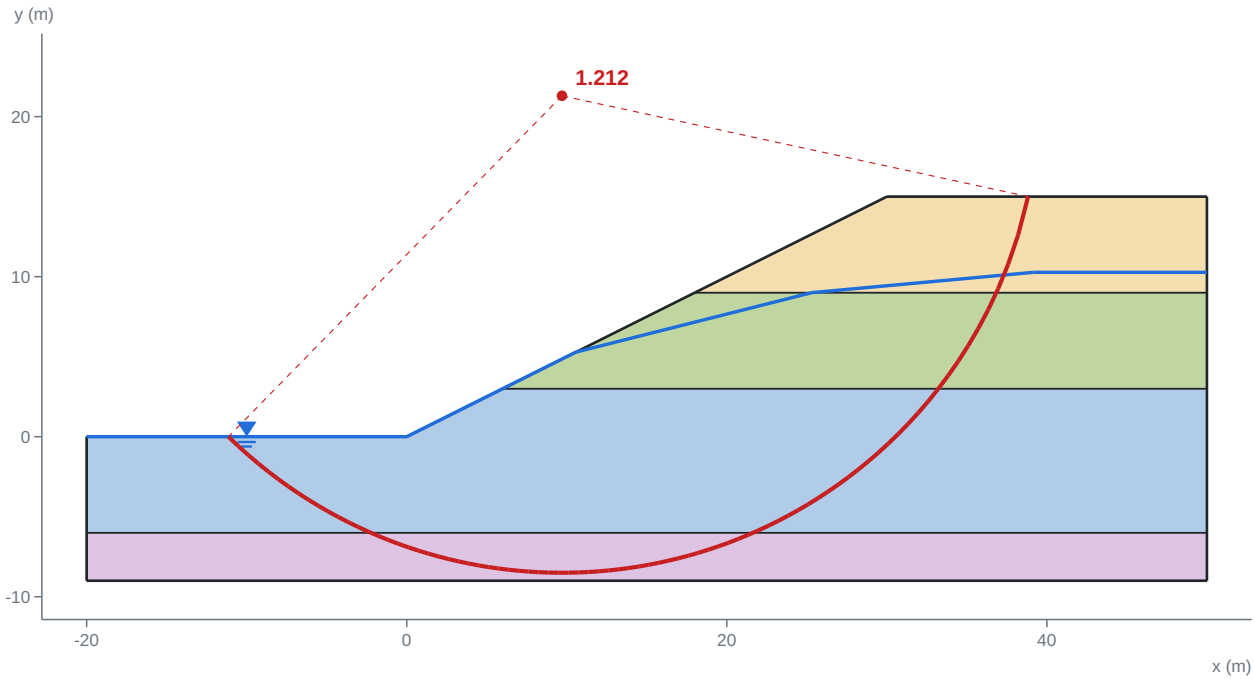


Figure 168. V096e: analysed circle (Spencer, FOS = 1.212).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Layer 1	Mohr-Coulomb	18.80	c' = 20.00	18.00°
Layer 2	Mohr-Coulomb	18.50	c' = 40.00	22.00°
Layer 3	Mohr-Coulomb	18.40	c' = 25.00	26.00°
Layer 4	Mohr-Coulomb	18.00	c' = 10.00	12.00°

Published value	Method	Source	cSlope
1.211	Spencer	Zhu & Lee	1.212
1.197	Morgenstern-Price (half-sine)	Zhu & Lee	1.197
1.166	explicit solution (the paper's method)	Zhu & Lee (2002)	–
1.079	Bishop simplified	Zhu & Lee (2002) (disputed: see the note)	1.199
0.922	Ordinary	Zhu & Lee (2002)	–

cSlope analysis	Other programs	cSlope
Spencer	–	1.212
Morgenstern-Price (half-sine)	–	1.197
Bishop simplified	–	1.199

On the circle fitted to the paper's figure cSlope reproduces the paper's Spencer and Morgenstern-Price values (1.211, 1.197) within 0.1 %. The paper's simplified Bishop value 1.079 lies 11 % below its own rigorous values for the same circle, as its dry value does; cSlope's Bishop gives 1.199. It is shown but not counted in the error statistics.

3.169 V097a — Eurocode 7 cutting with surcharge

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

Cutting in clay with an inclined piezometric line and a 35 kPa surcharge on the crest, analysed on the published circle.

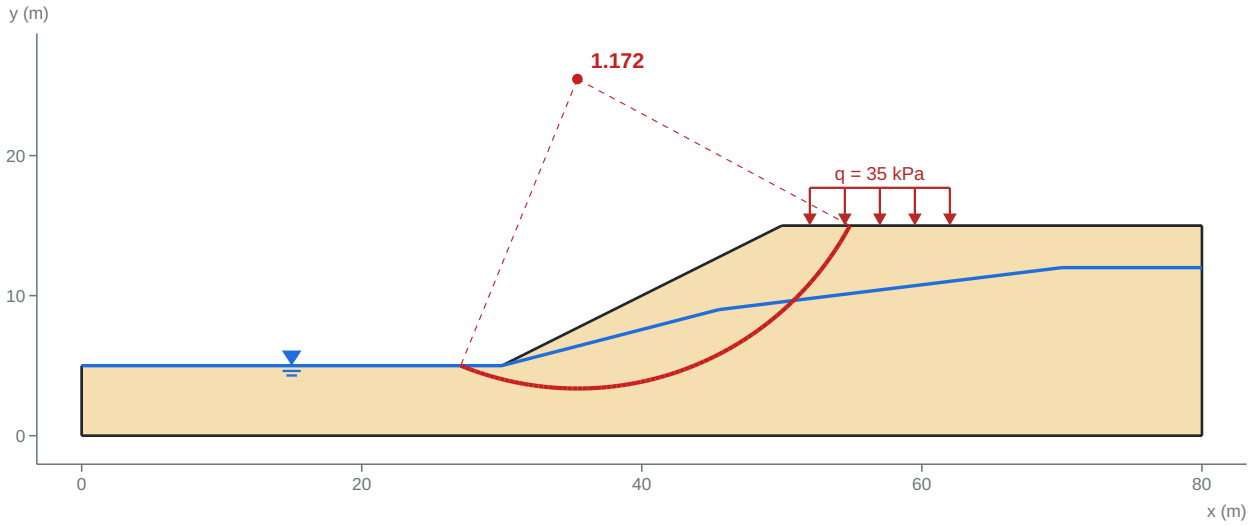


Figure 169. V097a: analysed circle (Bishop simplified, FOS = 1.172).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c' = 8.00	23.04°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.173	1.172

The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.

3.170 V097b — Eurocode 7 cutting, partial factors applied by cSlope

Verification manual of a commercial slope stability program.

Fixed circle · Units: SI (m, kPa, kN/m³)

As V097a, but with the characteristic strength c' = 10 kPa, φ' = 28° and Design Approach 3 material factors (set M2: γc' = γφ' = 1.25) applied by the analysis; the surcharge is permanent (factor 1.0). The FOS is the over-design factor on design values.

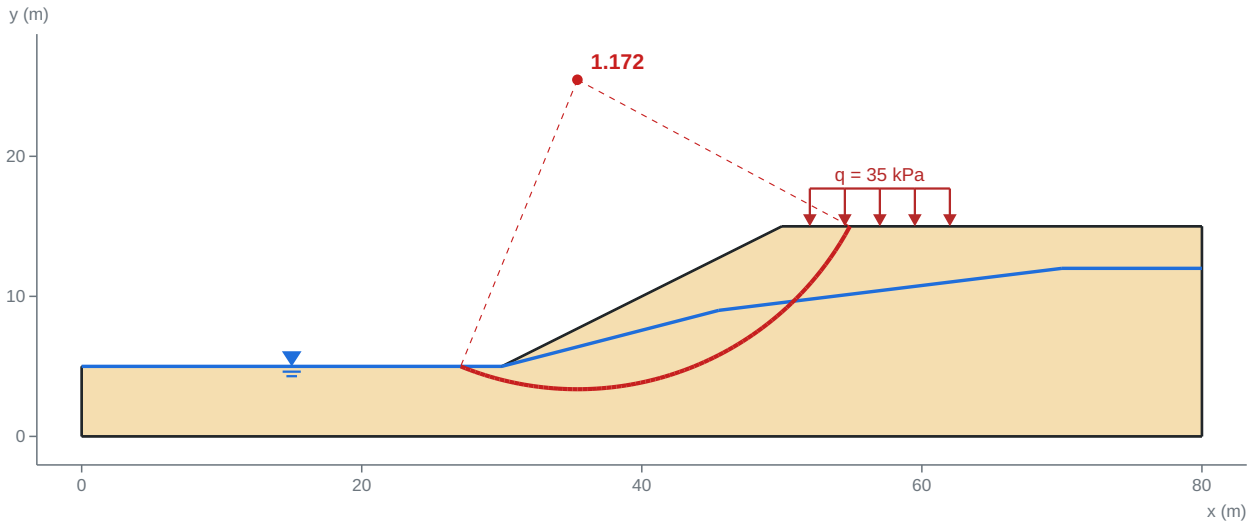


Figure 170. V097b: analysed circle (Bishop simplified, FOS = 1.172).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
Clay	Mohr-Coulomb	20.00	c' = 10.00	28.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Bishop simplified	1.173	1.172
Spencer	1.174	1.174

The problem comes from a program manual only. The manual cites 1.193 (Bishop) from the textbook example it is based on, but its water table is approximate because the book gives no coordinates, so that value is not used as a check.

3.171 V098 — Example slope without seismic load

Verification manual of a commercial slope stability program.

Circle search with non-circular optimisation · Units: SI (m, kPa, kN/m³)

10 m, 2:1 slope in three layers over a horizontal base, dry, no seismic load; Spencer search, circular and non-circular.

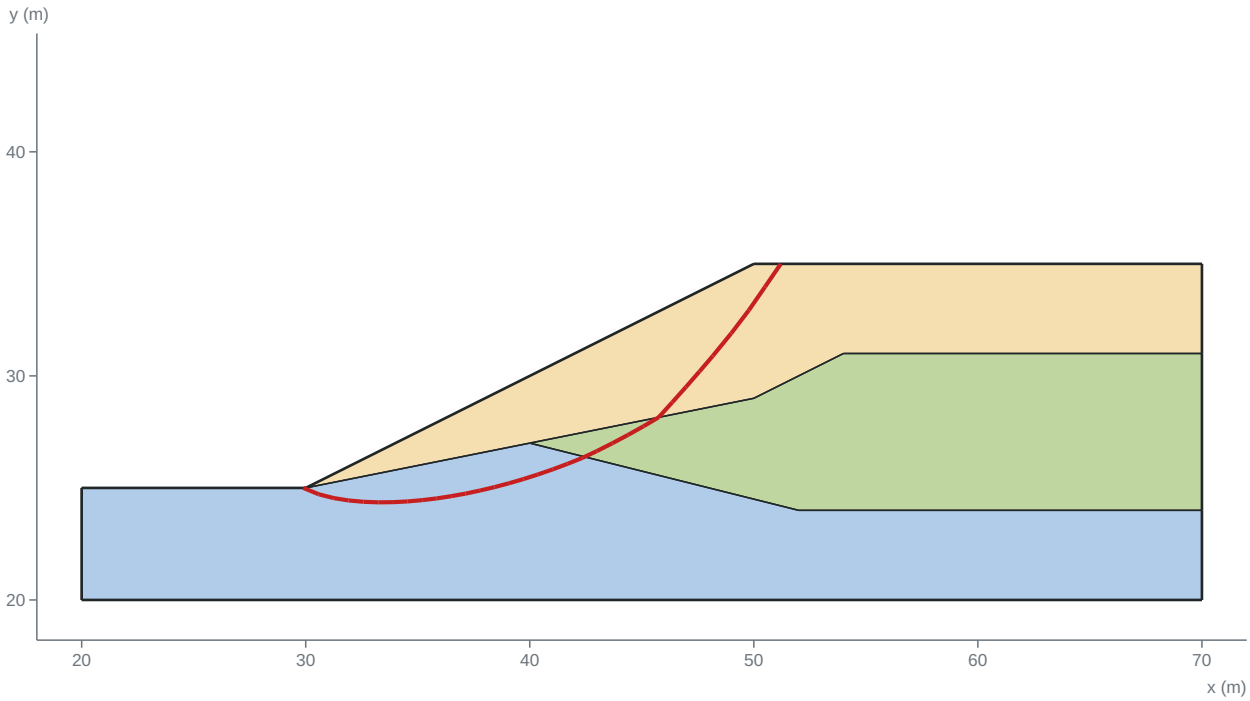


Figure 171. V098: critical surface (Spencer, non-circular, FOS = 1.357).

Material	Model	γ (kN/m³)	Strength (kPa)	φ
soil1	Mohr-Coulomb	19.50	c' = 0.00	38.00°
soil2	Mohr-Coulomb	19.50	c' = 5.30	23.00°
soil3	Mohr-Coulomb	19.50	c' = 7.20	20.00°

The source publishes no factor of safety for this problem.

cSlope analysis	Other programs	cSlope
Spencer, non-circular	1.360	1.357
Spencer	1.372	1.372

The problem comes from a program manual only; no independent value is published.

4 Strength reduction by finite elements

cSlope's finite element analysis reduces c and tan φ by a factor F until the elastic–perfectly plastic slope can no longer be brought into equilibrium. The mesh is of six-node triangles, refined adaptively where the plastic dissipation at failure is largest until F changes by less than 0.01 or twice. Each problem is run with its source's flow rule (Davis B, associated, or non-associated with ψ = 0); the source's elastic parameters do not change F appreciably. The finite element factor is compared with the source's finite element, finite difference, limit analysis or theoretical value: within 2 % of a theoretical value or a pair of bounds, and within 5 % of a published numerical value, which depends on the mesh, the elements and the failure criterion.

Problem	Published by	Published	Flow rule	Elements: F	cSlope
V002	limit analysis, lower and upper bound, Kim et al. (2002)	0.40–0.45	associated	2404: 0.417, 4482: 0.420	0.420
V013a	finite-element strength reduction (3D, without piles), Cai & Ugai (2000)	1.14	non-associated	1366: 1.141, 3382: 1.129, 8071: 1.125	1.125
V018	plasticity theory (Prandtl mechanism), Chen & Shao (1988)	1.0	davis-b	1543: 1.000, 2990: 0.999	0.999
V019	finite-element strength reduction, Cheng et al. (2007)	1.47	associated	1832: 1.426, 3340: 1.403, 6310: 1.408	1.407
V020	finite-element strength reduction, Cheng et al. (2007)	1.327	associated	1476: 1.367, 2844: 1.369	1.369
V026	finite-element strength reduction, Duncan, Wright & Brandon (2014)	1.19	non-associated	3026: 1.188, 6274: 1.175, 12705: 1.177	1.177
V038b	finite-element strength reduction, piezometric line, Duncan, Wright & Brandon (2014)	1.07	non-associated	2978: 1.078, 5826: 1.067, 11338: 1.058	1.058
V039	finite-element strength reduction, Duncan, Wright & Brandon (2014)	1.26	non-associated	5251: 1.234, 10872: 1.242	1.242
V055	finite elements, Griffiths & Lane (1999)	1.4	non-associated	2200: 1.348, 5361: 1.350	1.350
V056	finite elements (read from fig. 7), Griffiths & Lane (1999)	0.6	non-associated	2738: 0.528, 5546: 0.487, 11177: 0.476	0.476
V057a	finite elements, Griffiths & Lane (1999)	1.45	non-associated	2296: 1.445, 4671: 1.456, 10335: 1.457	1.457
V057b	finite elements, Griffiths & Lane (1999)	2.03	non-associated	2296: 2.090, 4623: 2.046, 10043: 2.057	2.057
V058	finite elements, Griffiths & Lane (1999)	1.9	non-associated	3566: 1.855, 6871: 1.836, 13609: 1.839	1.839
V059a	finite elements (read from fig. 2), Guo & Griffiths (2020)	1.28	non-associated	2046: 1.270, 4324: 1.283	1.283
V059b	finite elements, Guo & Griffiths (2020)	1.35	non-associated	2046: 1.348, 4322: 1.352	1.352
V059c	finite elements (read from fig. 2), Guo & Griffiths (2020)	1.36	non-associated	2046: 1.387, 4397: 1.365, 10411: 1.372	1.372
V063	finite-element strength reduction, Huang & Jia (2009)	2.5	non-associated	3500: 2.411, 8298: 2.418	2.418
V068	limit analysis, lower and upper bound, Kim et al. (2002)	1.13–1.27	associated	2024: 1.156, 4135: 1.160	1.160
V077	plasticity theory, Prandtl (1920)	1.0	davis-b	4162: 1.000, 8099: 1.003	1.003